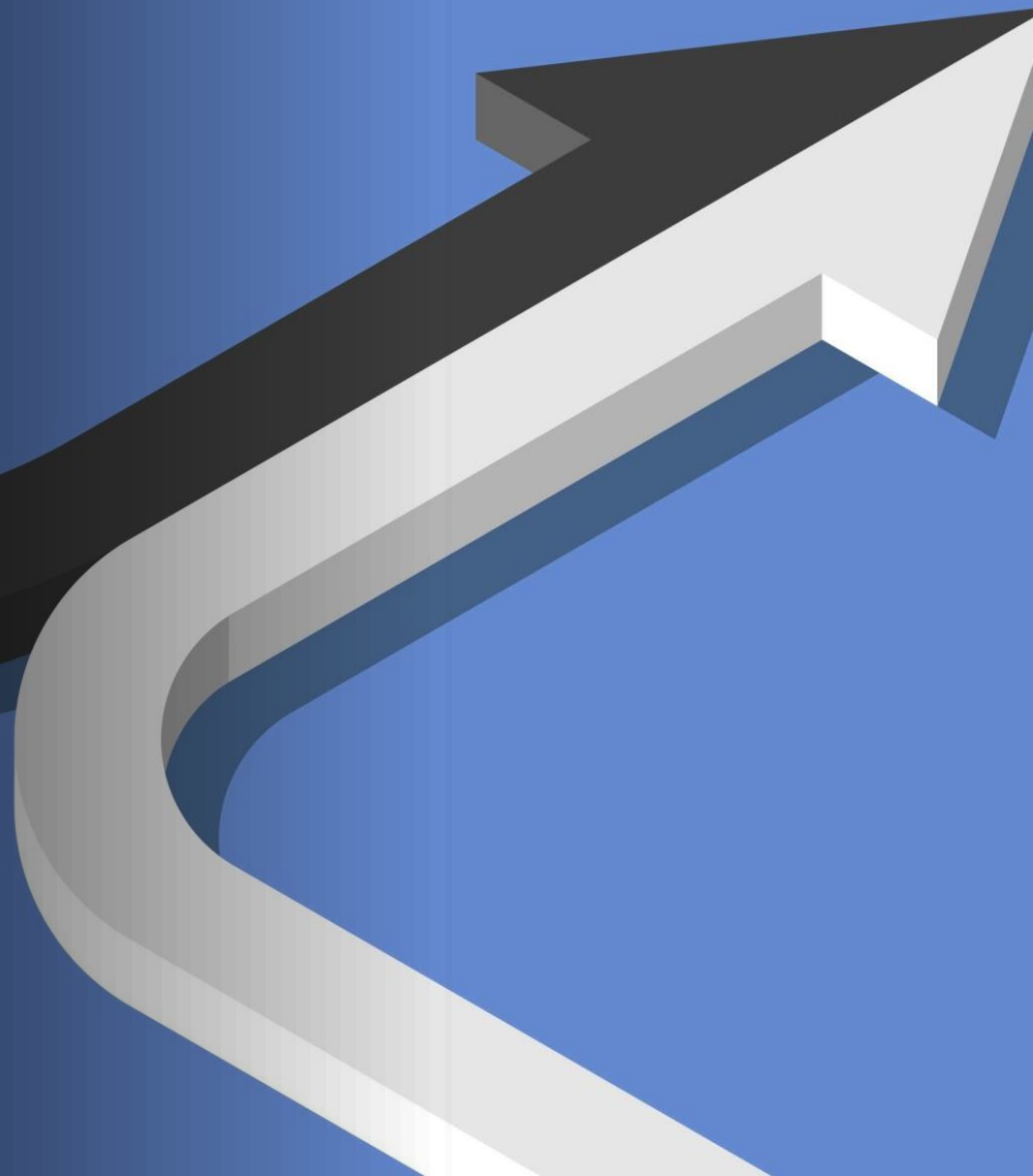
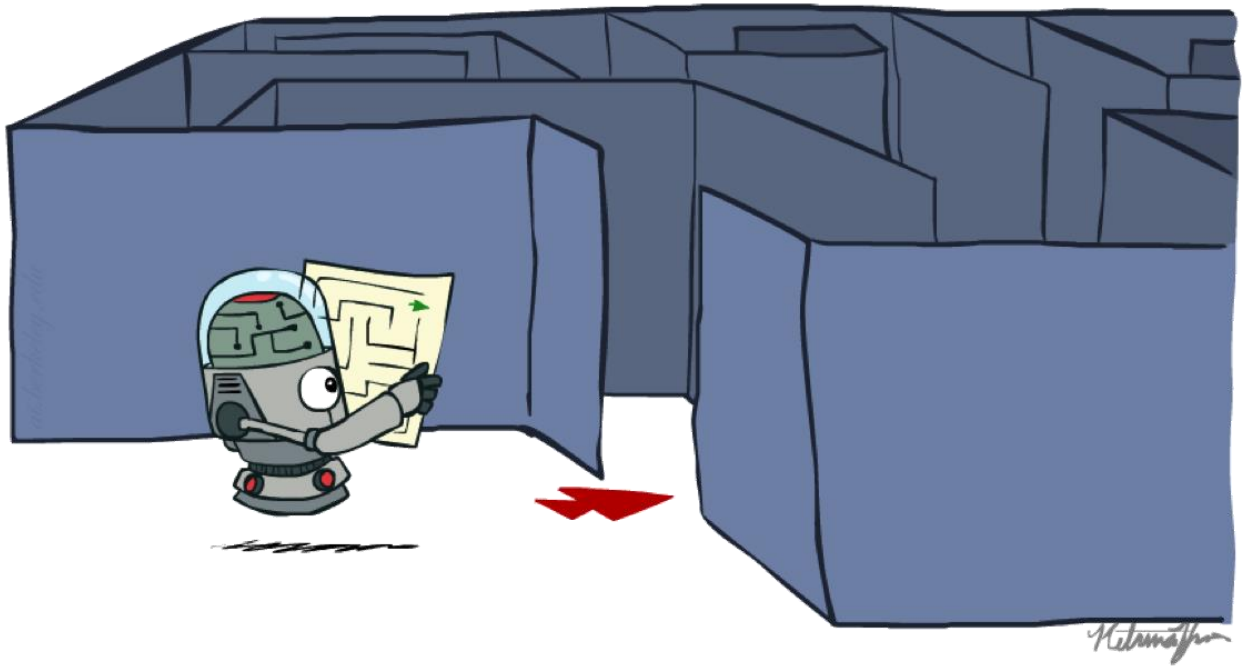
Estrutura de dados

Jean Nunes

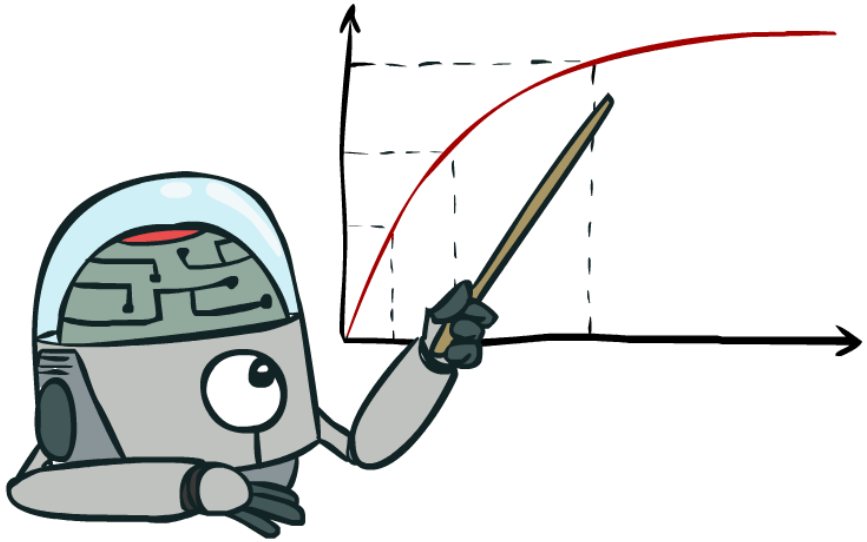


Estrutura de Dados

Busca em grafos (Search)



Maximizar a utilidade esperada



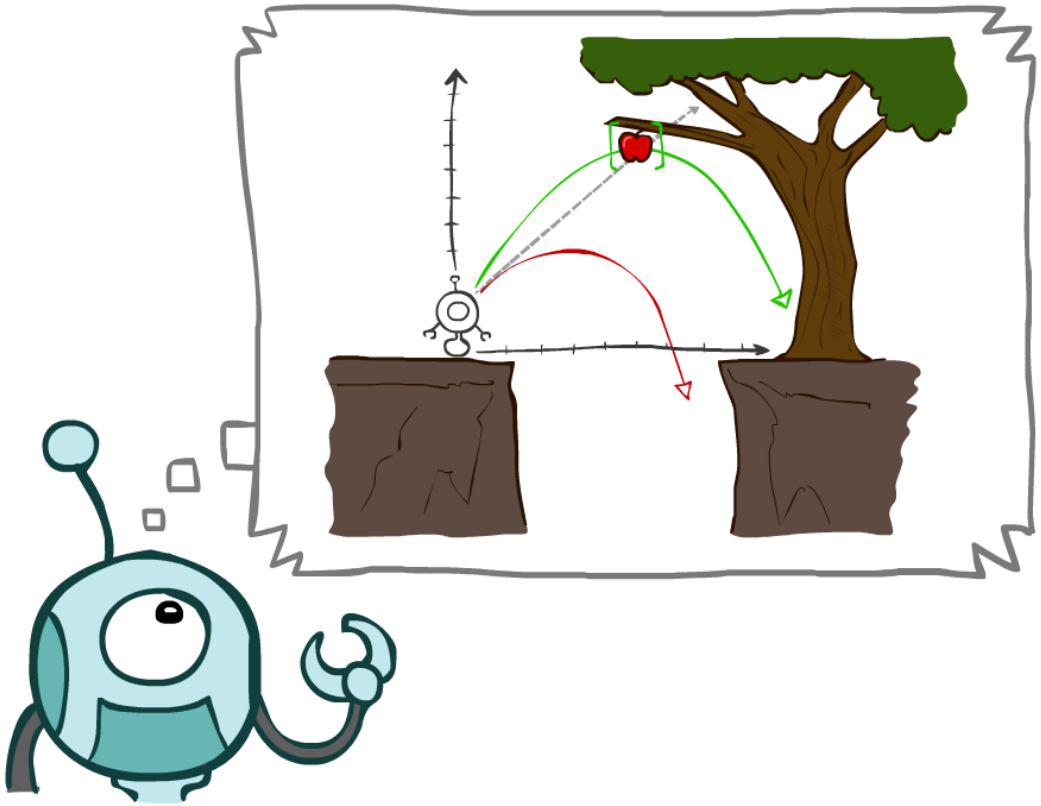
Busca

Agentes que planejam com antecedência

Problemas de busca

Busca não informada

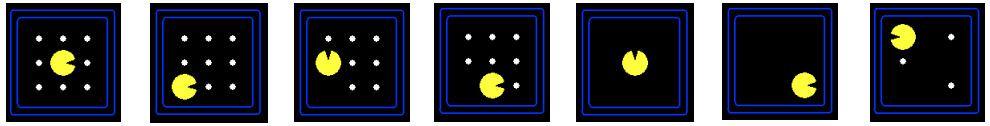
- Depth-First Search
- Breadth-First Search
- Uniform-Cost Search



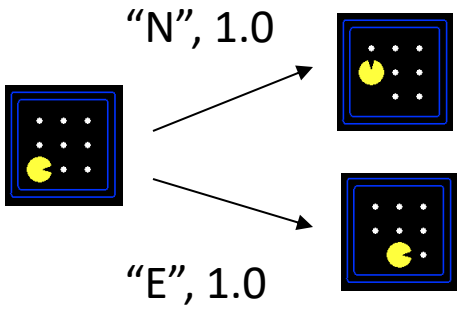
Problemas de Busca

Problemas de busca

- Uma problema de busca consiste de
  - Um espaço de estados



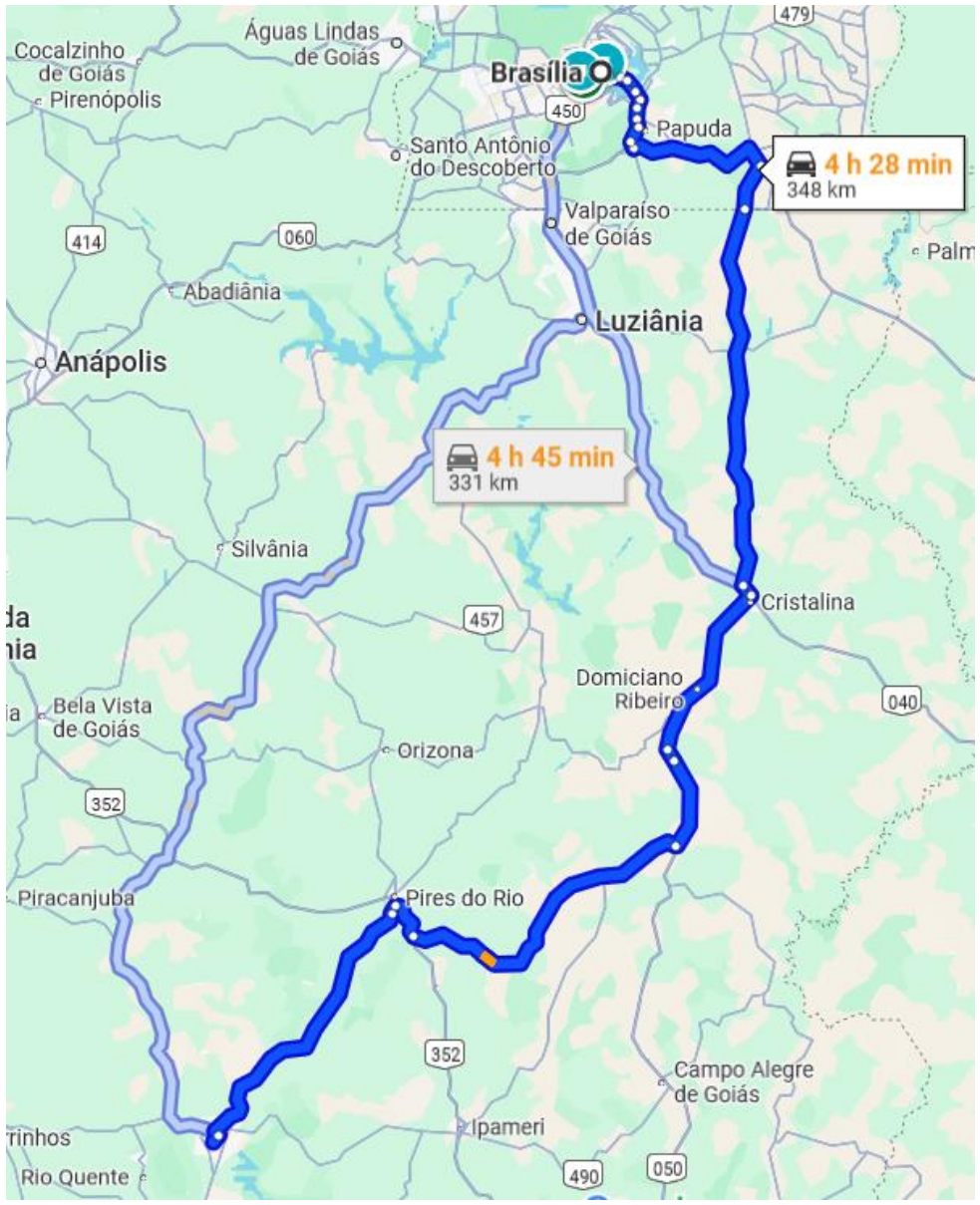
  - Uma função sucessora (com ações, custos)



  - Um estado inicial e um teste de objetivo
- Uma solução é uma sequência de ações (um plano) que transforma o estado inicial em um estado objetivo

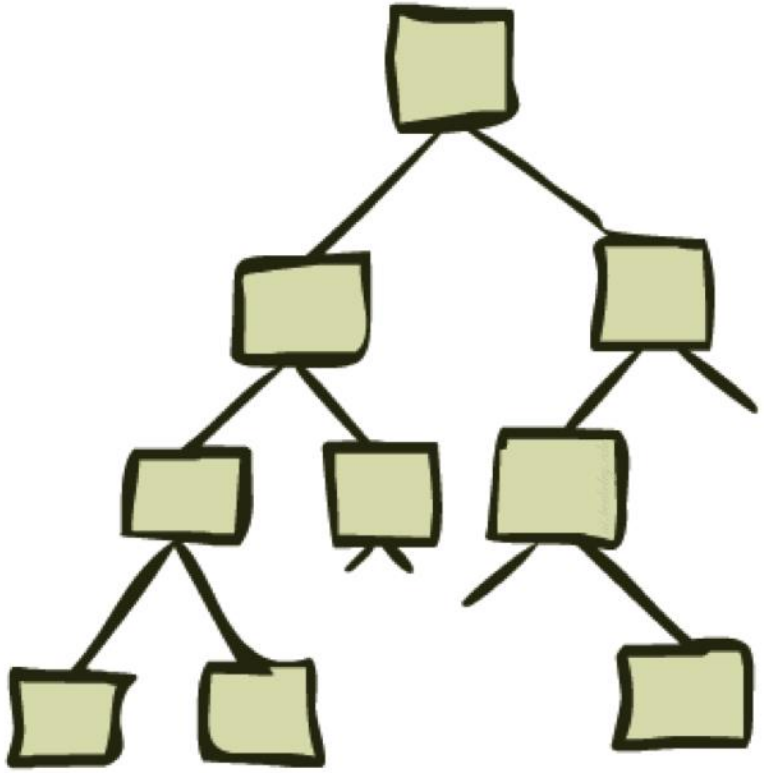
Problemas de busca são modelos

Exemplo



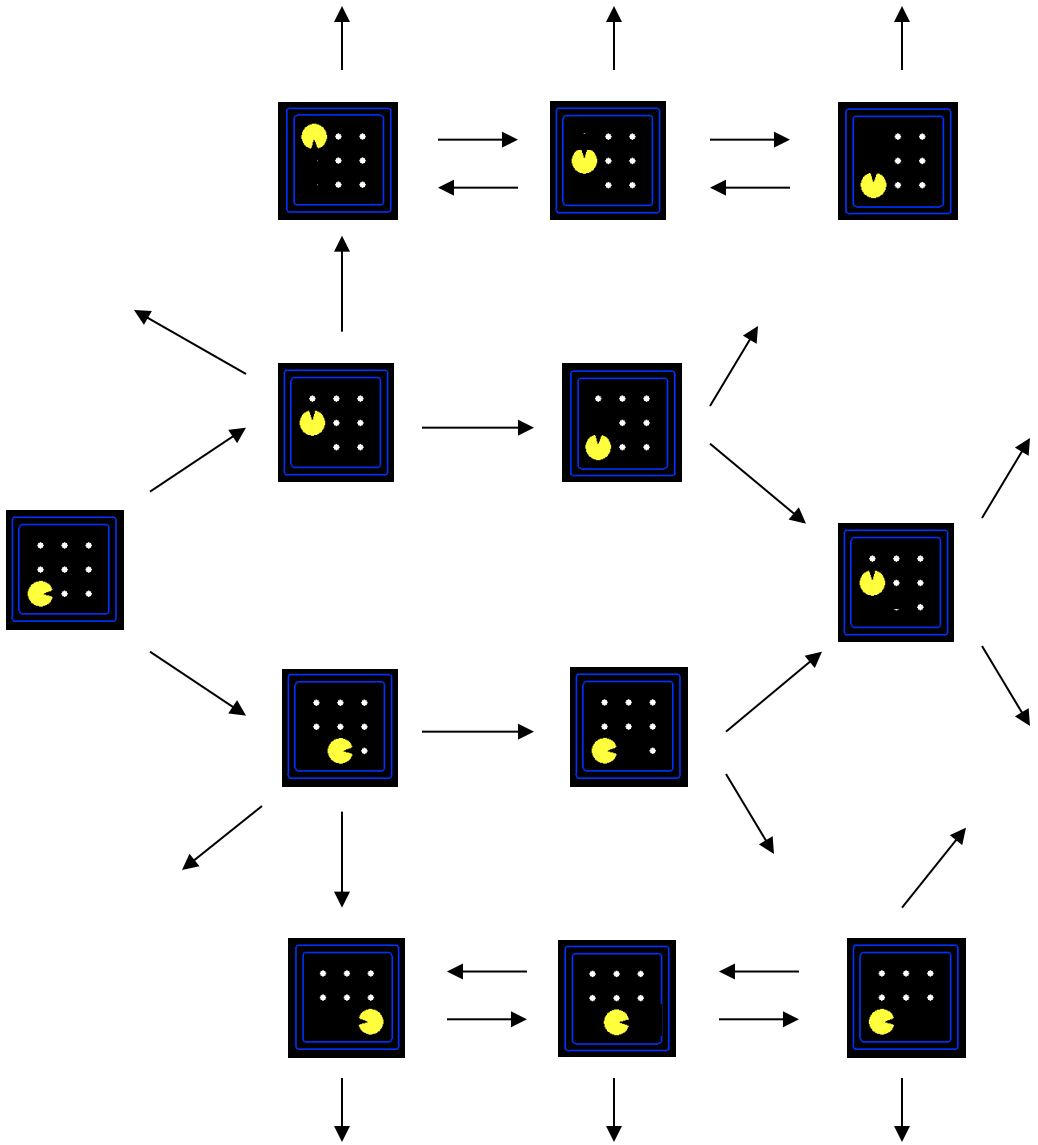
- Espaço de estados:
  - Cidades
- Função sucessora:
  - Estradas: vá para a cidade adjacente com custo = distância
- Estado inicial:
  - Brasília
- Teste de objetivo:
  - O estado == Caldas Novas?
- Solução?

Grafos de espaço de estados e árvores de busca

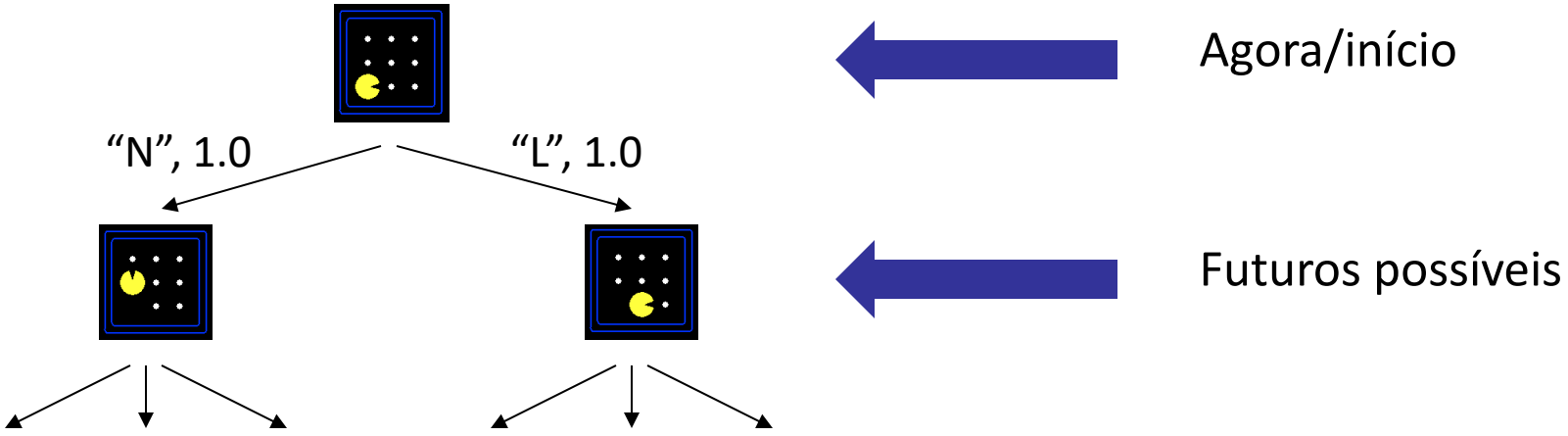


Grafos de espaço de estados

- Uma representação matemática de um problema de busca
  - Os vértices são configurações globais (abstratas)
  - As arestas representam sucessores (resultados da ação)
  - O teste de objetivo é um conjunto de nós de objetivo (talvez apenas um)
- Em um grafo de espaço de estado, cada estado ocorre apenas uma vez!
- Raramente podemos construir esse grafo completo na memória (é muito grande), mas é uma ideia útil

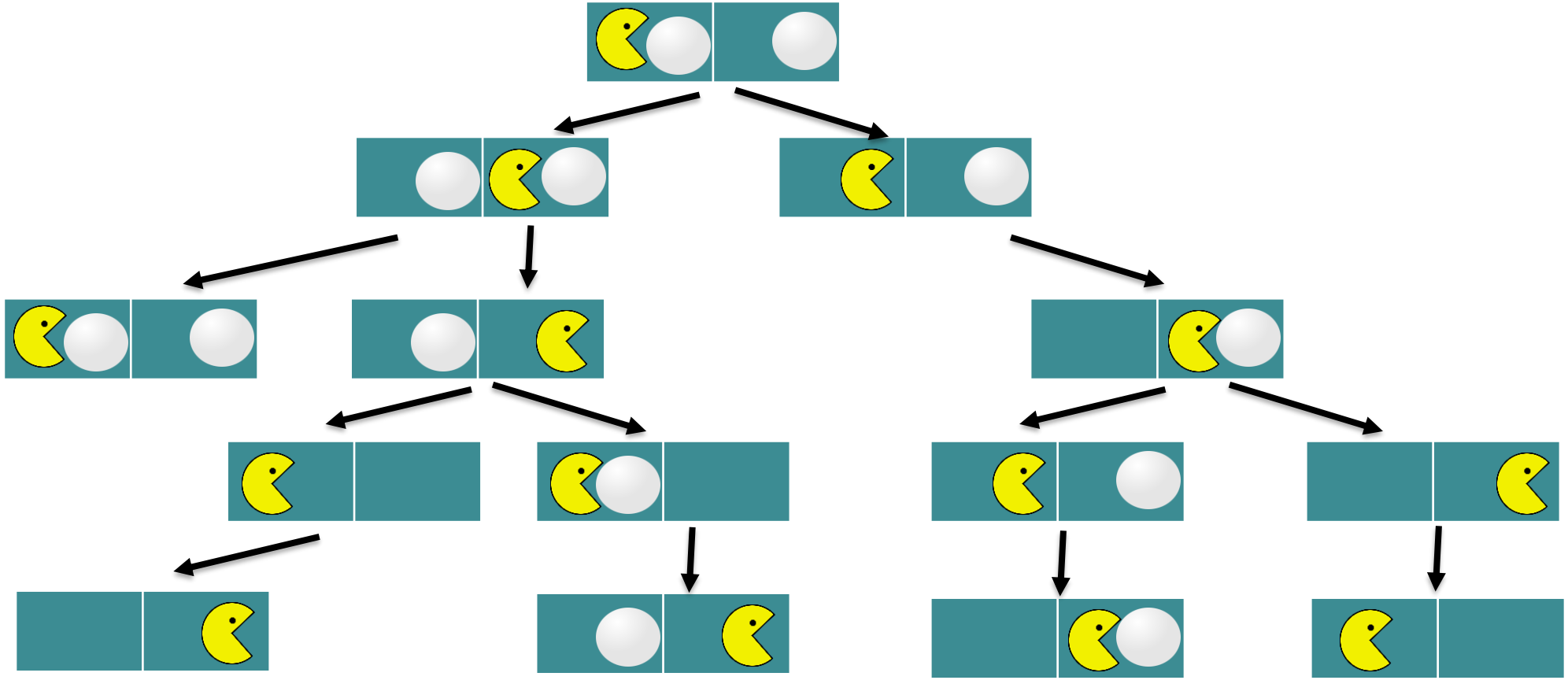


Árvores de busca



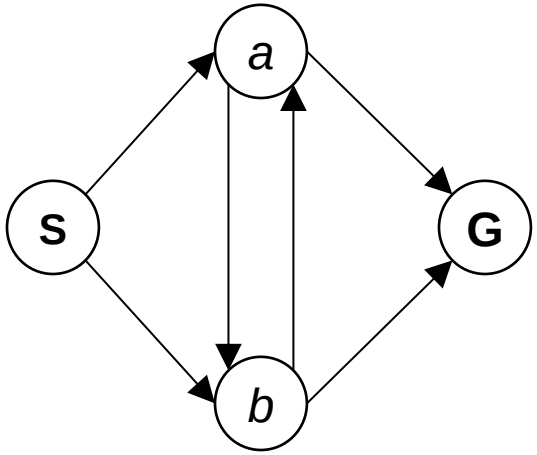
- Uma árvore de busca:
  - Uma árvore de planos "e se" e seus resultados
  - O estado inicial é o nó raiz
  - Os nós filhos correspondem aos sucessores
  - Os nós mostram estados, mas correspondem a PLANOS que alcançam esses estados
  - Para a maioria dos problemas, nunca podemos realmente construir a árvore inteira

Árvores de busca



Quiz: Grafos de espaço de estado x árvores de busca

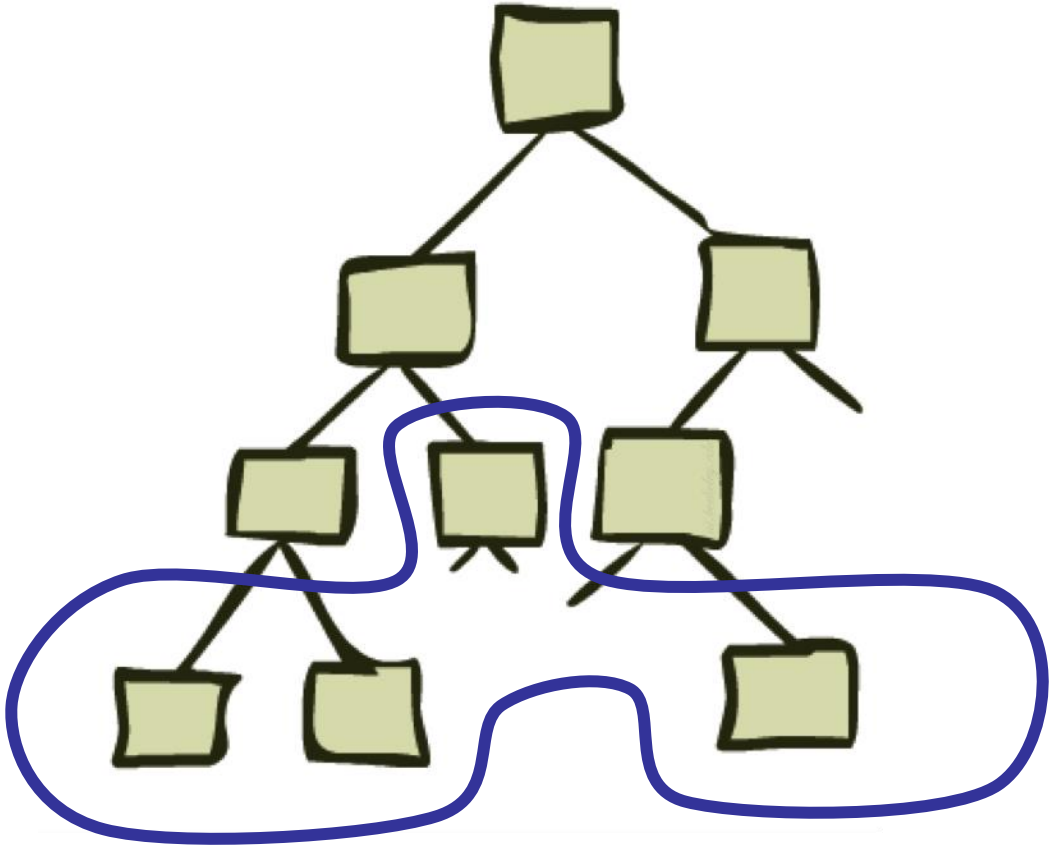
Considere este grafo de 4 estados:



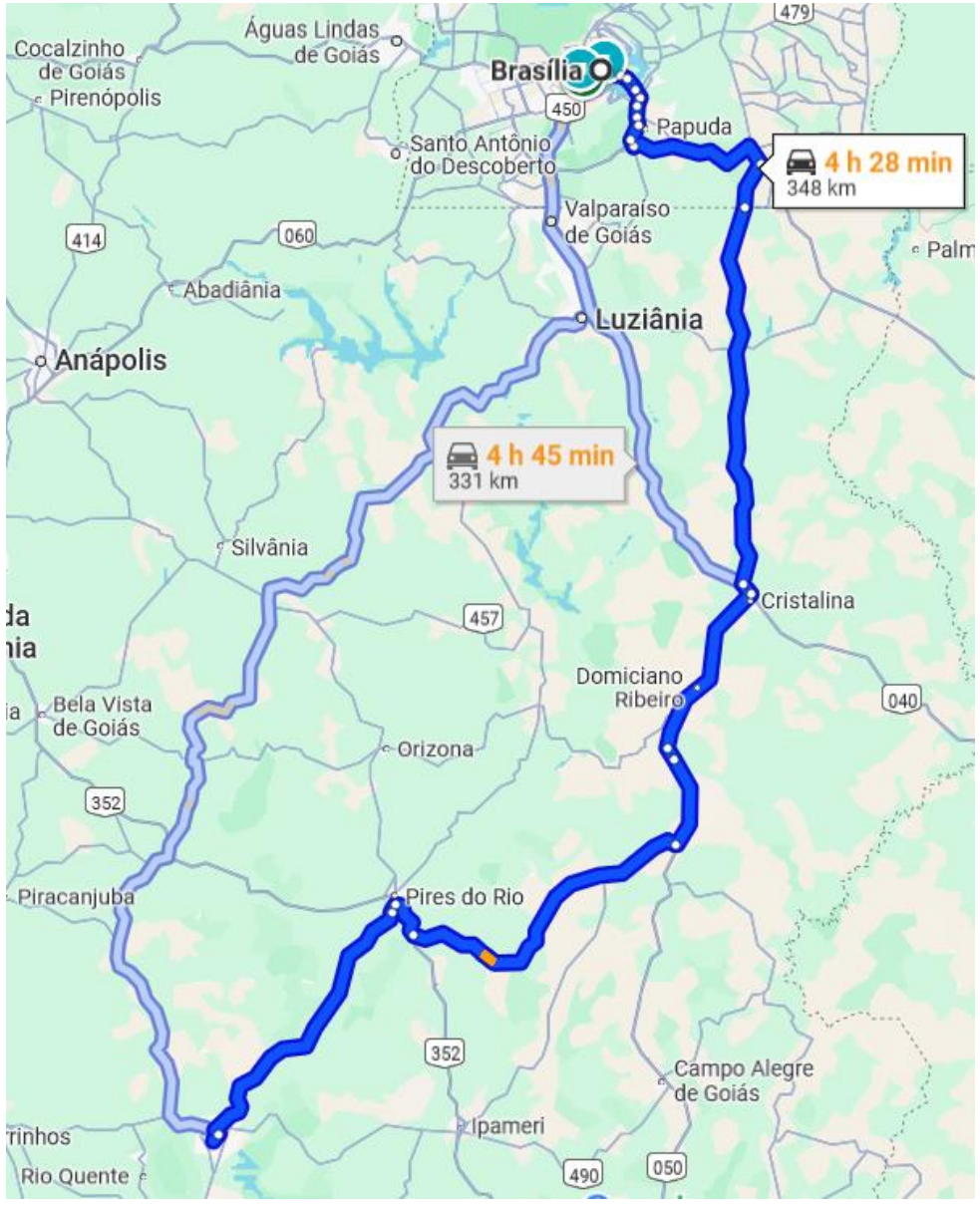
Qual é o tamanho da sua árvore de busca (de S)?

Importante: Muita estrutura repetida na árvore de busca!

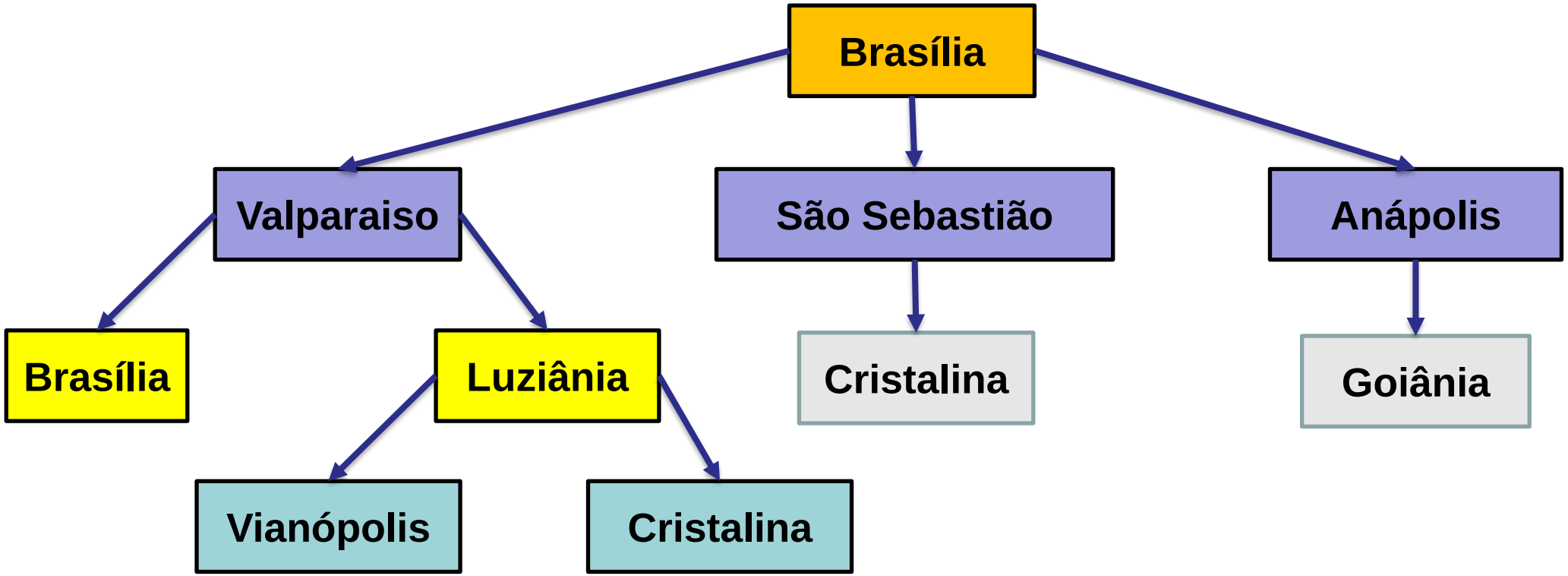
Árvore de busca



Exemplo

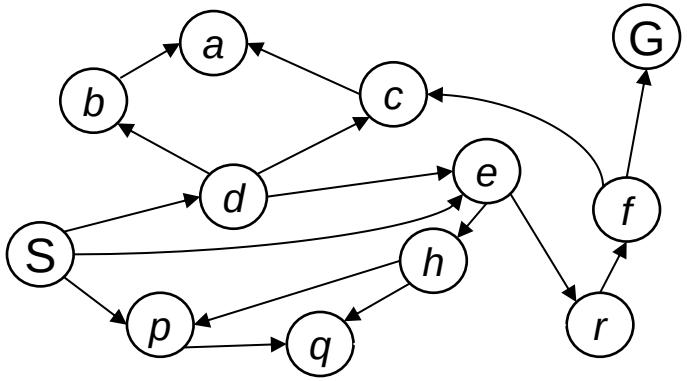

Águas Lindas de Goiás
Cocalzinho de Goiás
Pirenópolis
Brasília
479
450
Papuda
Santo Antônio do Descoberto
4 h 28 min
348 km
Valparaíso de Goiás
414
060
Abadiânia
Luziânia
Anápolis
4 h 45 min
331 km
Silvânia
Cristalina
457
Domiciano Ribeiro
040
Bela Vista de Goiás
Orizona
352
Piracanjuba
Pires do Rio
352
Campo Alegre de Goiás
Ipameri
Rio Quente
490
050

Buscando com uma árvore de busca



- Busca:
  - Expanda os planos em potencial (nós da árvore)
  - Manter uma **margem** de planos parciais em consideração
  - Tente expandir o menor número possível de nós da árvore

Exemplo: Árvore de busca



Estratégia

- Comece com uma **margem** que contenha o **estado inicial**.
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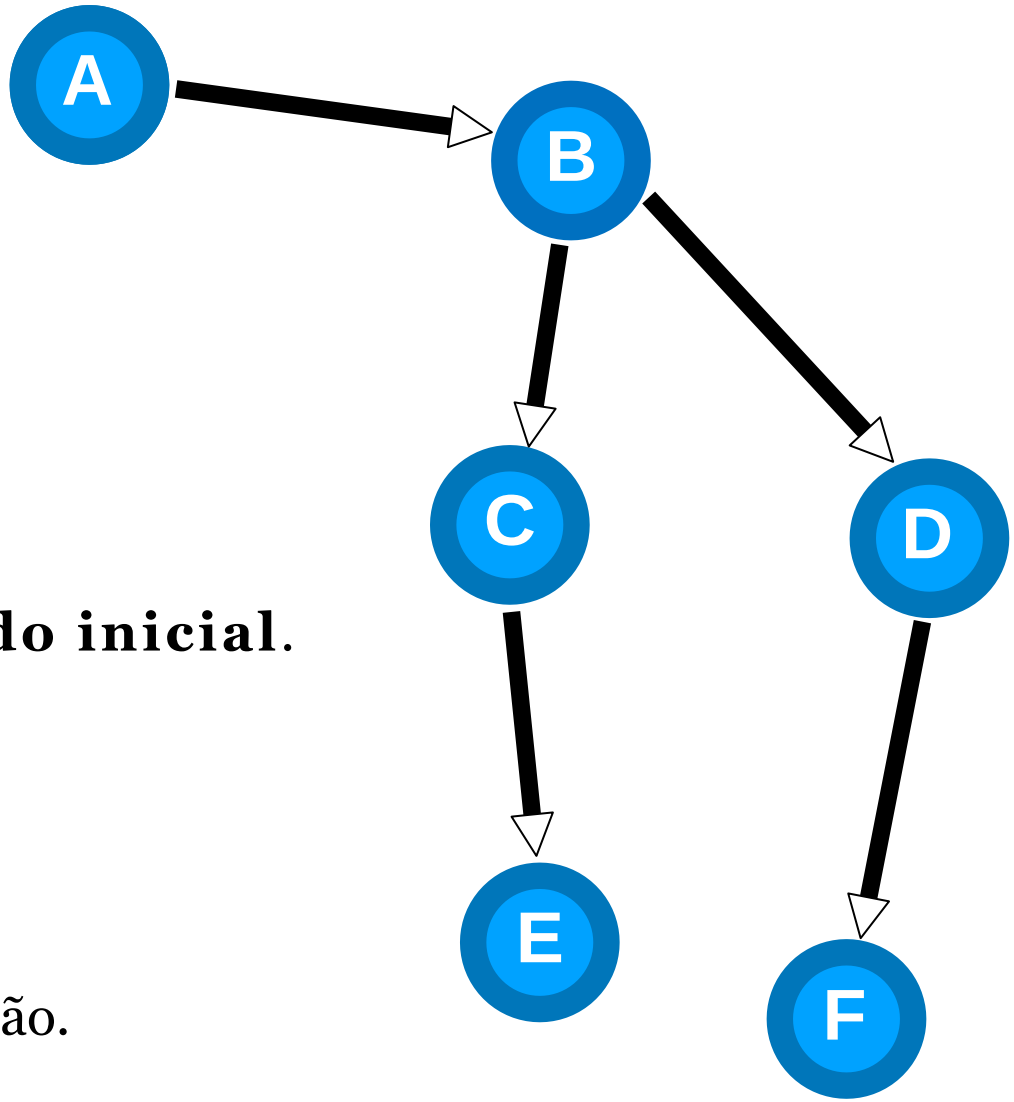
Encontre um caminho de A para E

Margem



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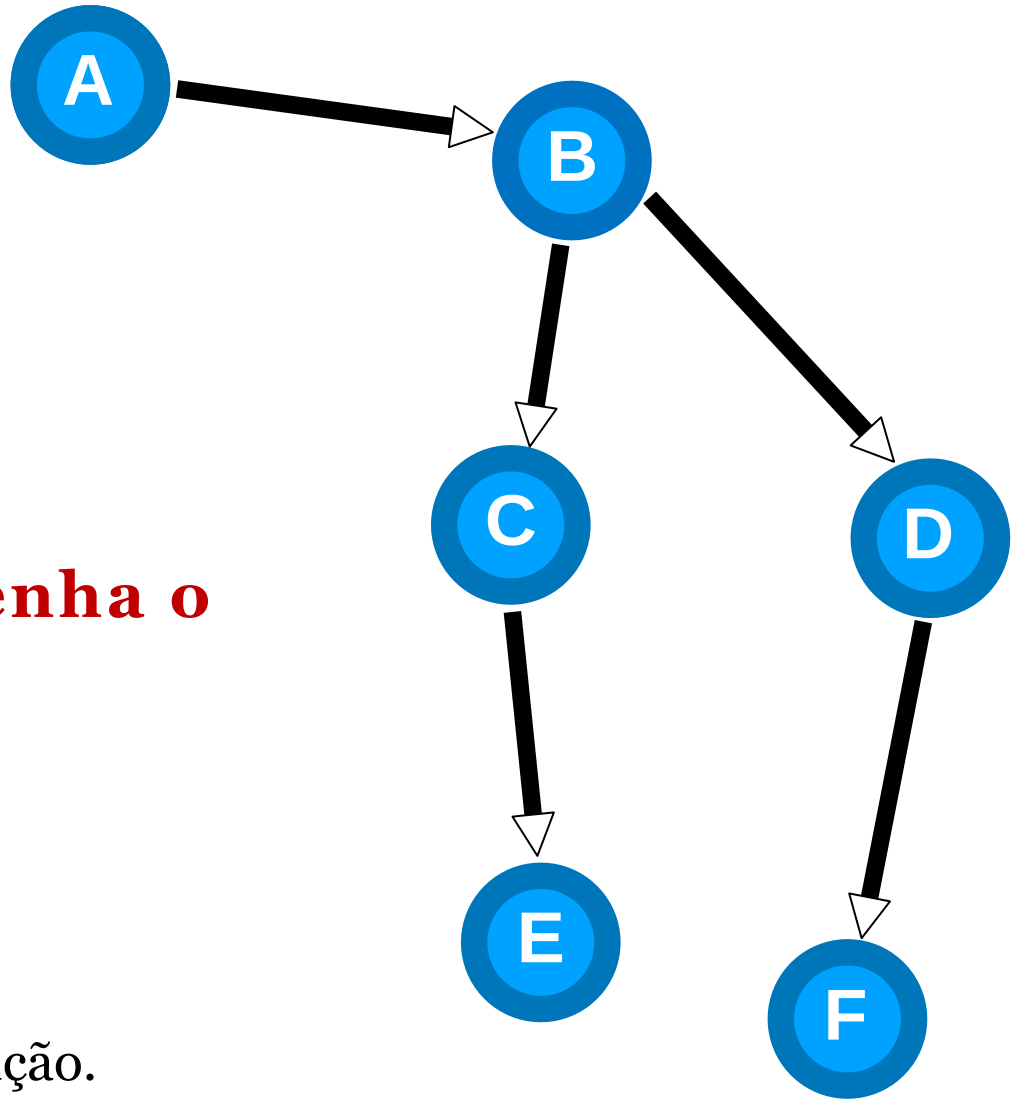

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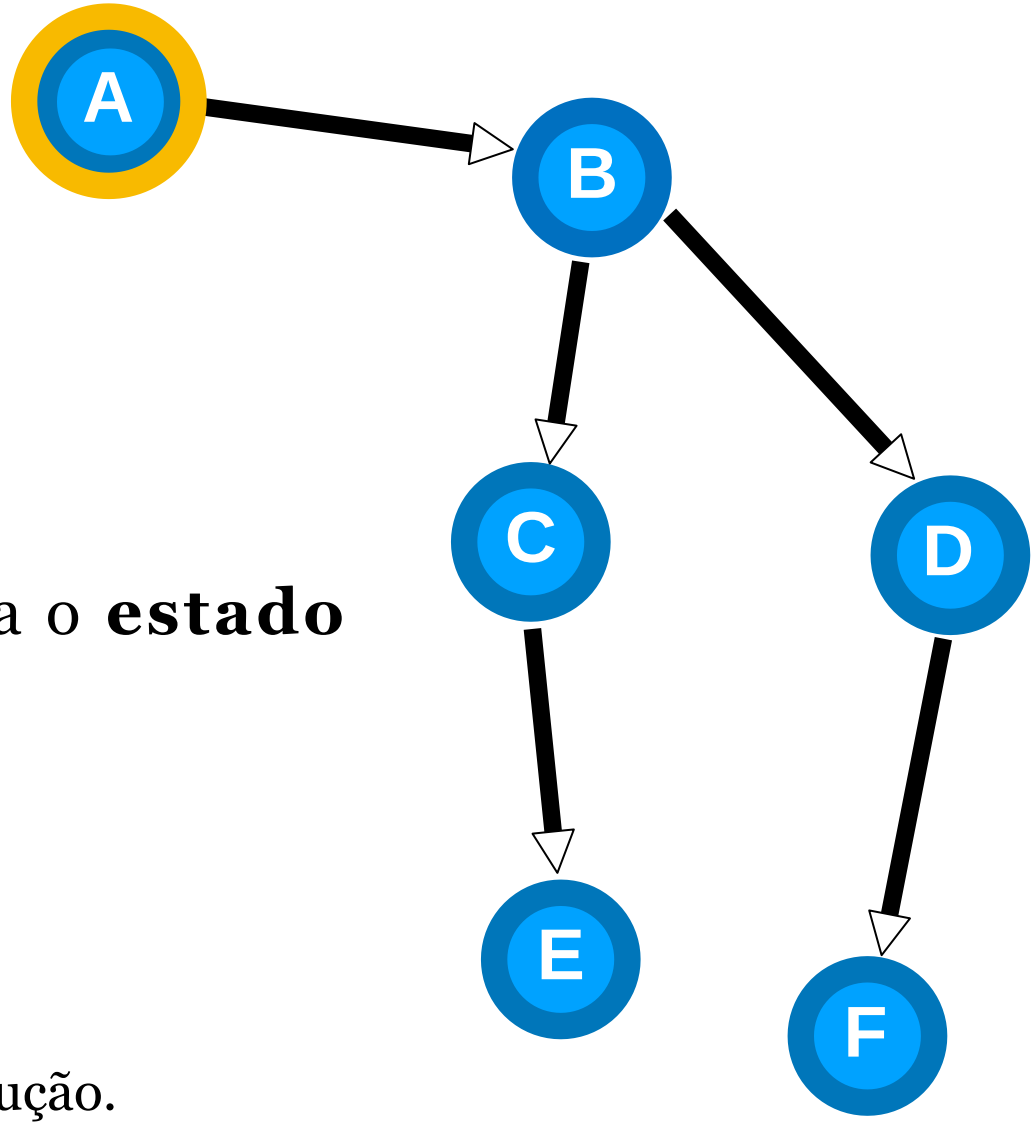

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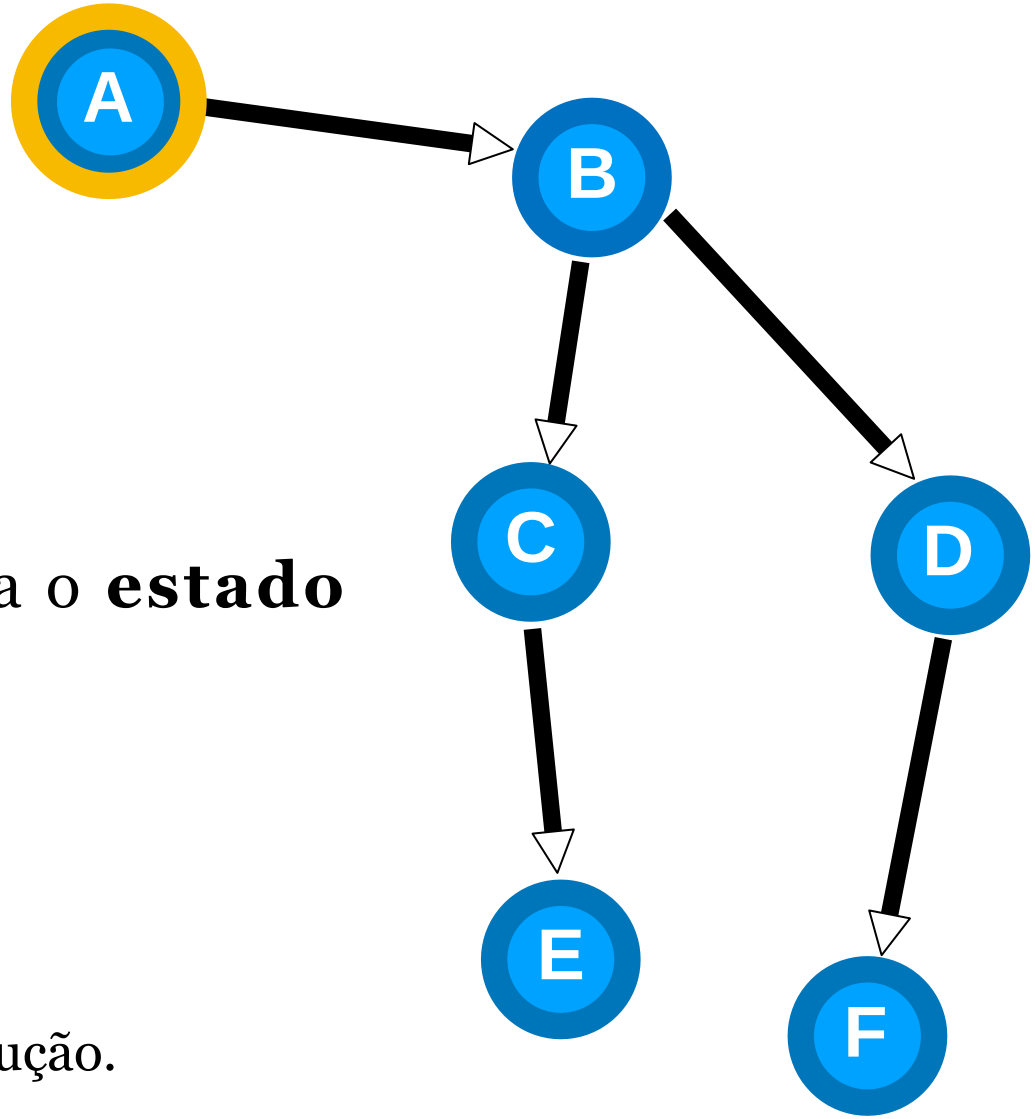

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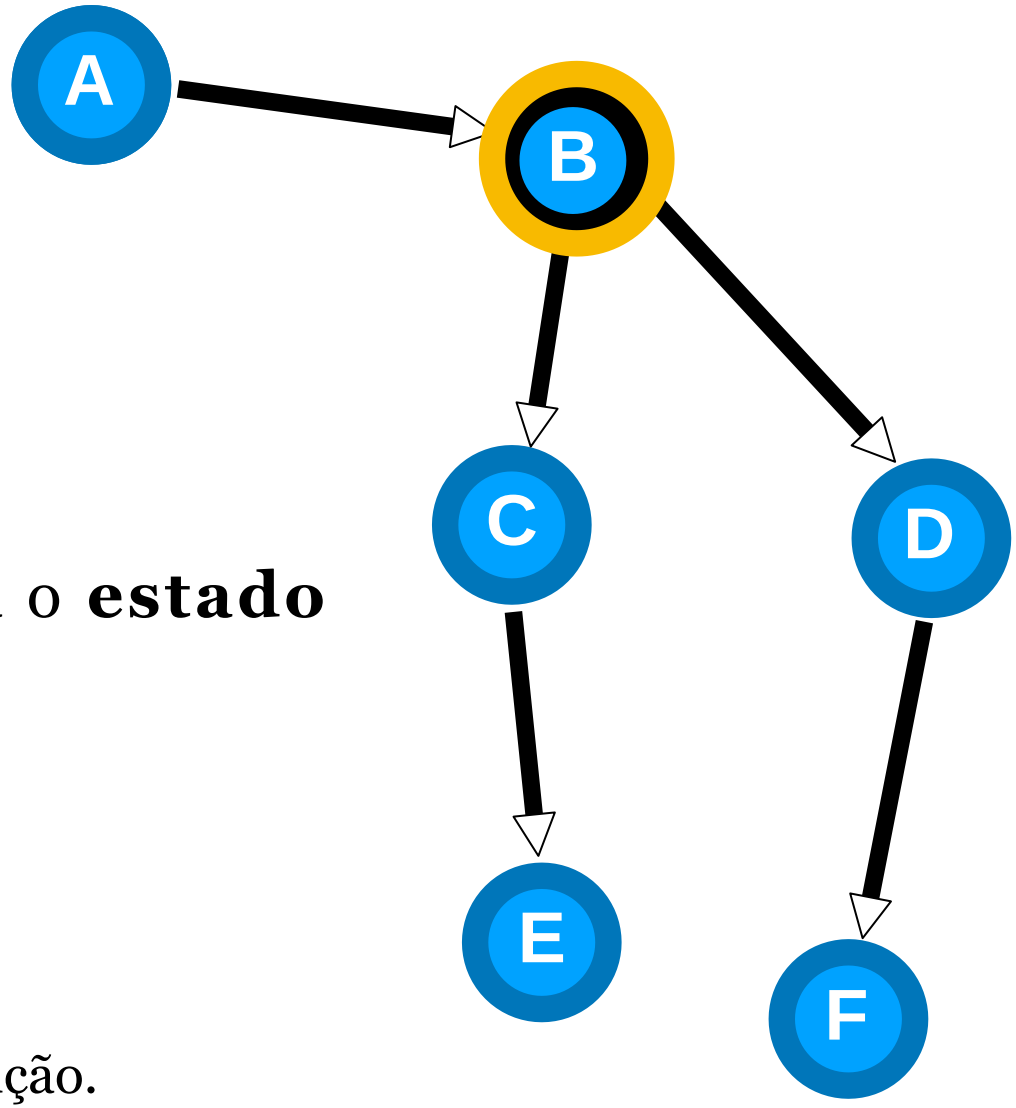

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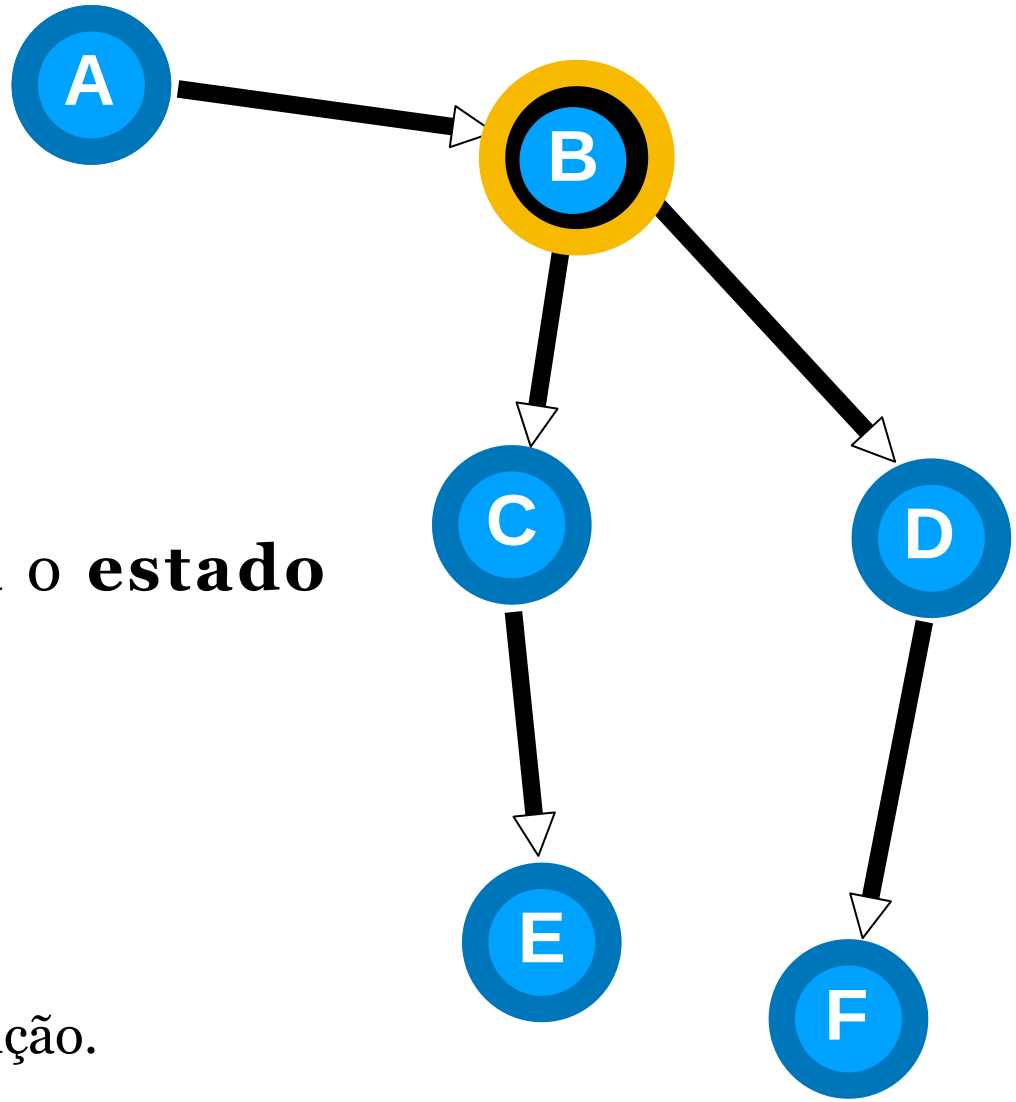

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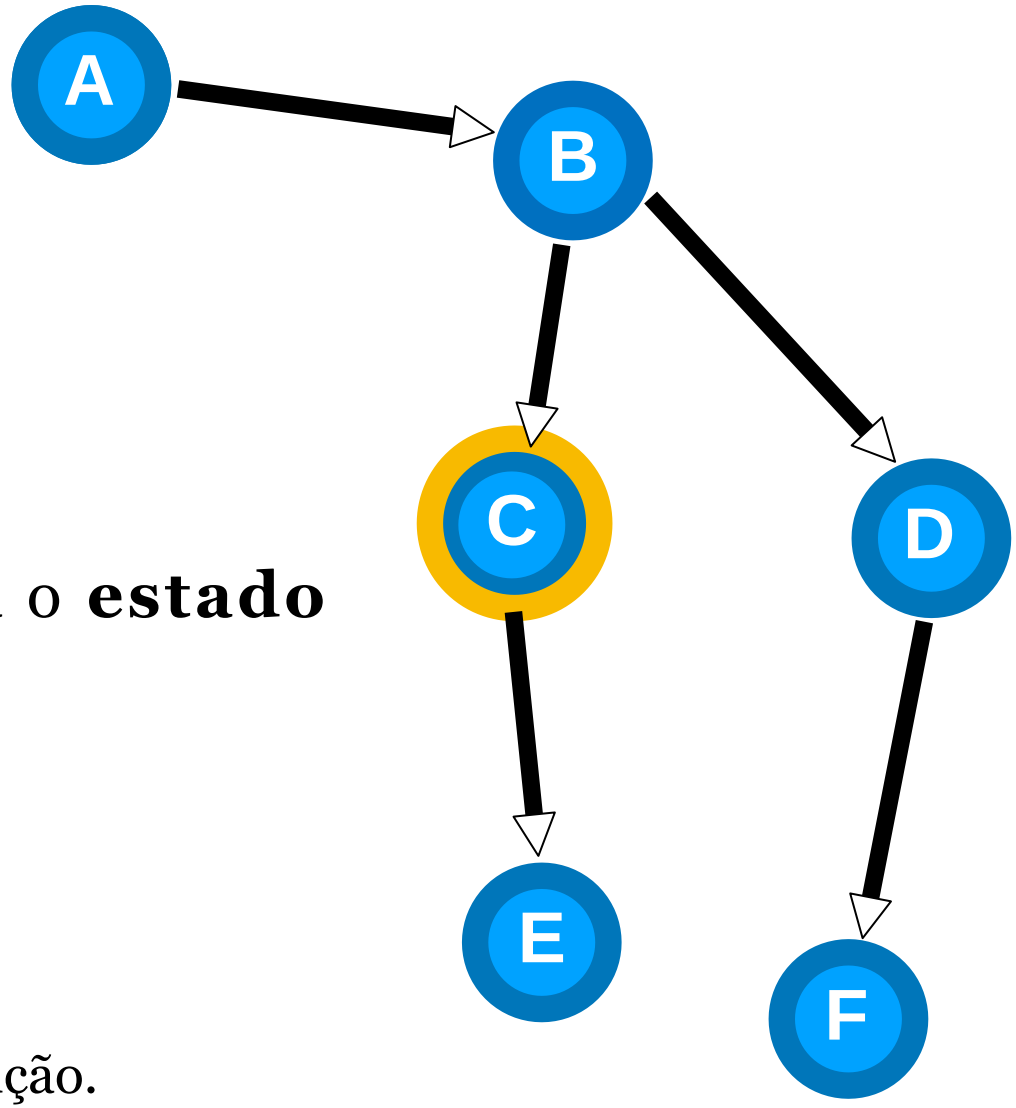

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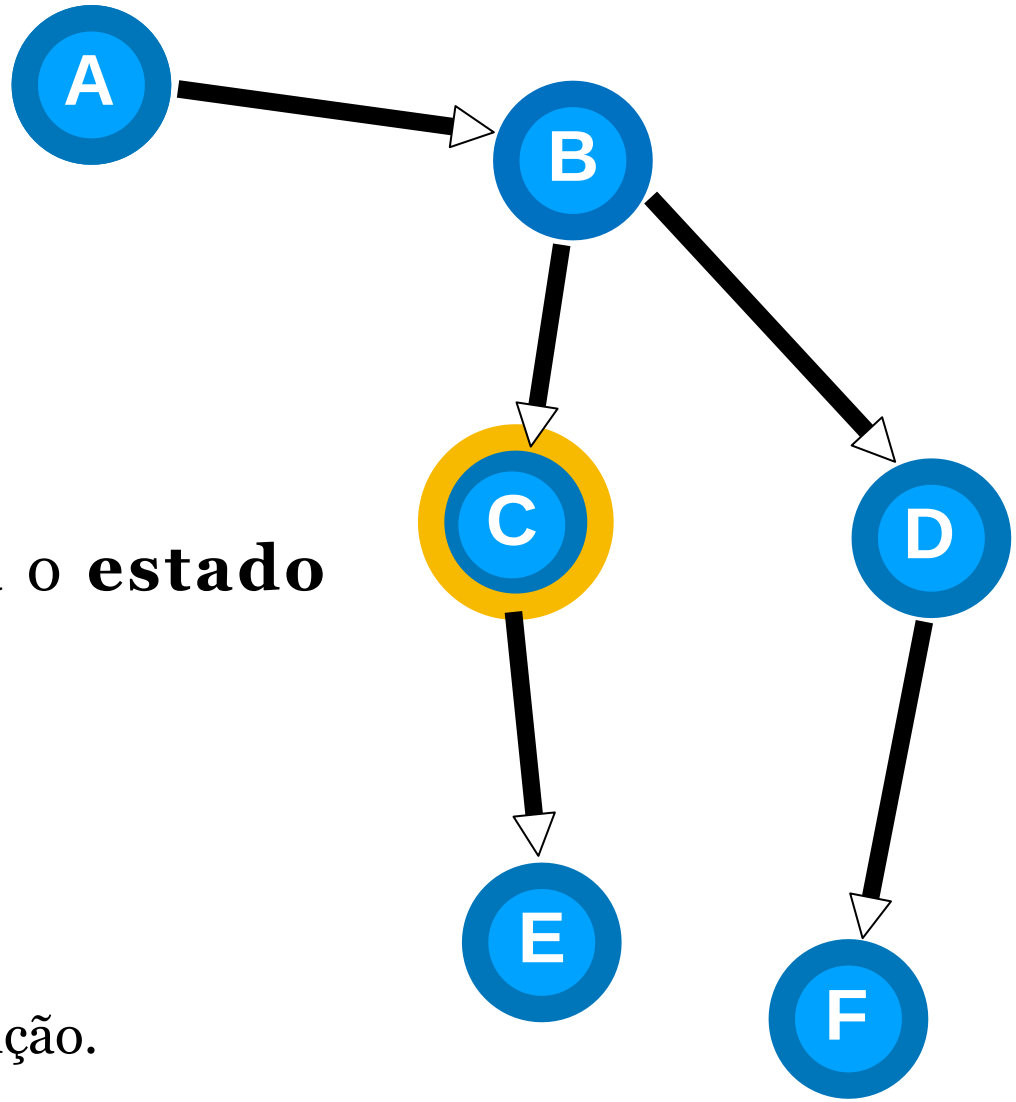

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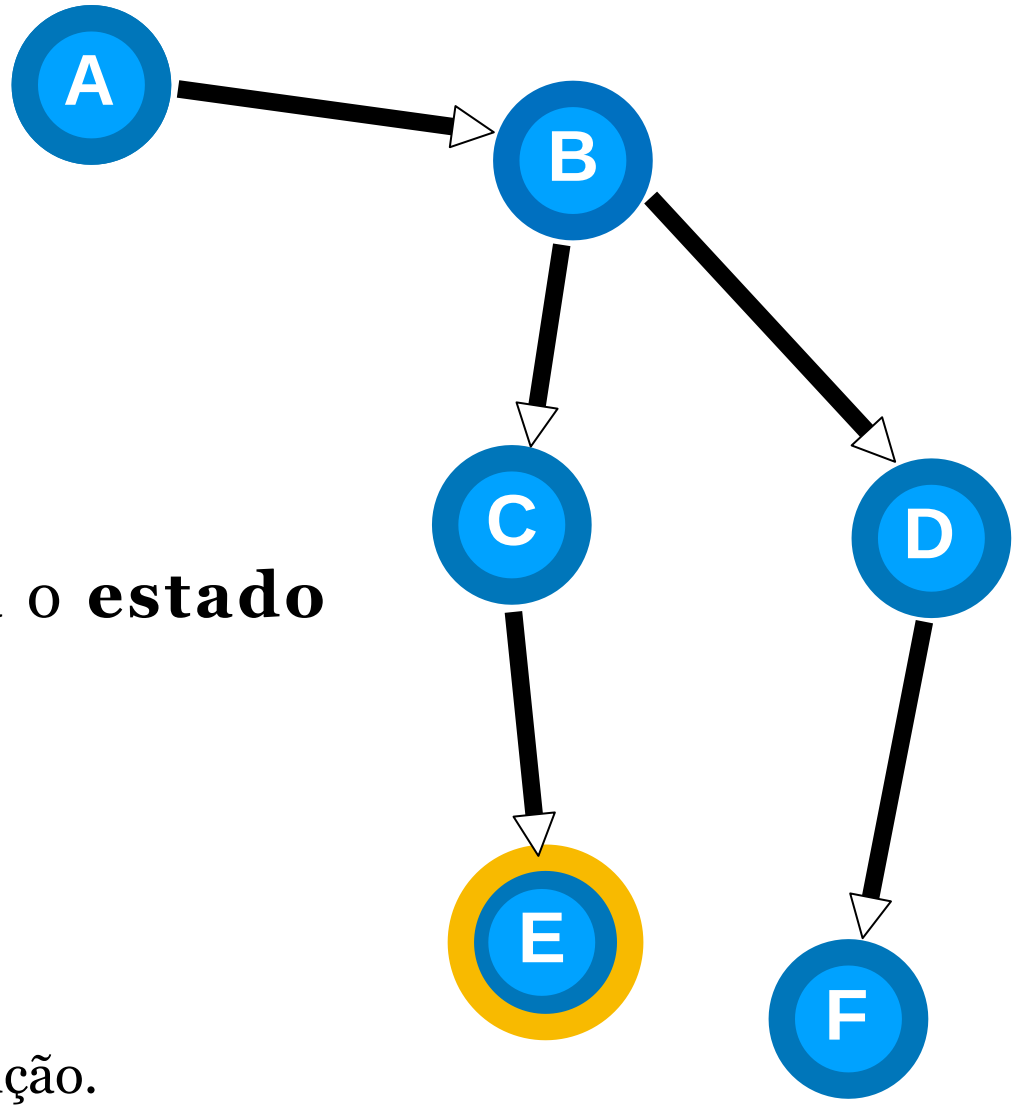

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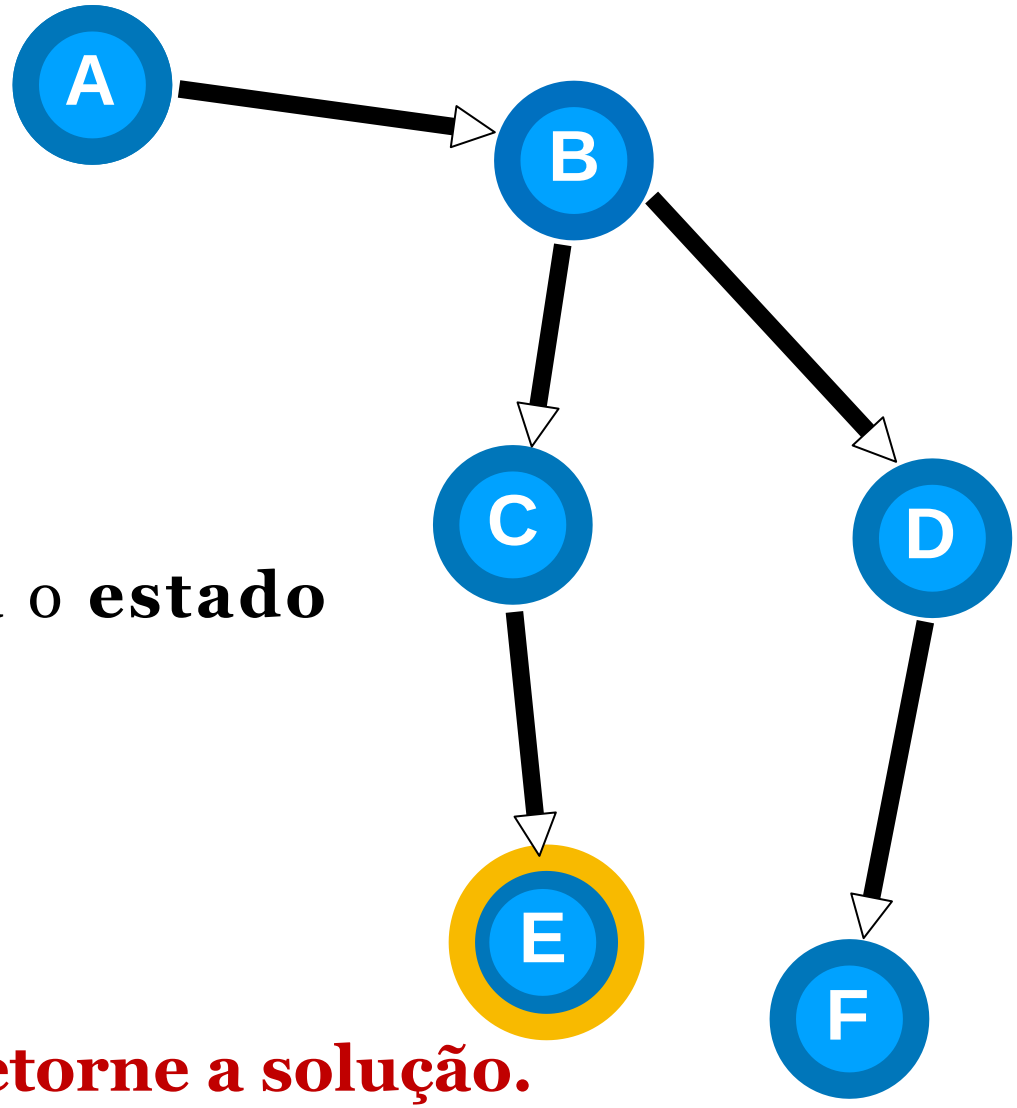

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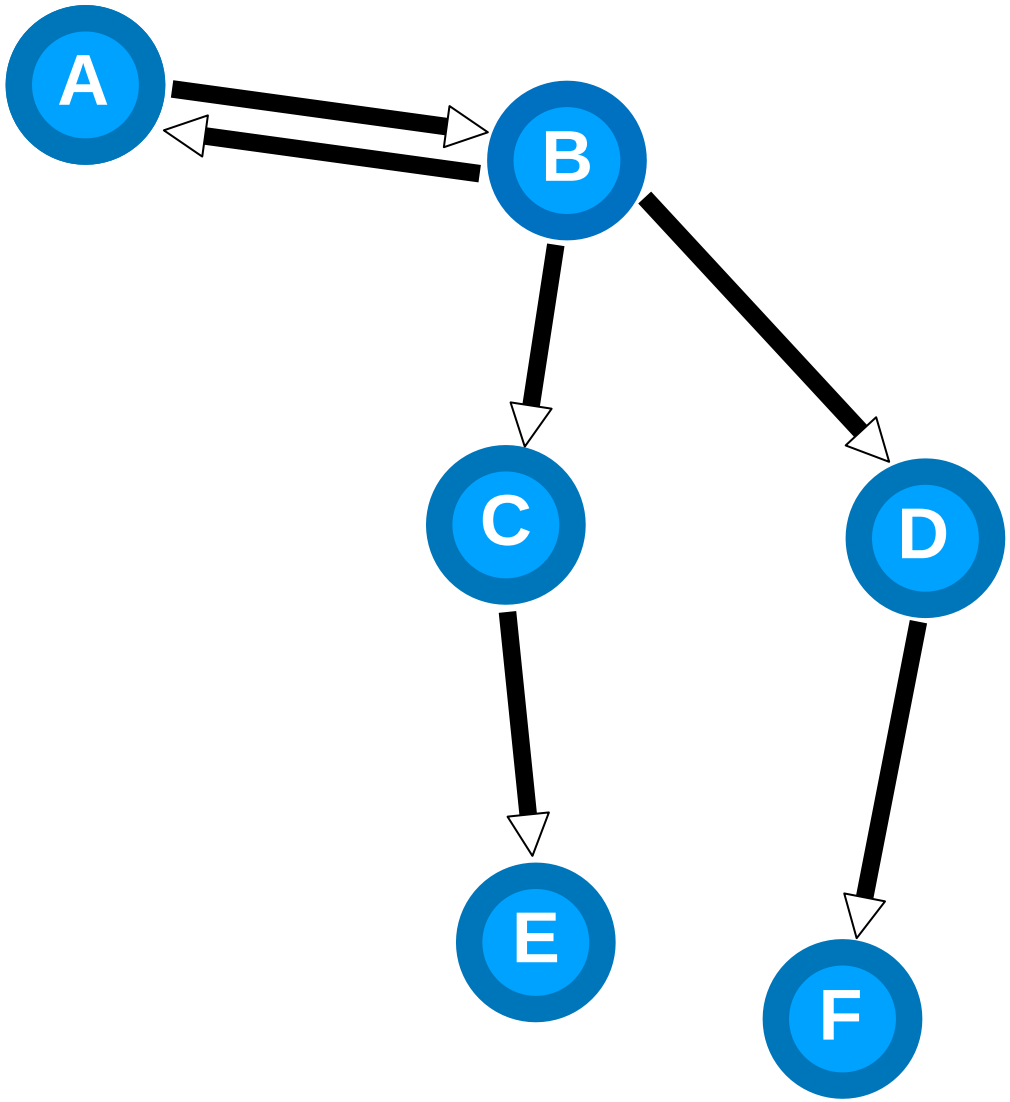
O que poderia dar errado?

Encontre um caminho de A para E

Margem



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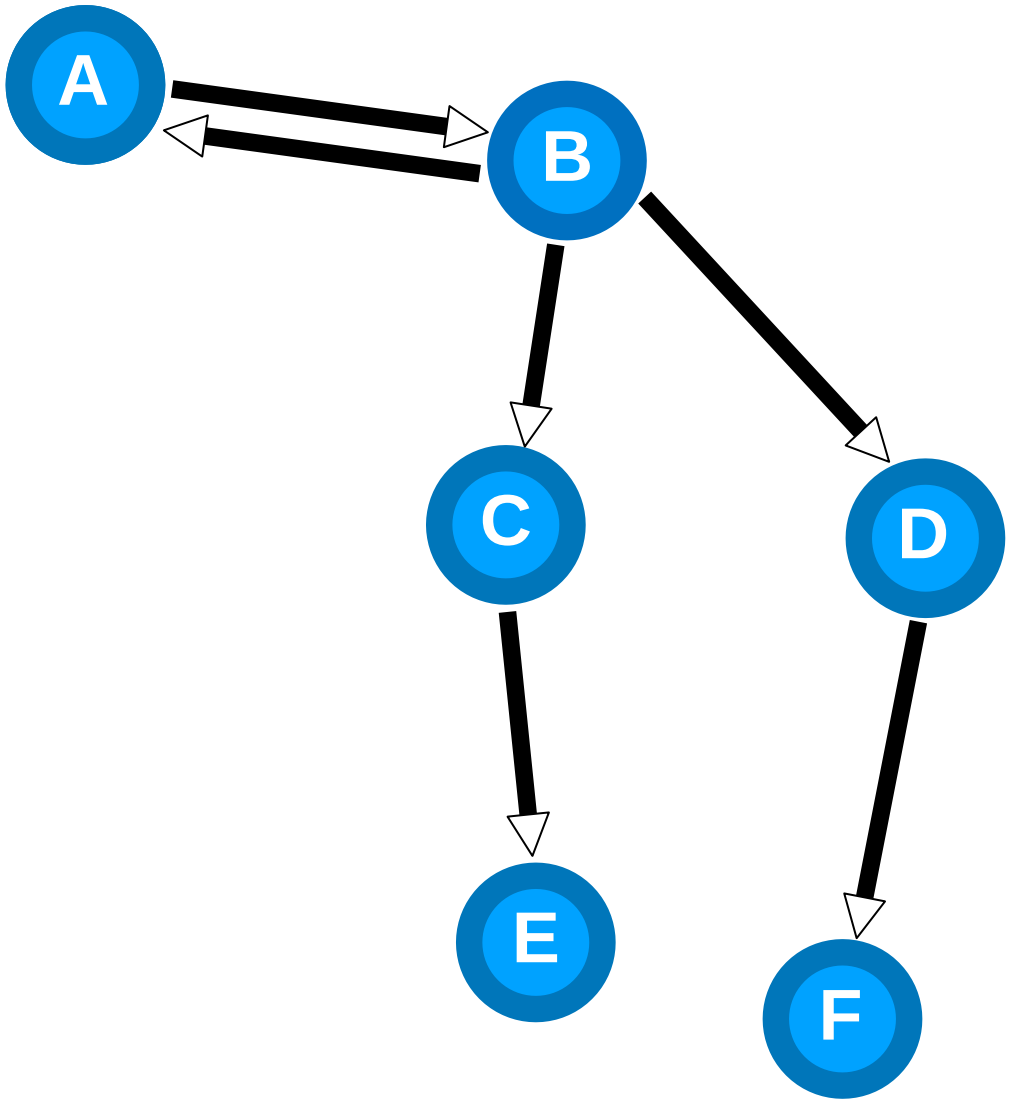

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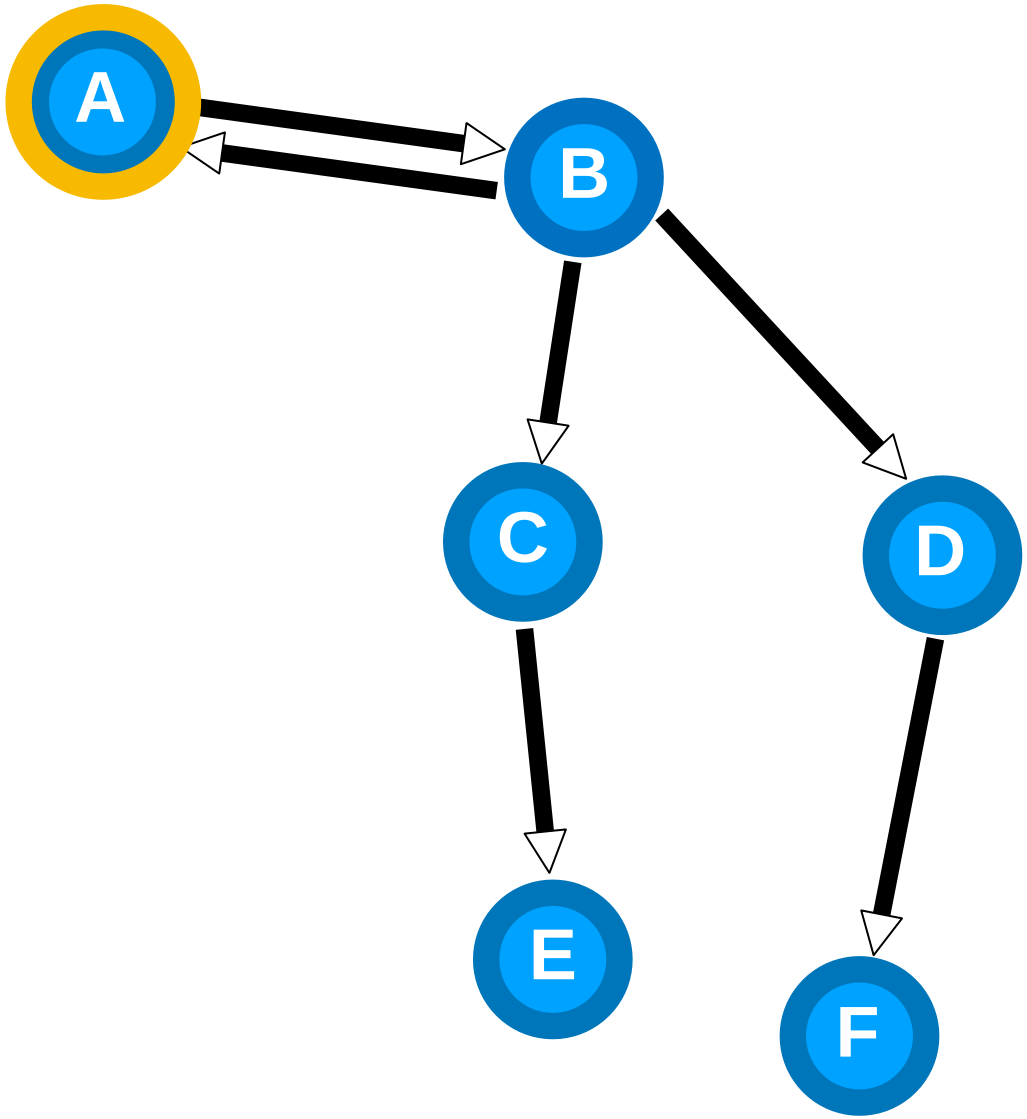

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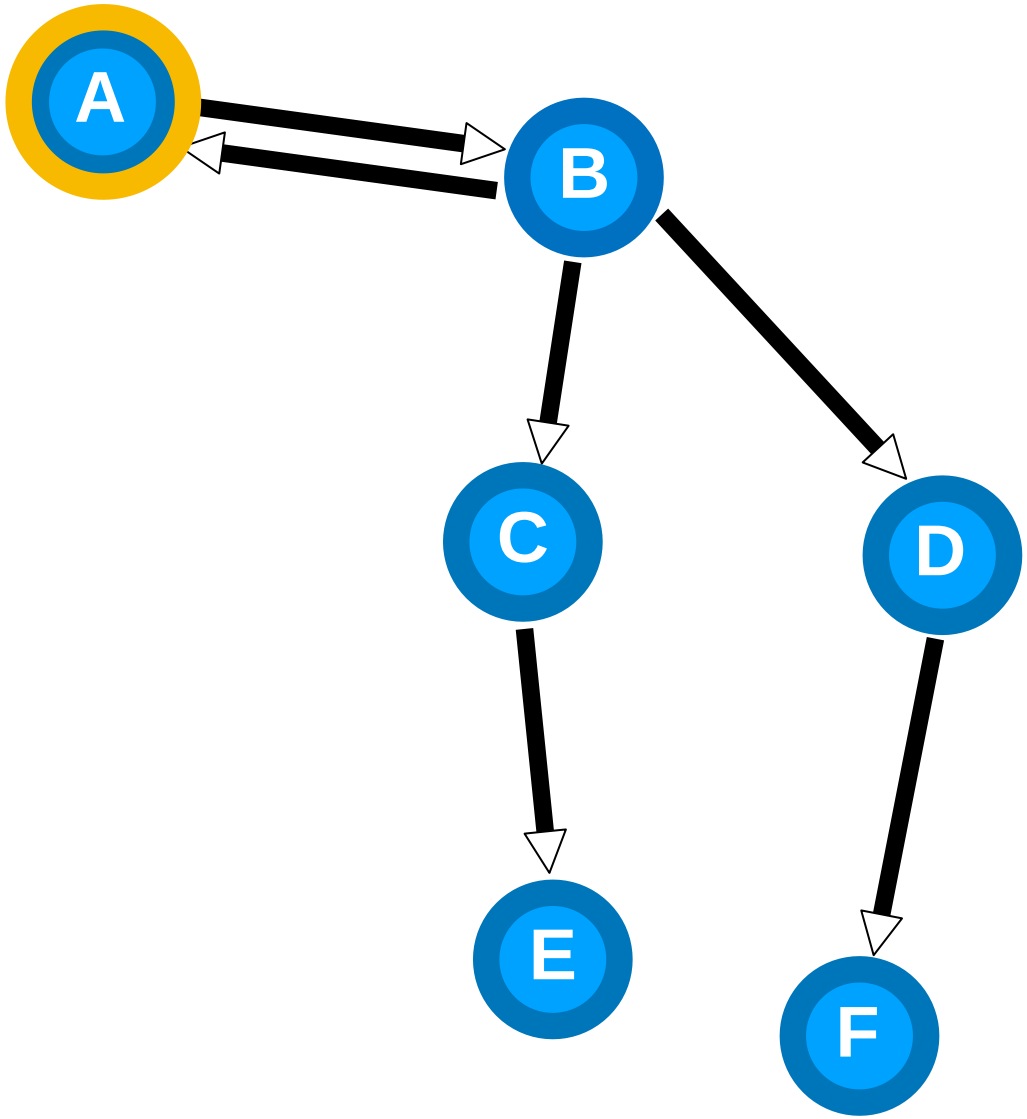

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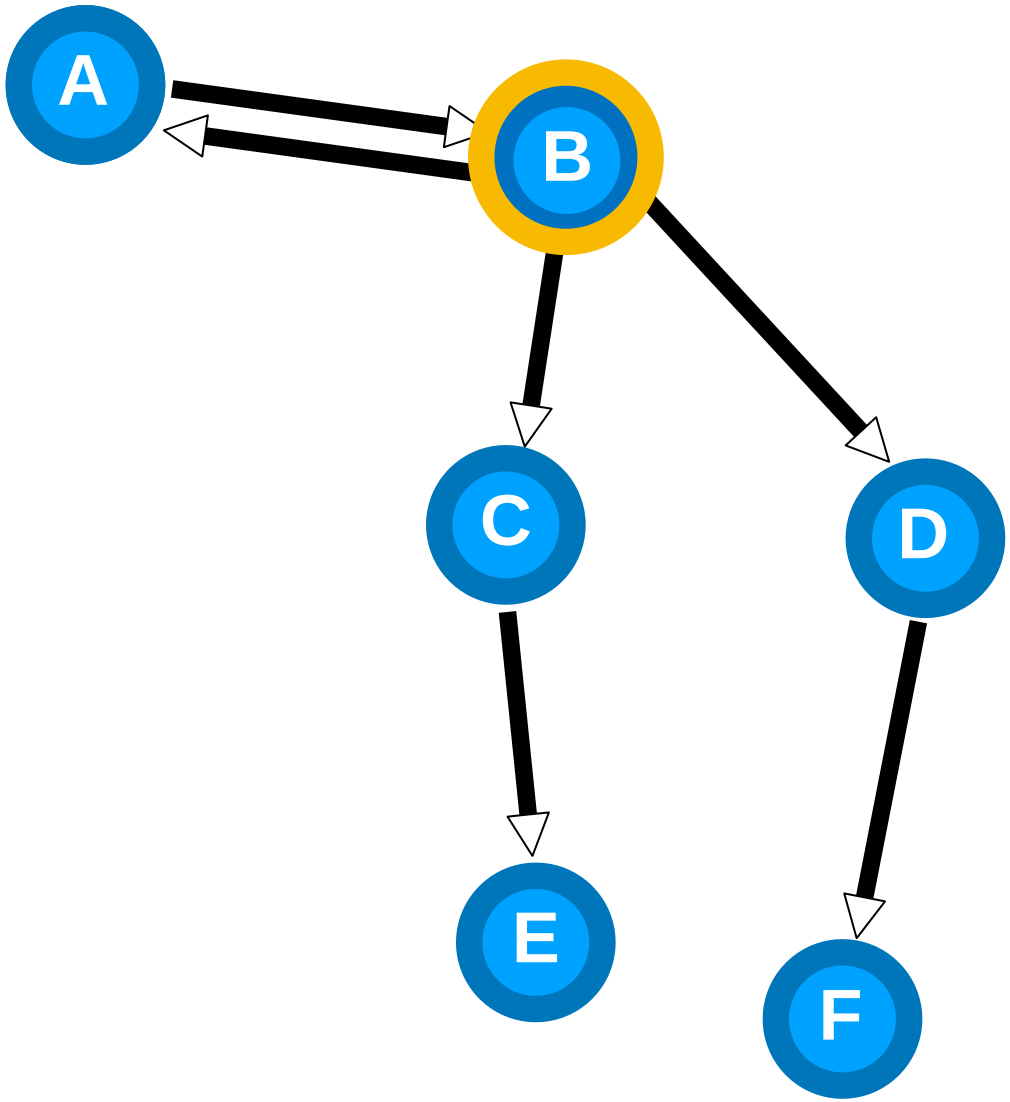

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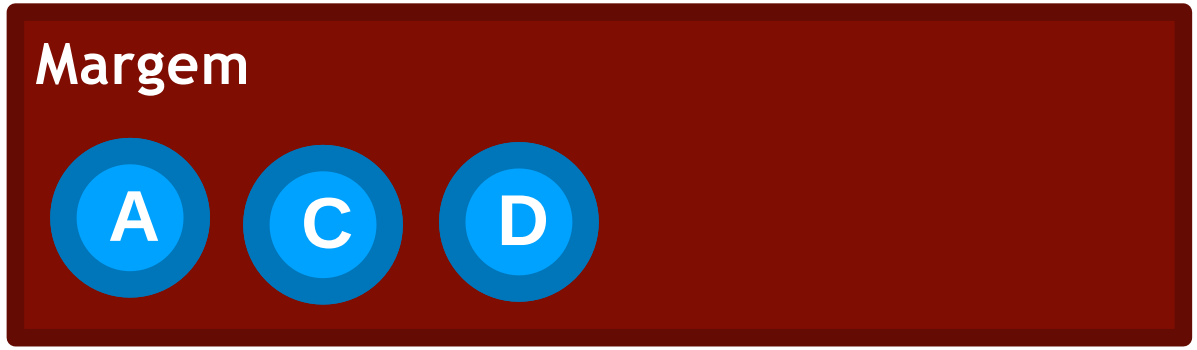

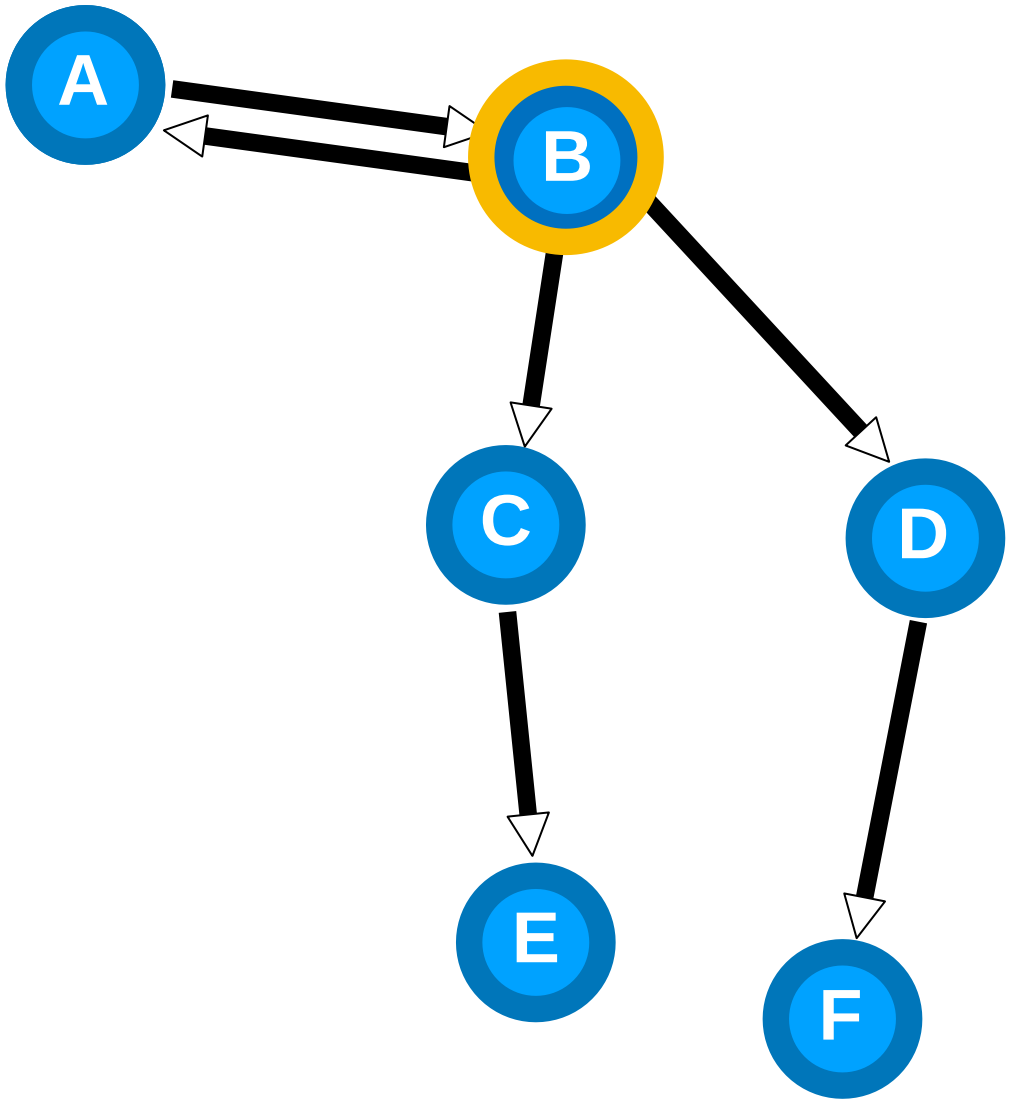

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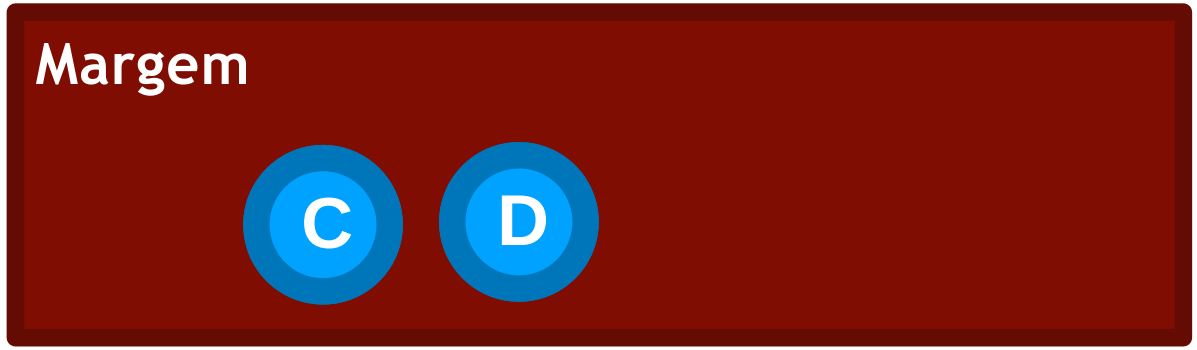

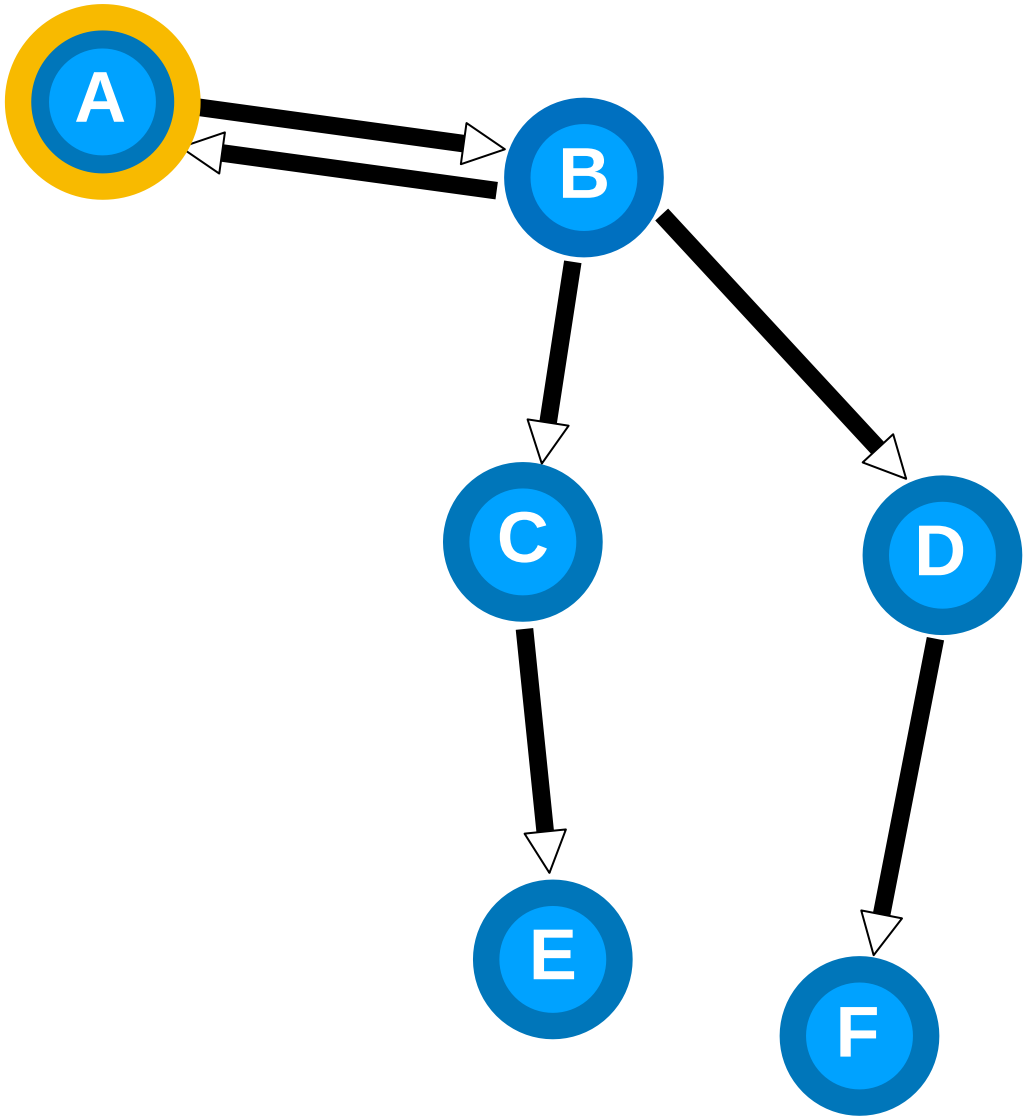

Estratégia Revisada

- Comece com uma **margem** que contenha o **estado inicial**.
- Comece com um **conjunto explorado** vazio.
- Repete:
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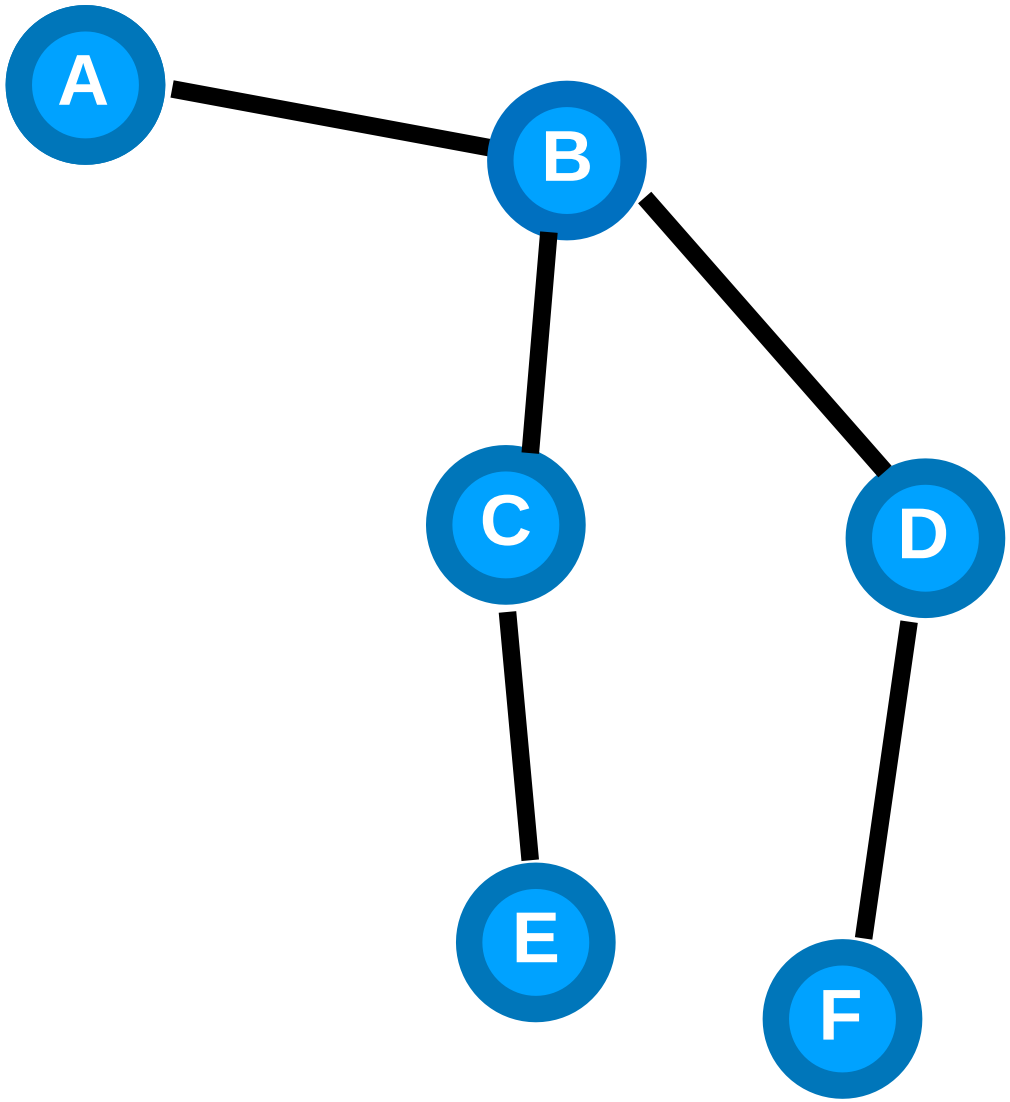
pilha

last-in first-out data structure type

Encontre um caminho de A para E

Margem

Conjunto Explorado

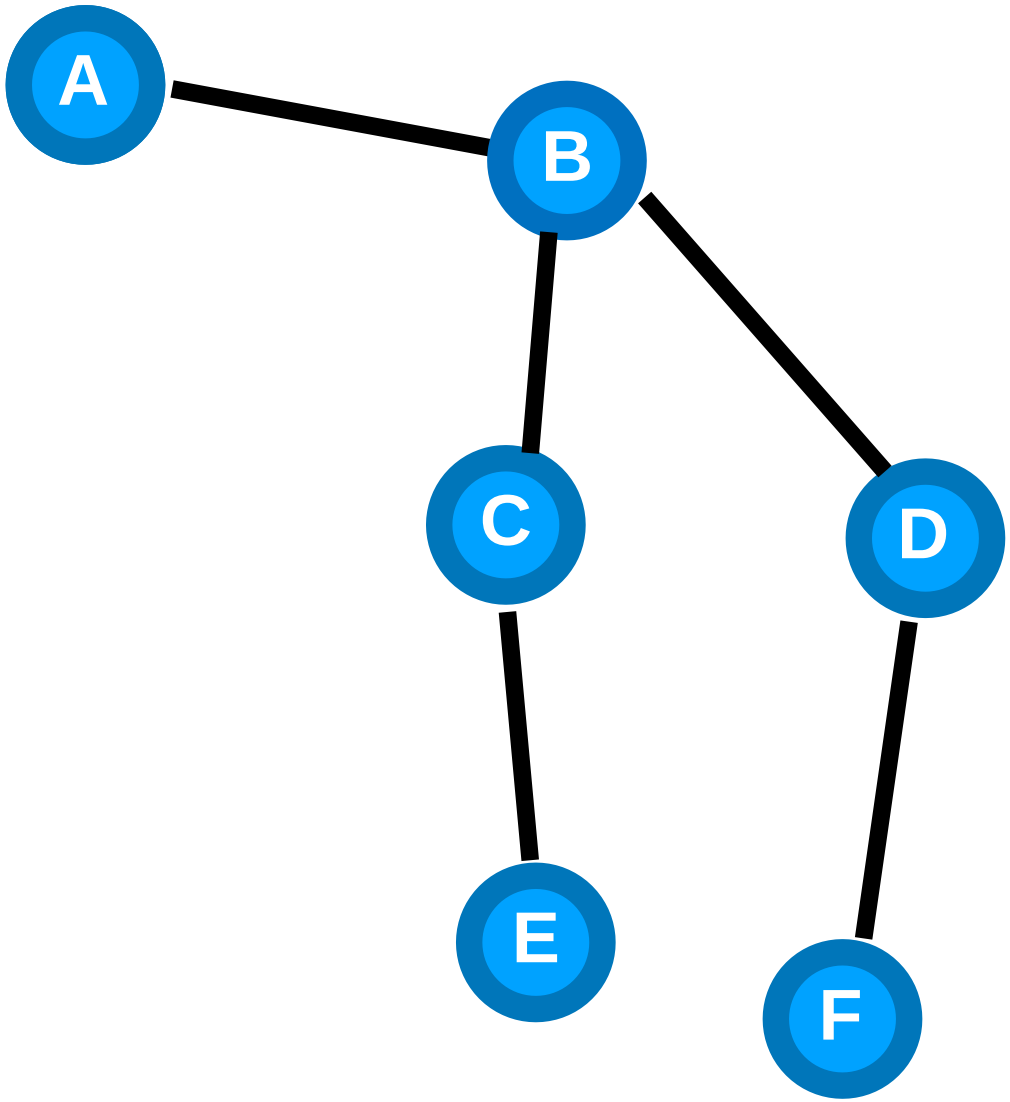


Encontre um caminho de A para E

Margem

A

Conjunto Explorado

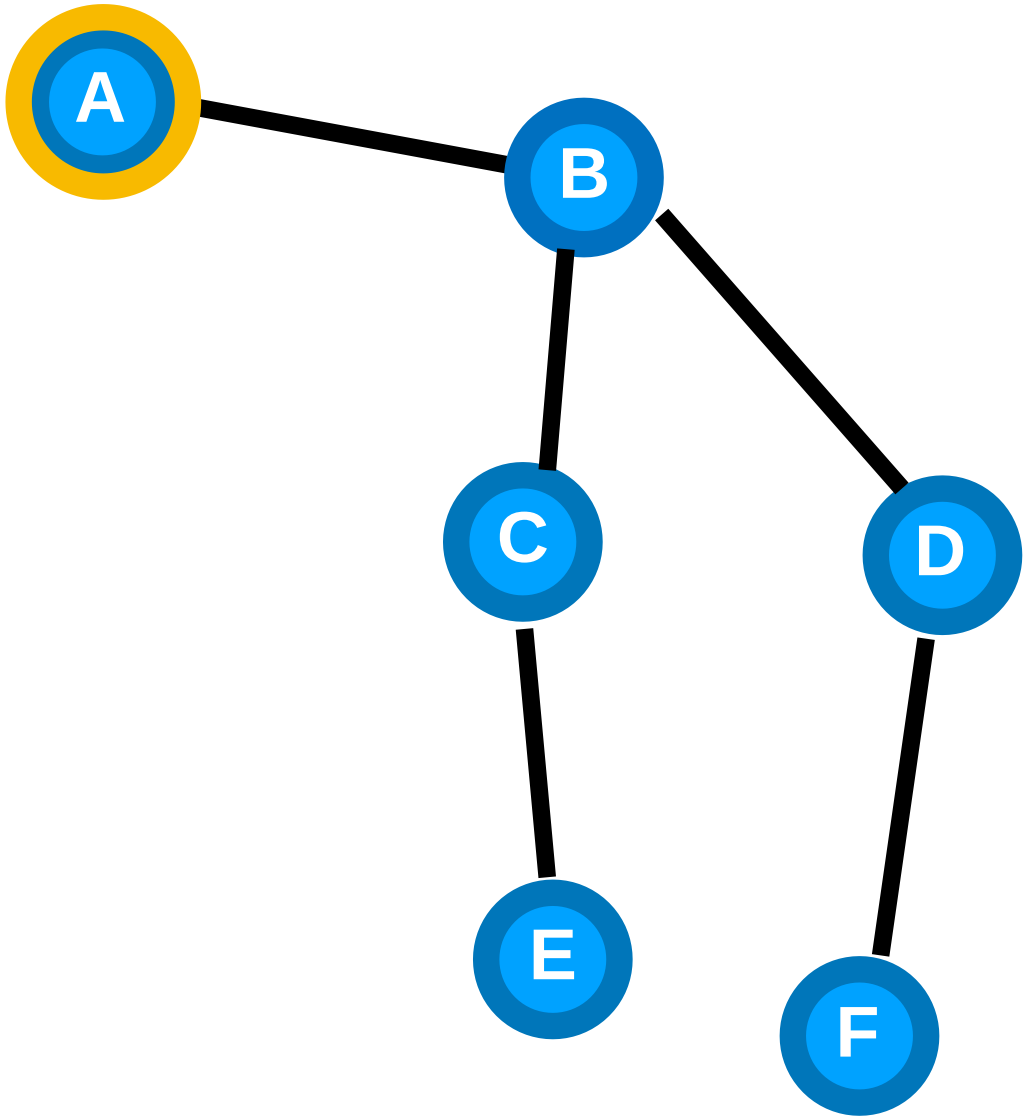


Encontre um caminho de A para E

Margem

Conjunto Explorado

A



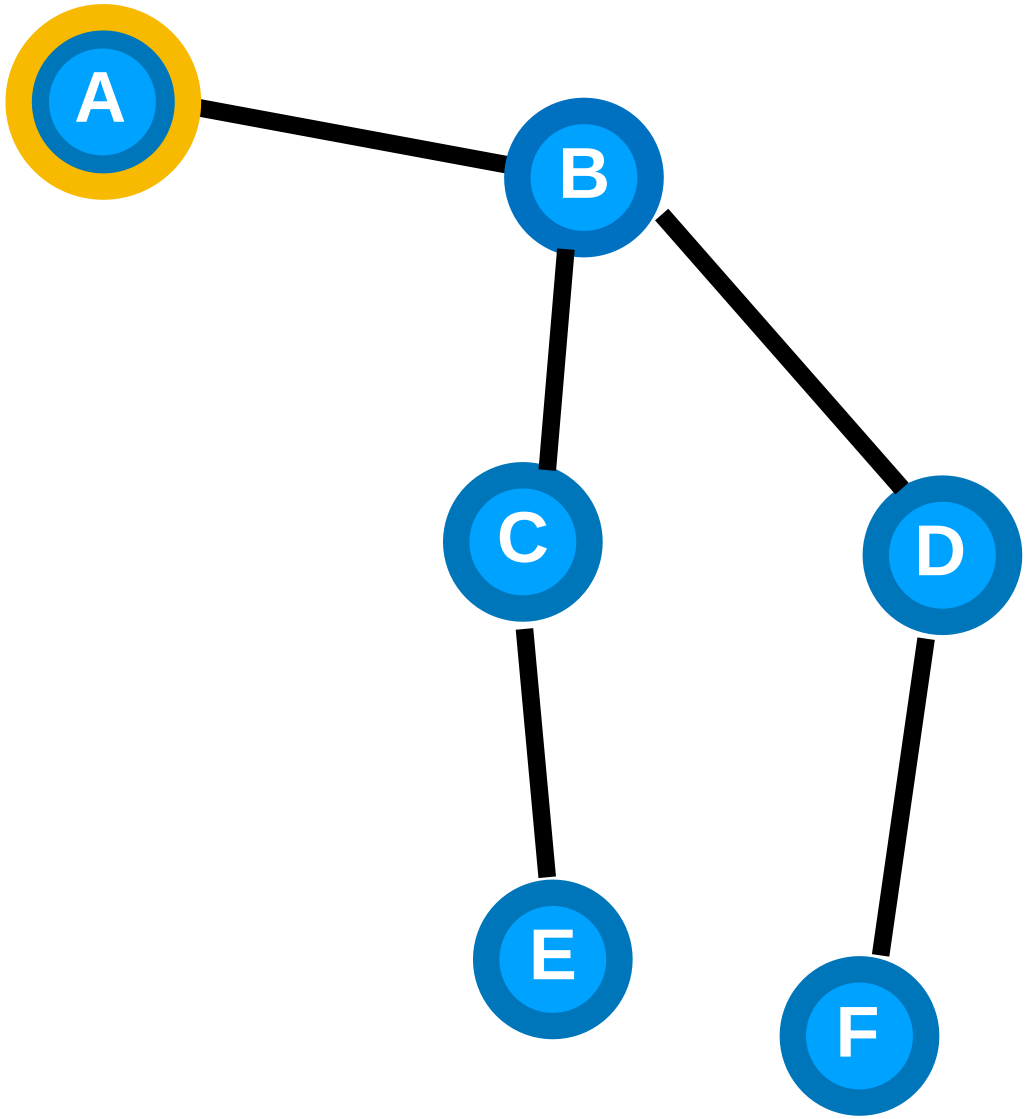
Encontre um caminho de A para E

Margem

B

Conjunto Explorado

A

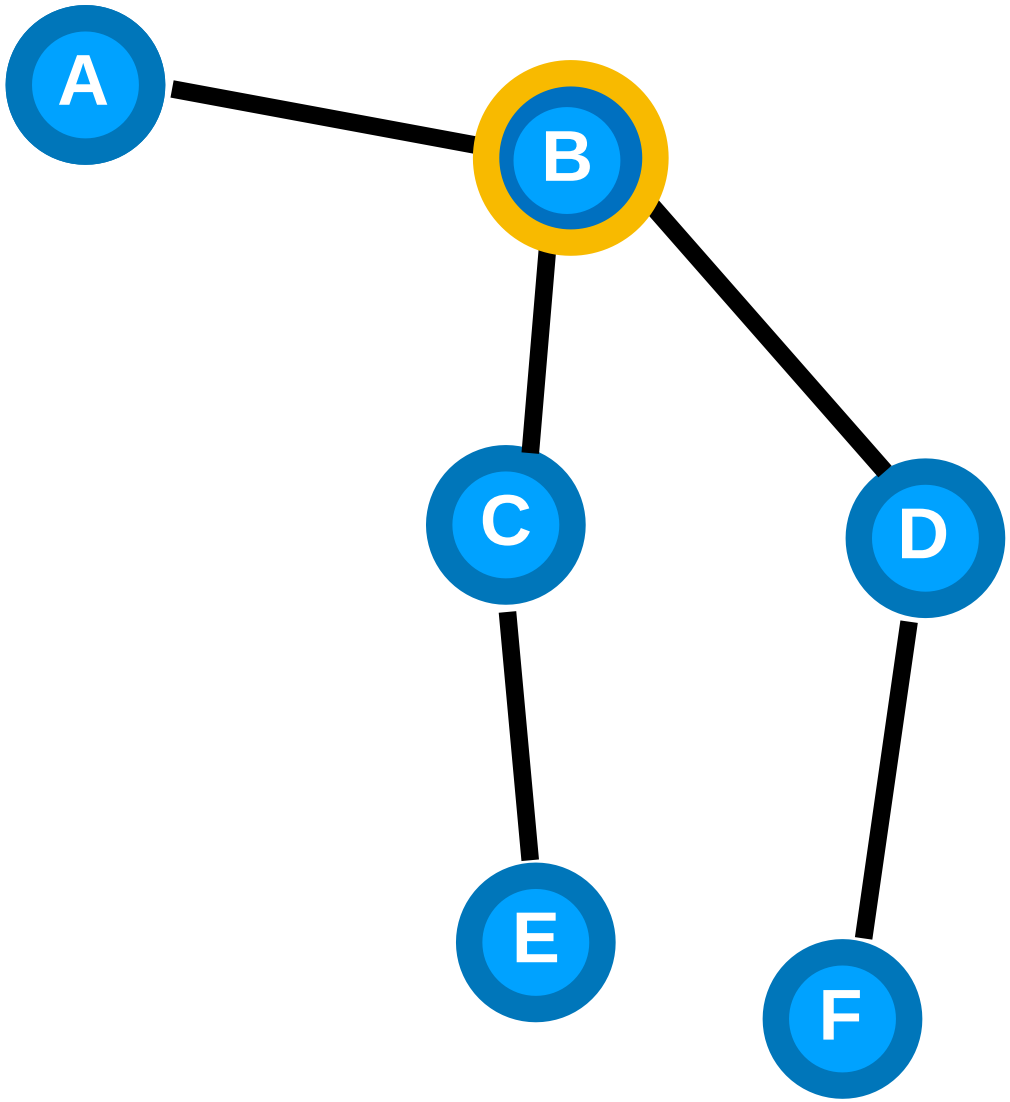


Encontre um caminho de A para E

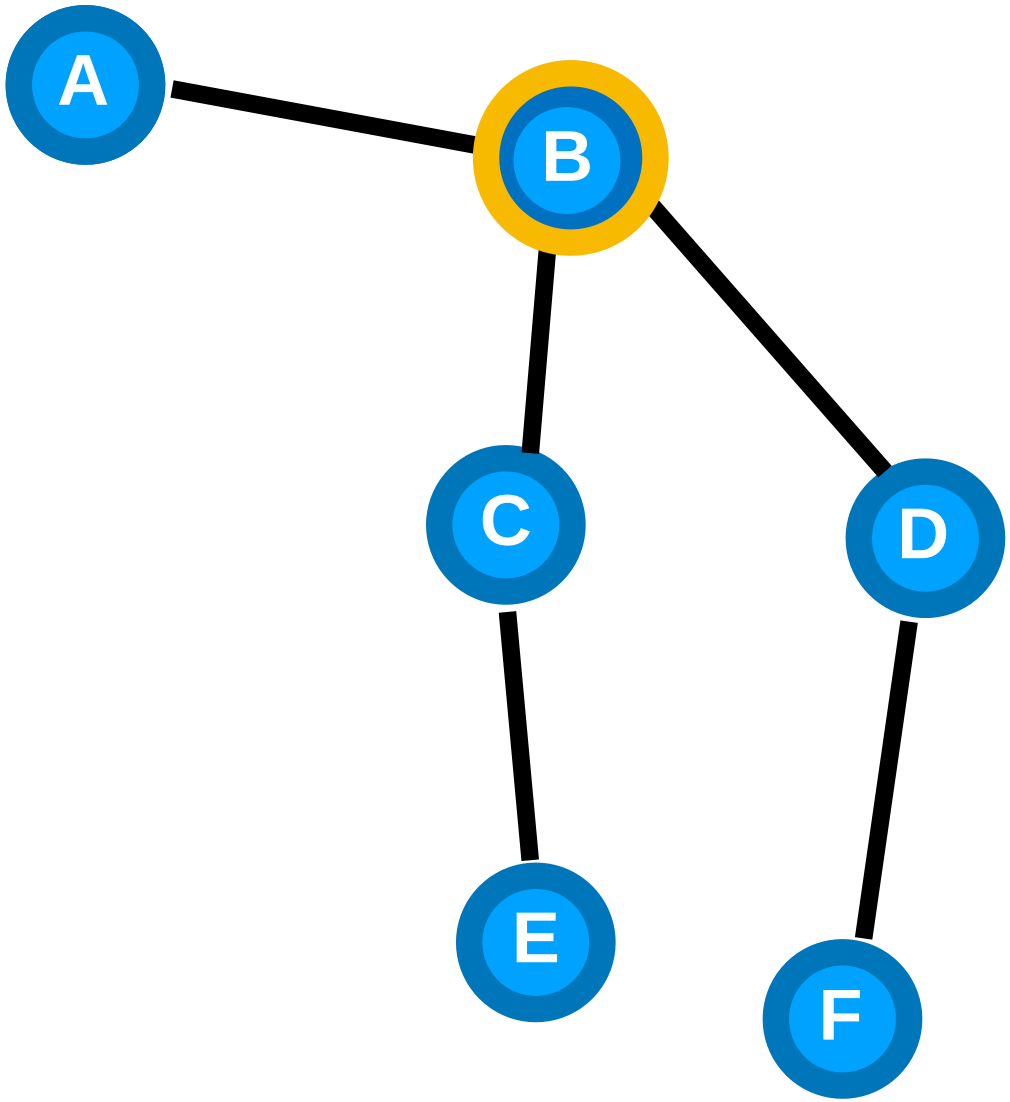
Margem

Conjunto Explorado

A B



Encontre um caminho de A para E



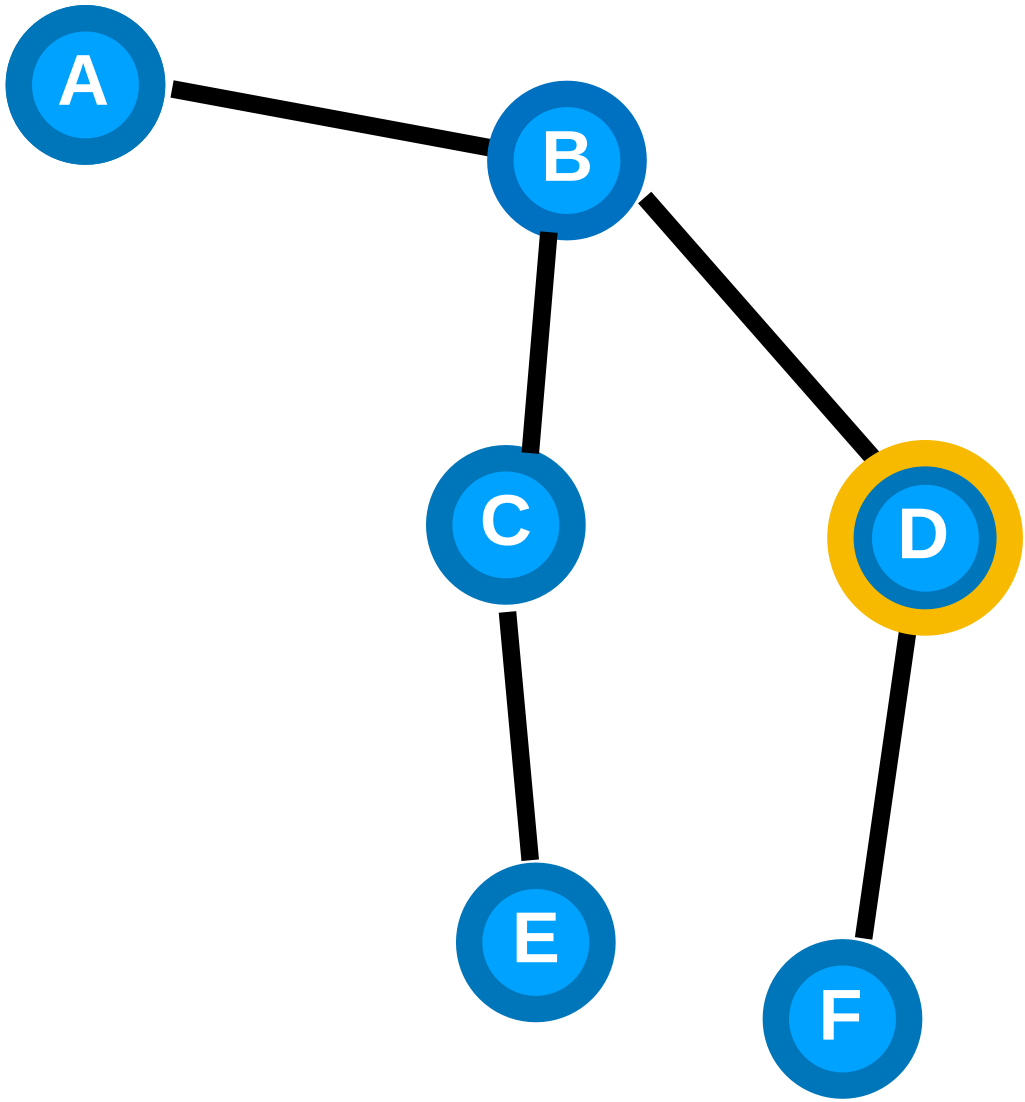
Encontre um caminho de A para E

Margem

C

Conjunto Explorado

A B D



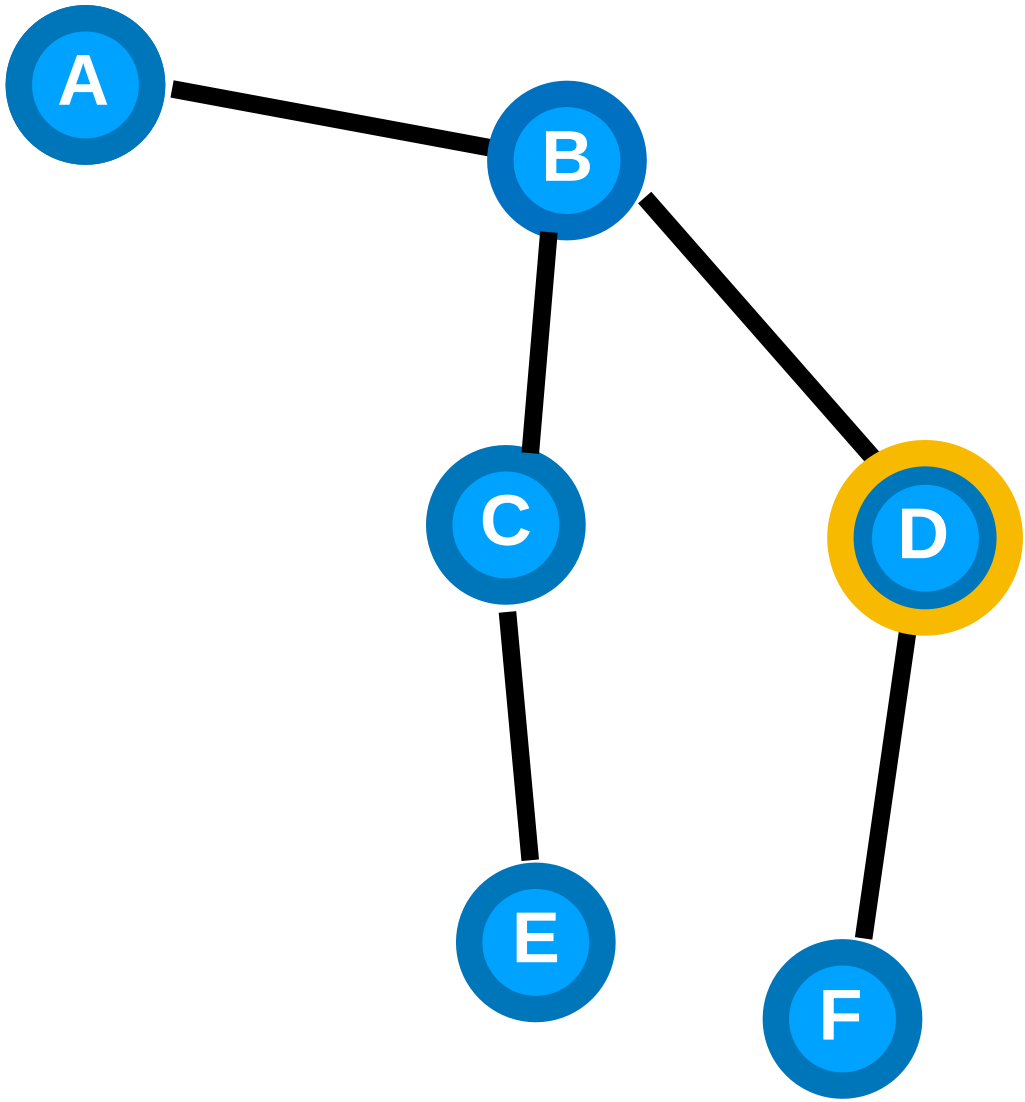
Encontre um caminho de A para E

Margem

C F

Conjunto Explorado

A B D



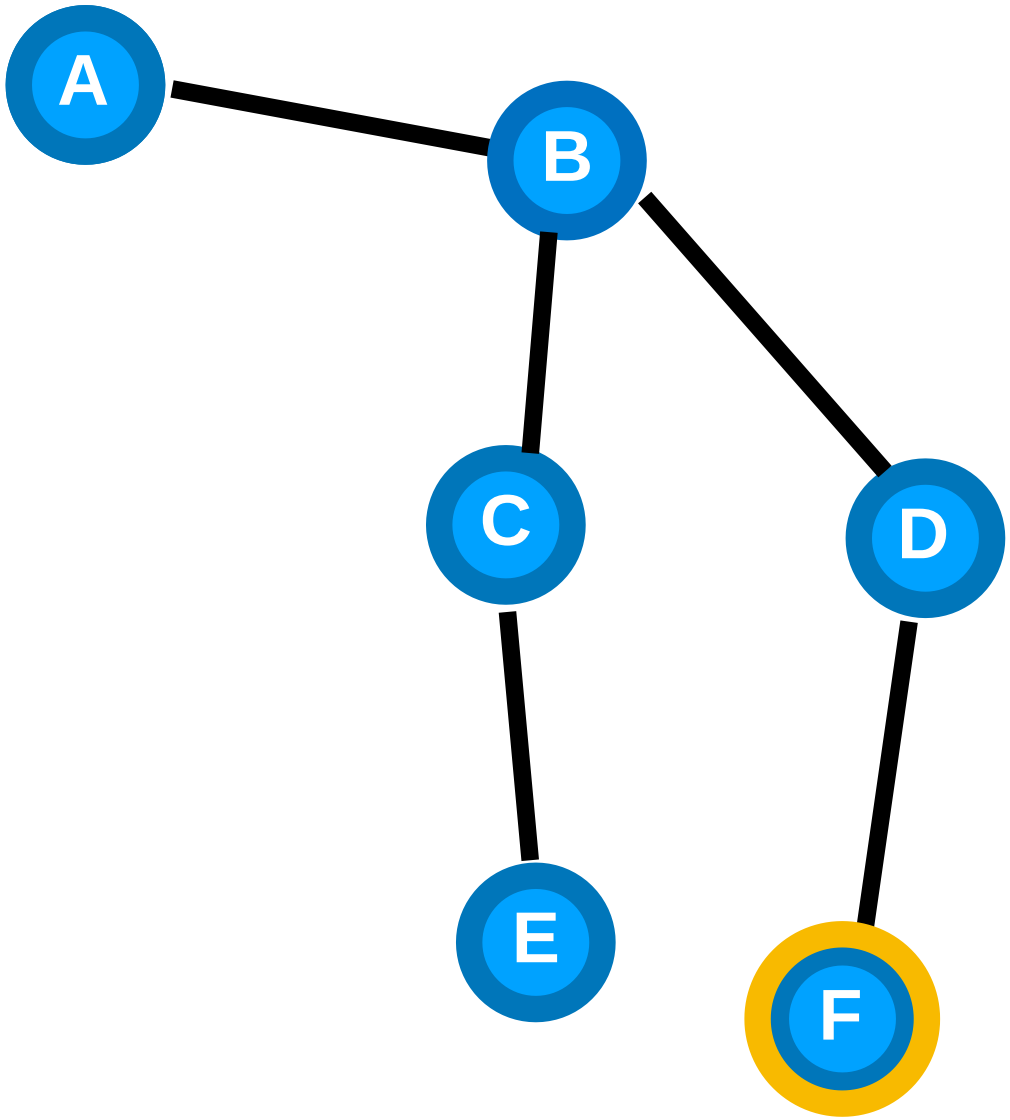
Encontre um caminho de A para E

Margem

C

Conjunto Explorado

A B D F

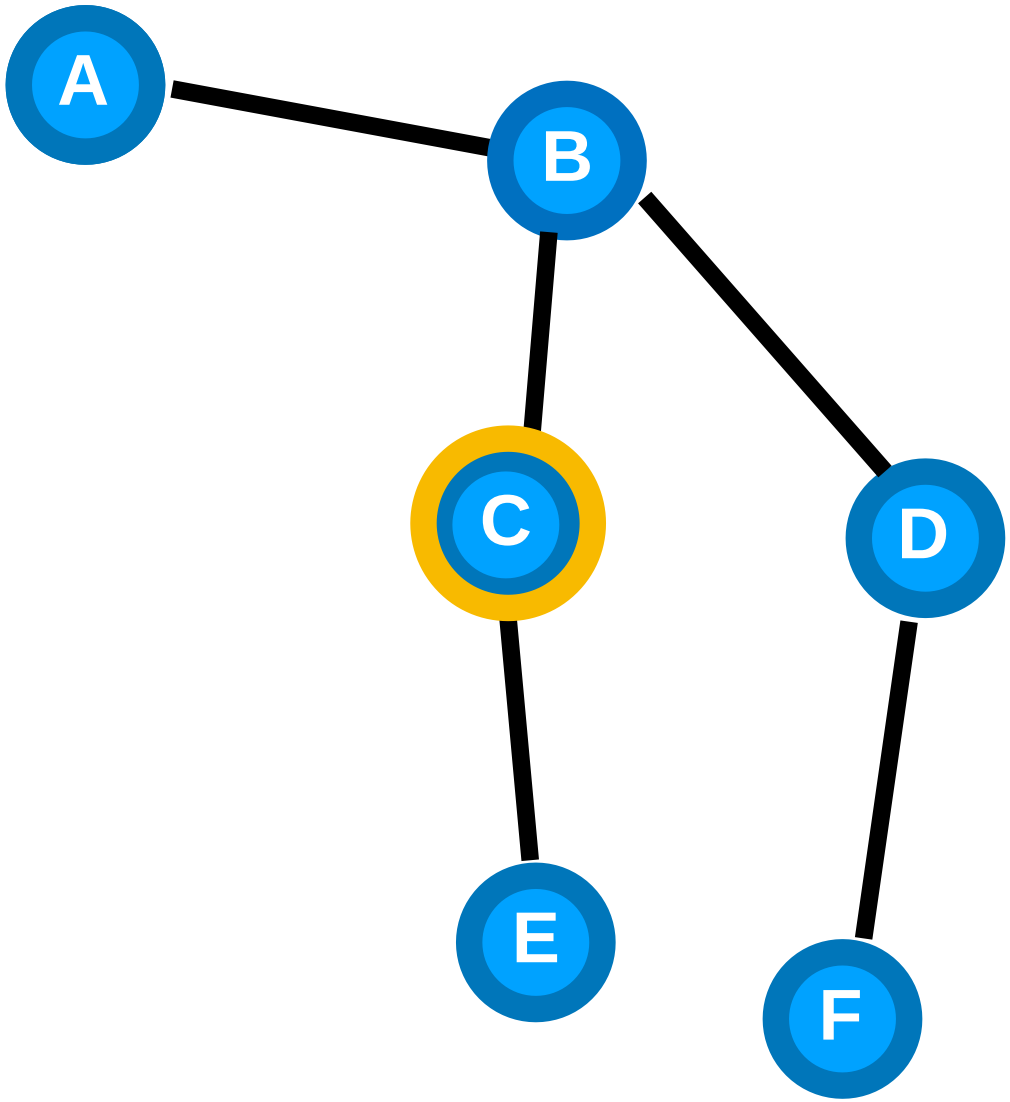


Encontre um caminho de A para E

Margem

Conjunto Explorado

A B D F C



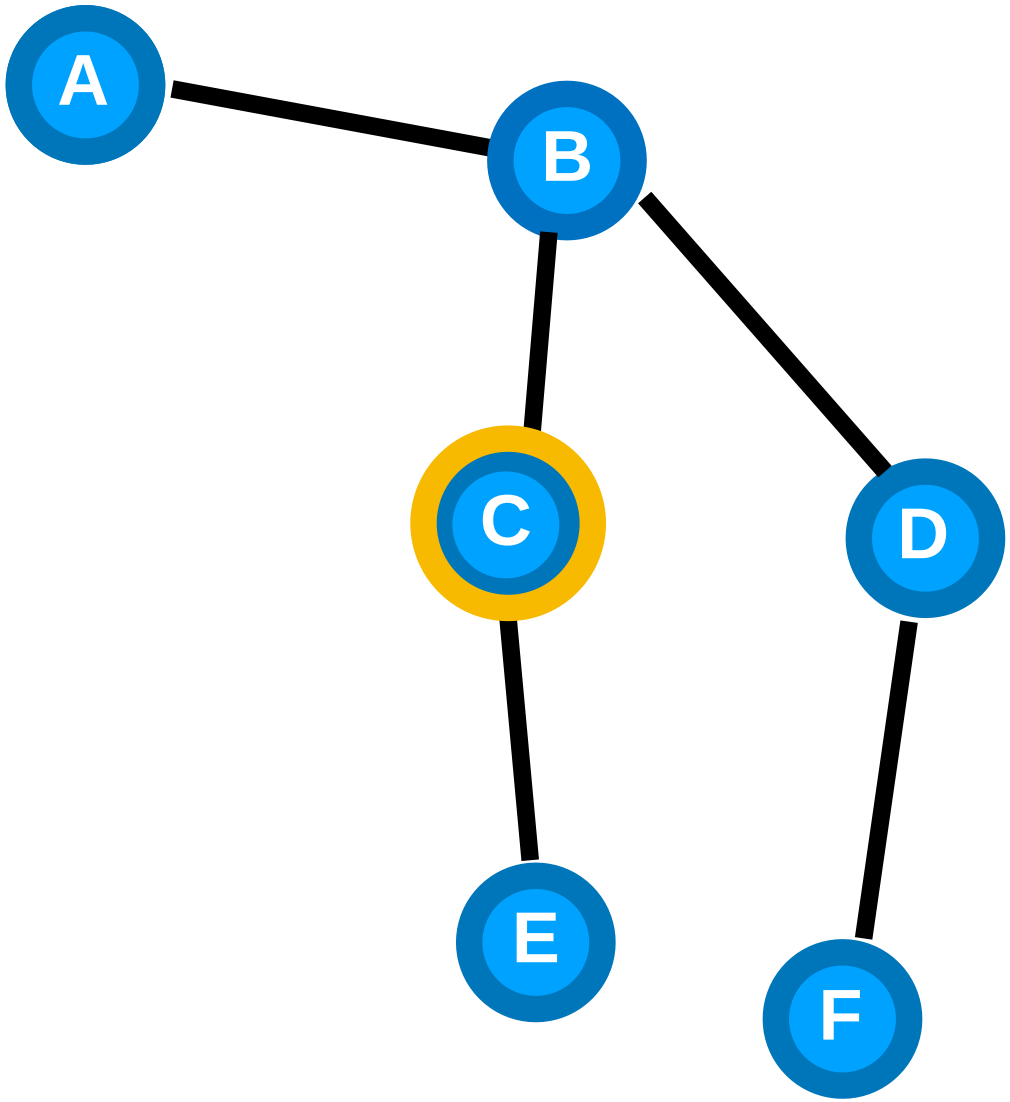
Encontre um caminho de A para E

Margem

E

Conjunto Explorado

A B D F C

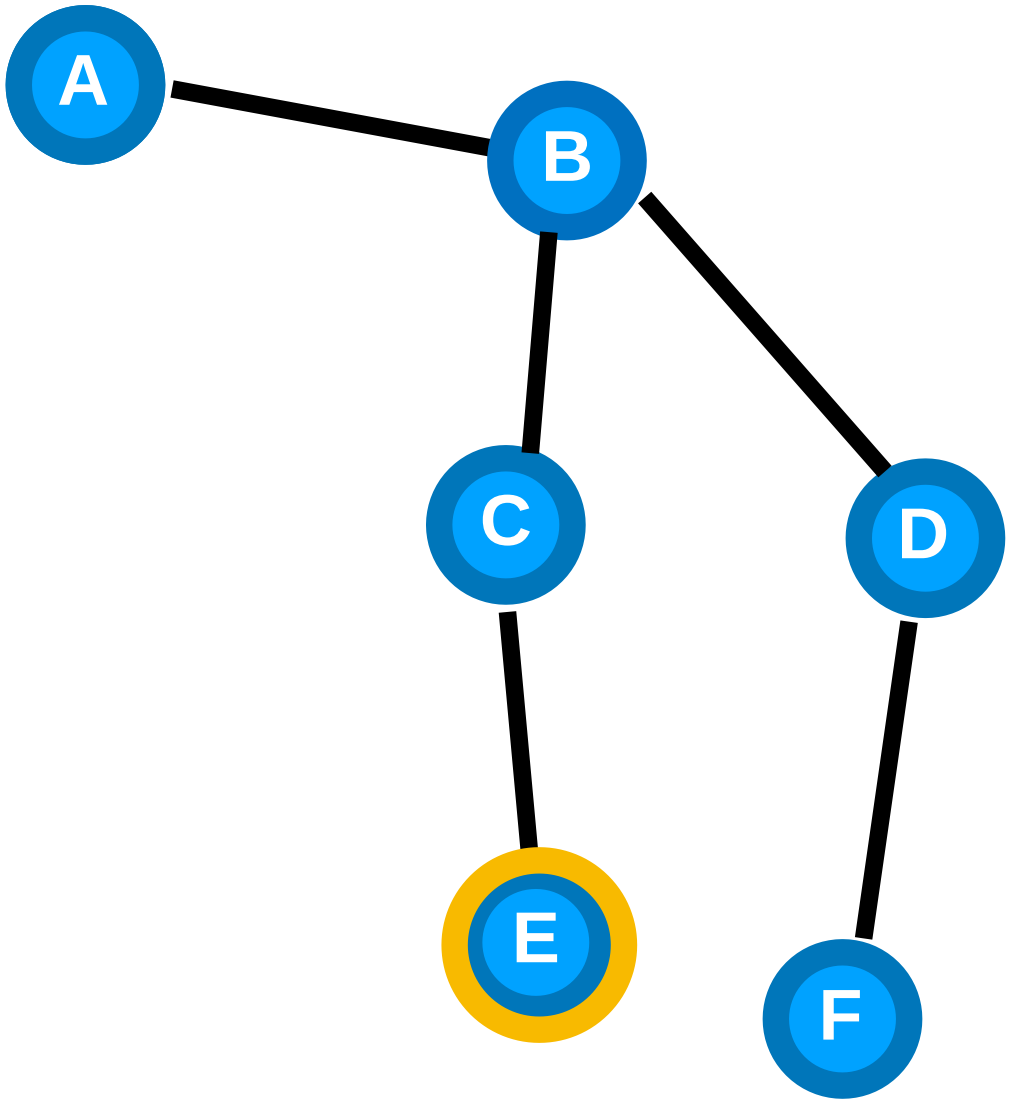


Encontre um caminho de A para E

Margem

Conjunto Explorado

A B D F C



depth-first search

algoritmo de busca que sempre expande o nó mais profundo na margem

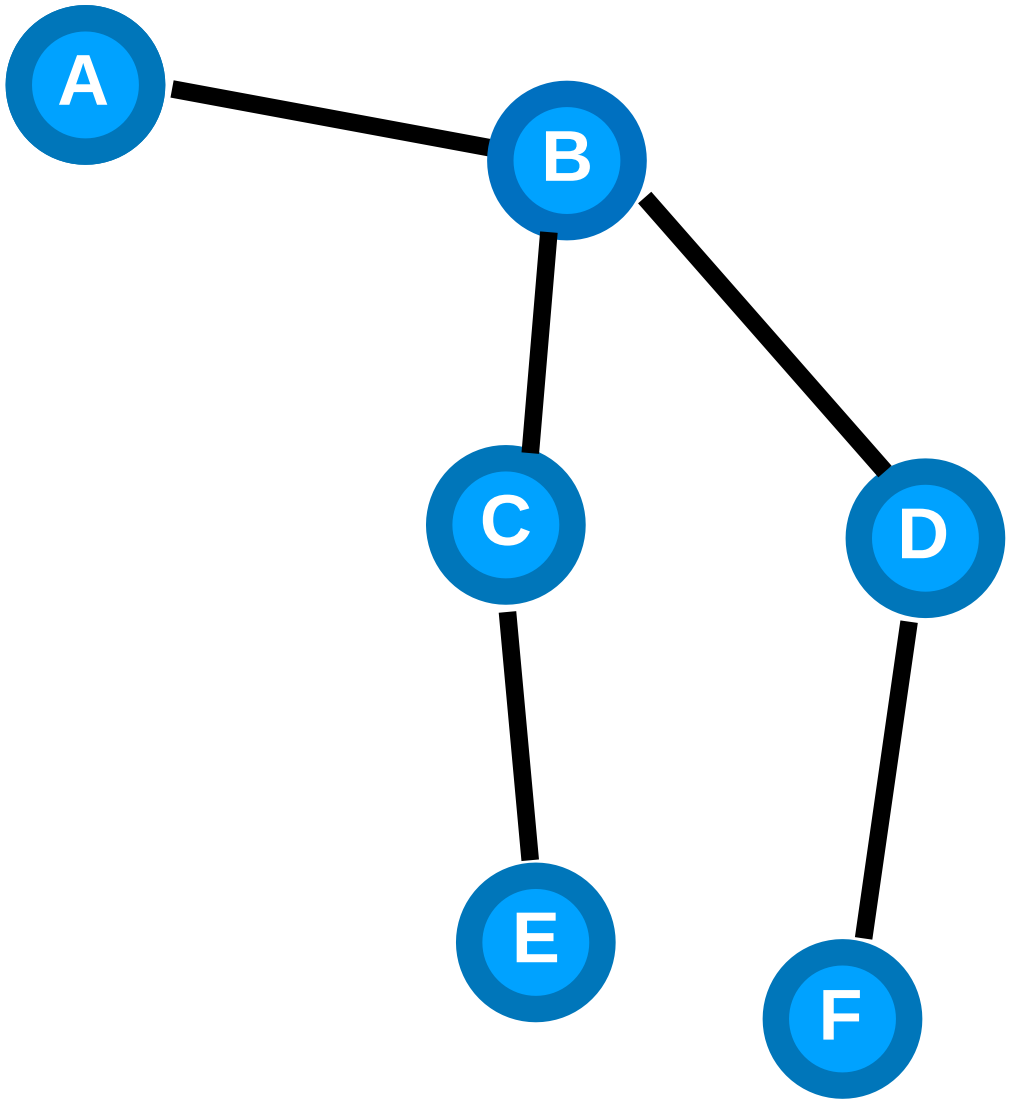
fila

first-in first-out data structure type

Encontre um caminho de A para E

Margem

Conjunto Explorado

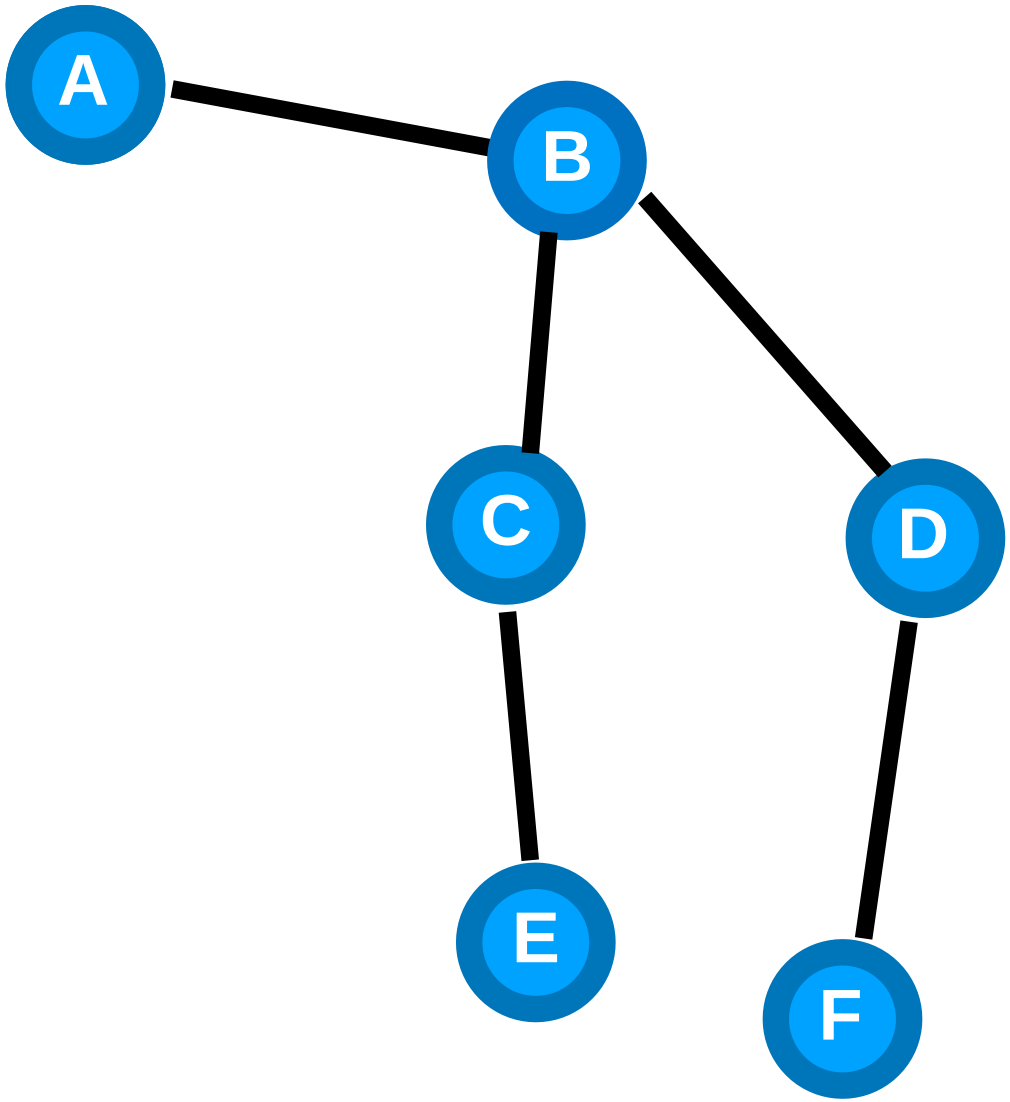


Encontre um caminho de A para E

Margem

A

Conjunto Explorado

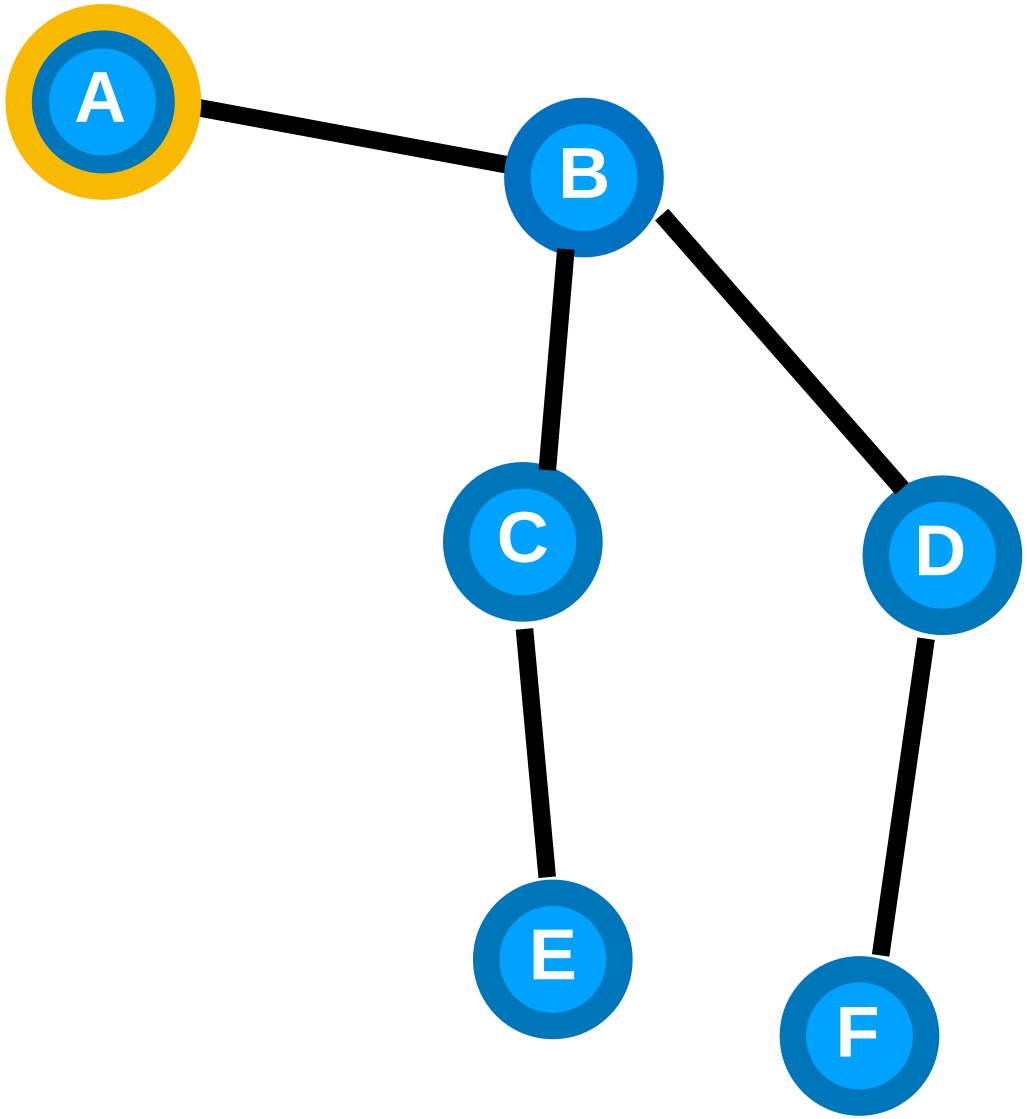


Encontre um caminho de A para E

Margem

Conjunto Explorado

A



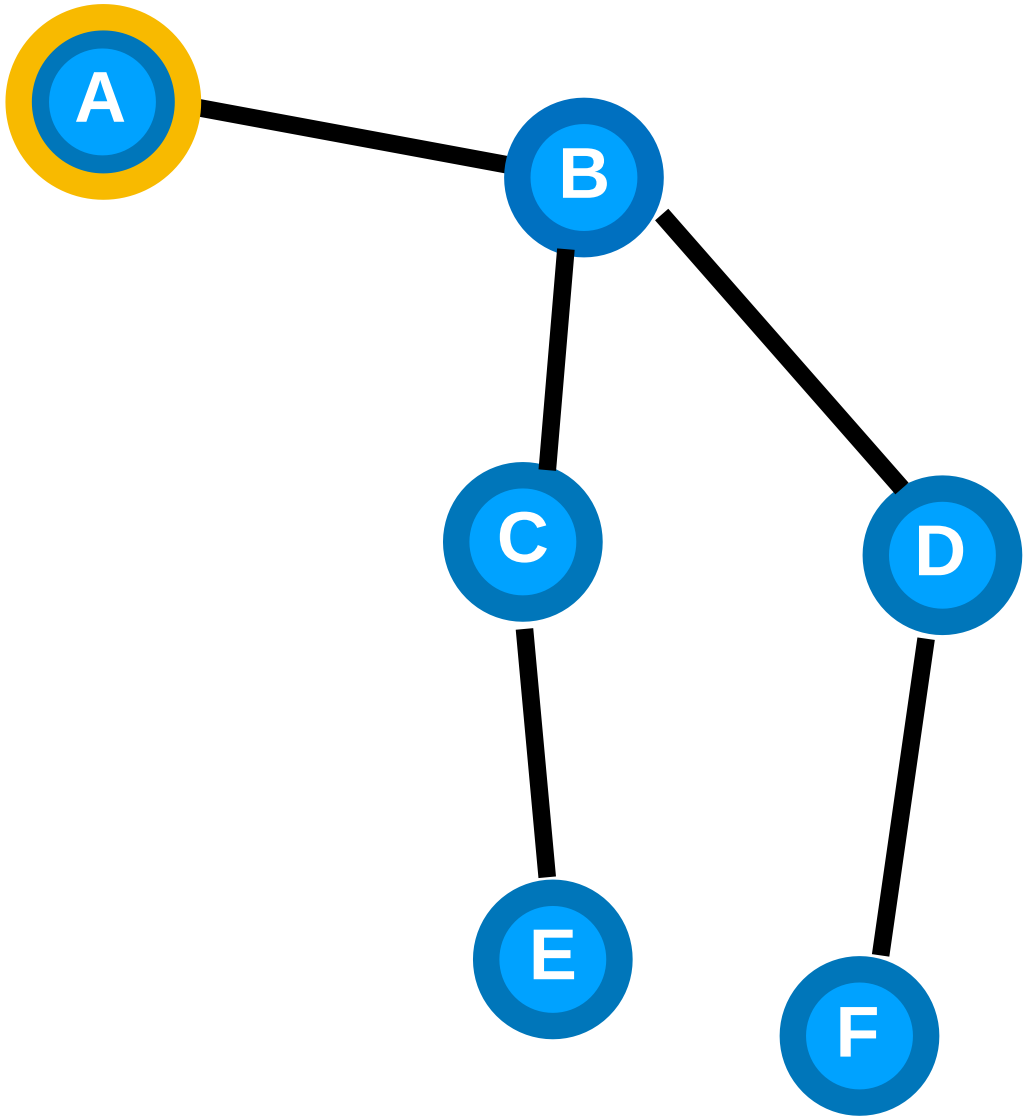
Encontre um caminho de A para E

Margem

B

Conjunto Explorado

A

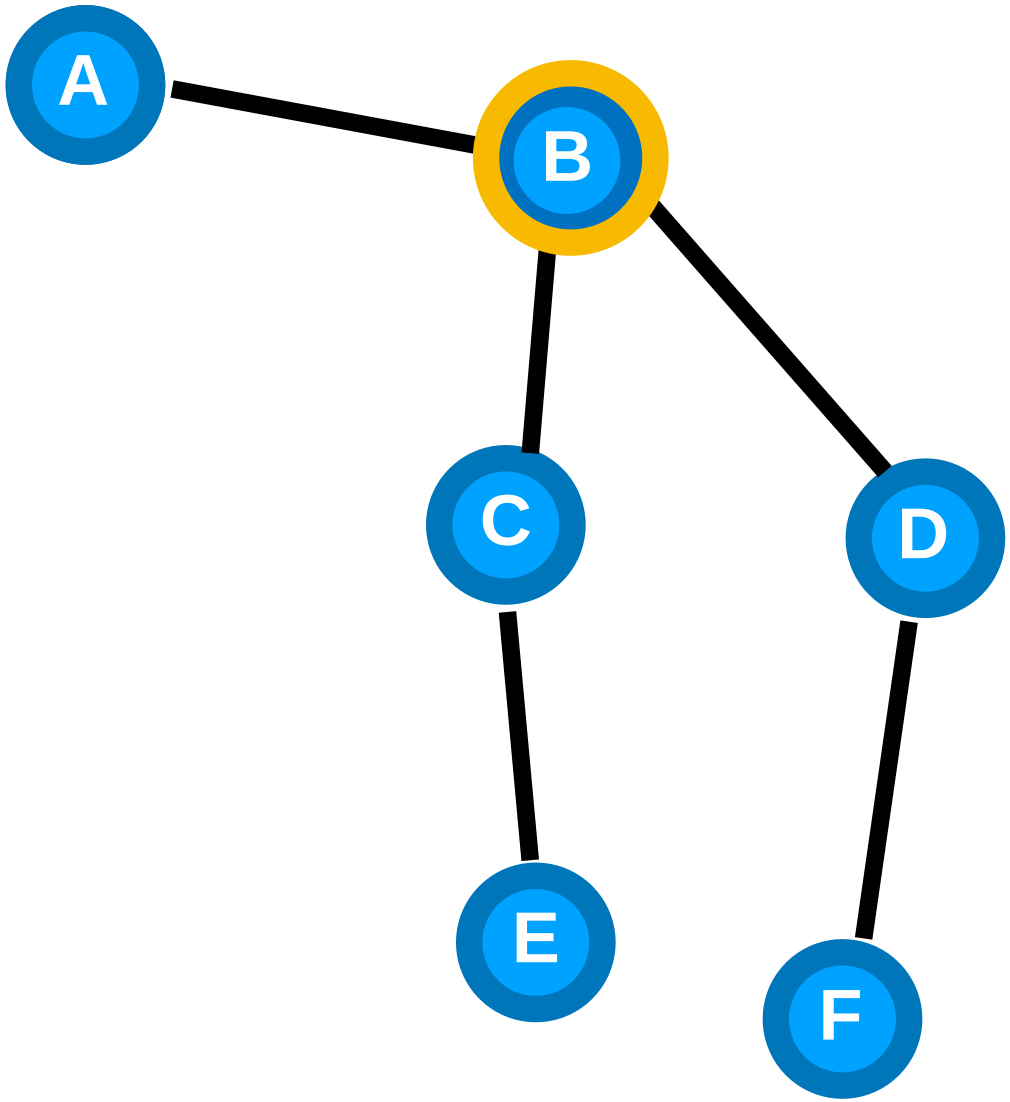


Encontre um caminho de A para E

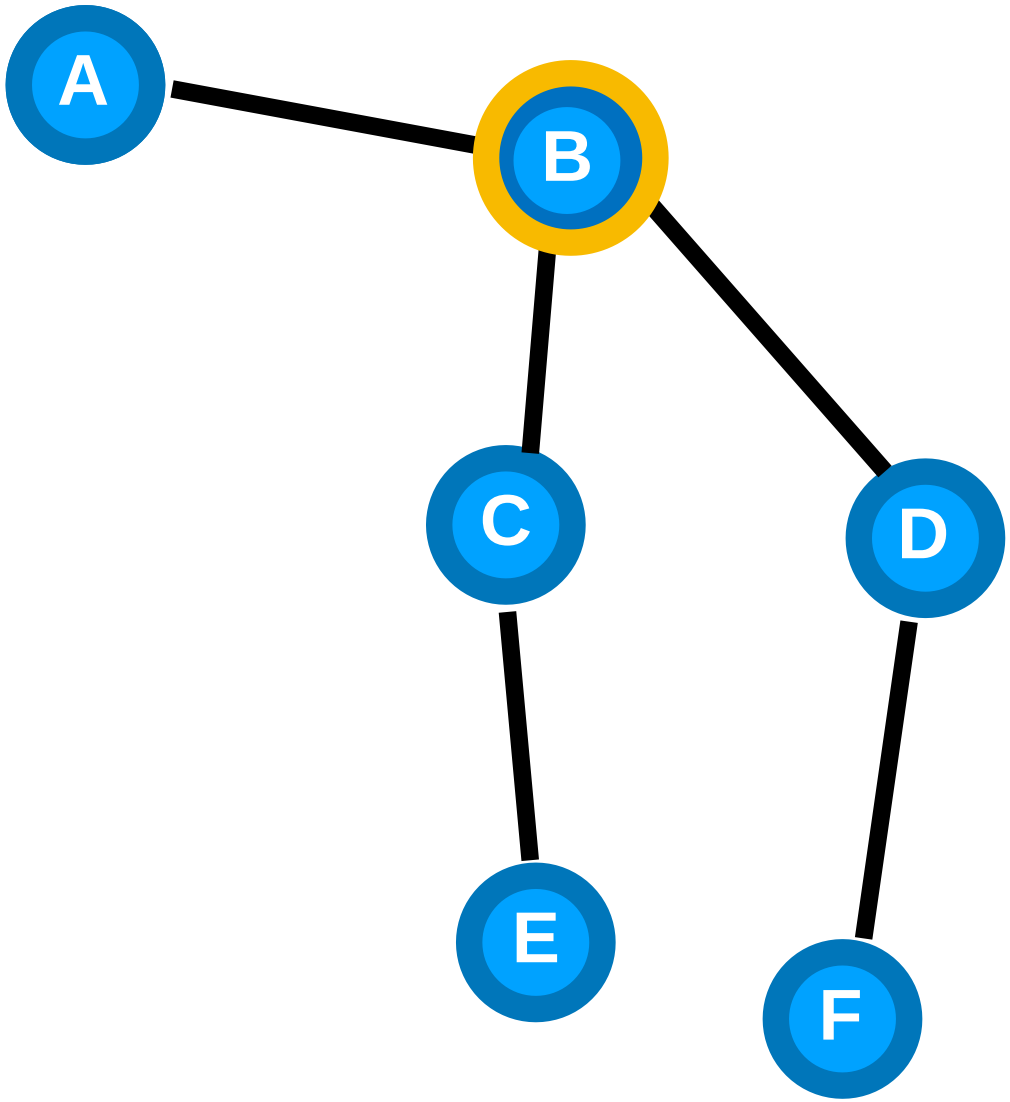
Margem

Conjunto Explorado

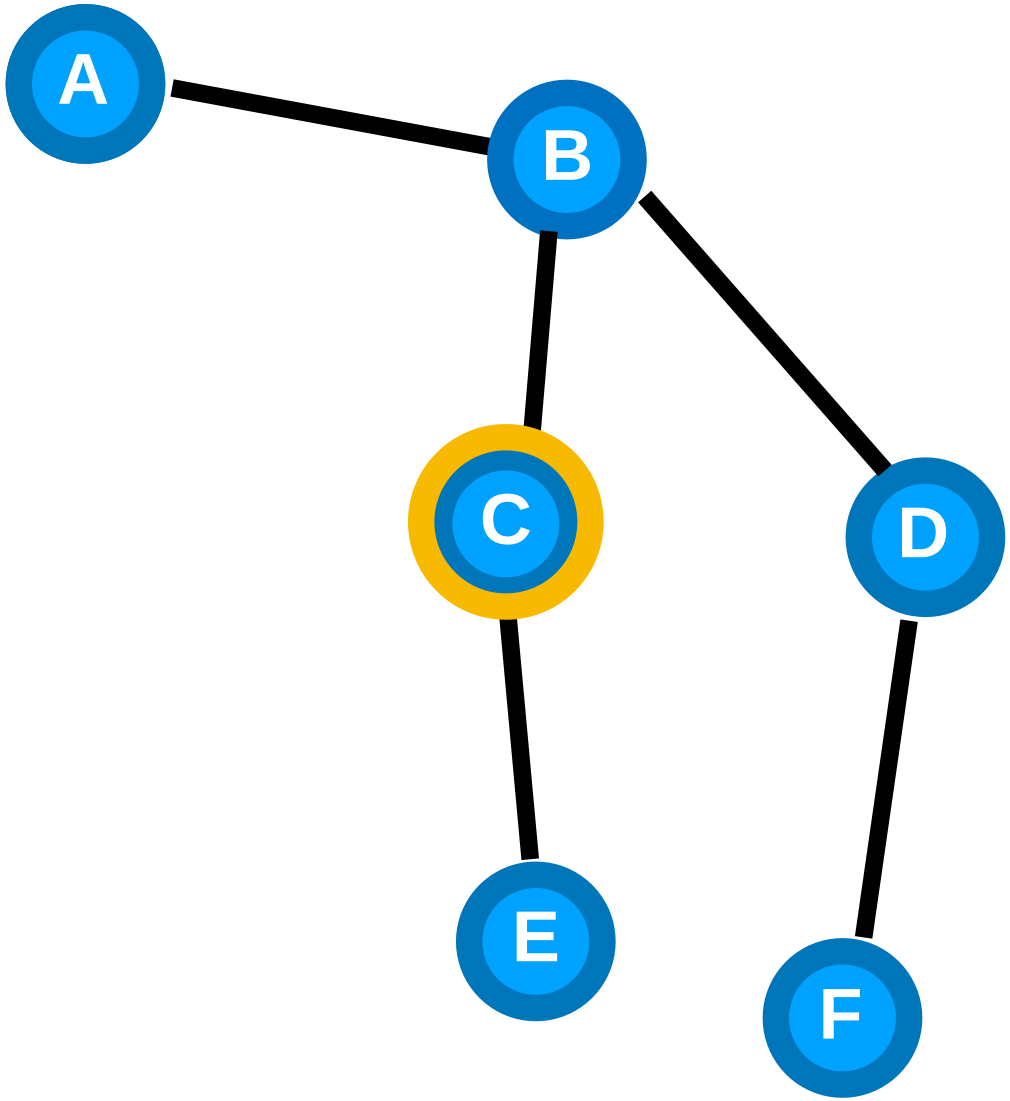
A B



Encontre um caminho de A para E



Encontre um caminho de A para E



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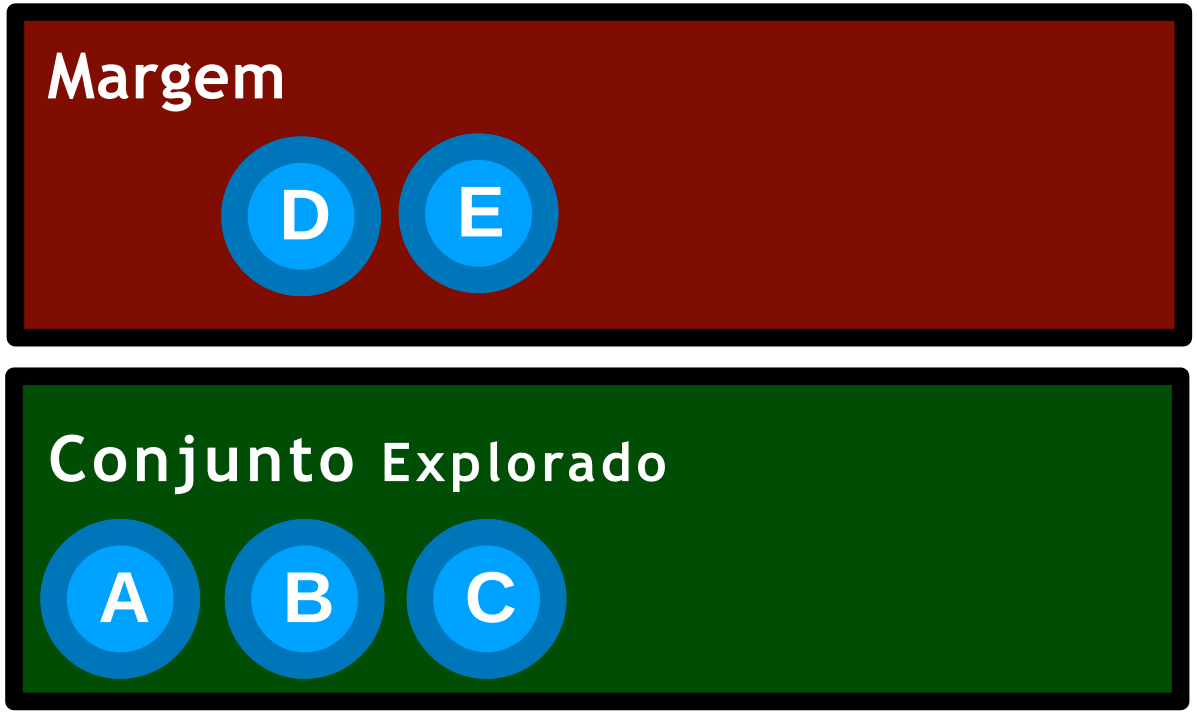
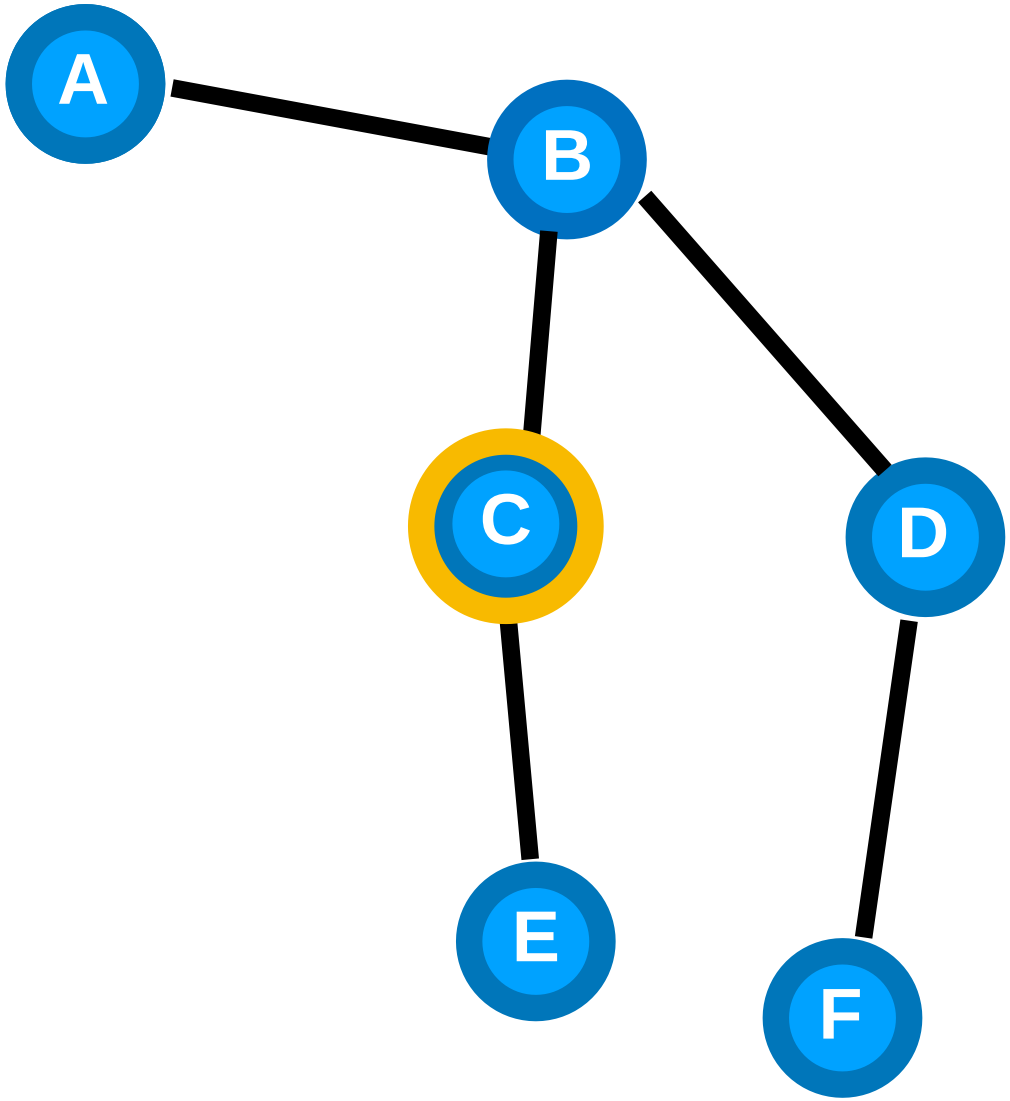
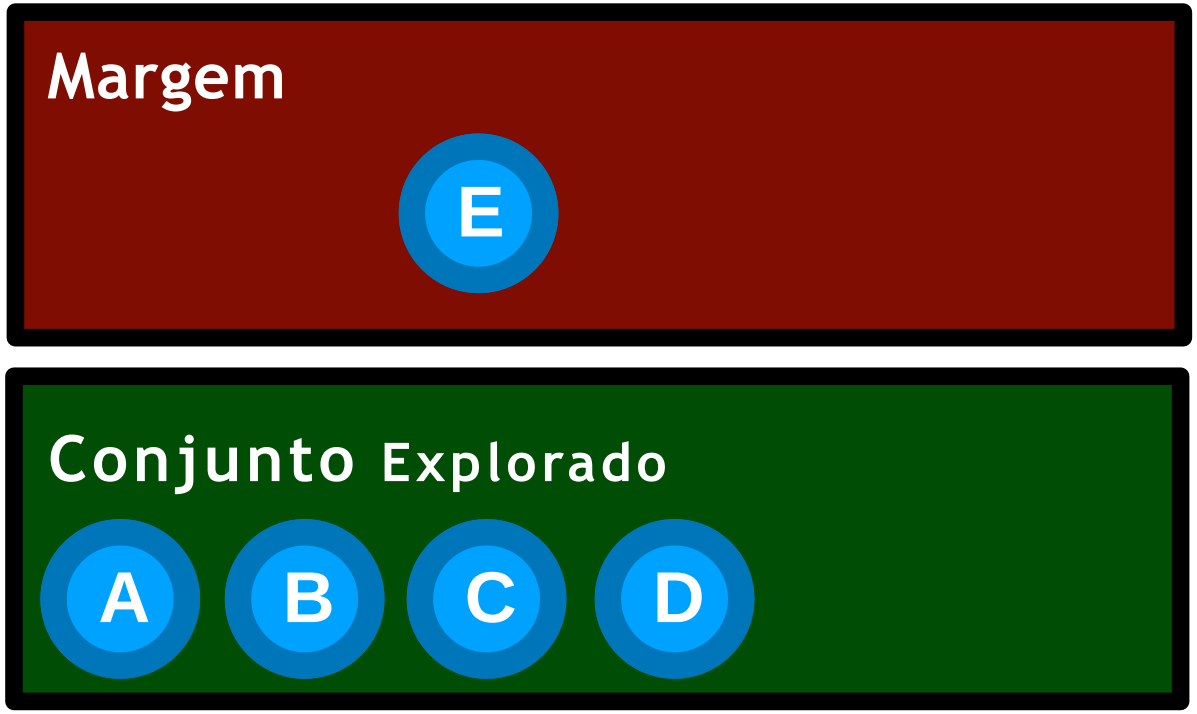
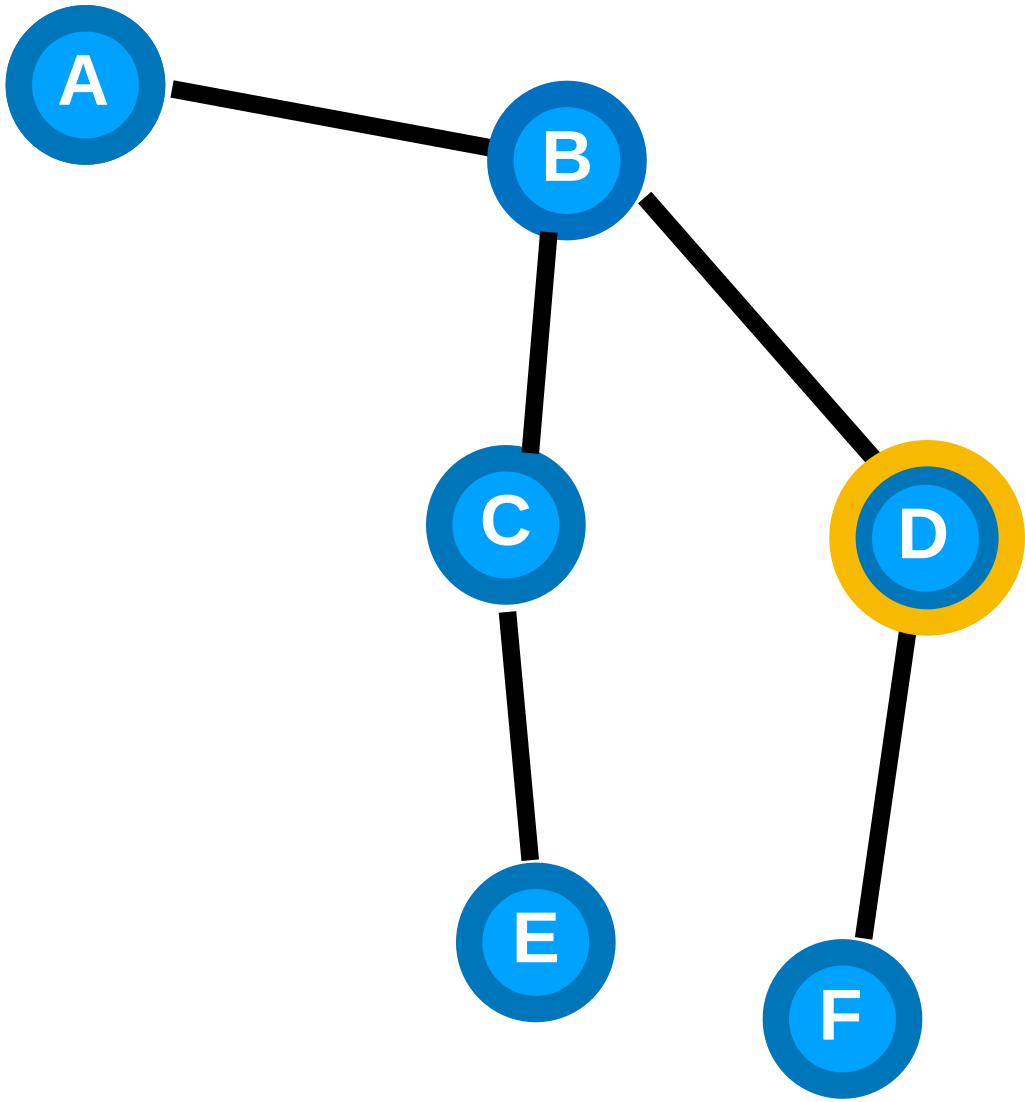




Encontre um caminho de A para E





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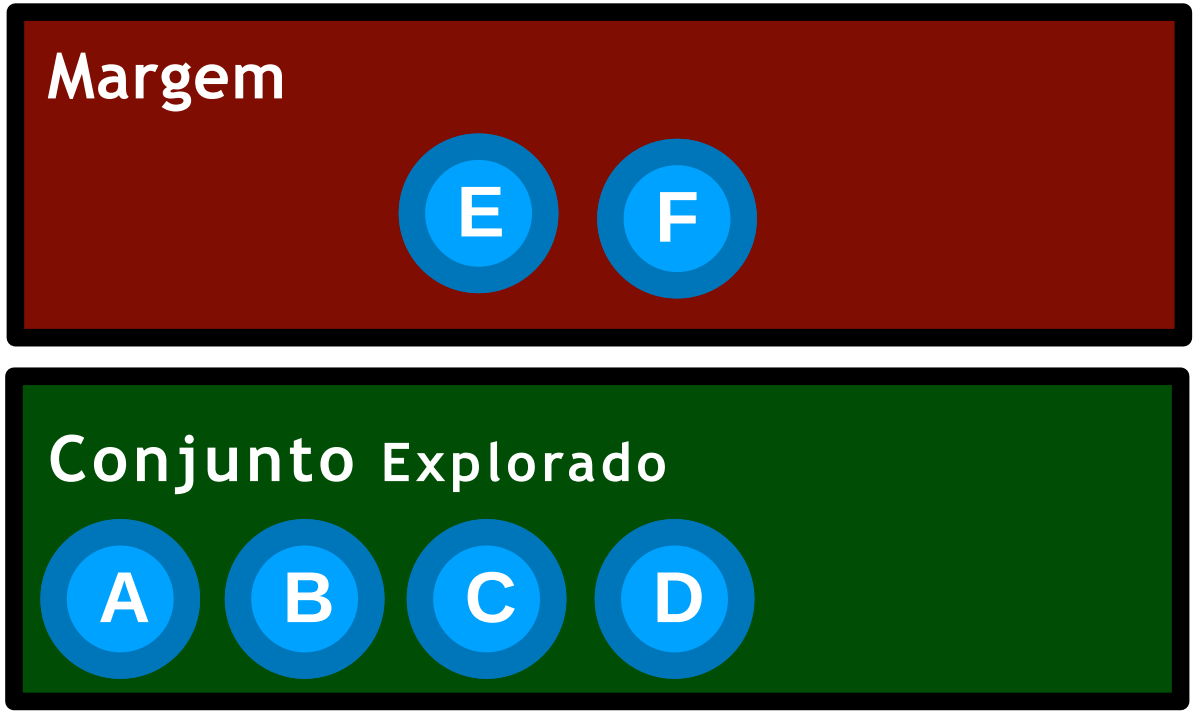

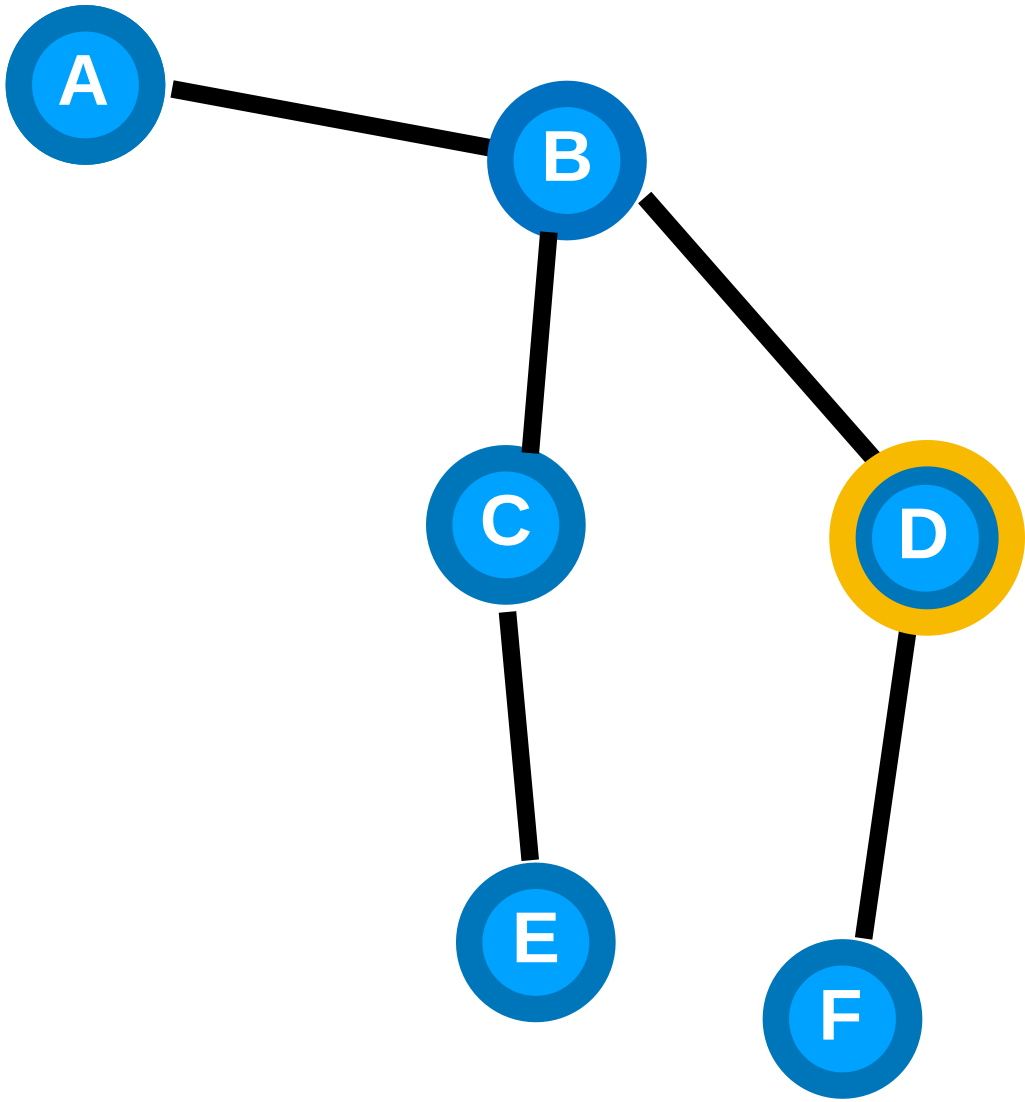

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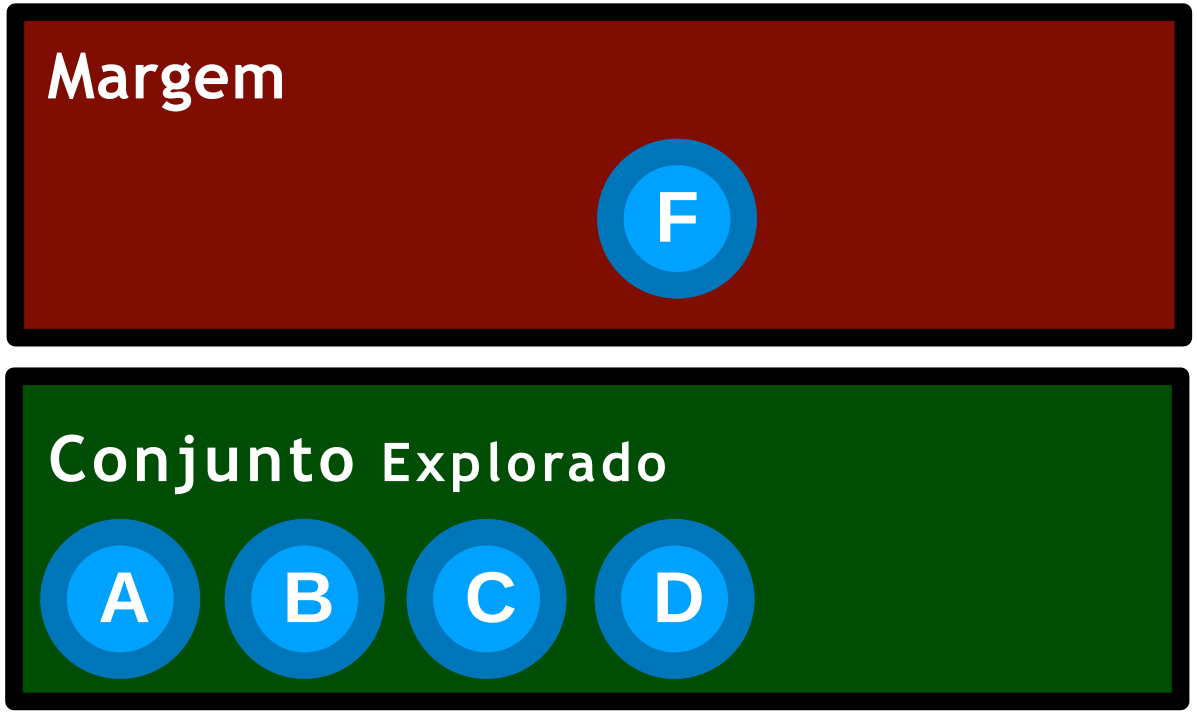
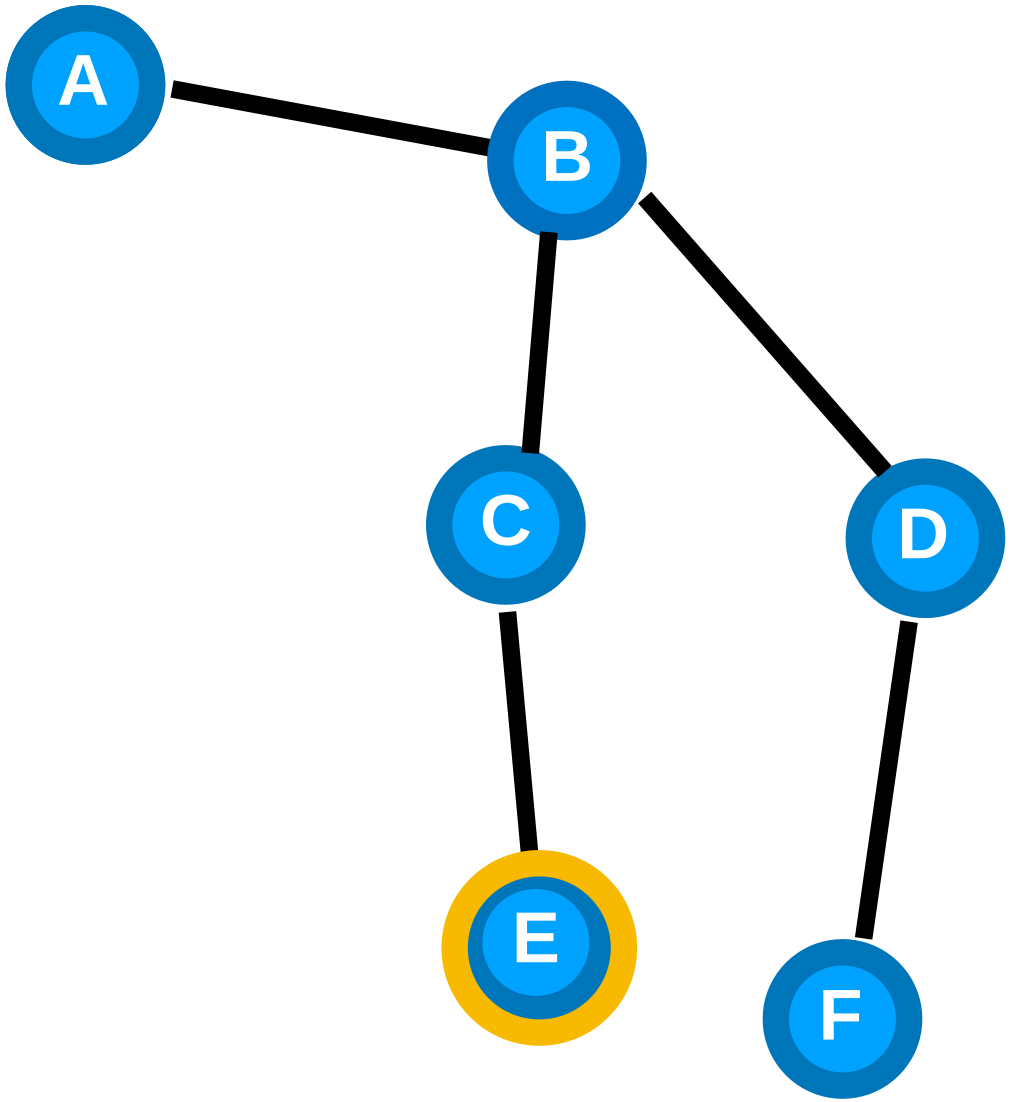




breadth-first search

algoritmo de busca que sempre expande o nó mais raso na margem

deaph-first search

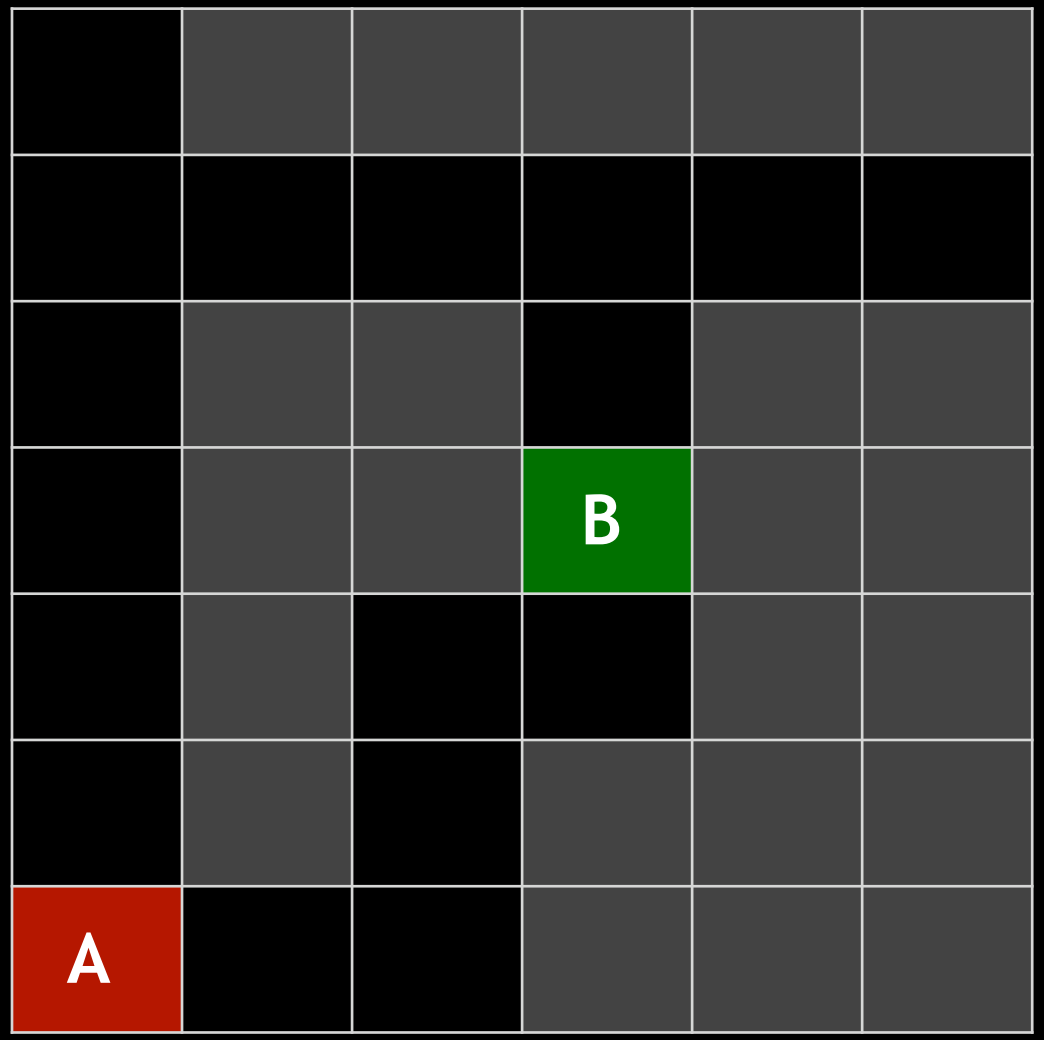
x

breadth-first search

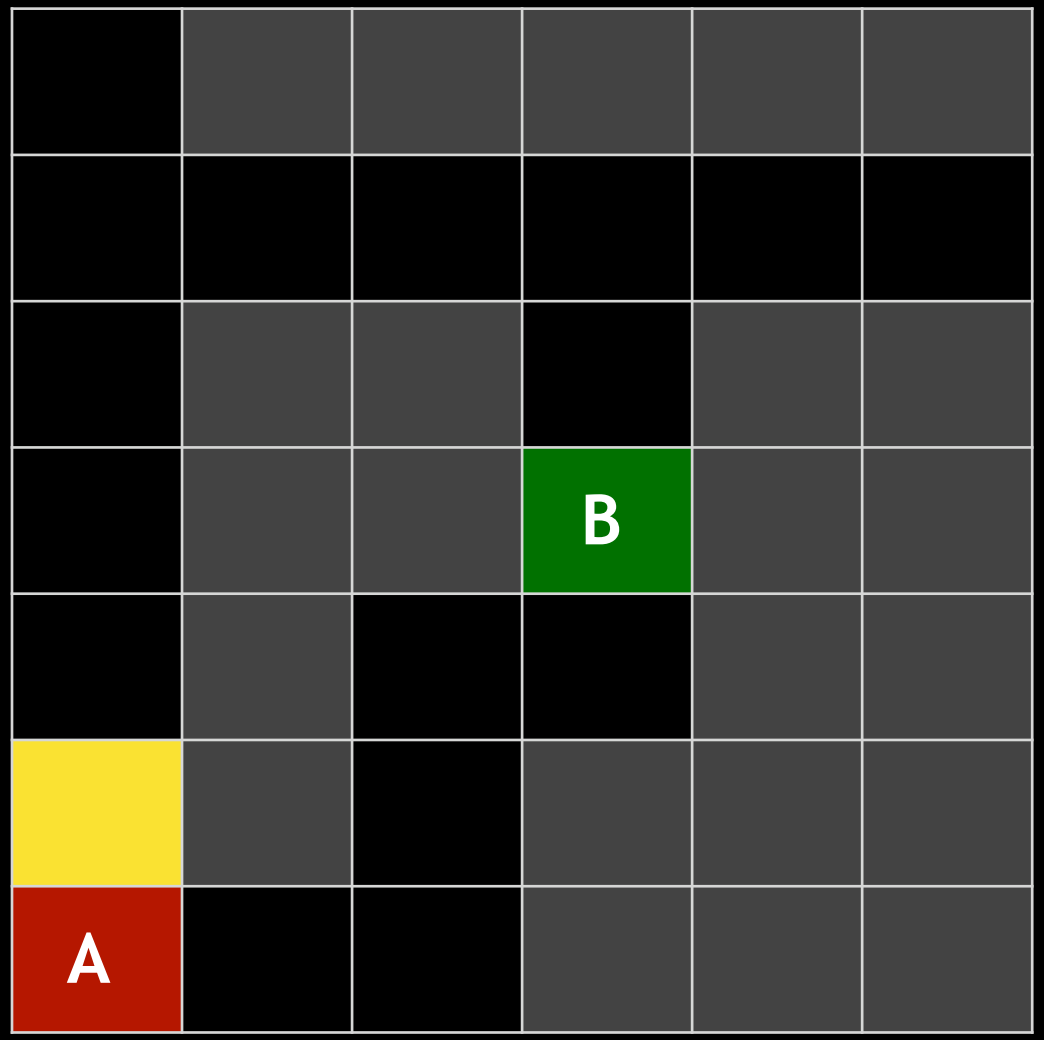
depth-first search

algoritmo de busca que sempre expande o nó mais profundo na margem

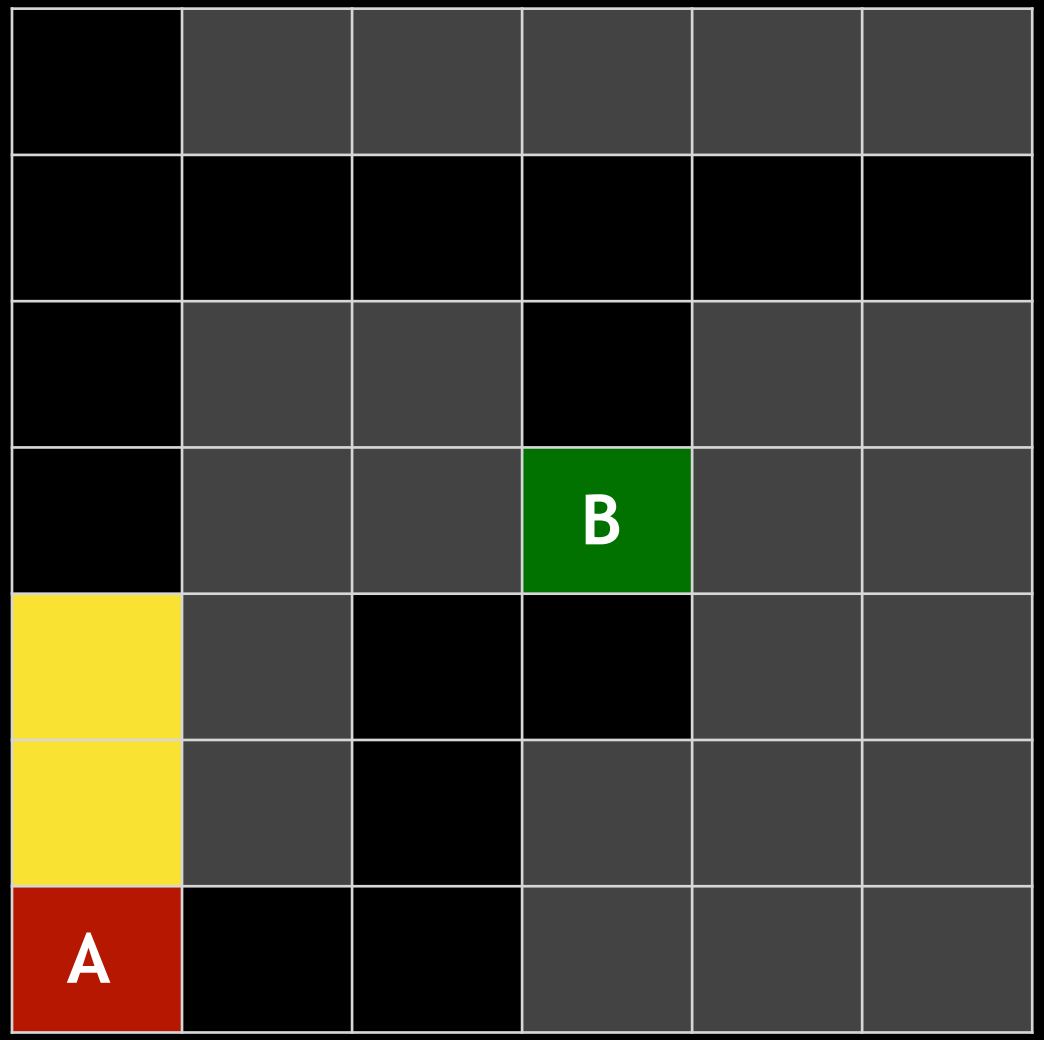
Depth-First Search
turn to left



Depth-First Search



Depth-First Search



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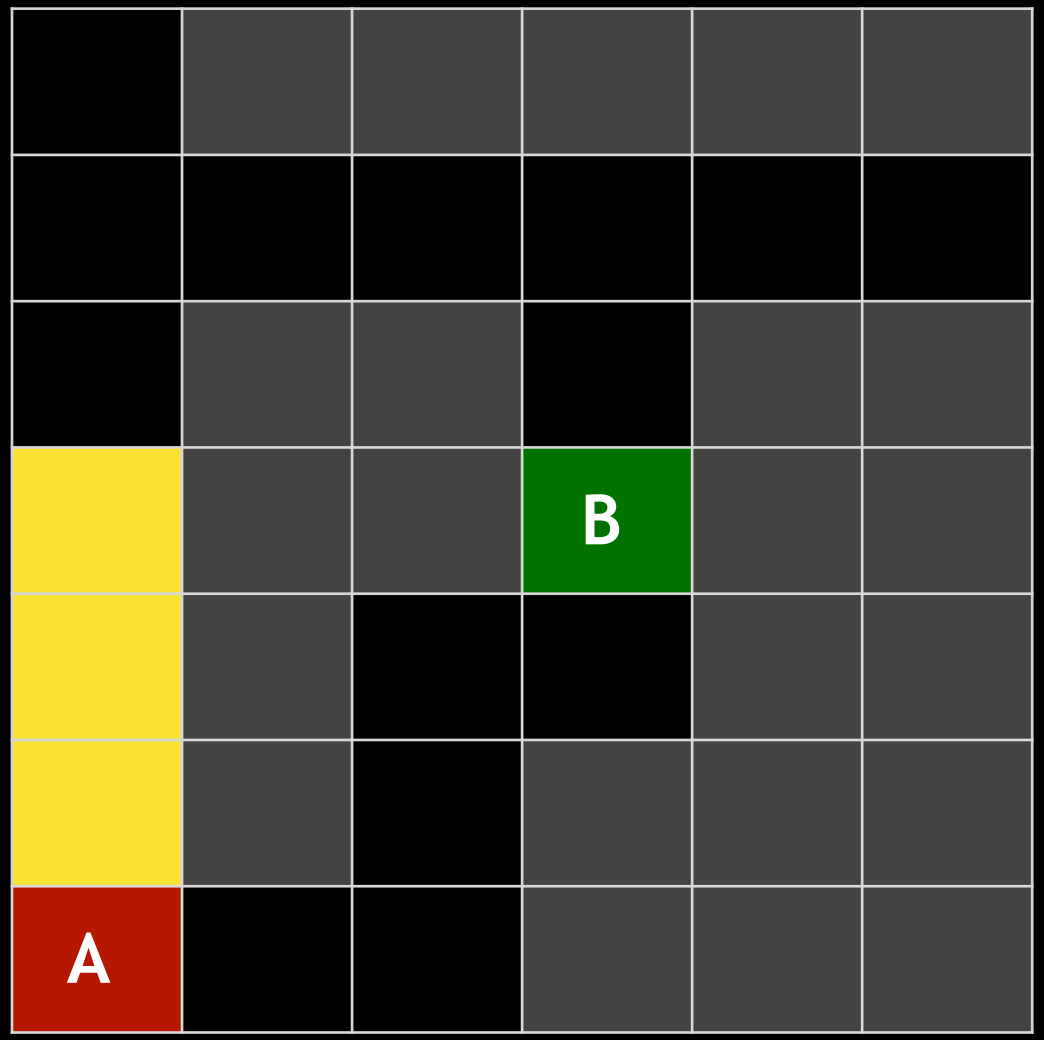
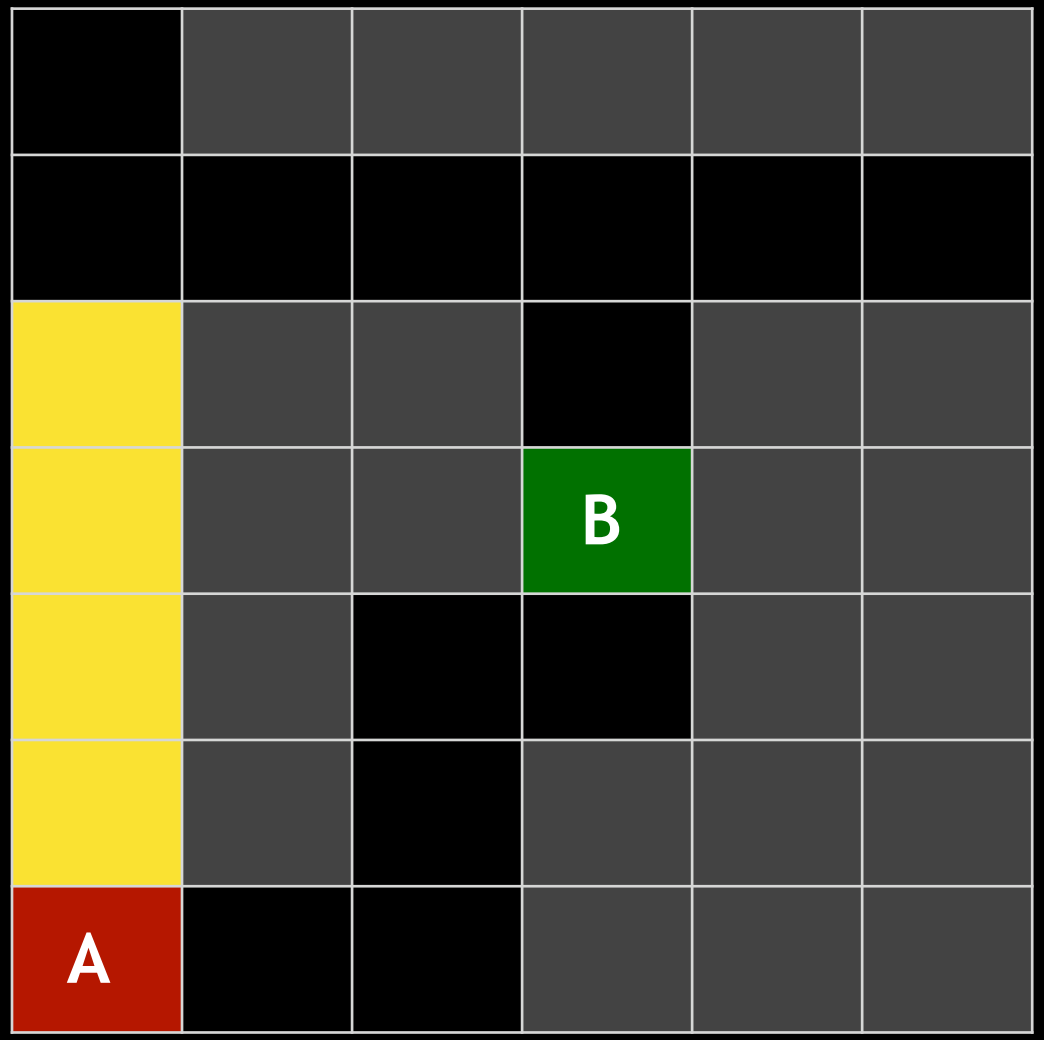
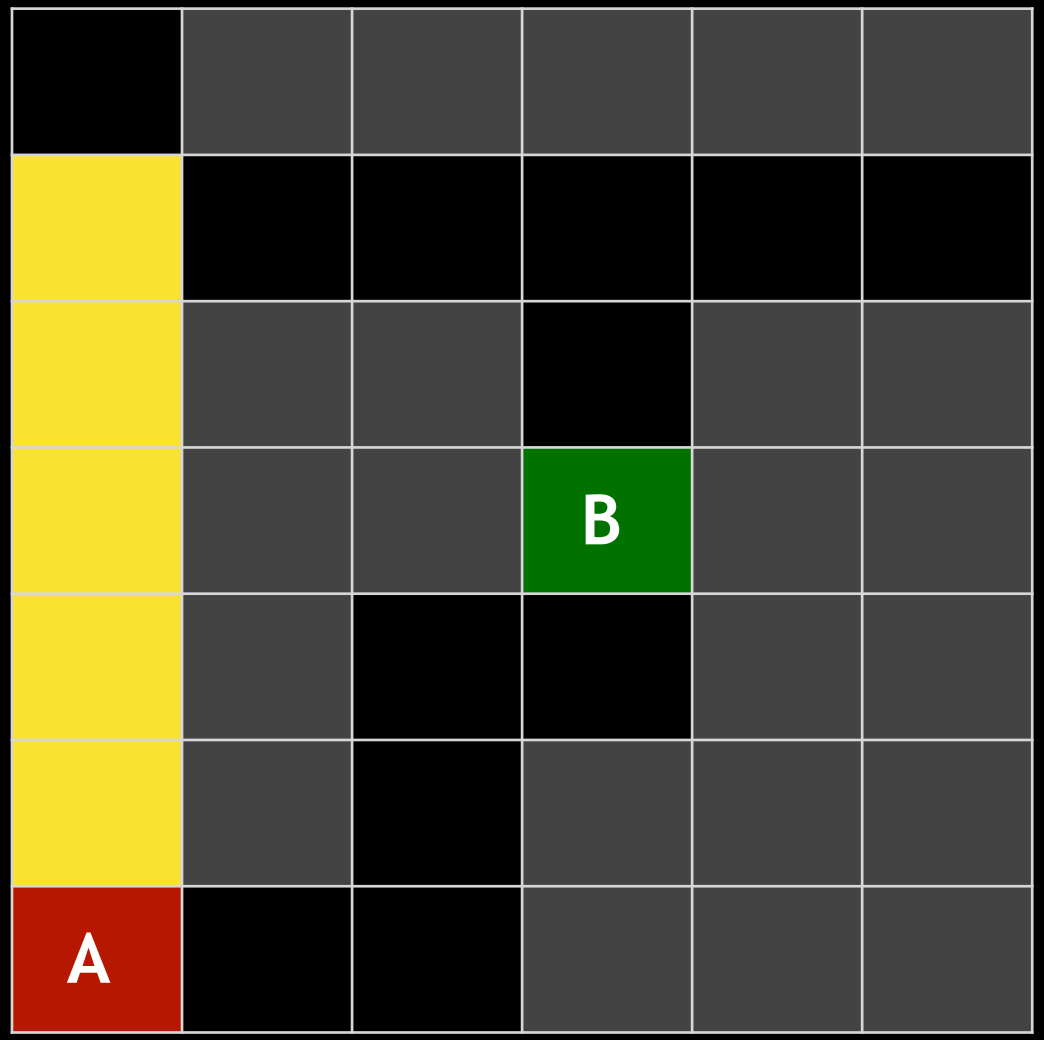


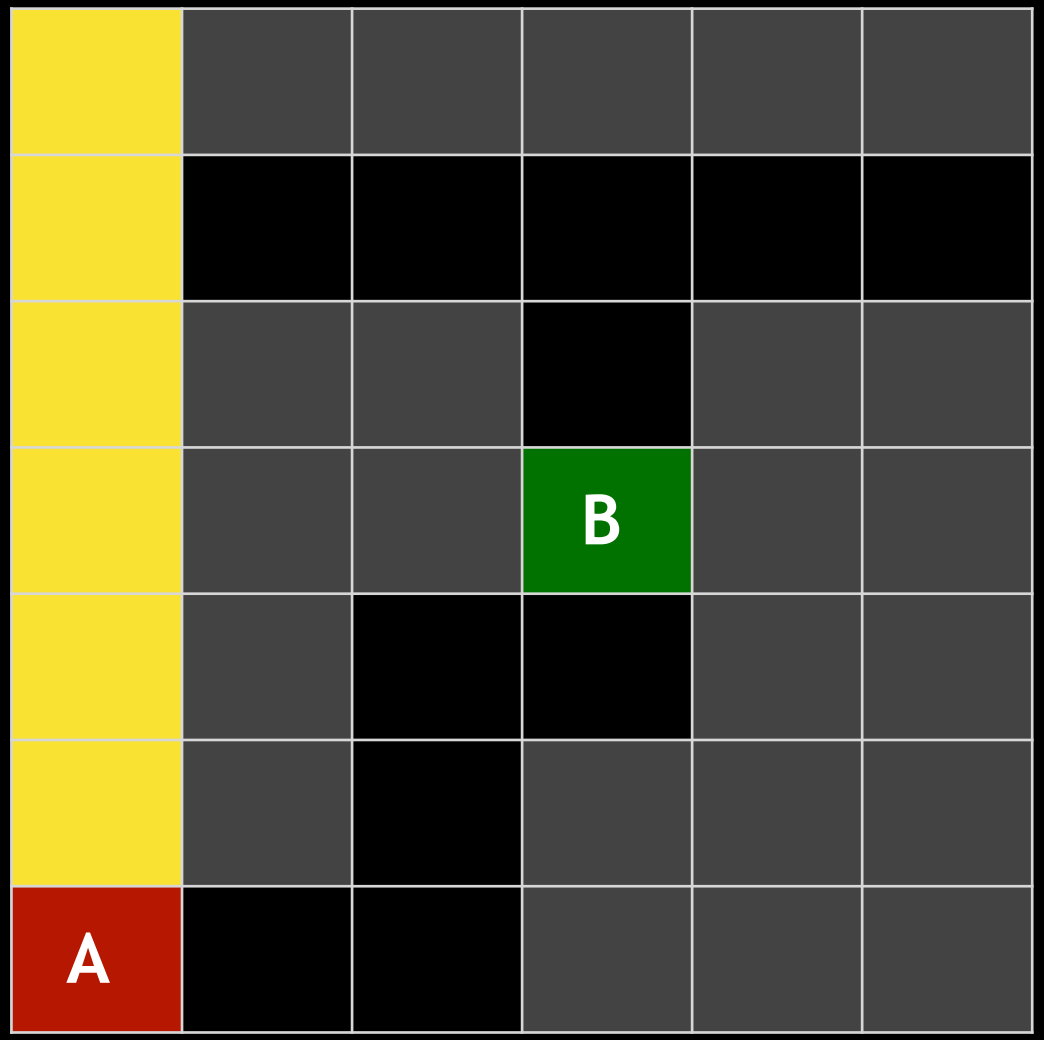
Depth-First Search



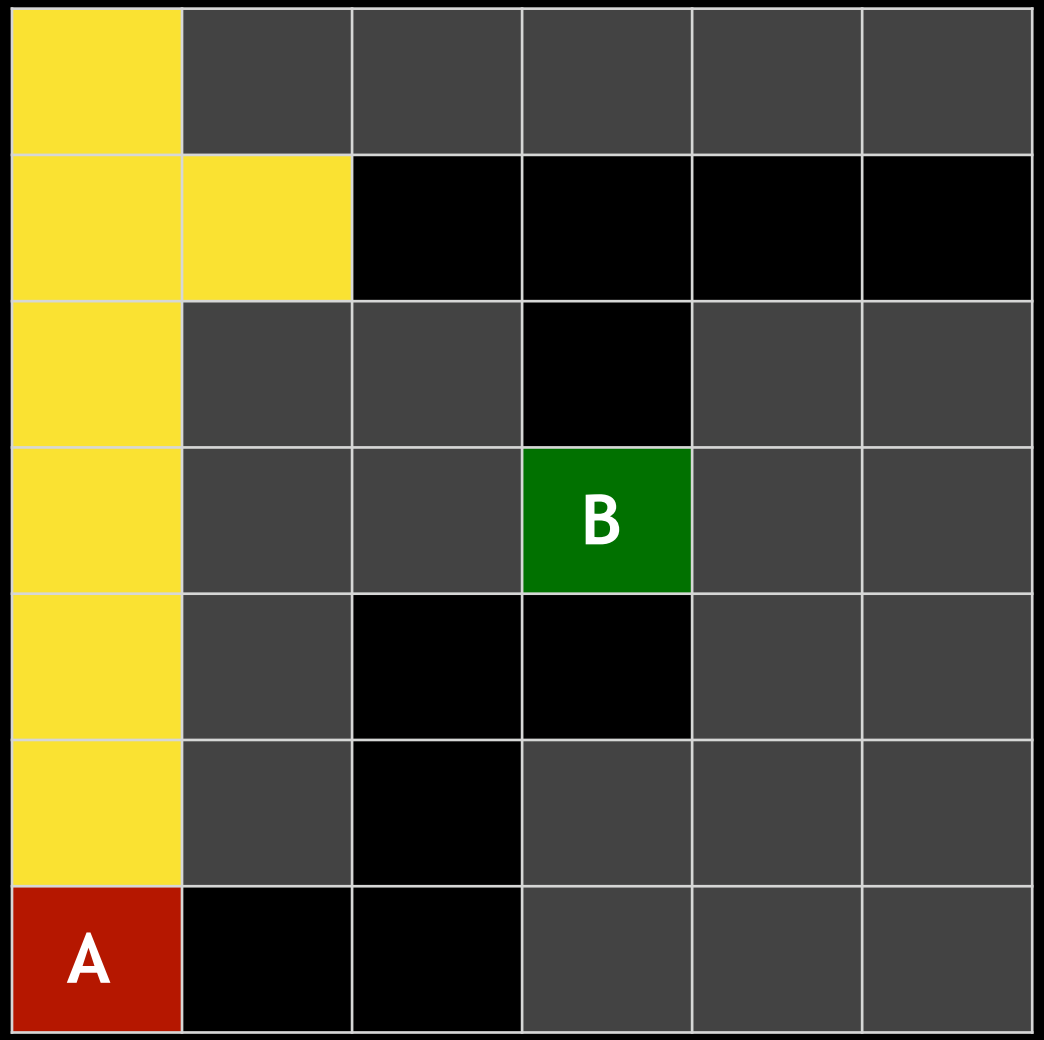
Depth-First Search



Depth-First Search



Depth-First Search



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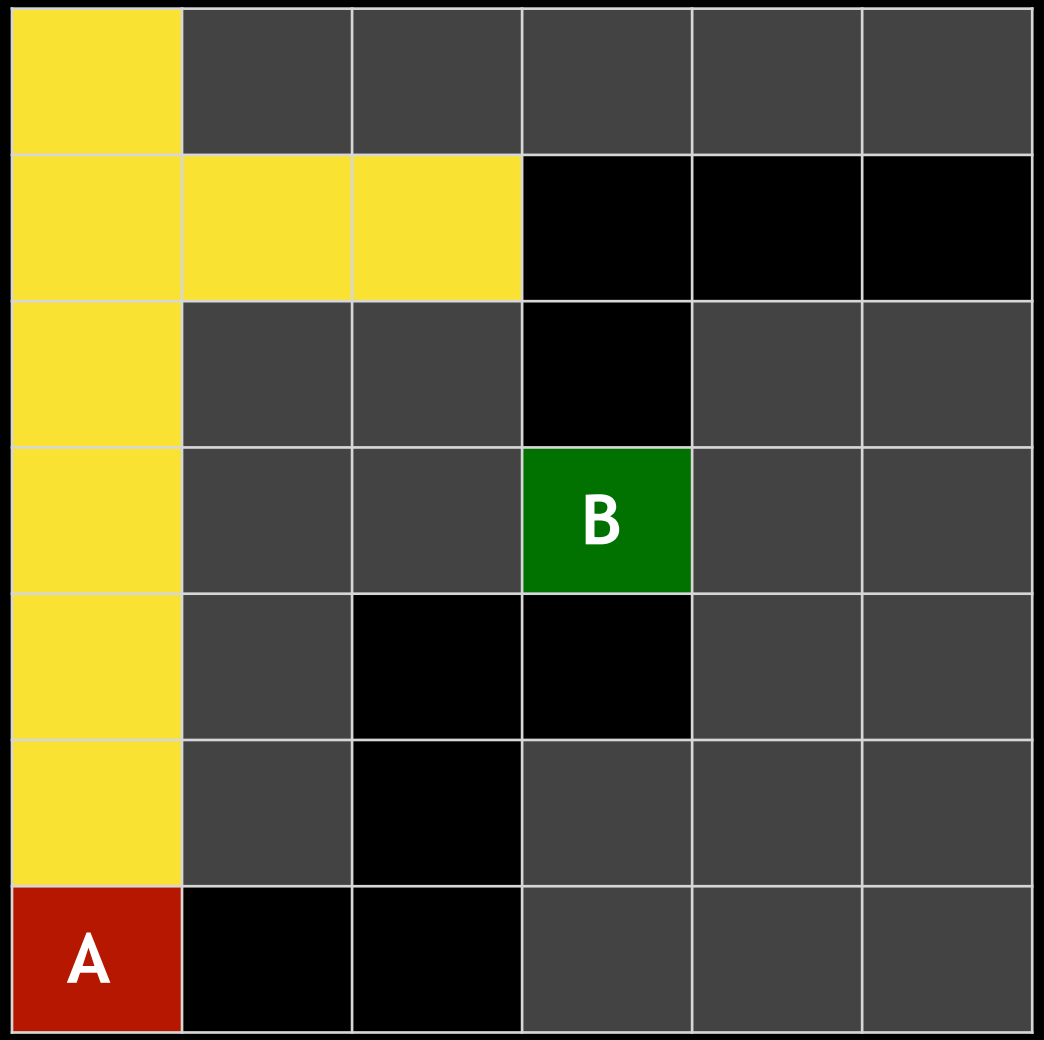
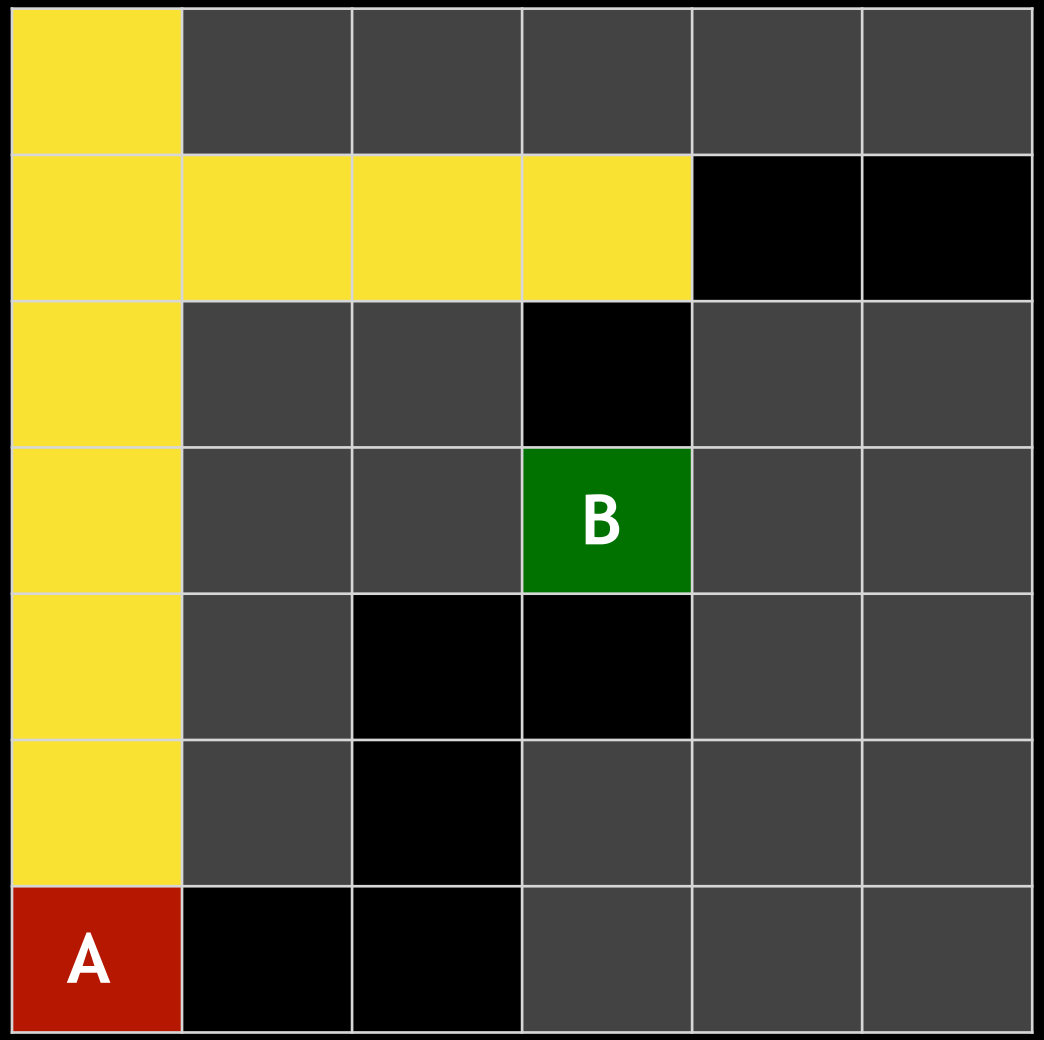
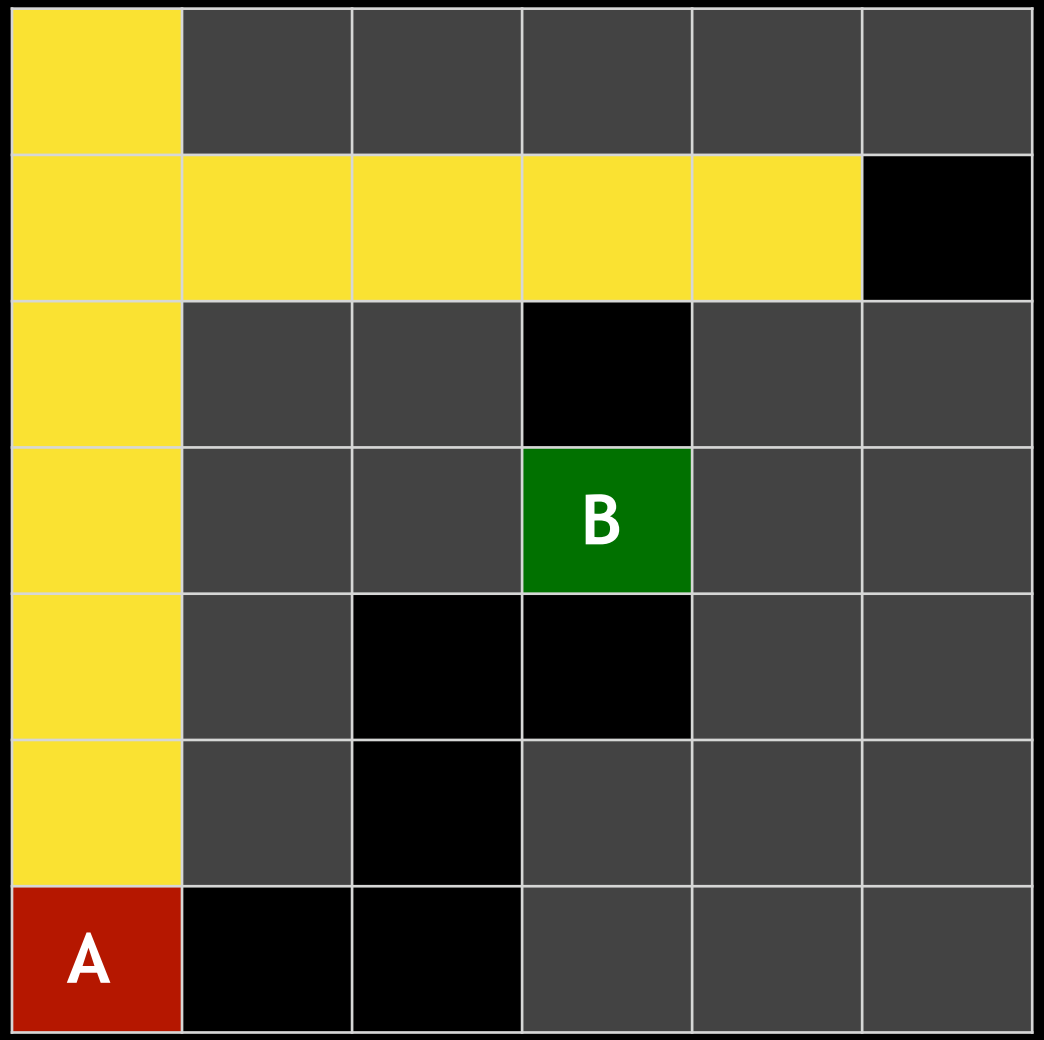


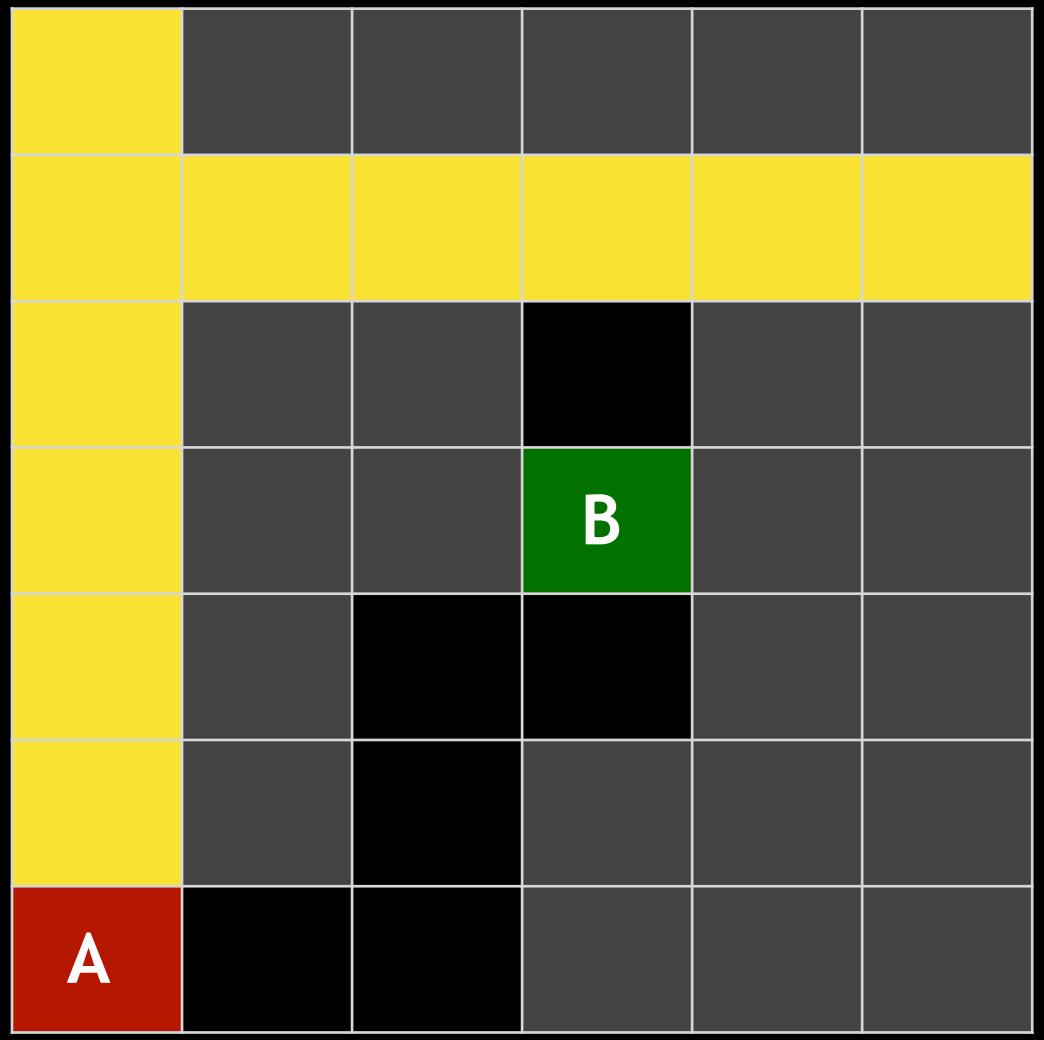
Depth-First Search



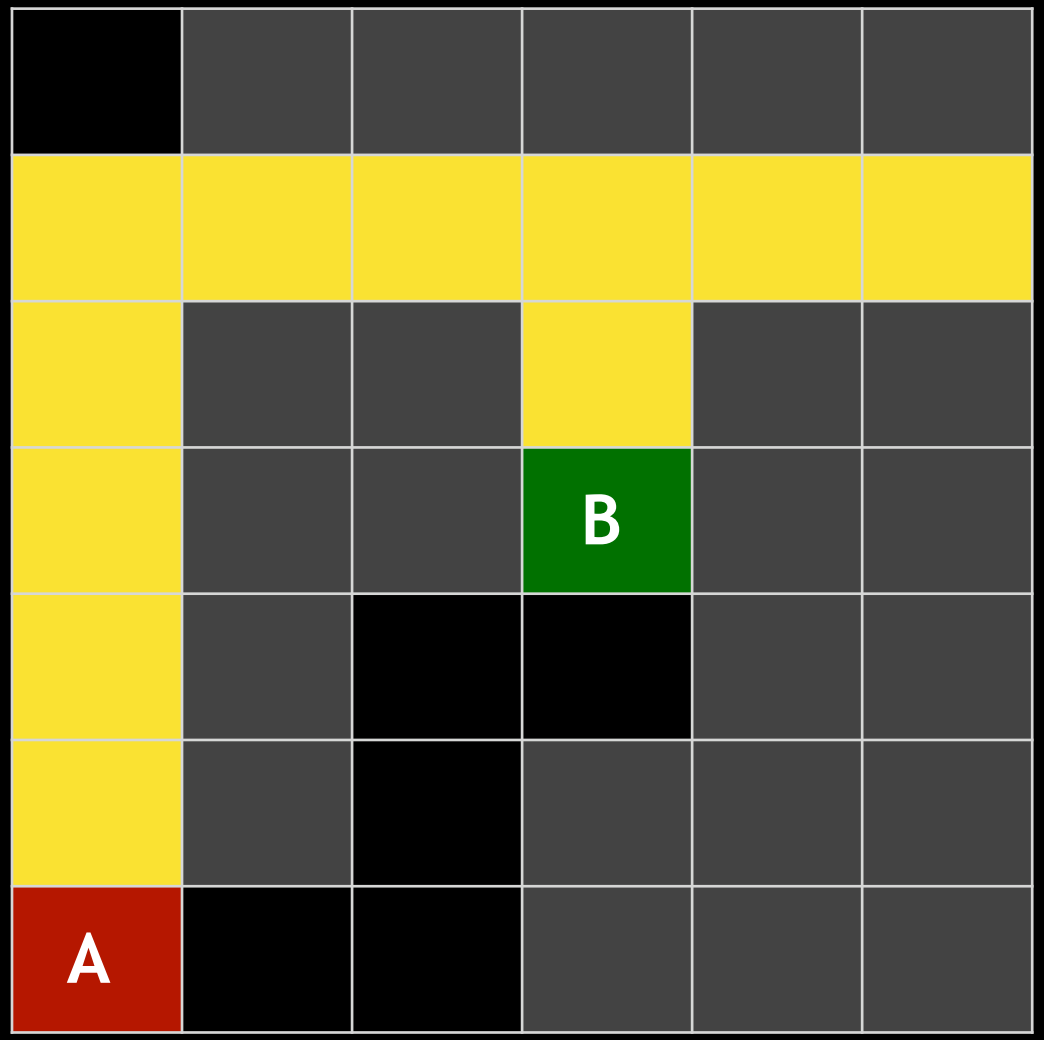
Depth-First Search



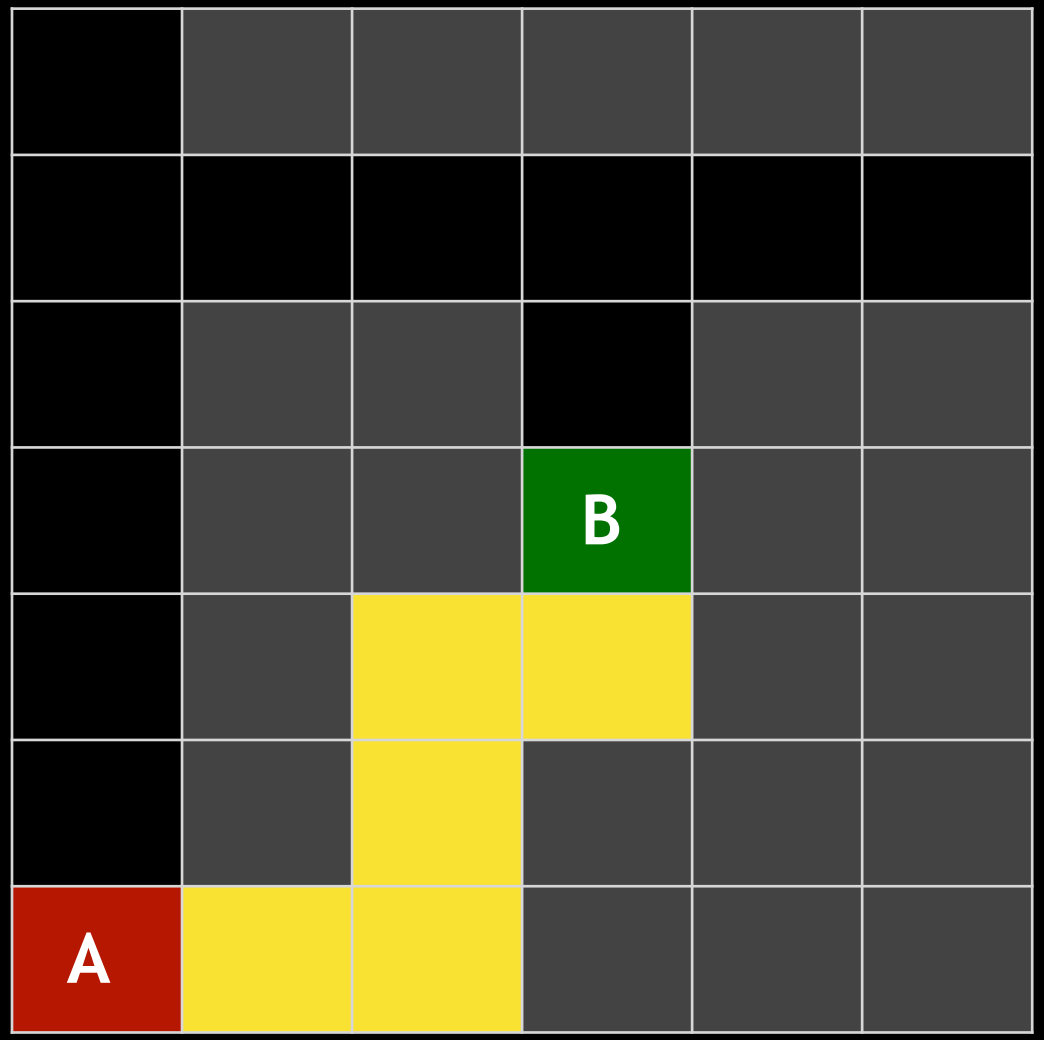
Depth-First Search



Depth-First Search



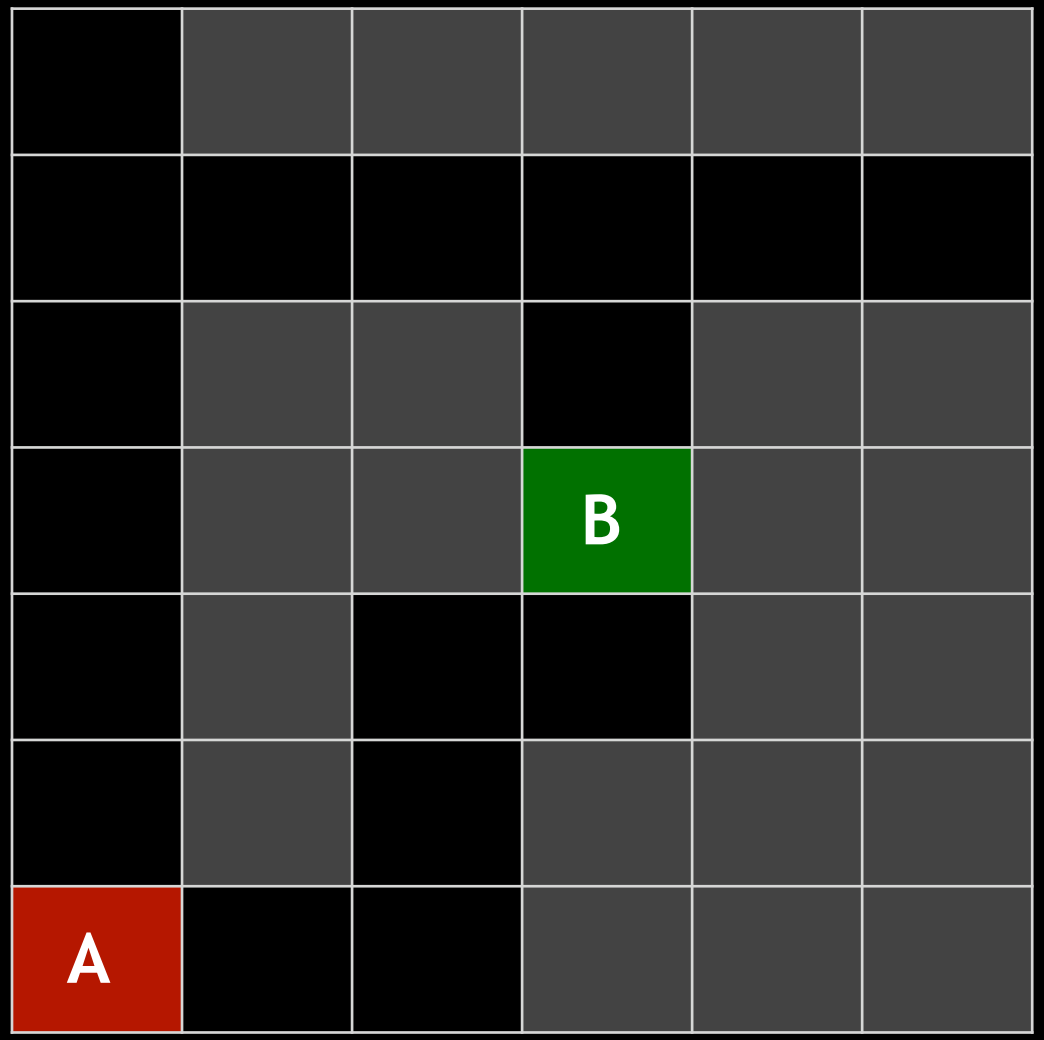
Depth-First Search
turn to right



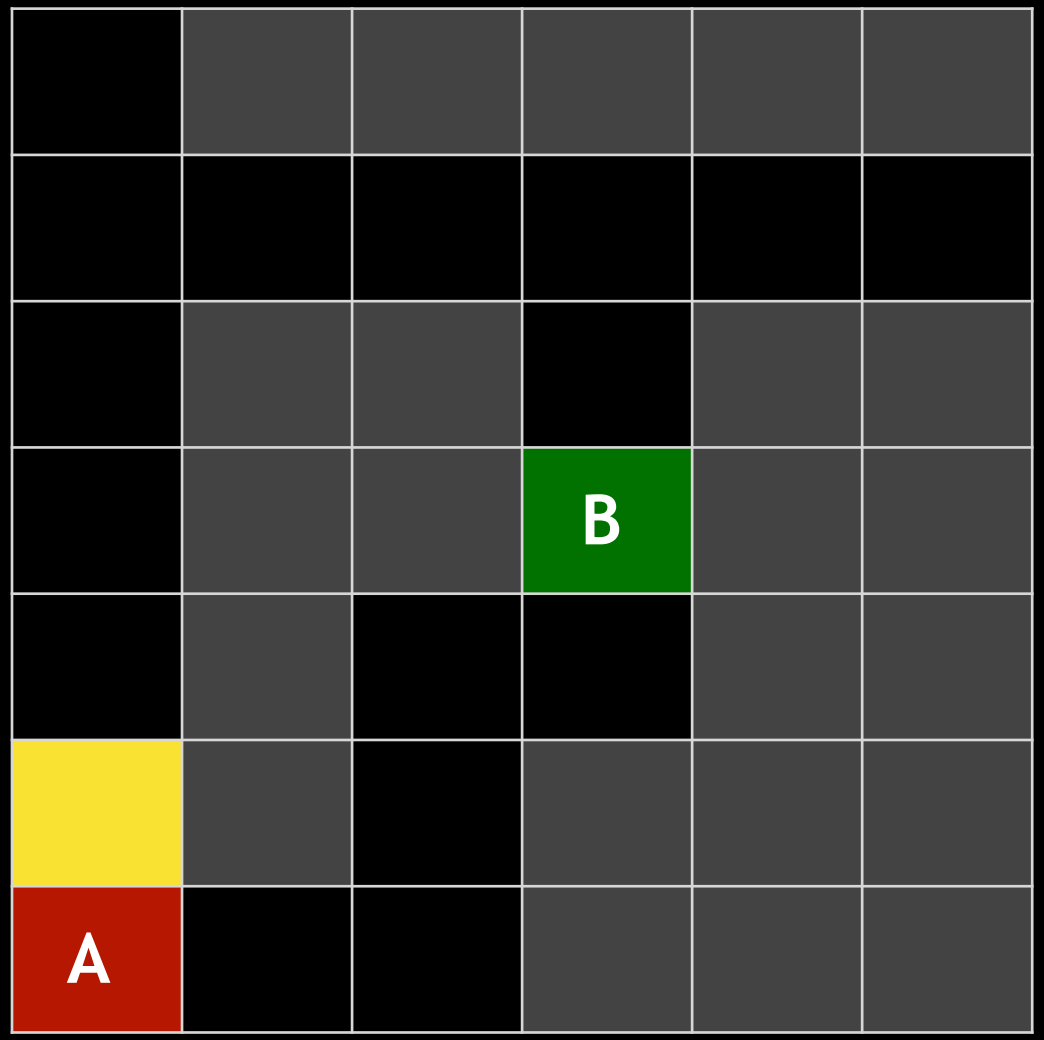
breadth-first search

algoritmo de busca que sempre expande o nó mais raso na margem

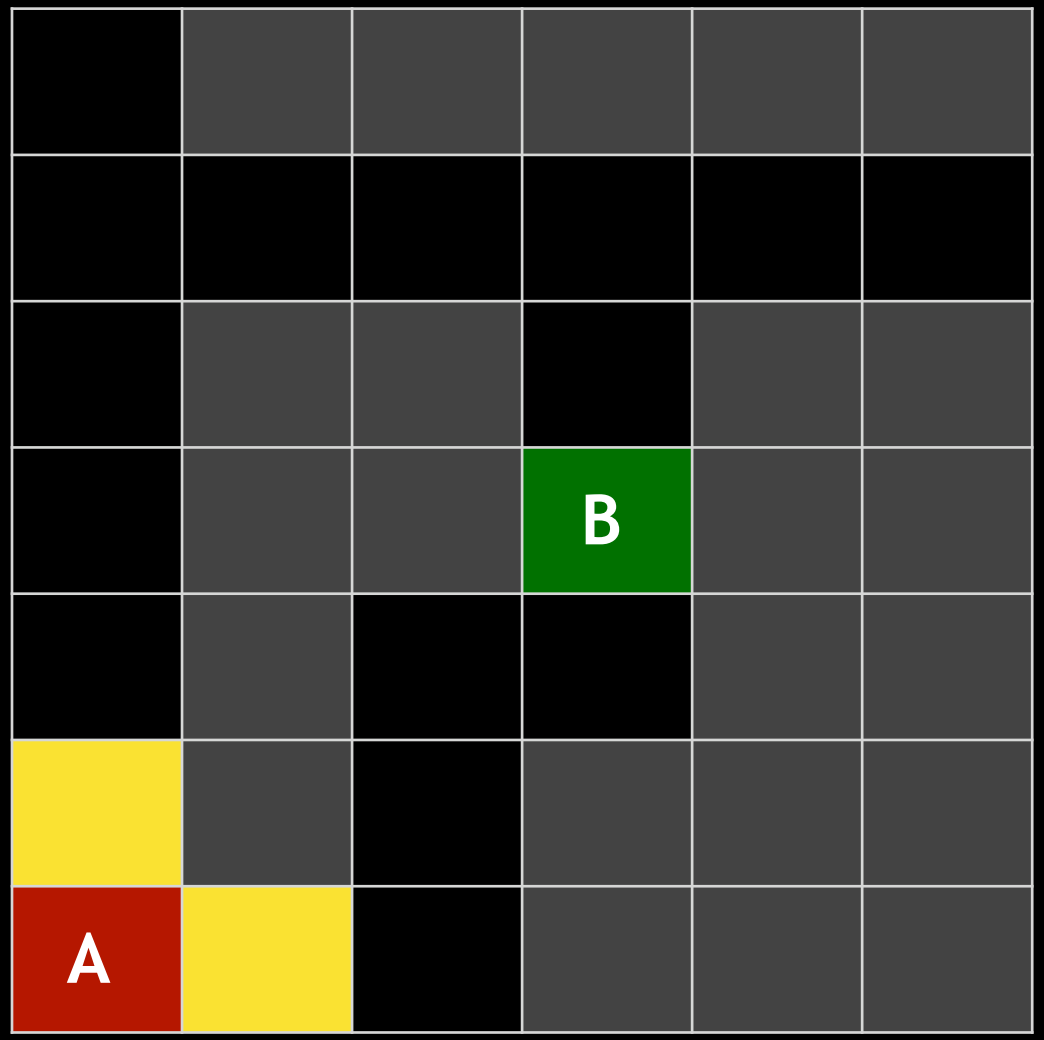
Breadth-First Search



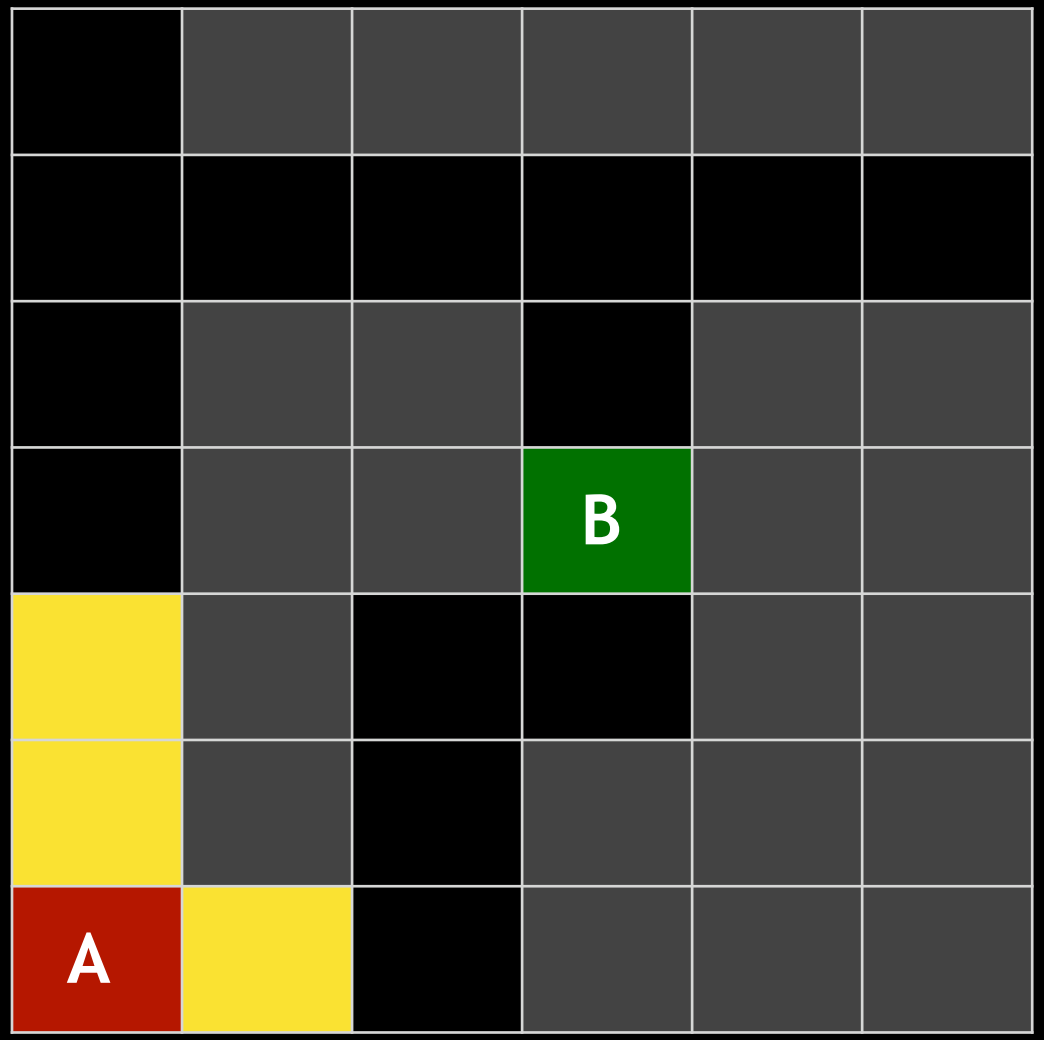
Breadth-First Search



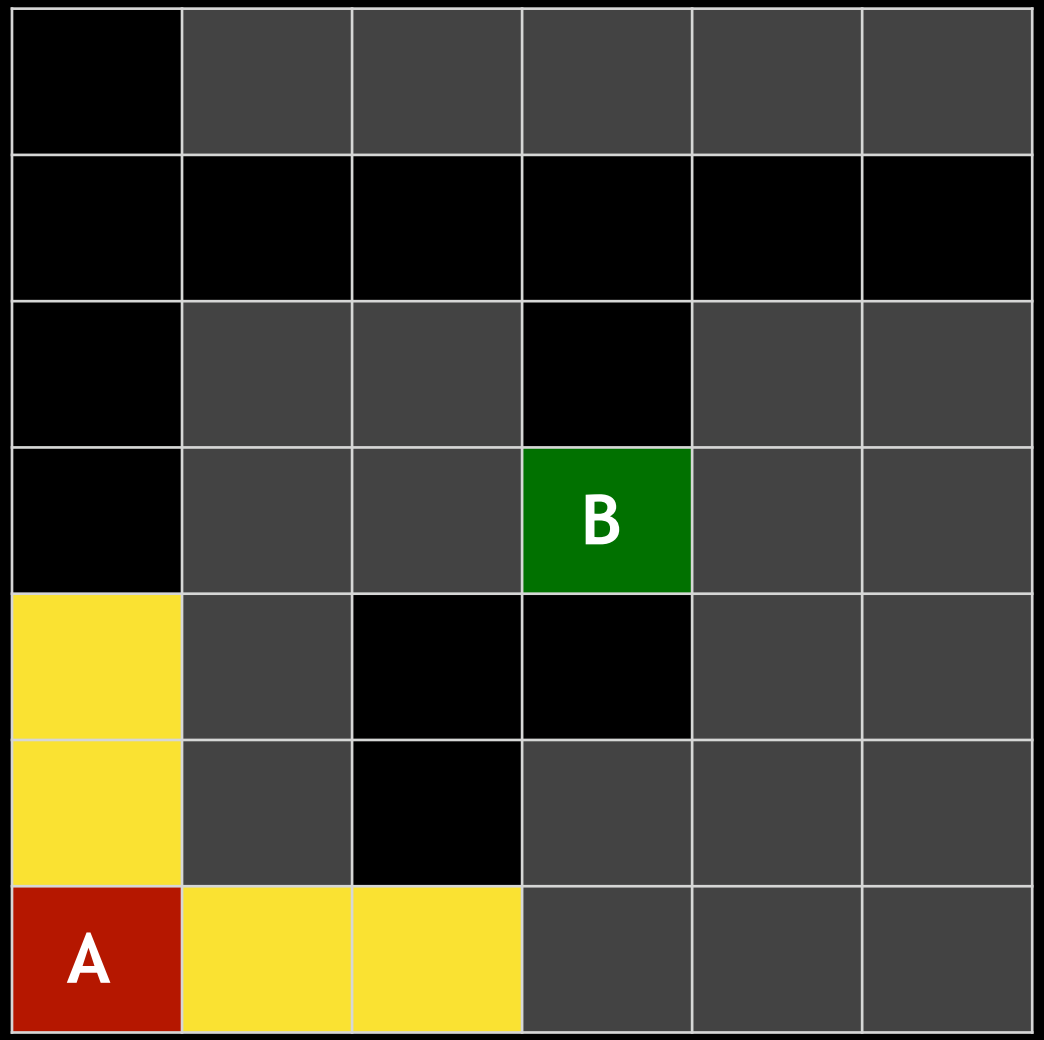
Breadth-First Search



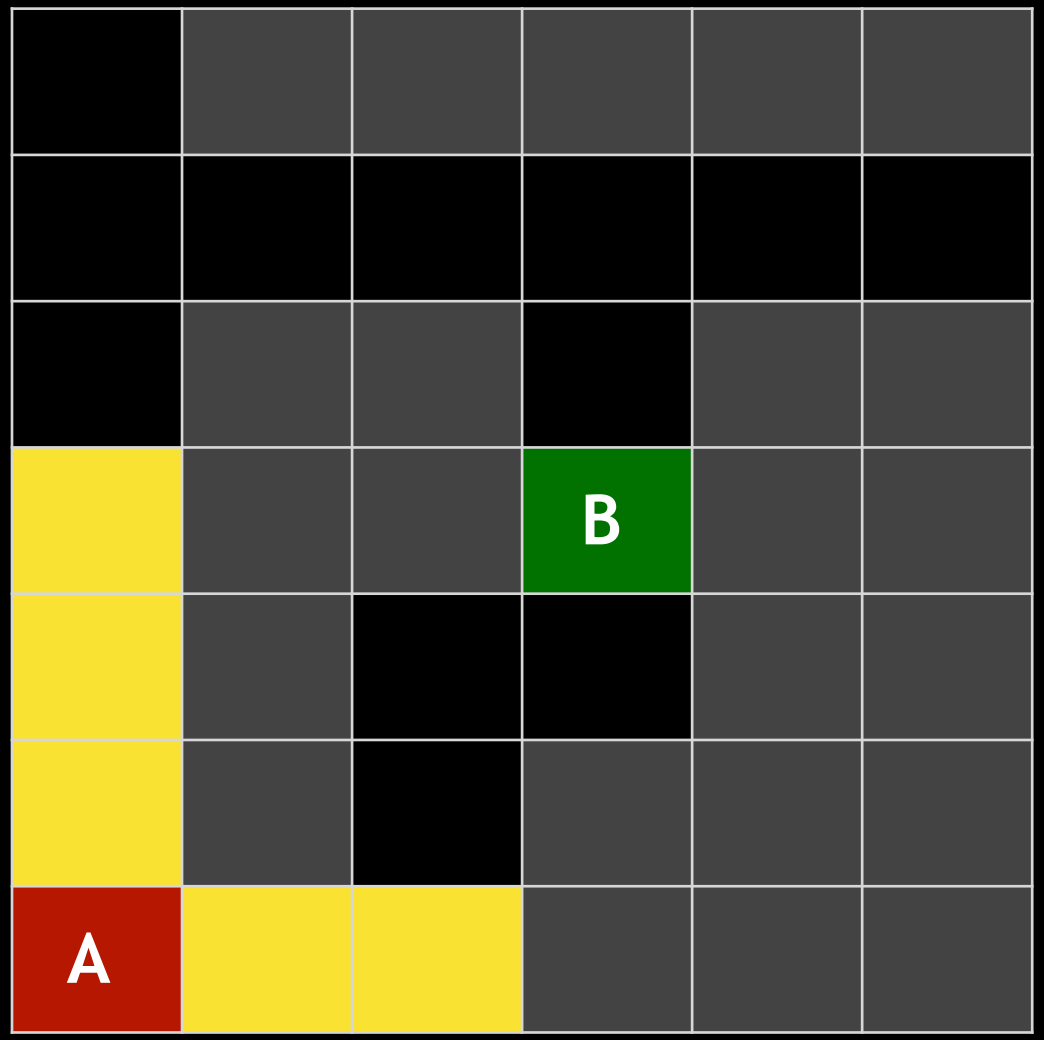
Breadth-First Search



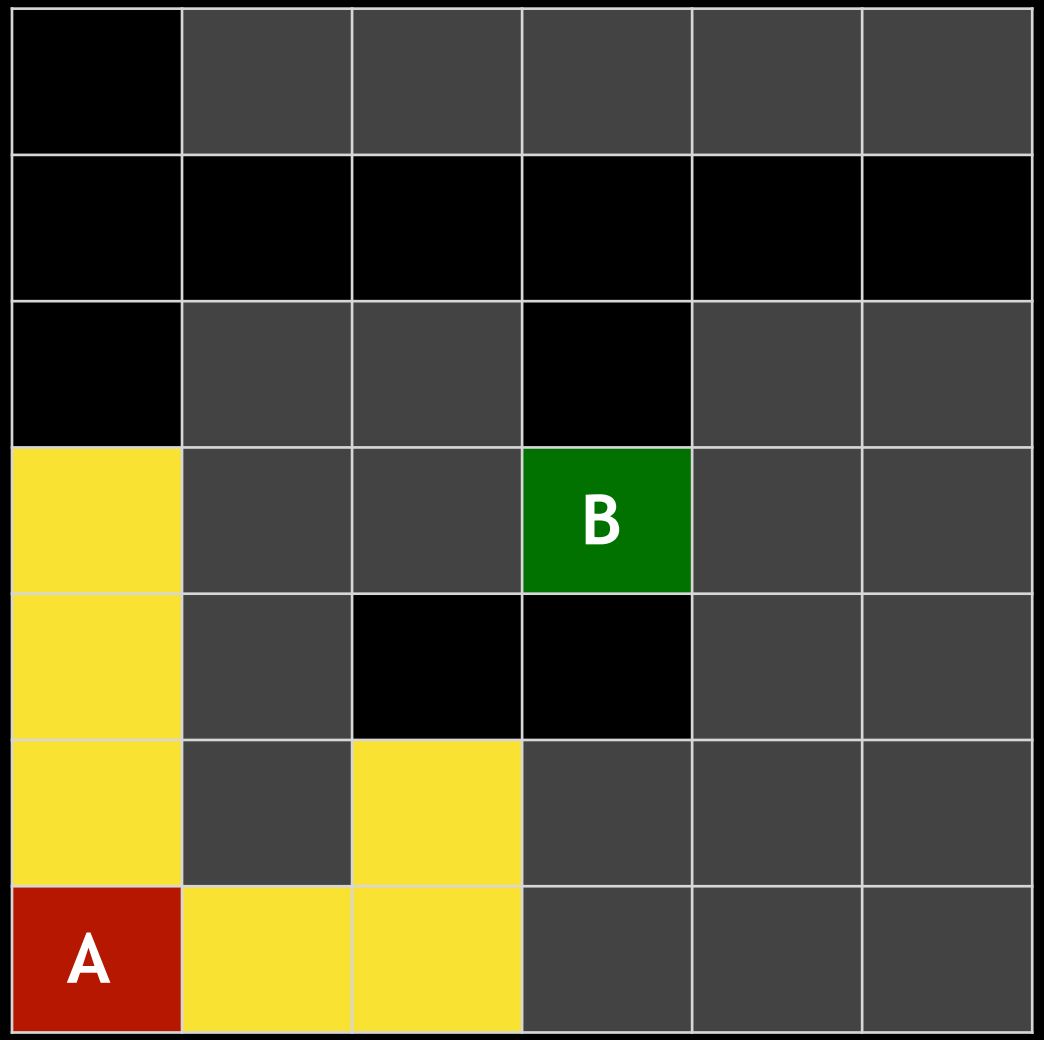
Breadth-First Search



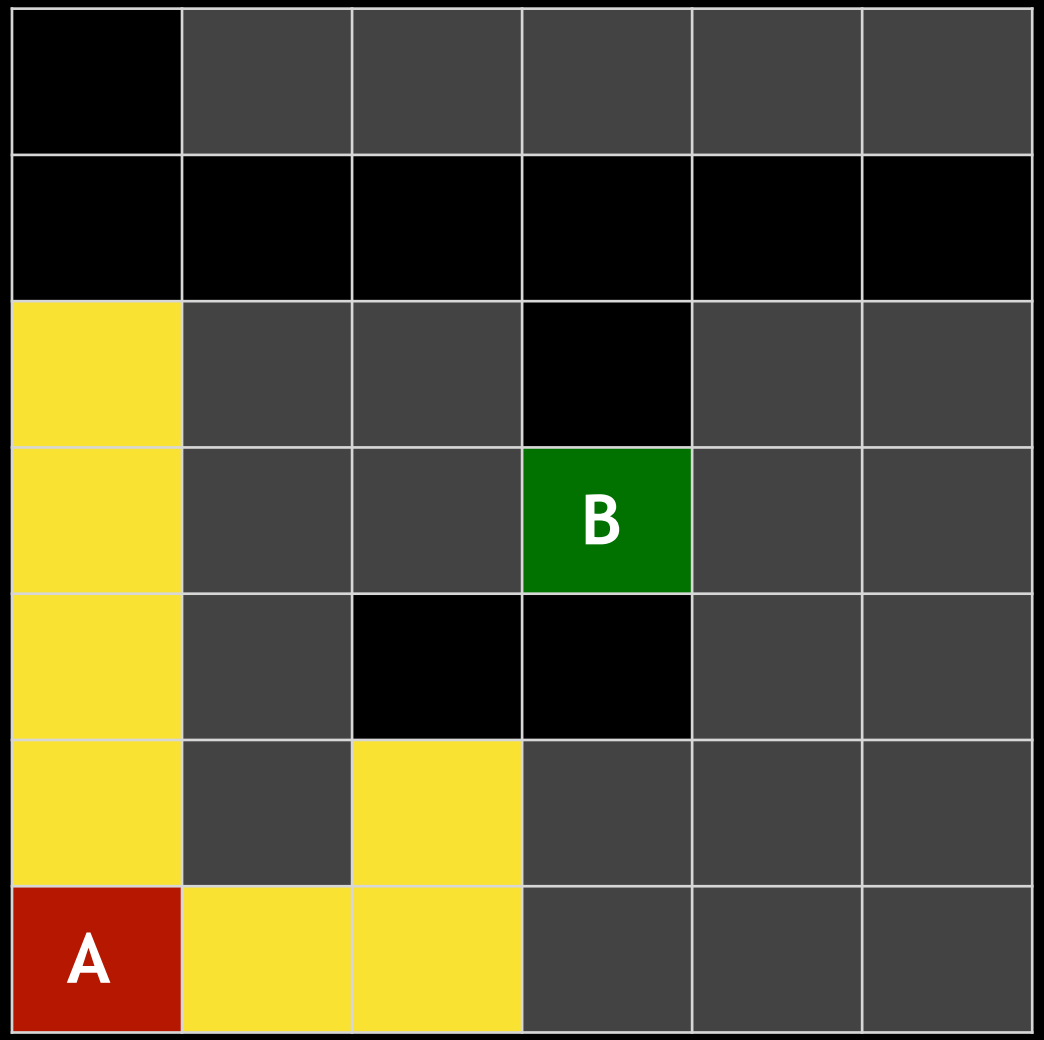
Breadth-First Search



Breadth-First Search



Breadth-First Search



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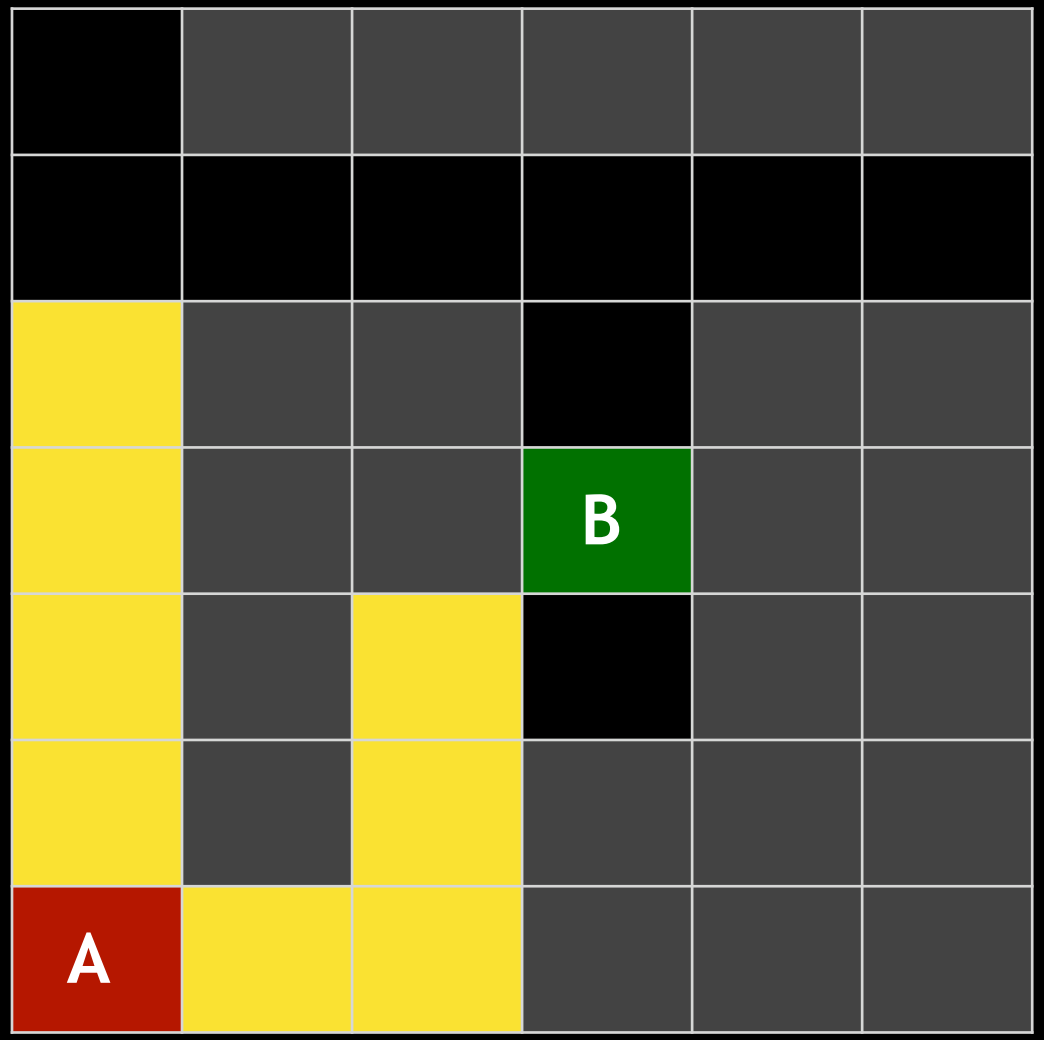

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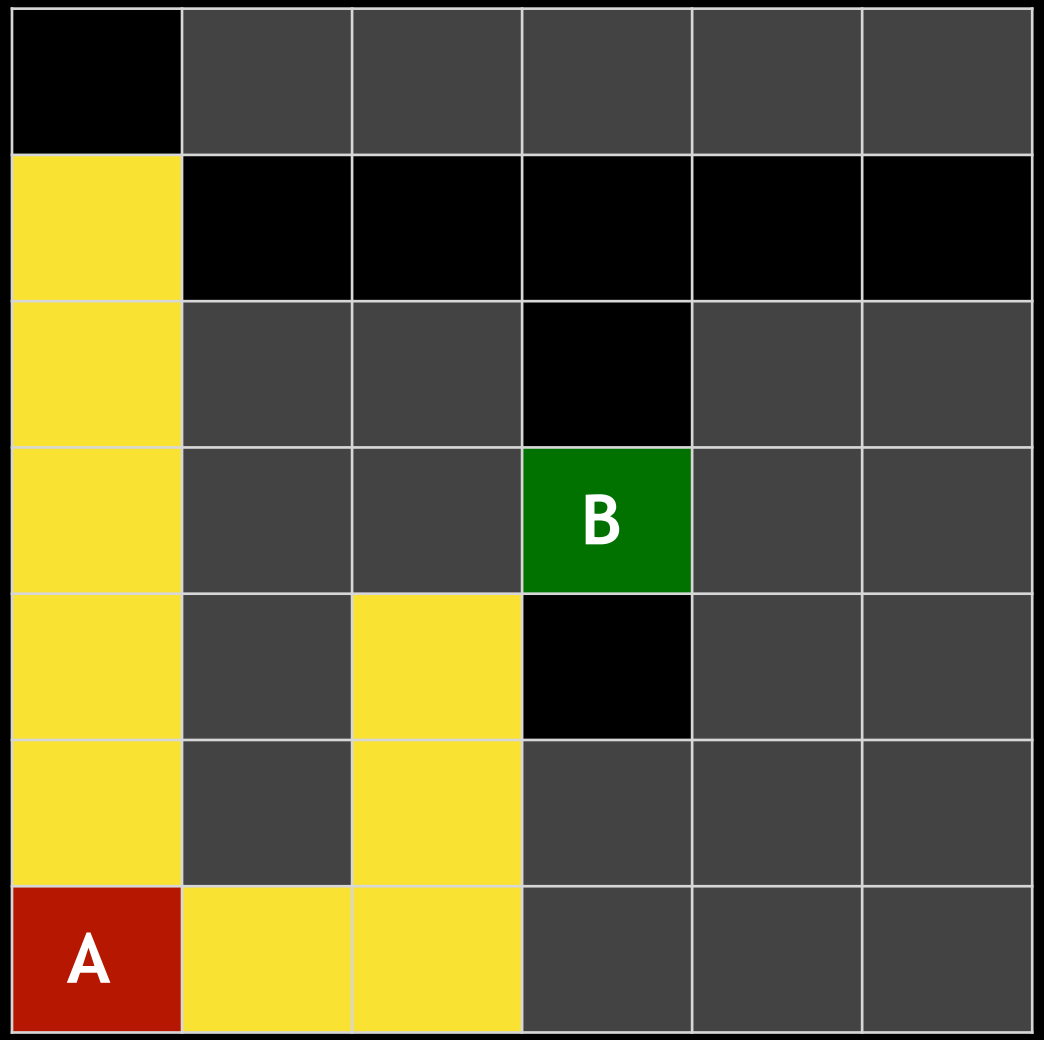

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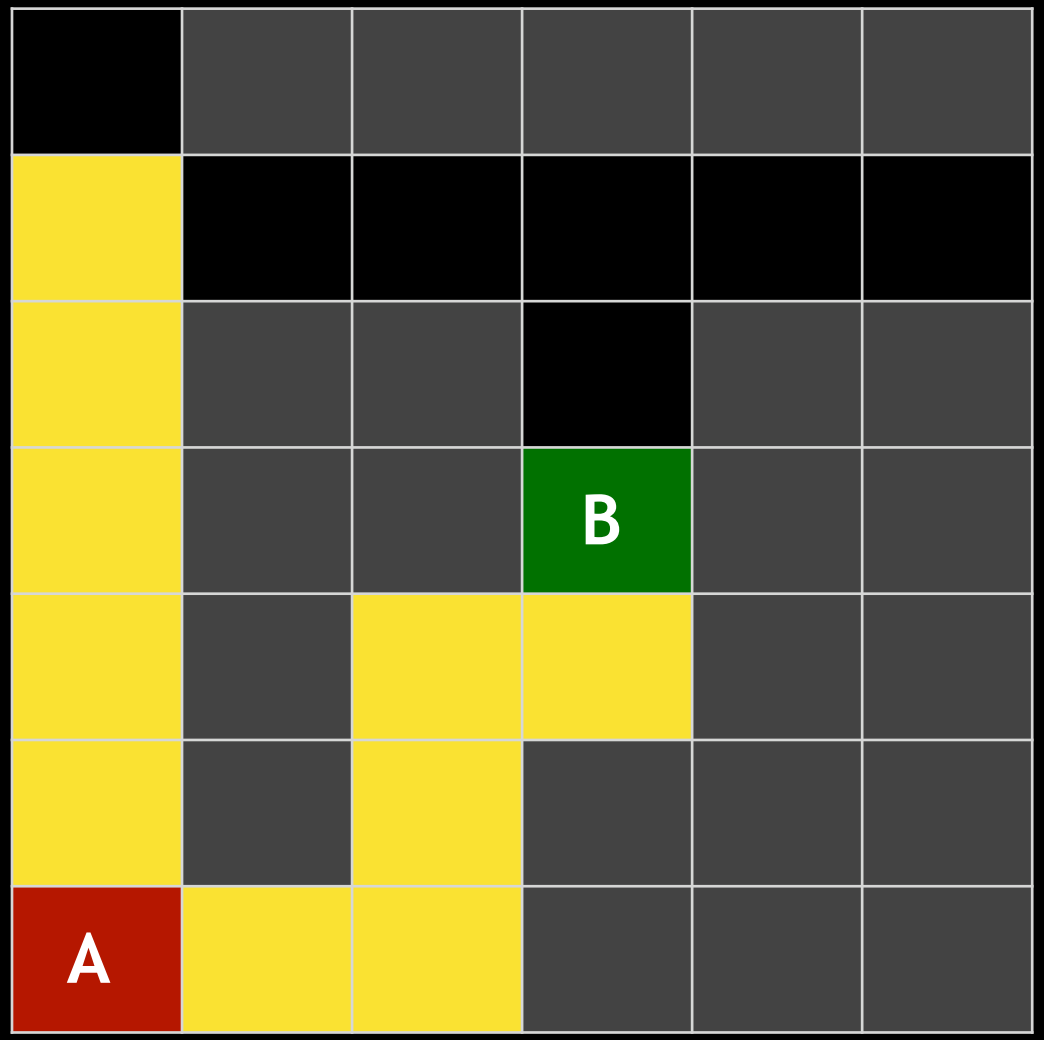
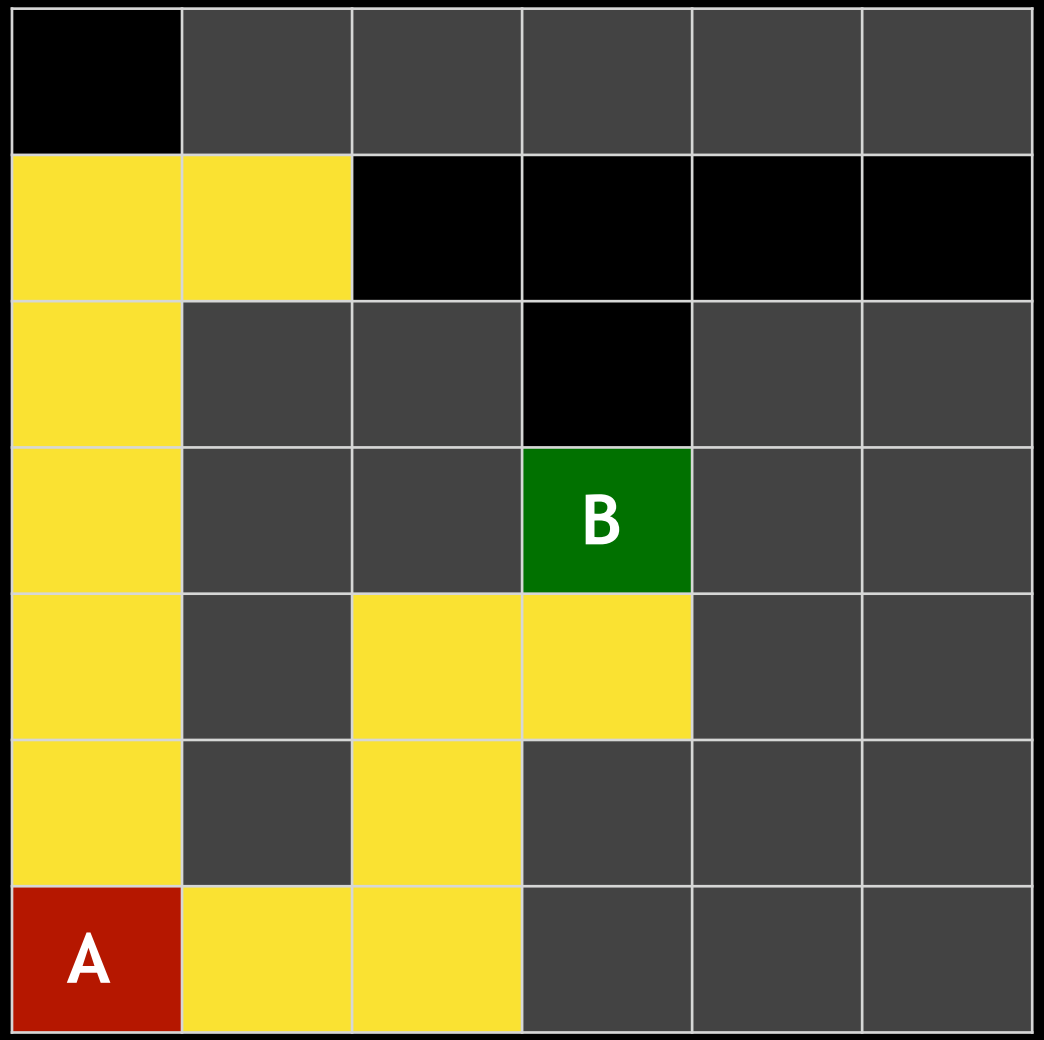


Breadth-First Search



deaph-first search

x

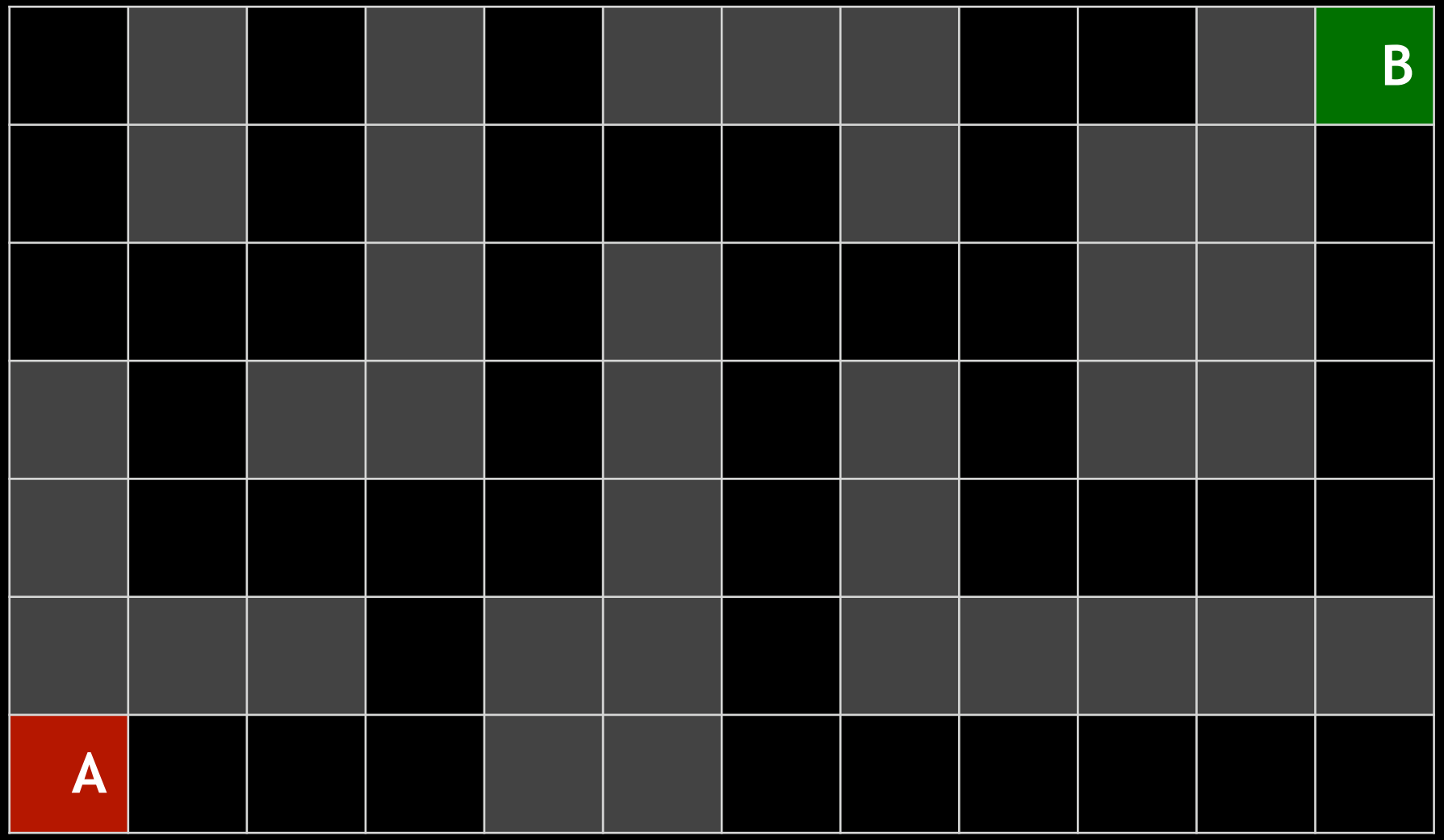
breadth-first Search

qual o melhor???

depth-first search

algoritmo de busca que sempre expande o nó mais profundo na margem

Depth-First Search



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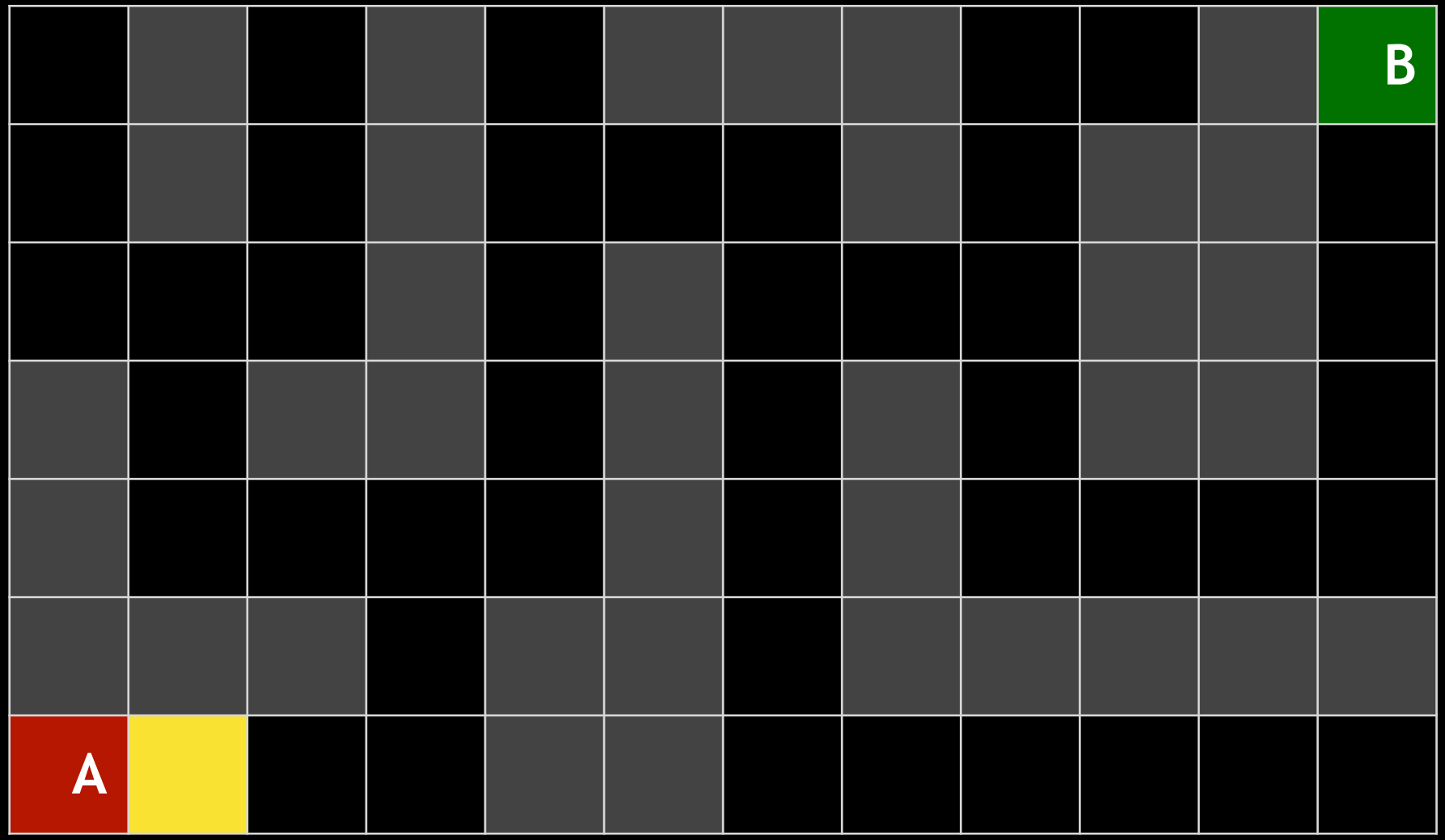

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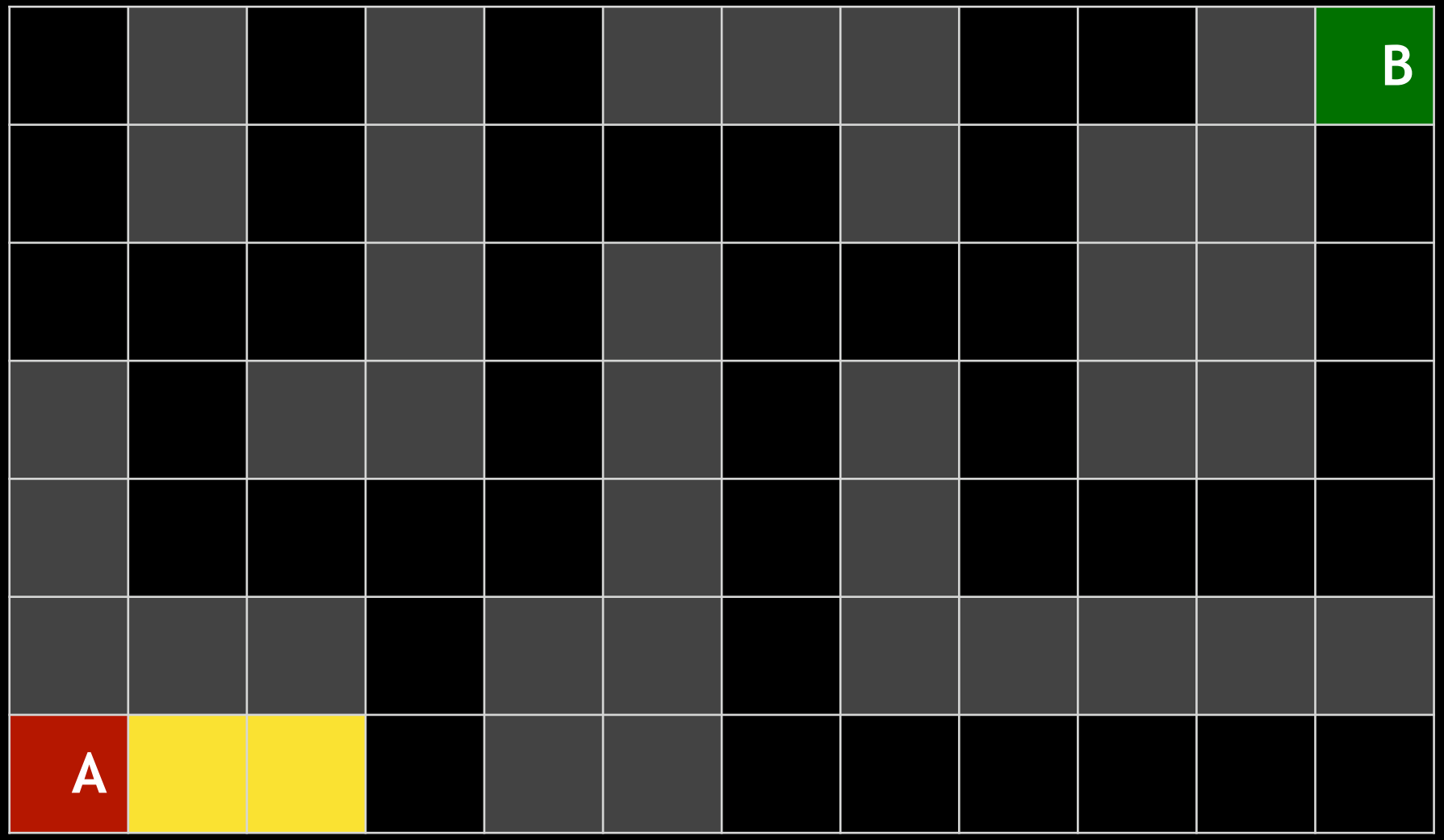

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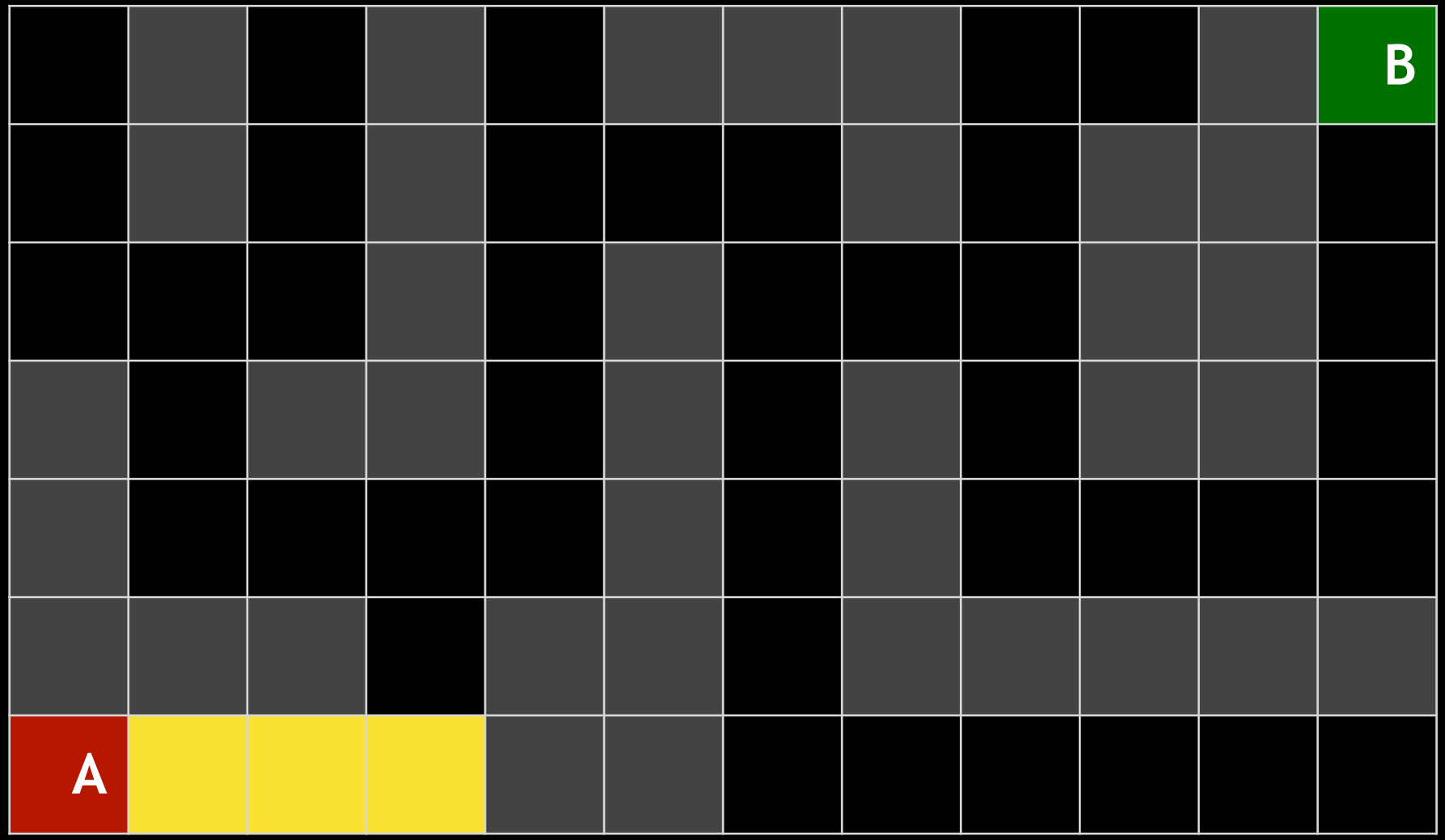

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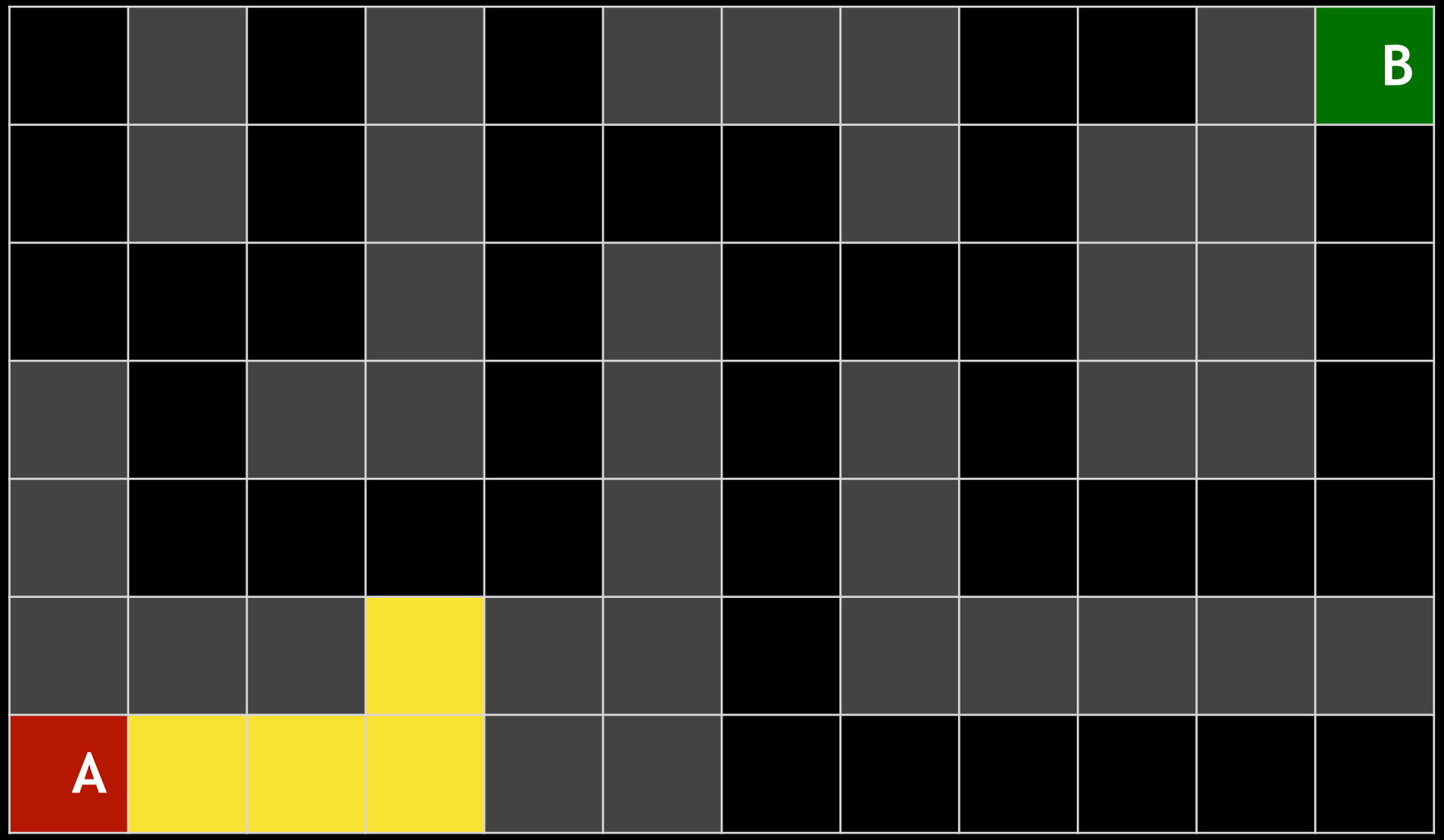

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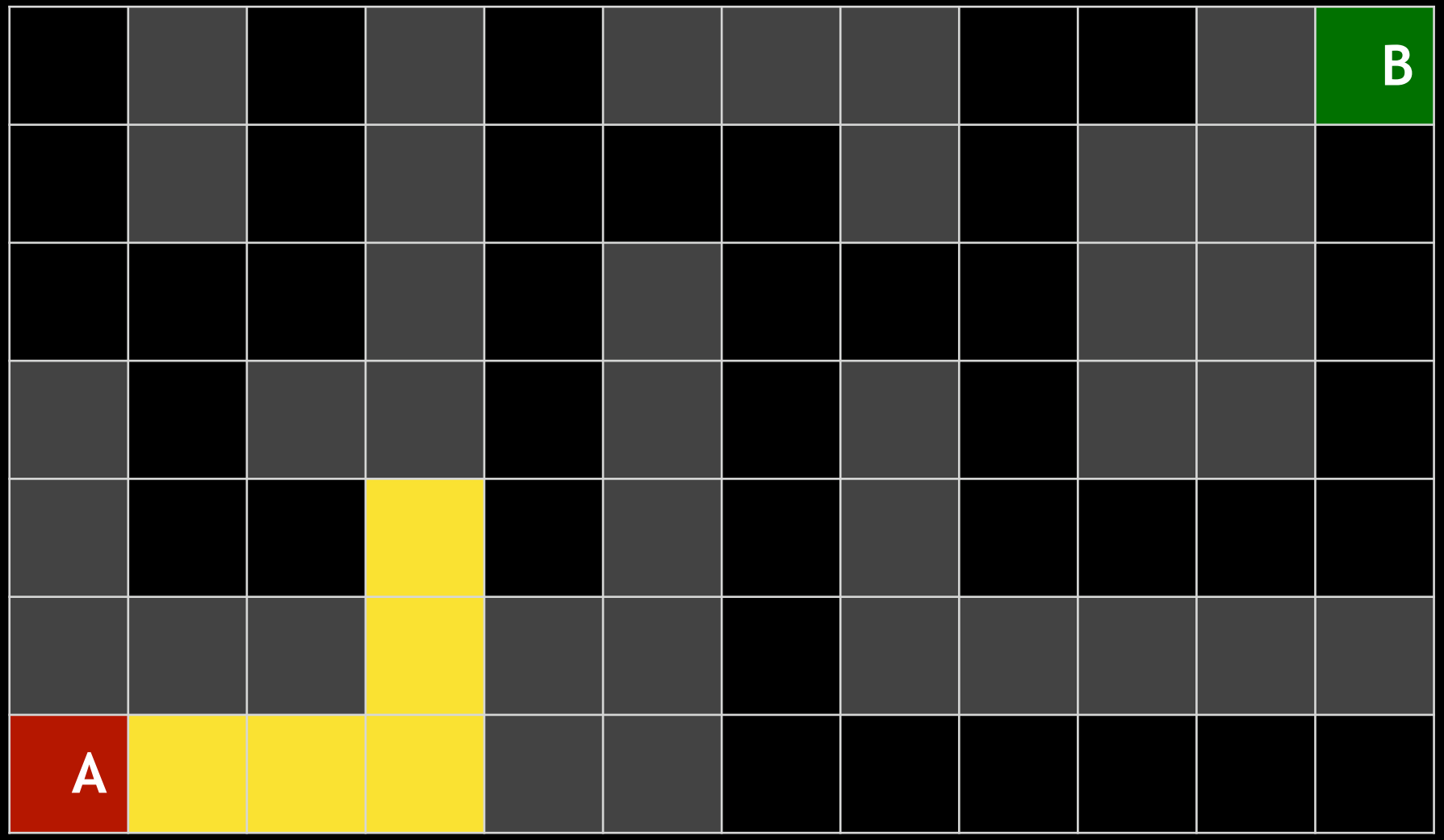

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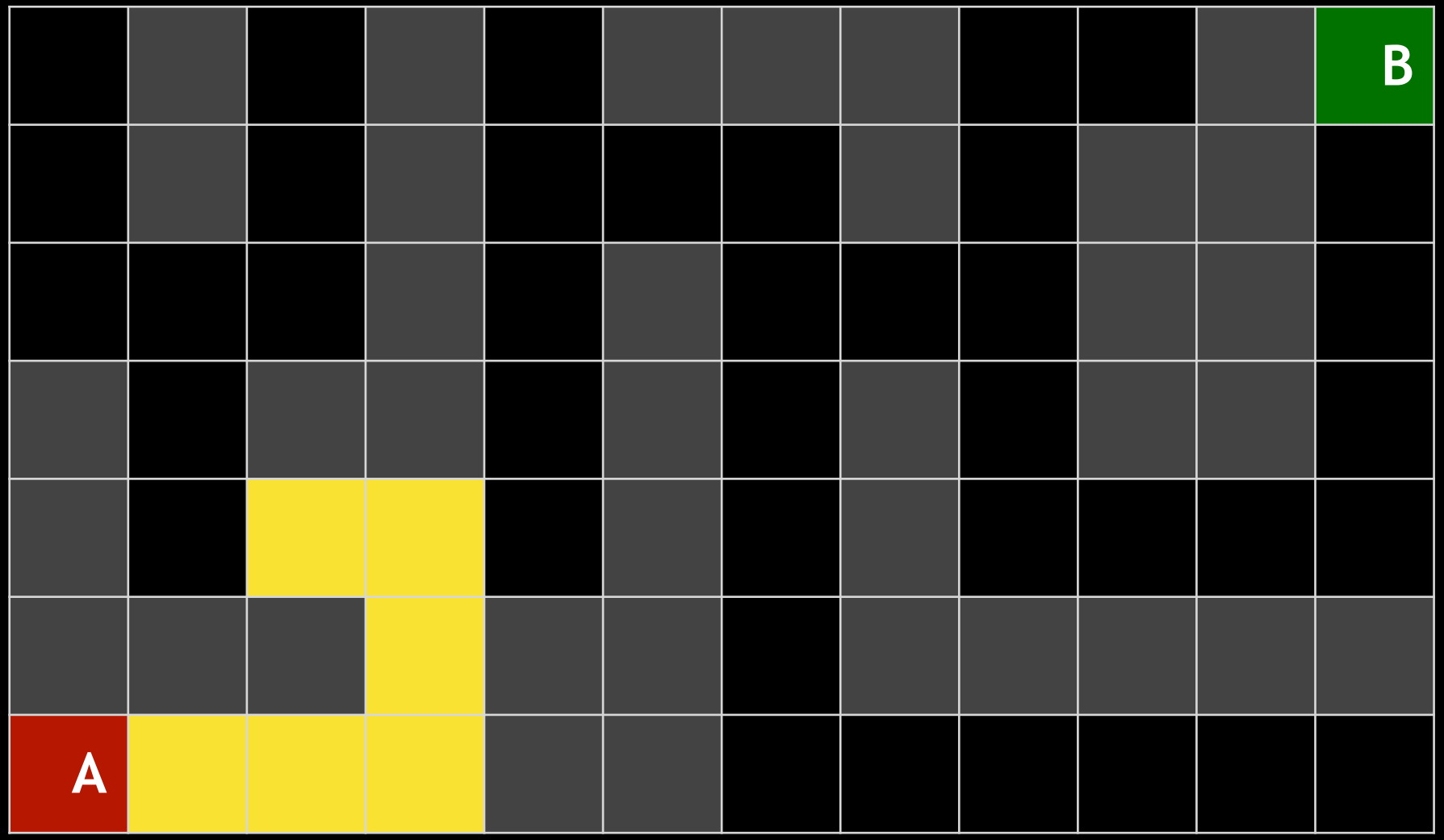

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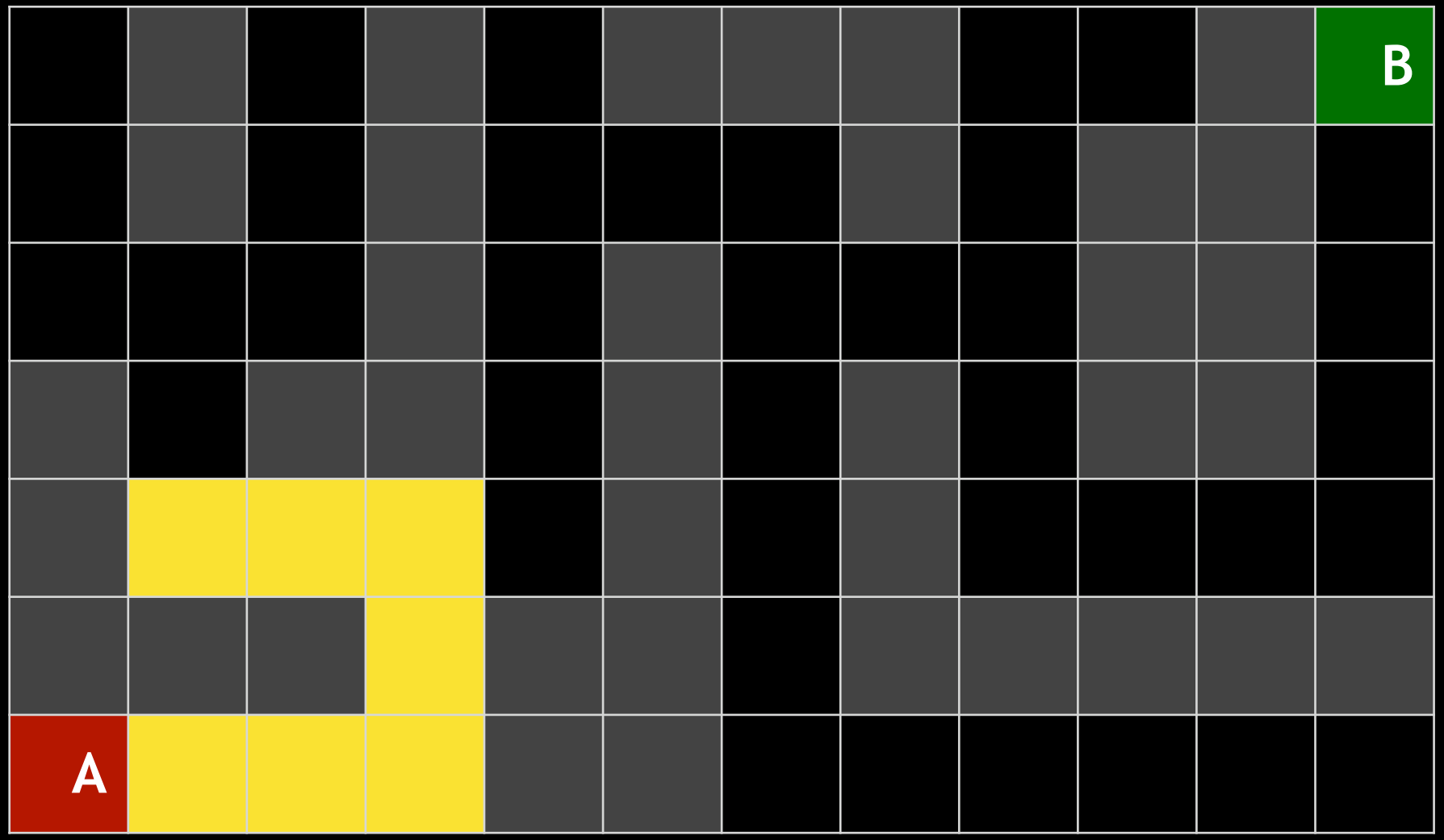

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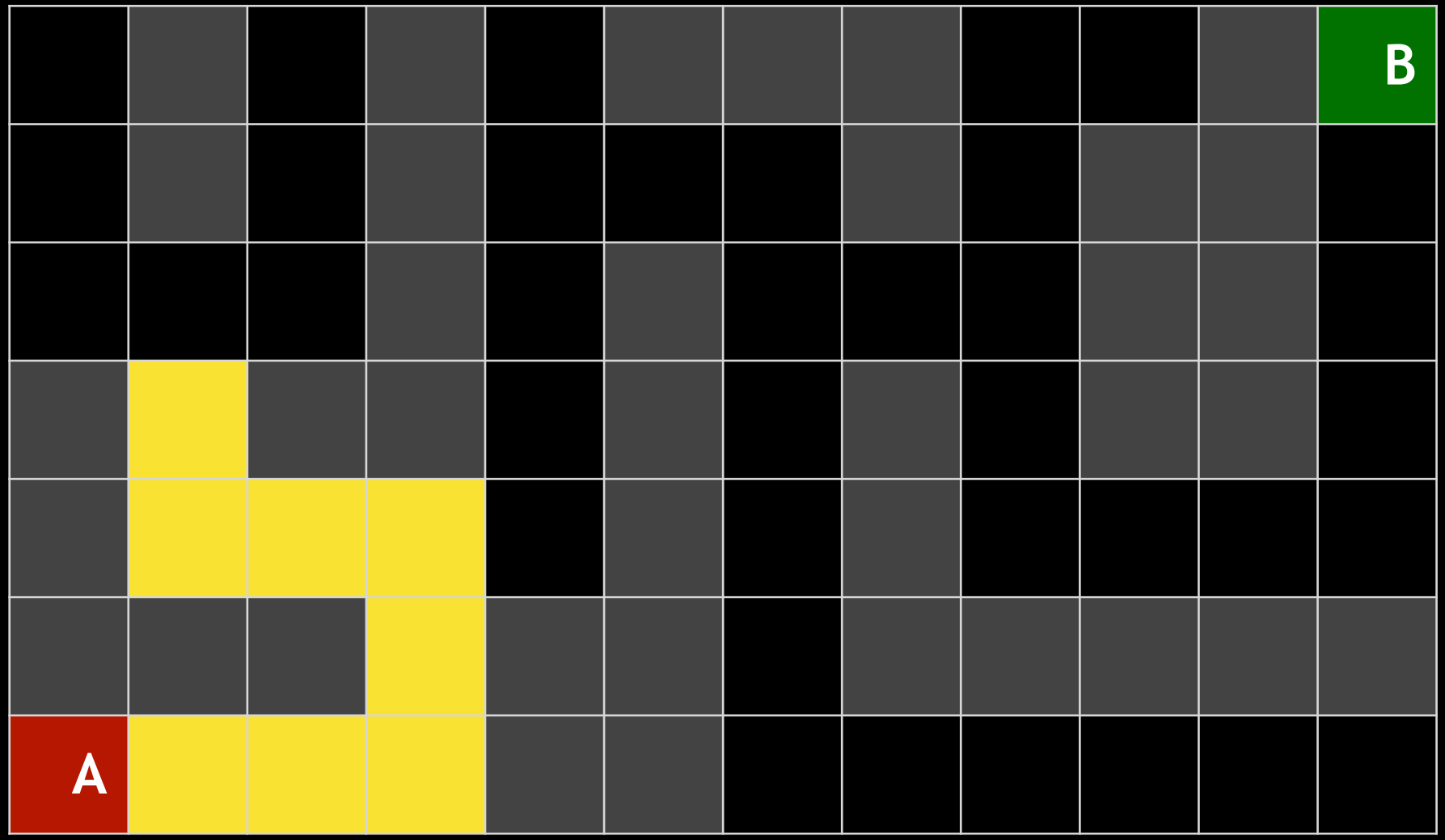

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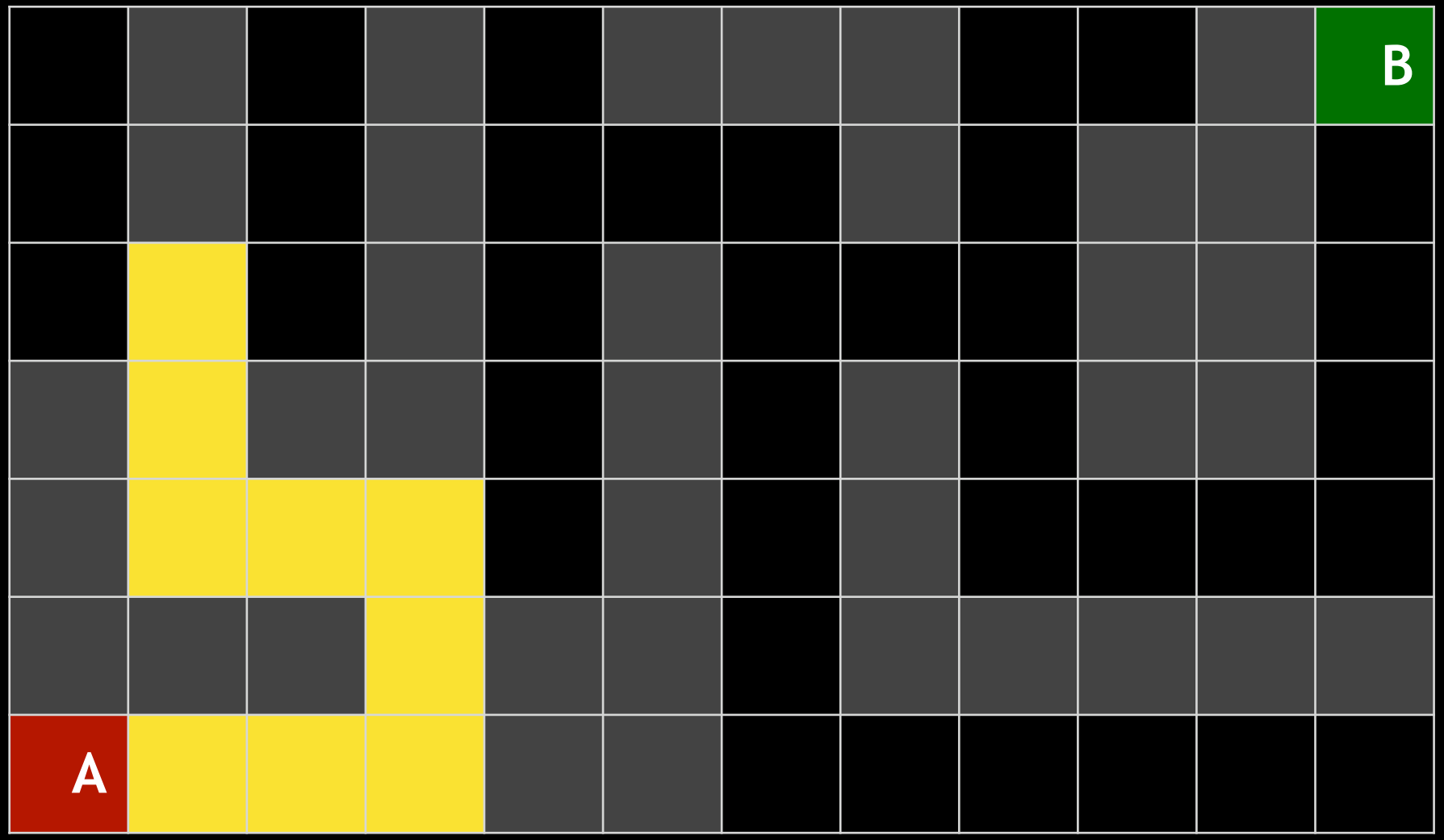

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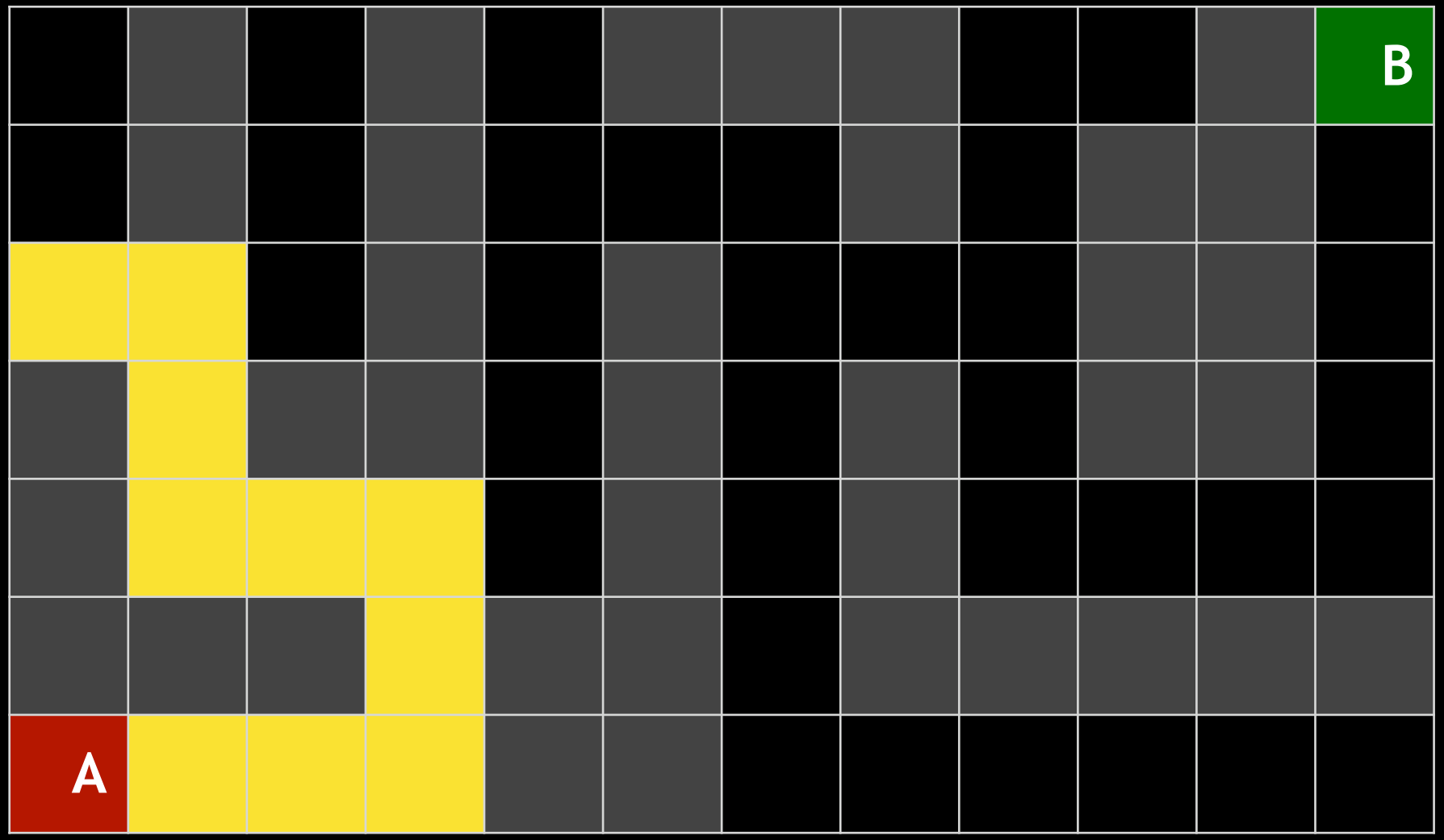

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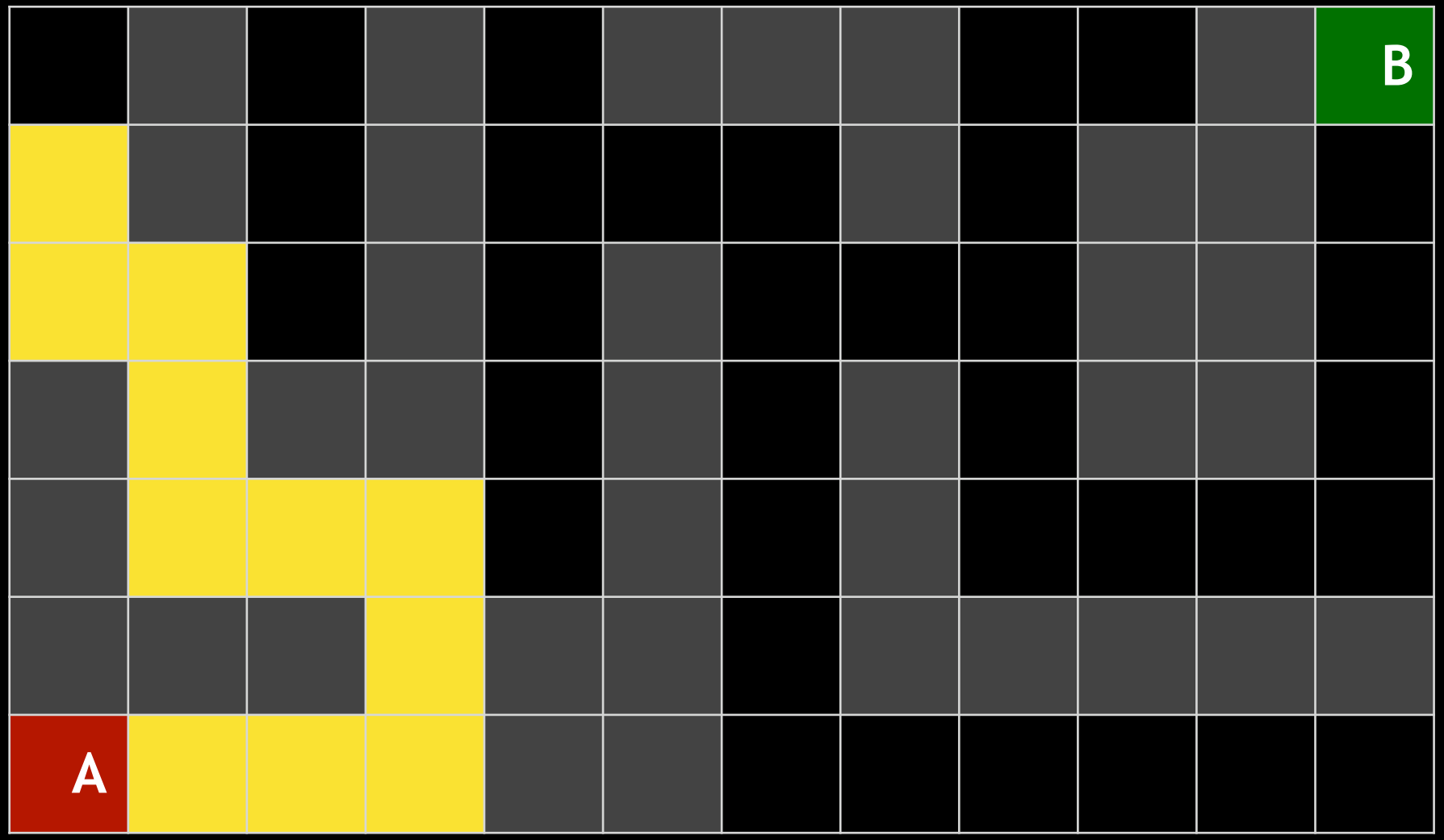

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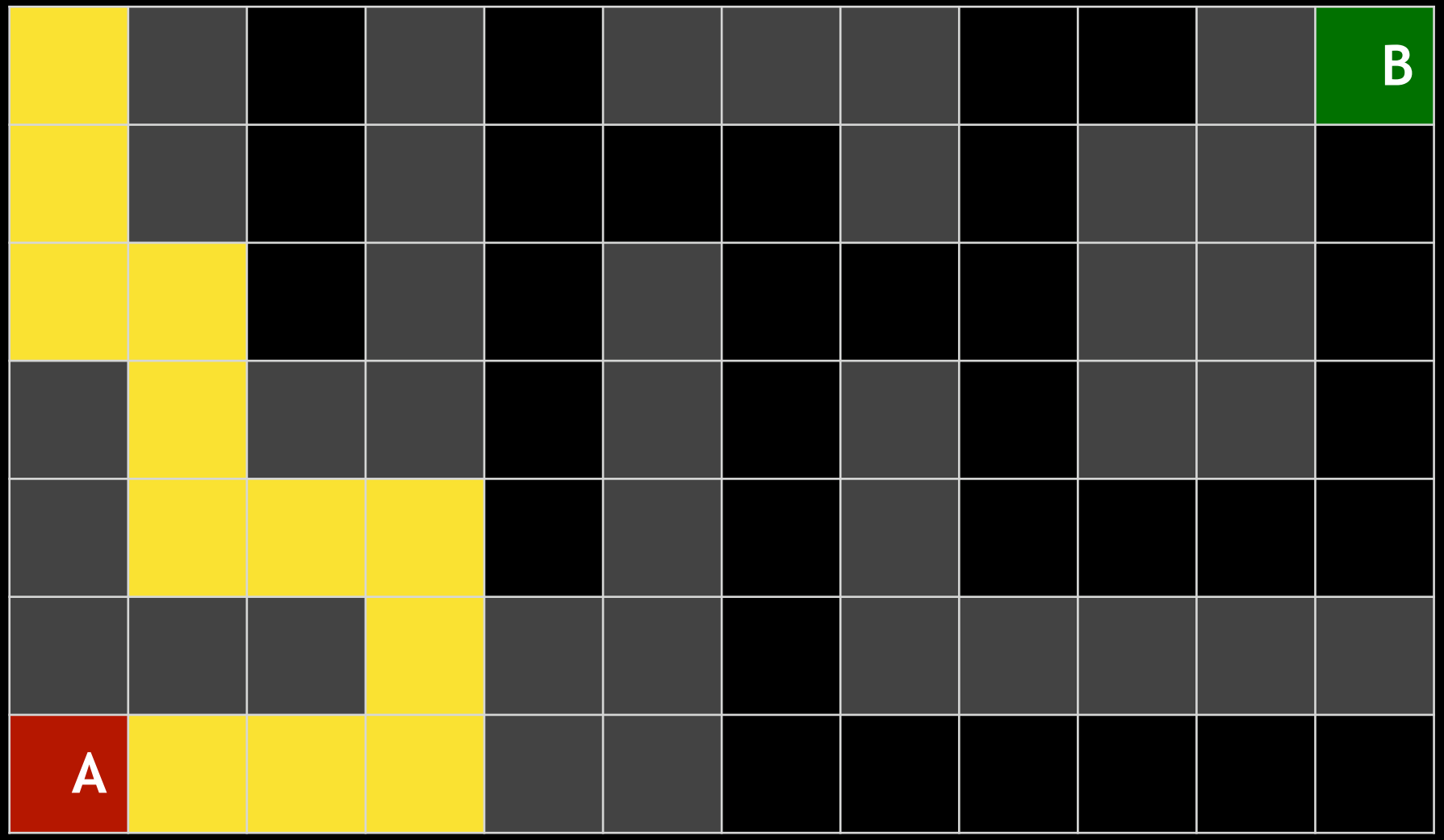

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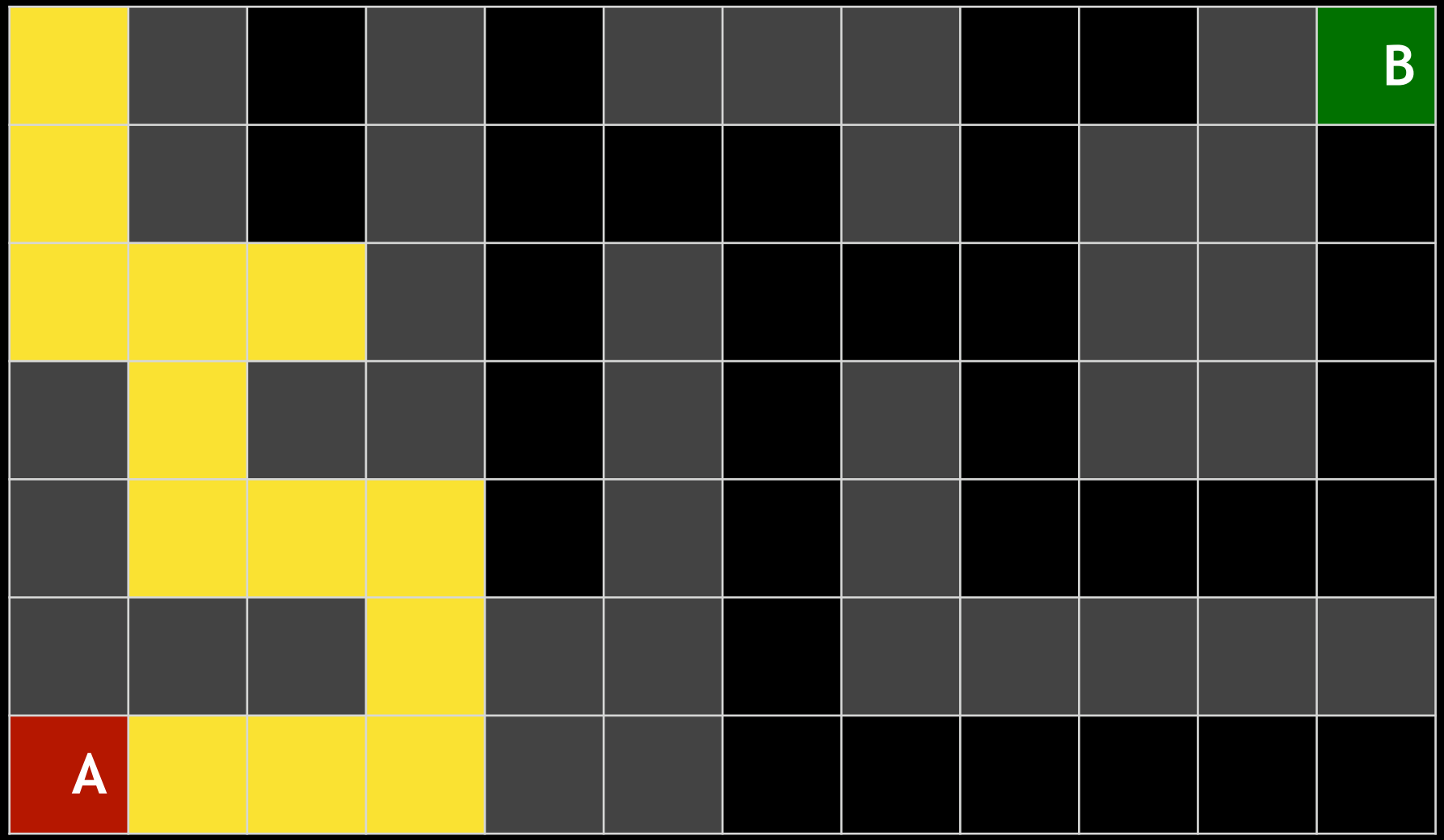

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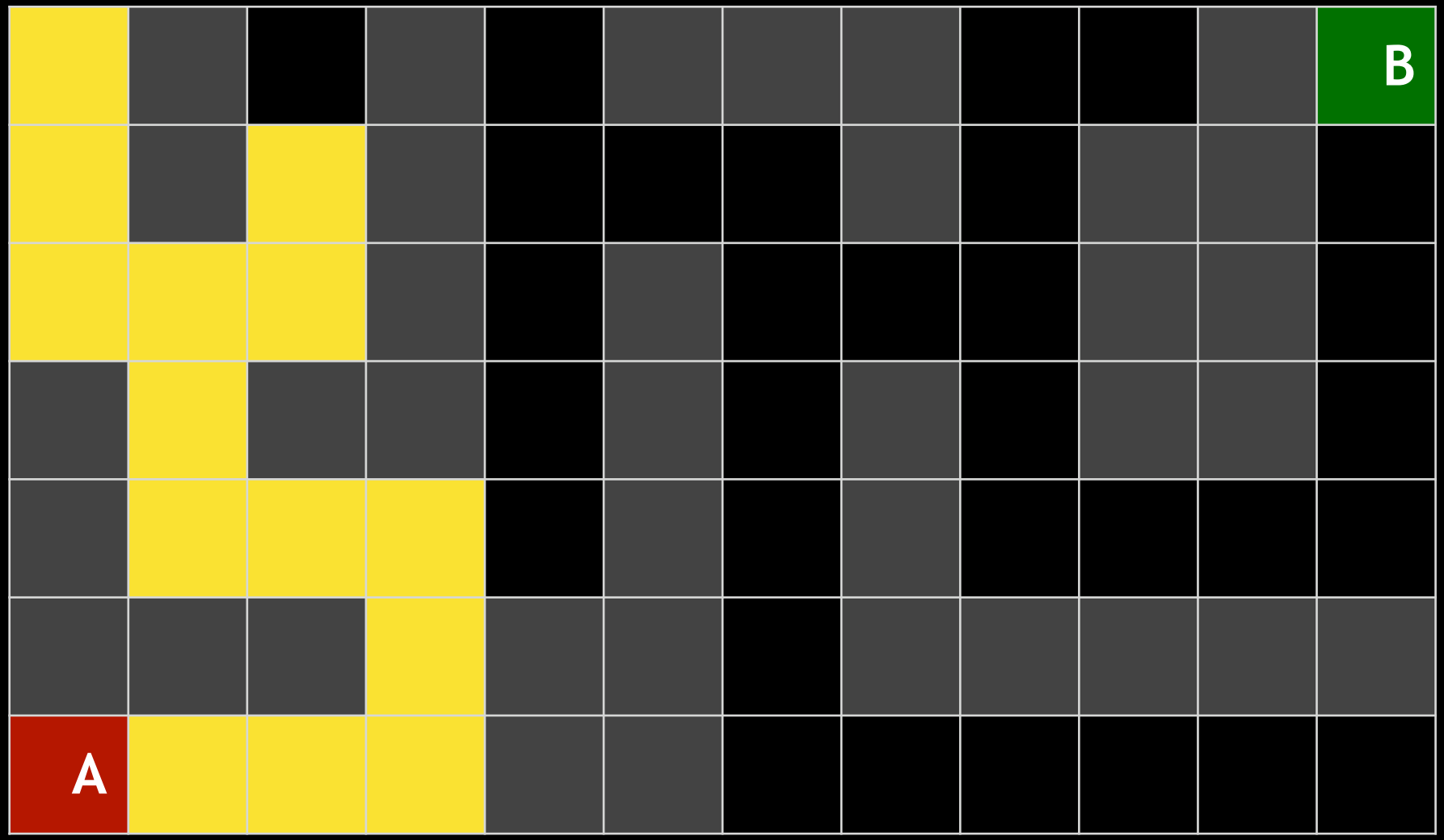

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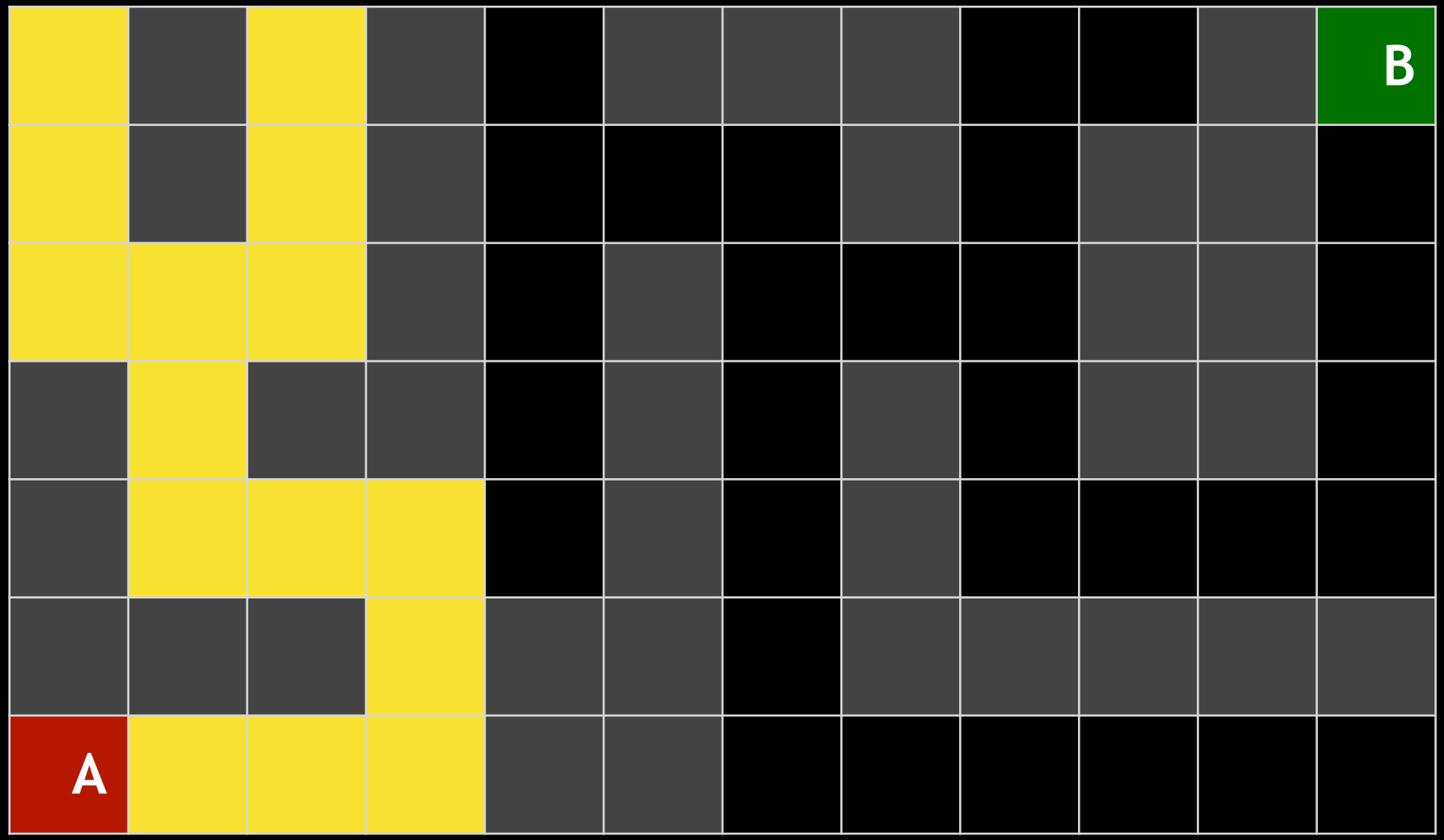

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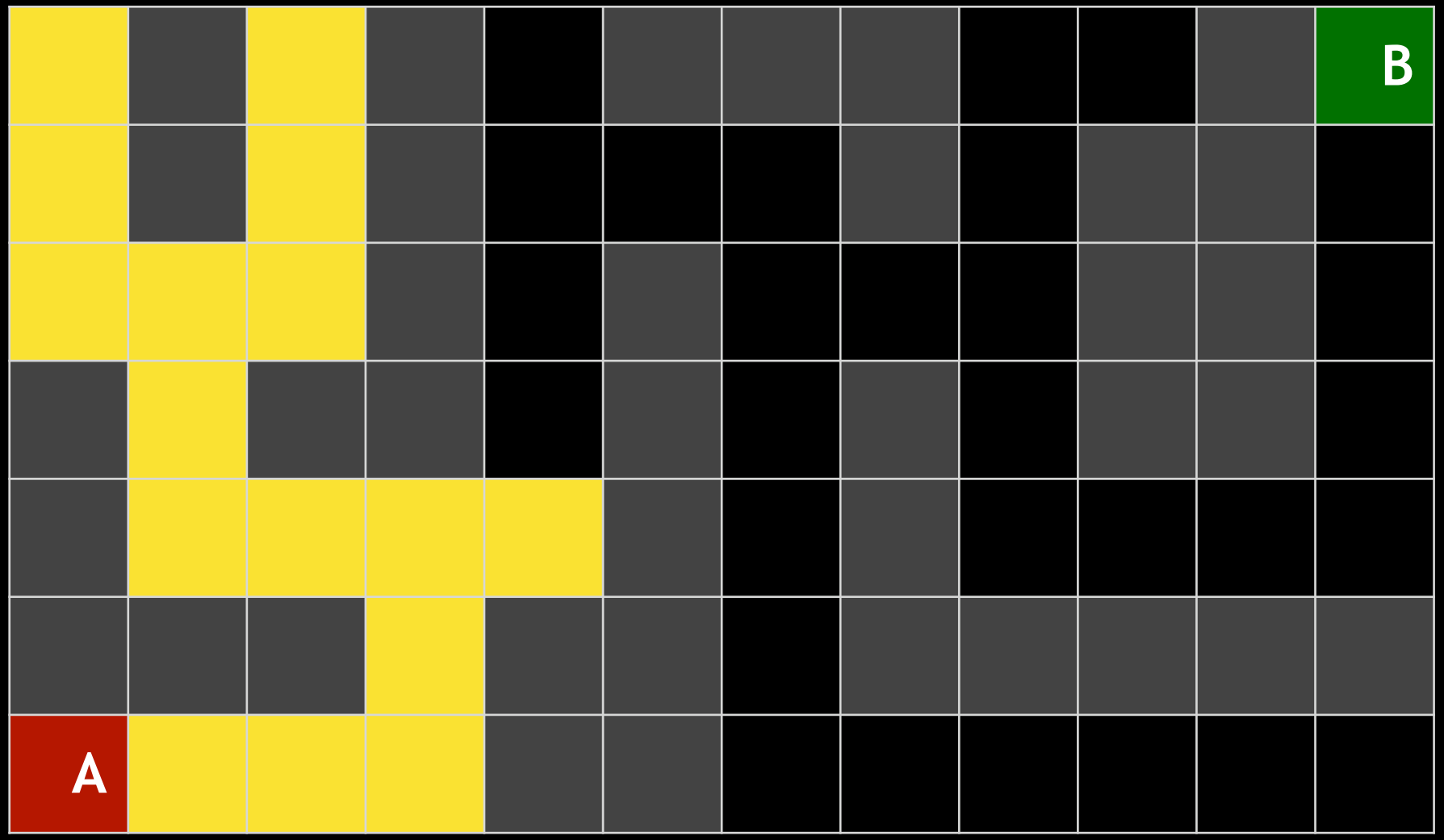

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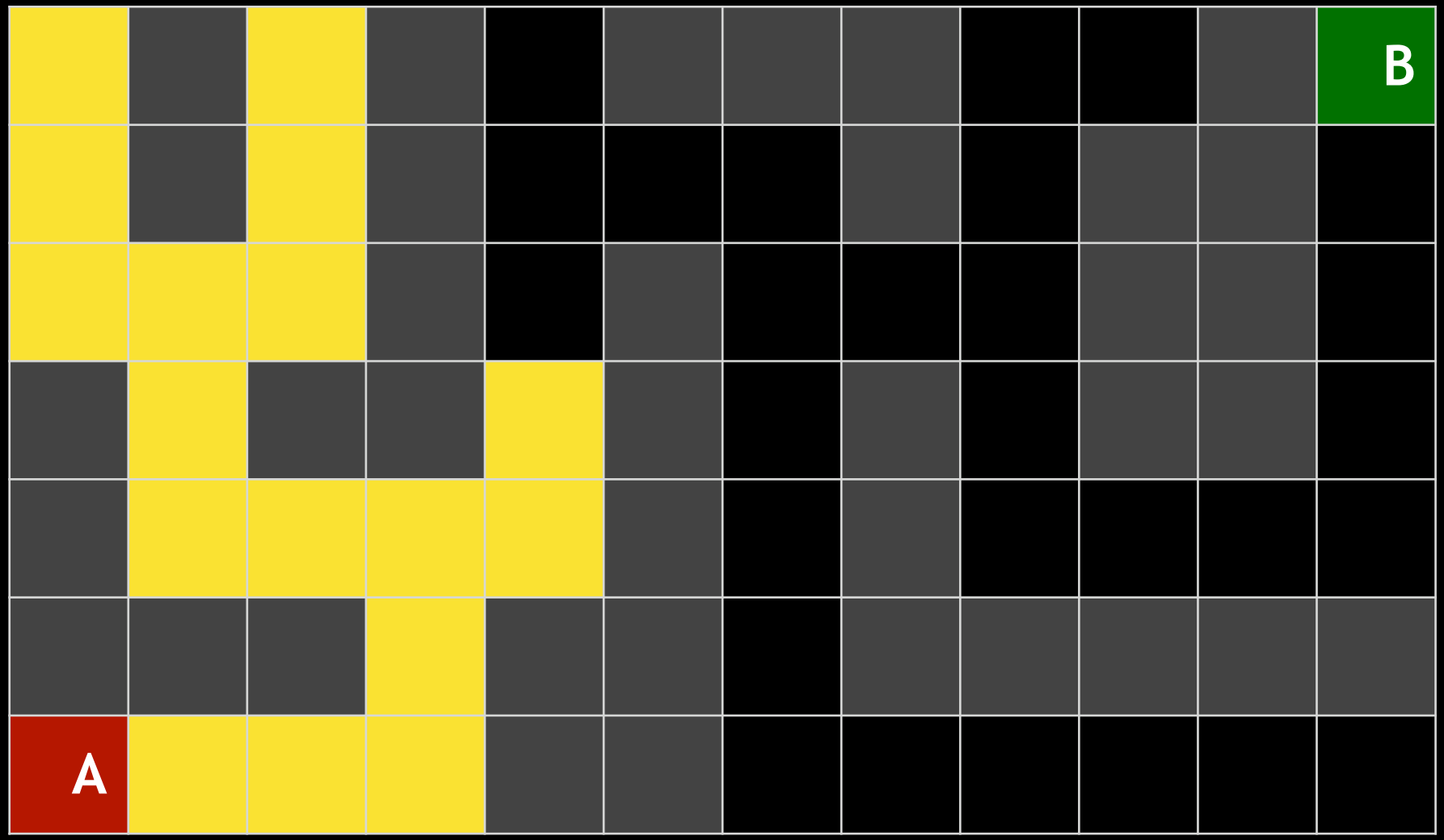

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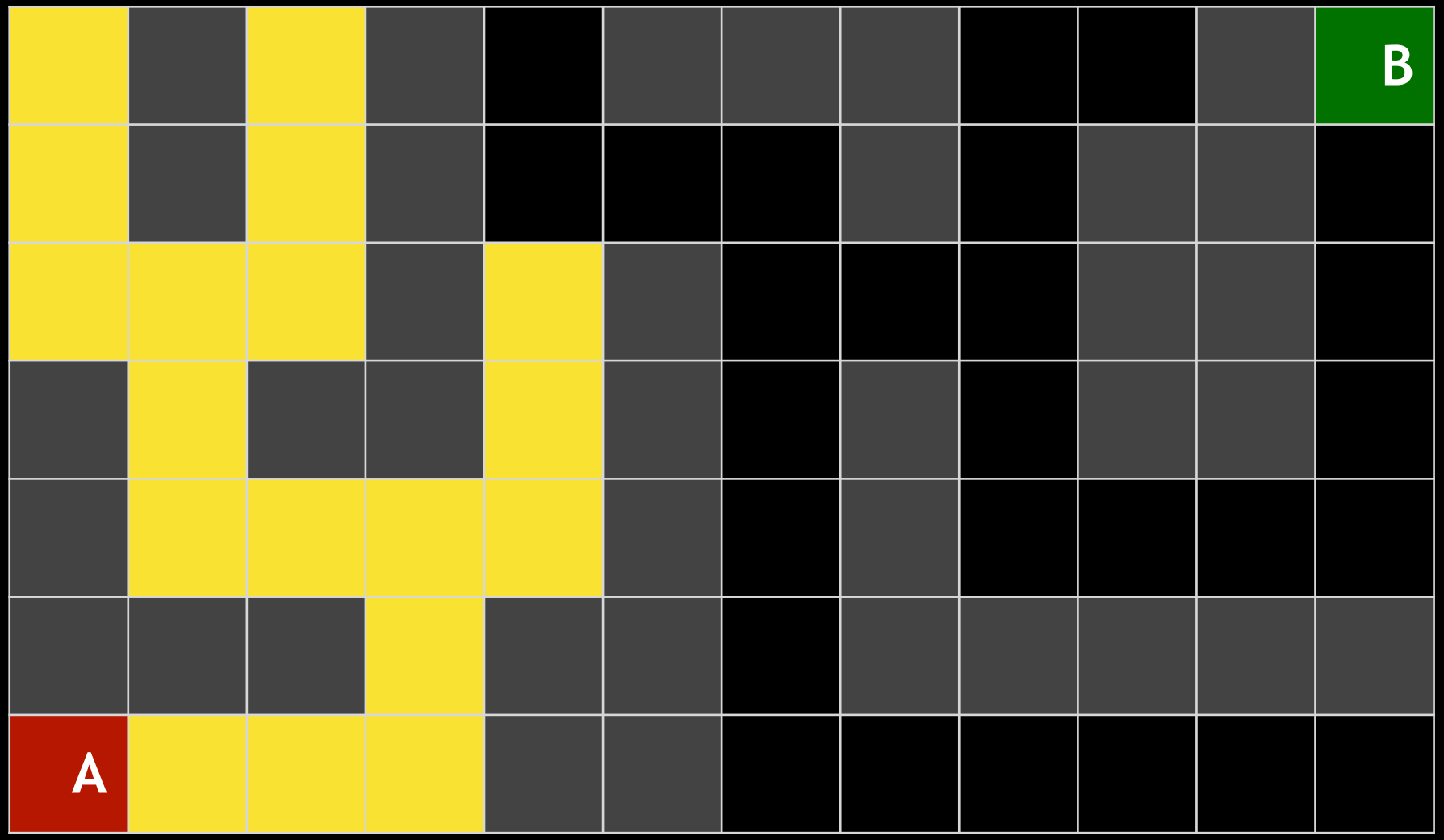

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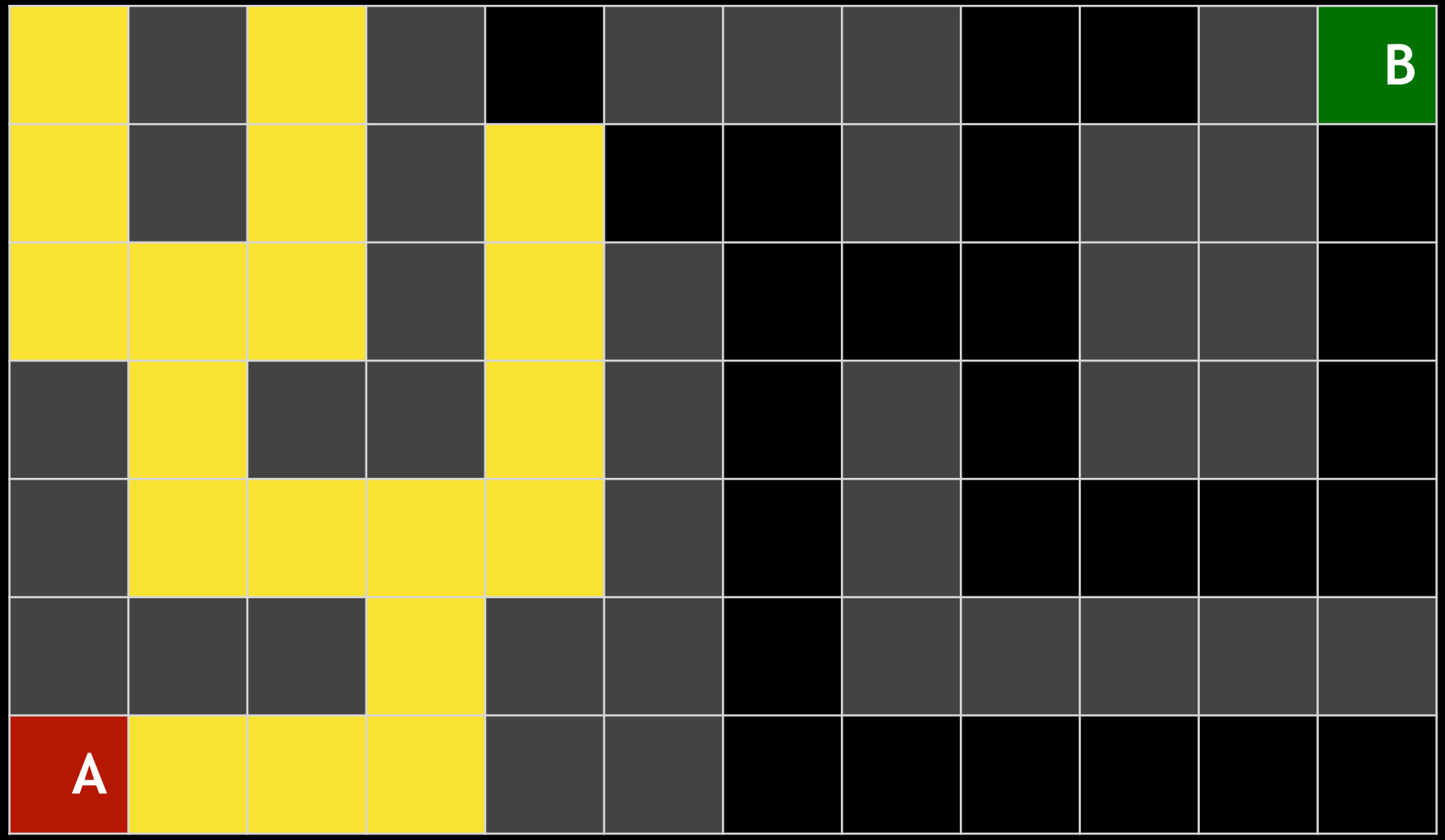

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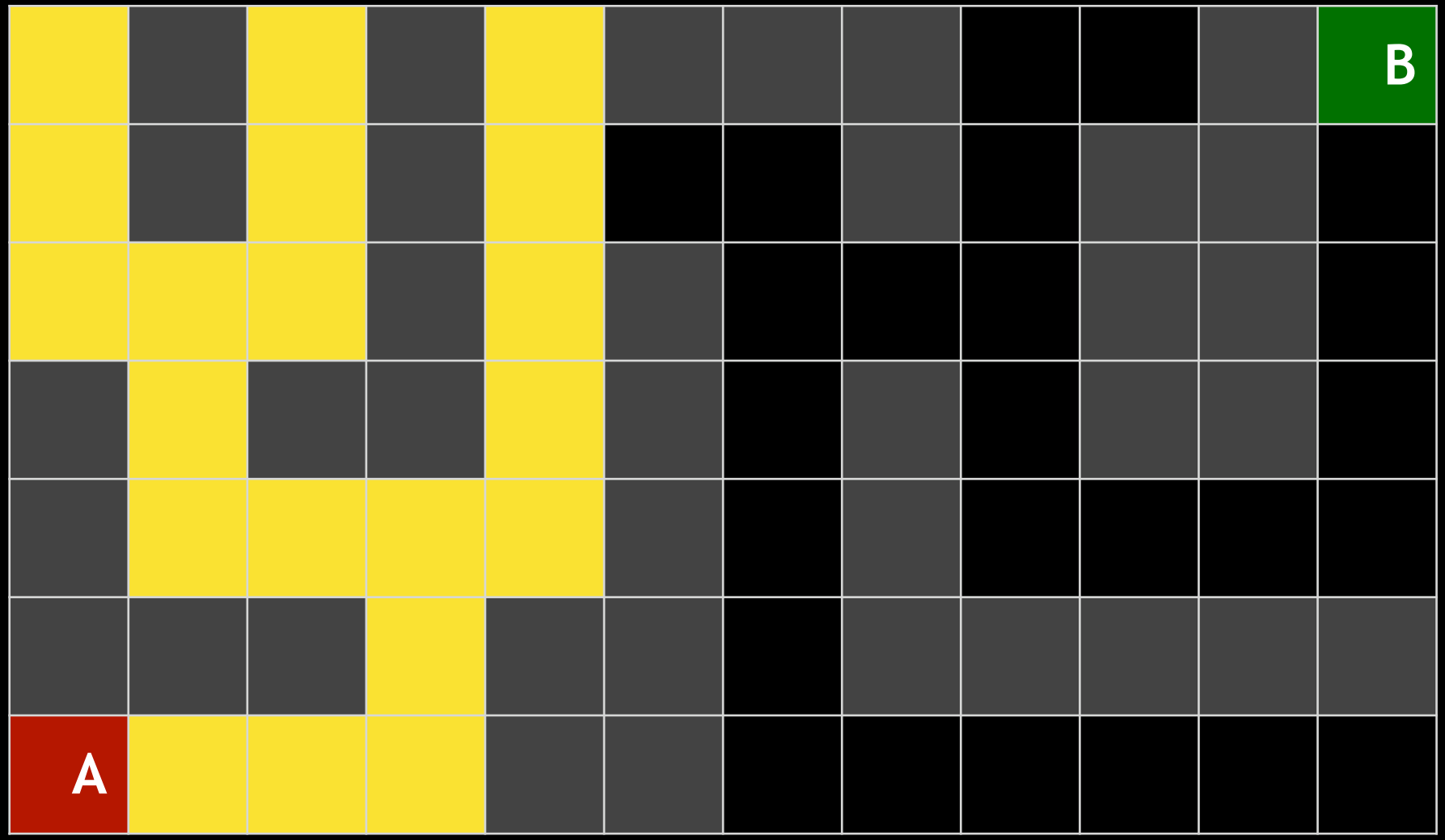

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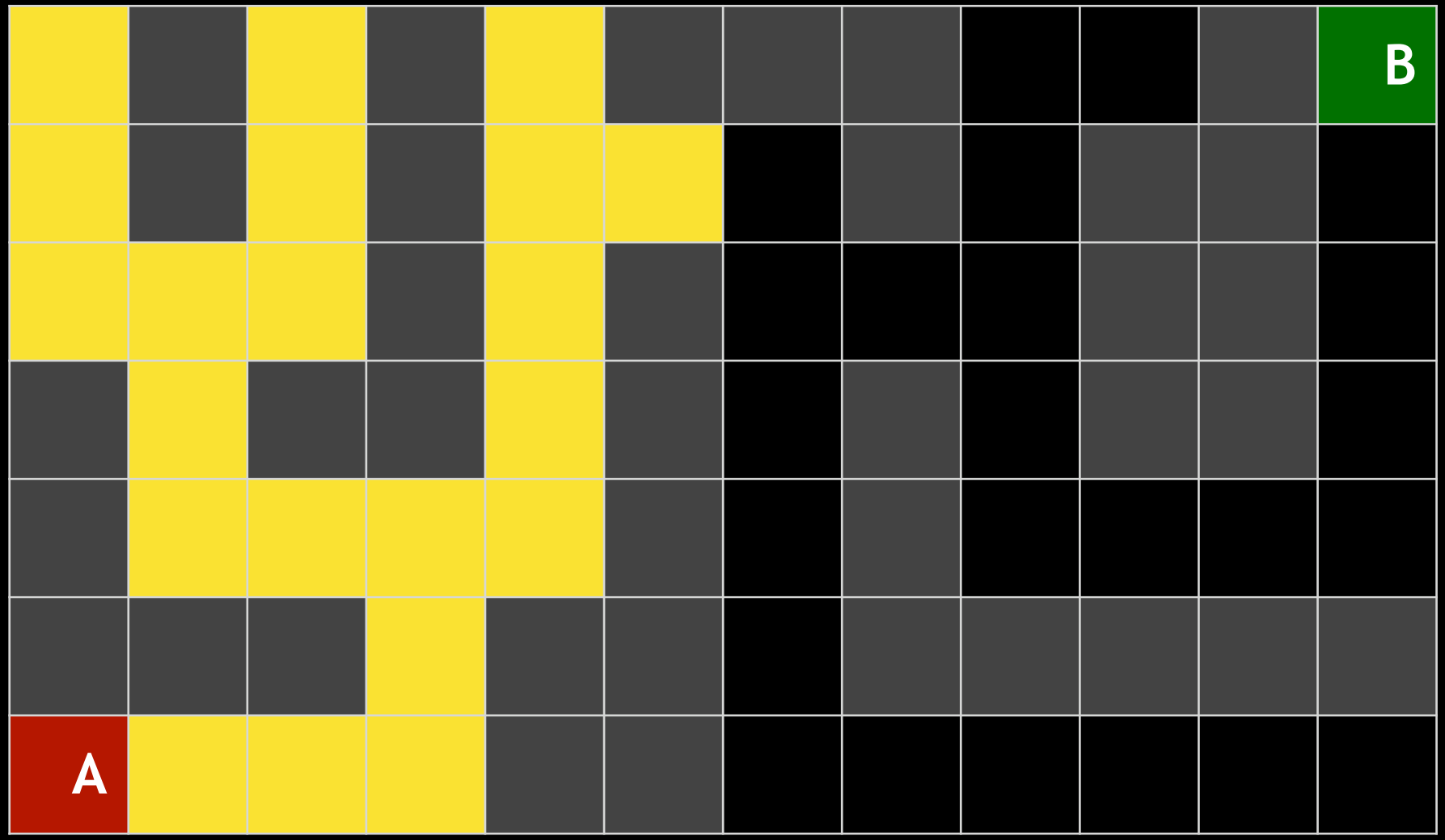

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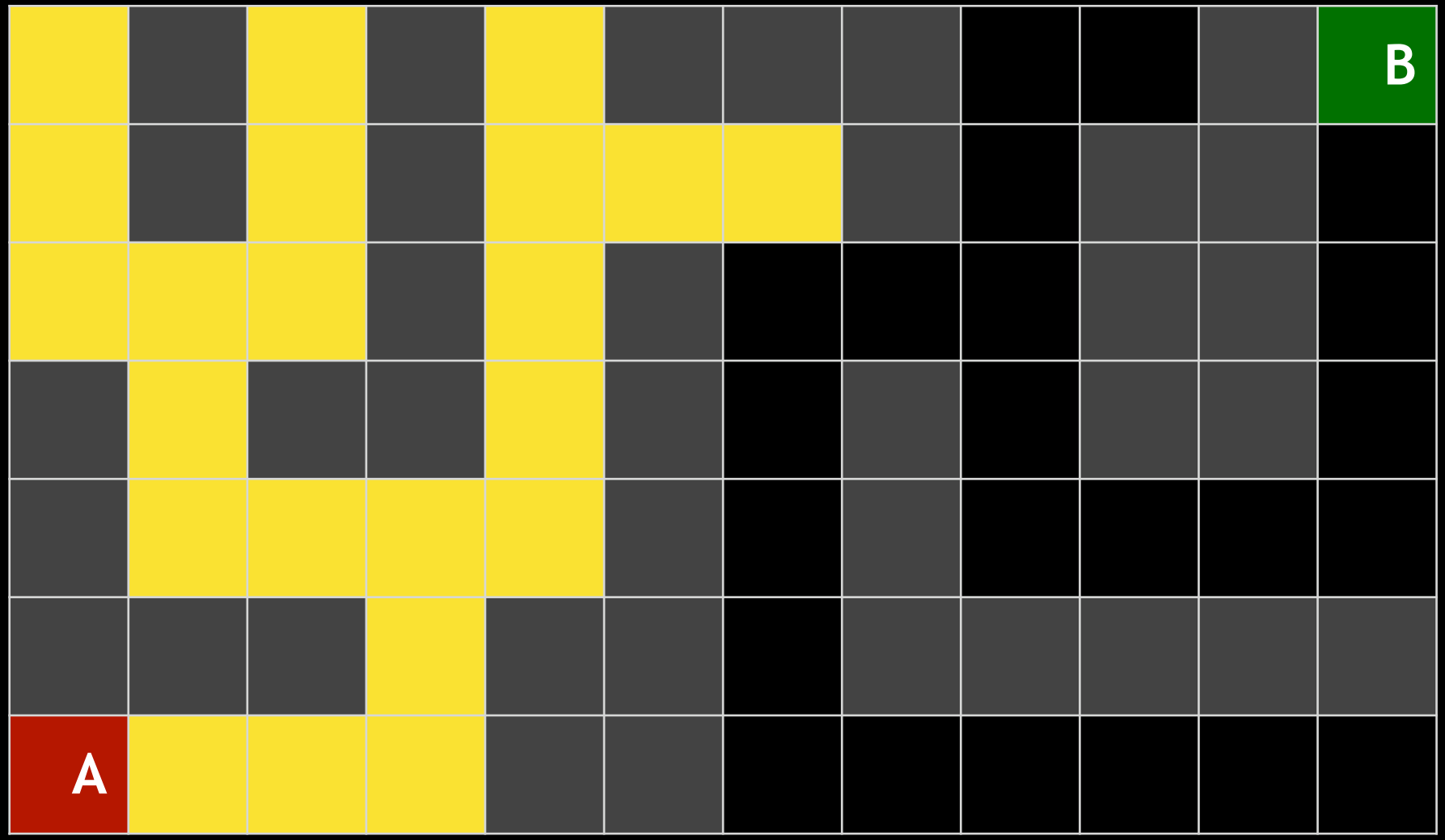

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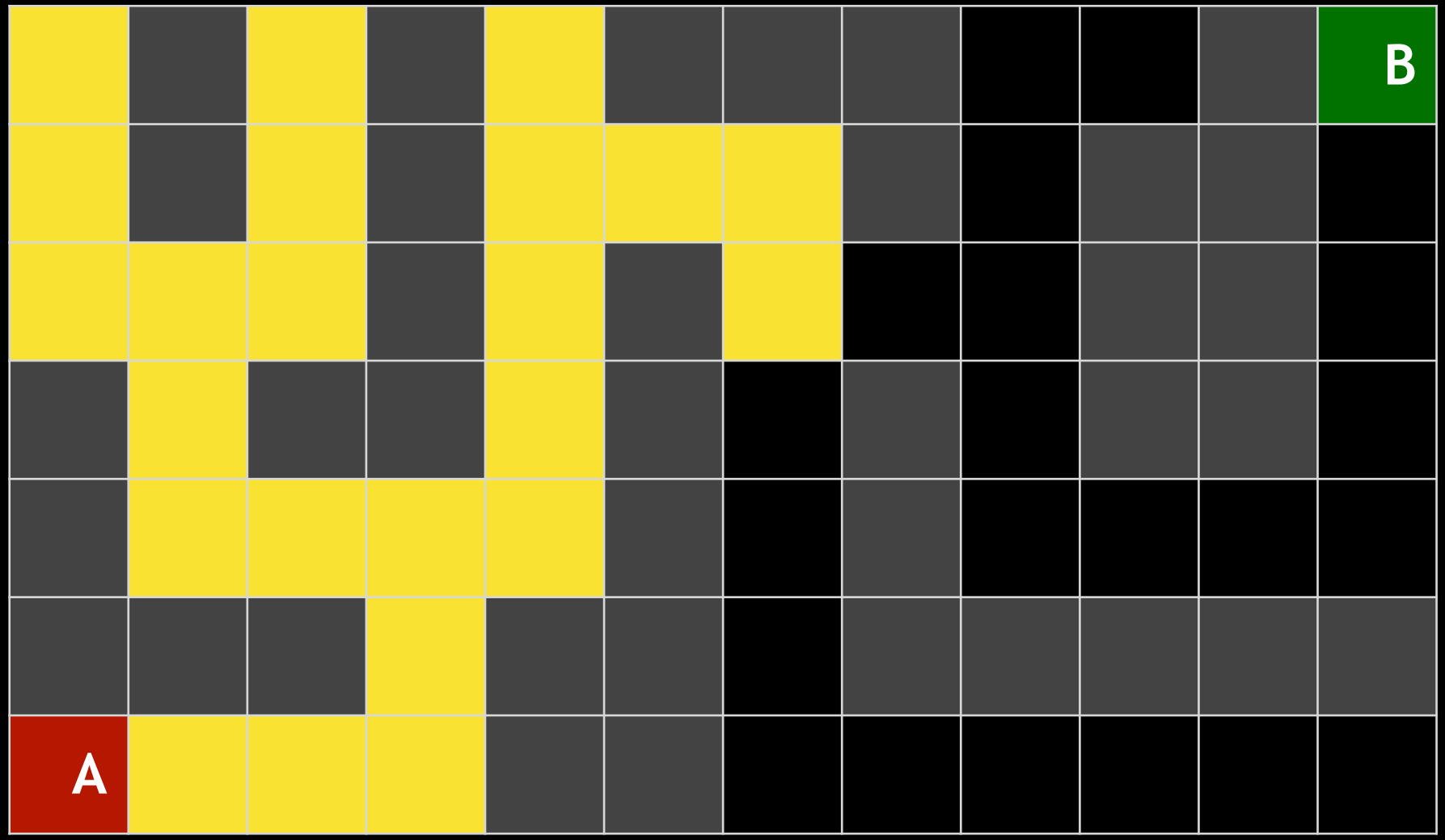

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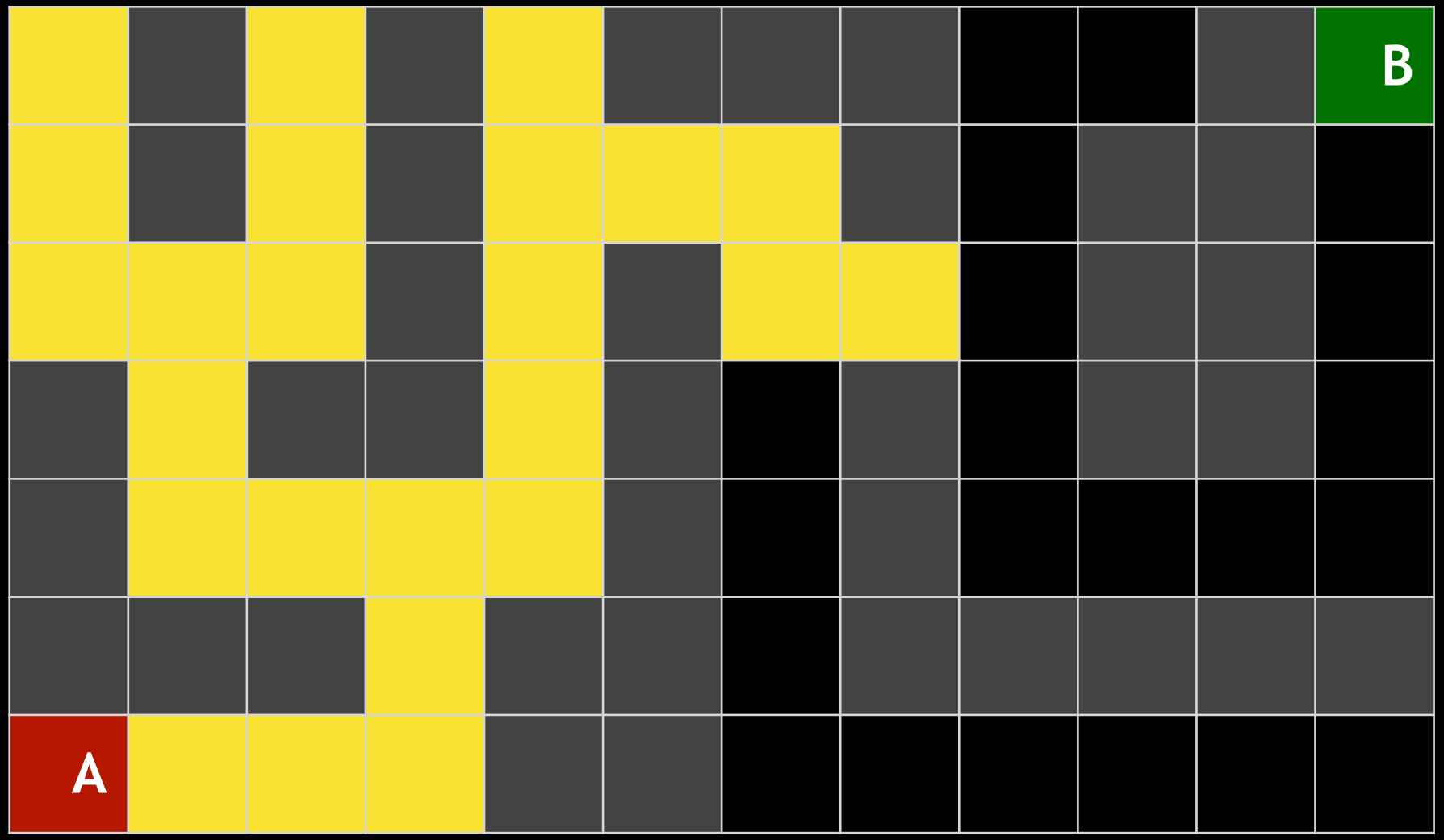

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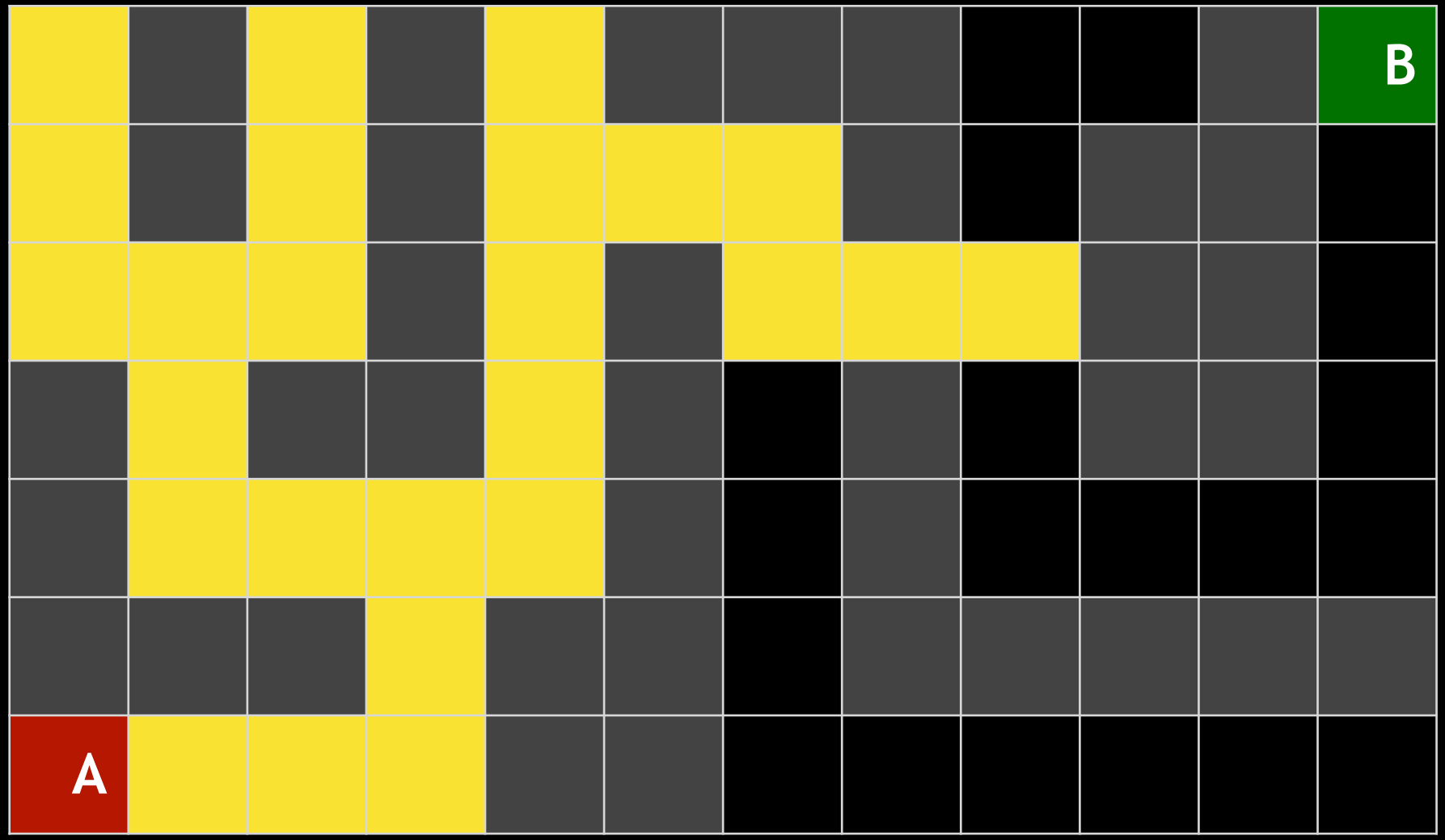

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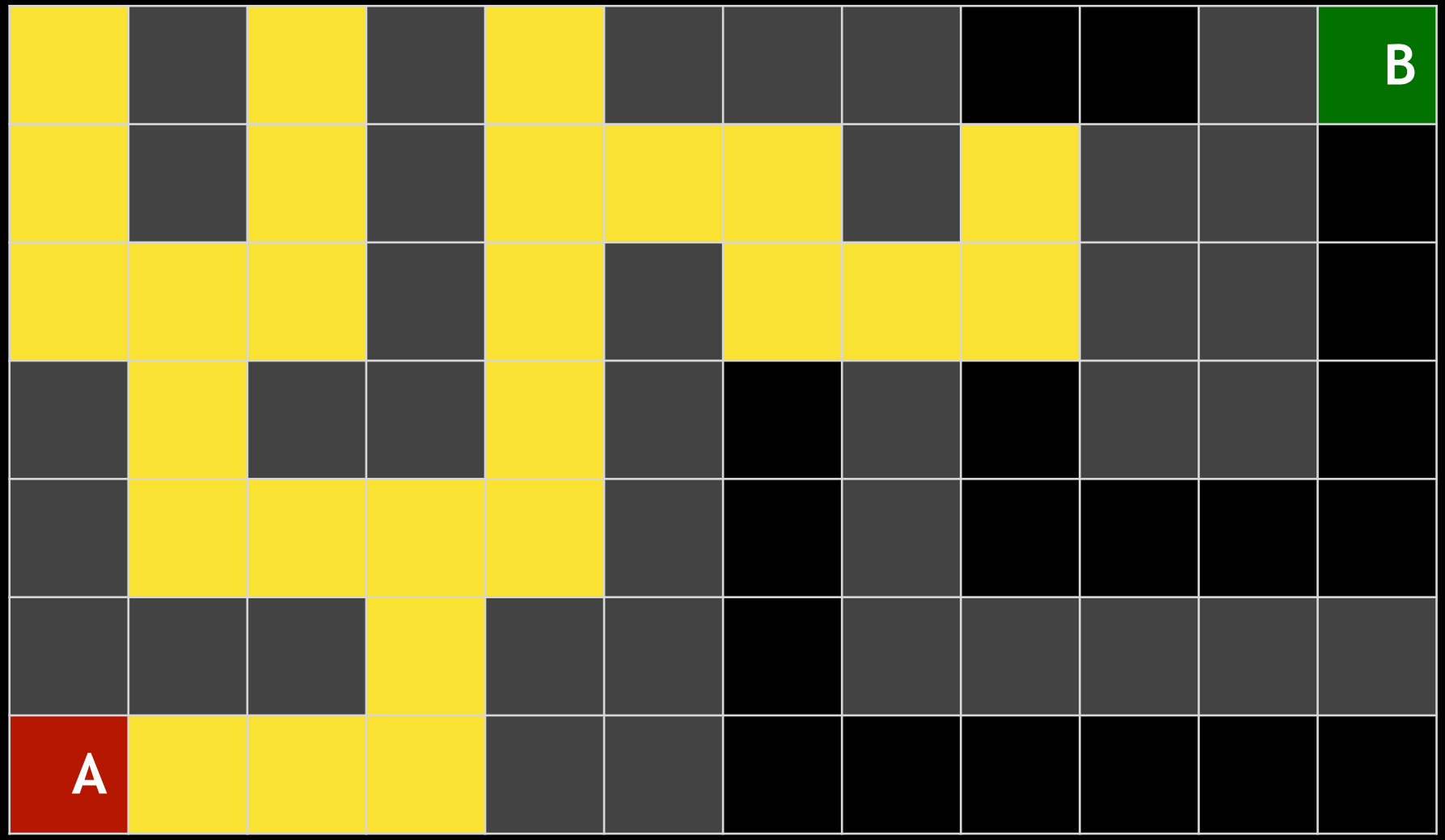

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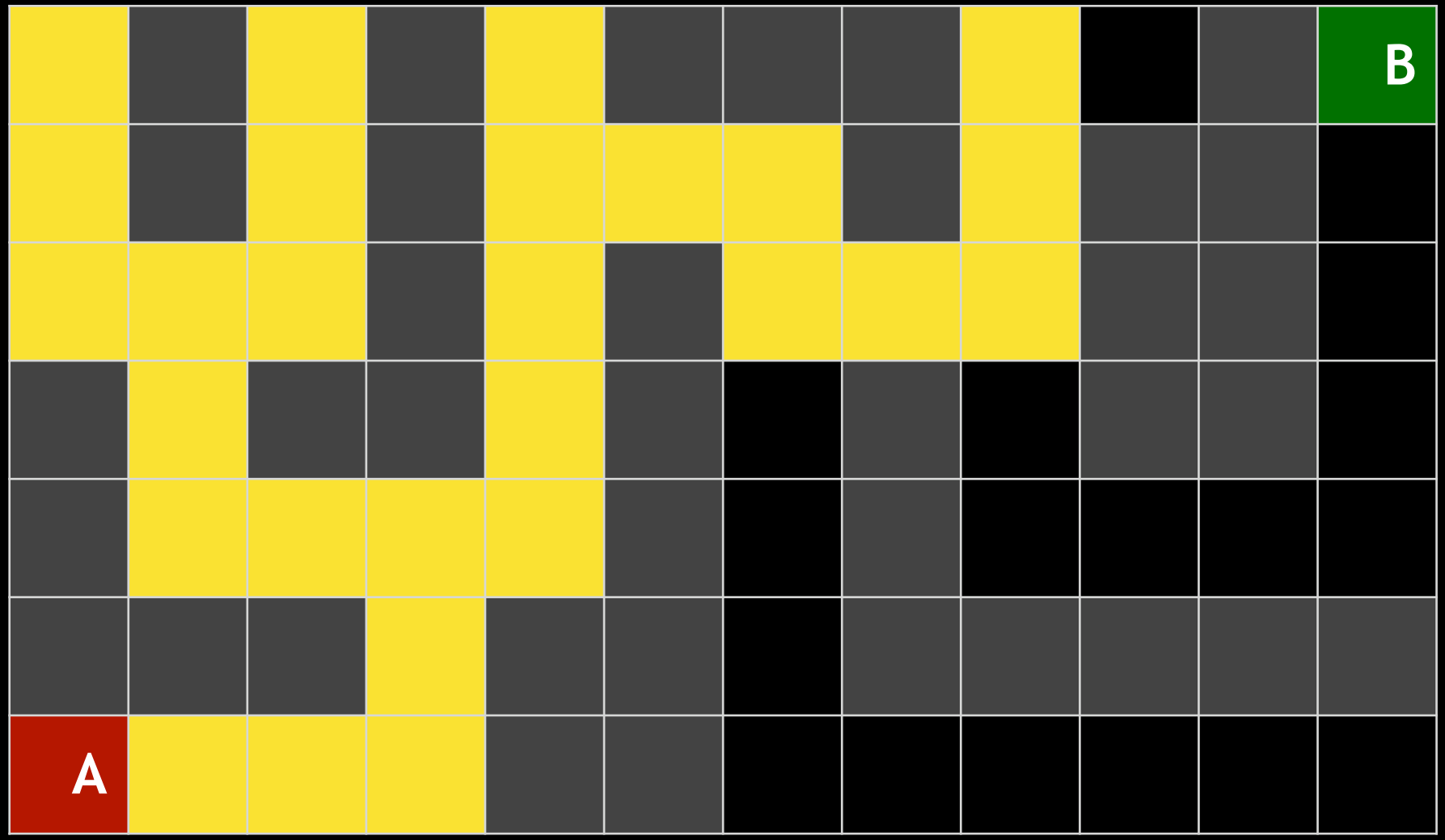

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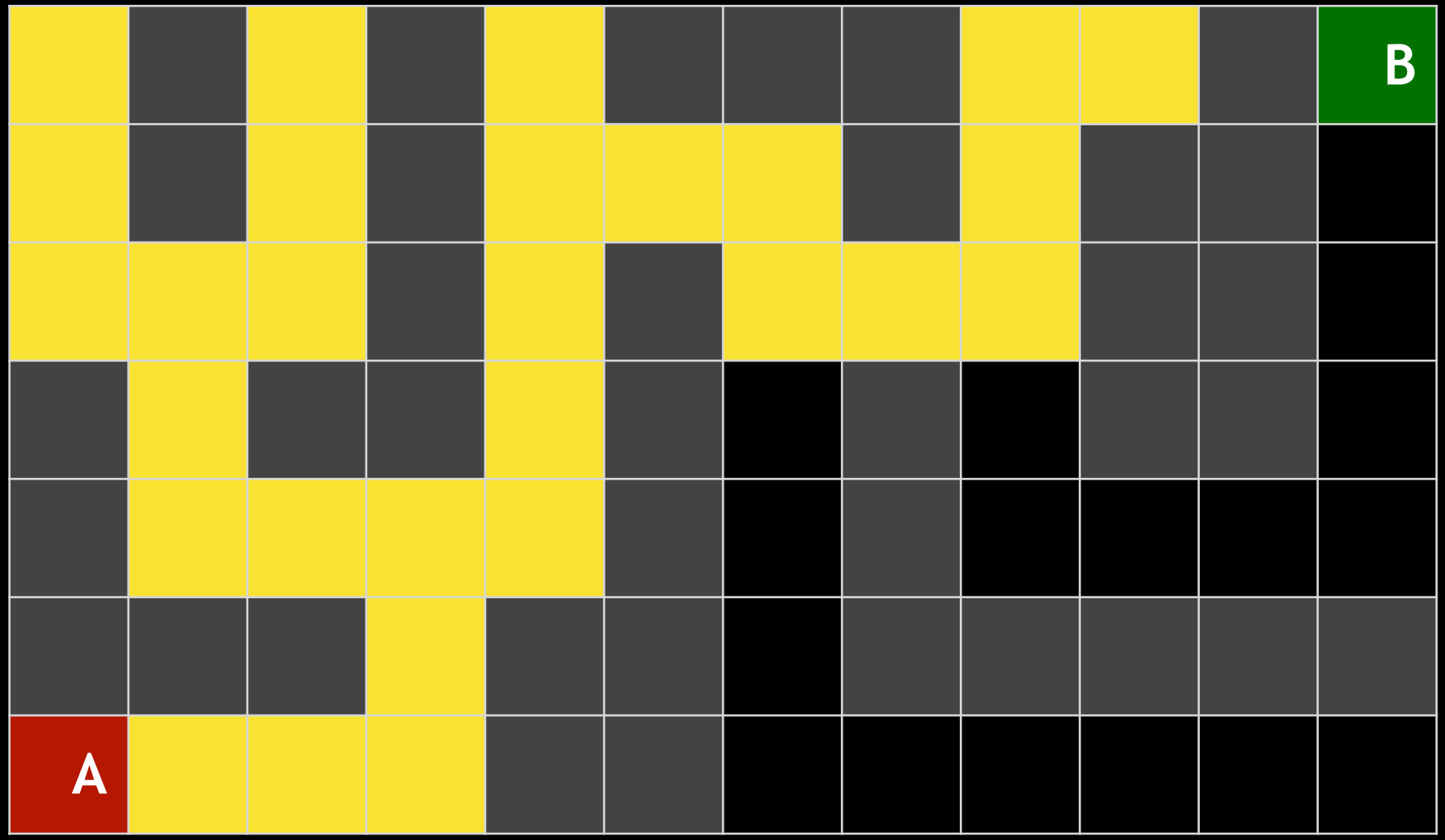

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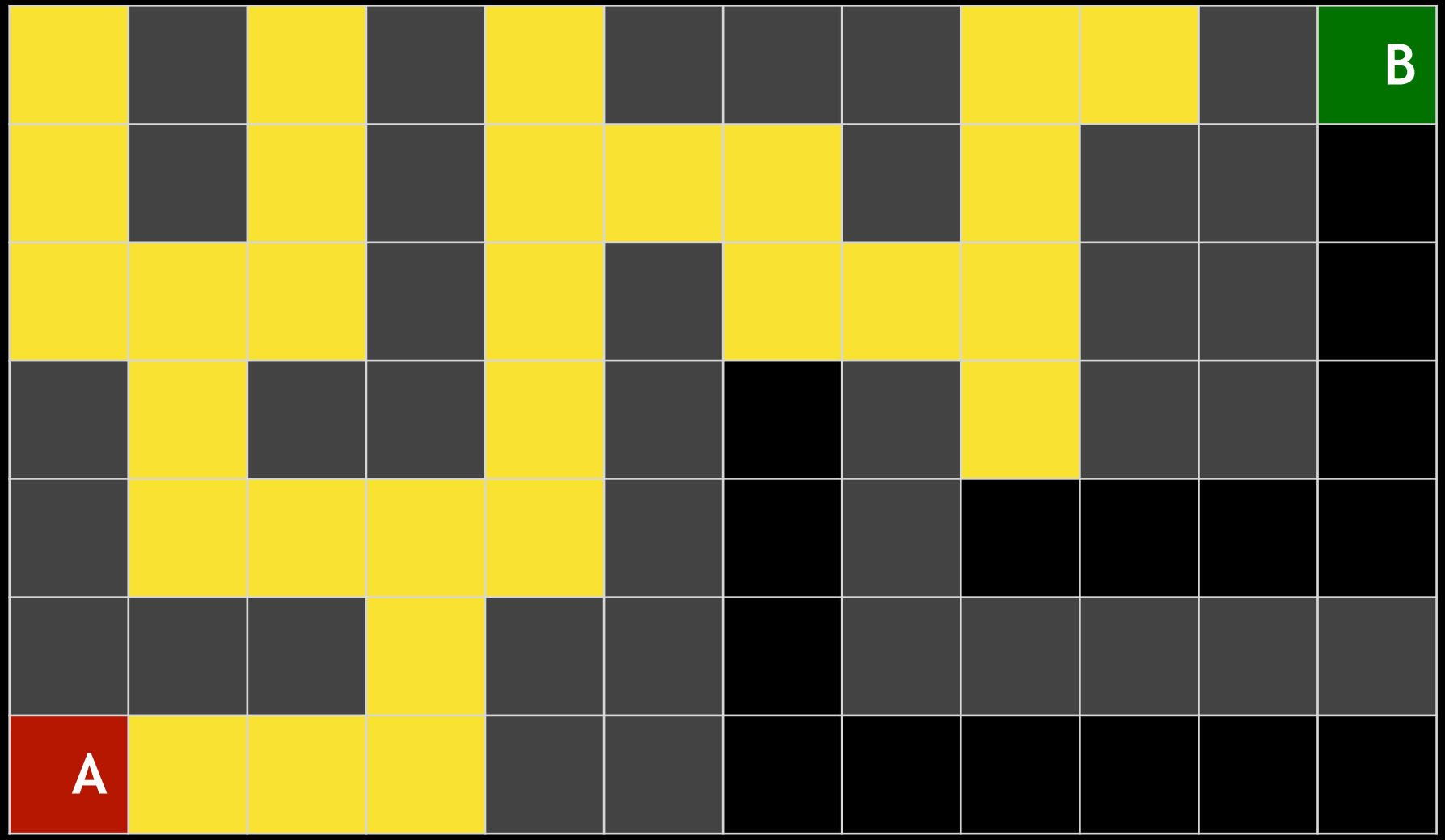

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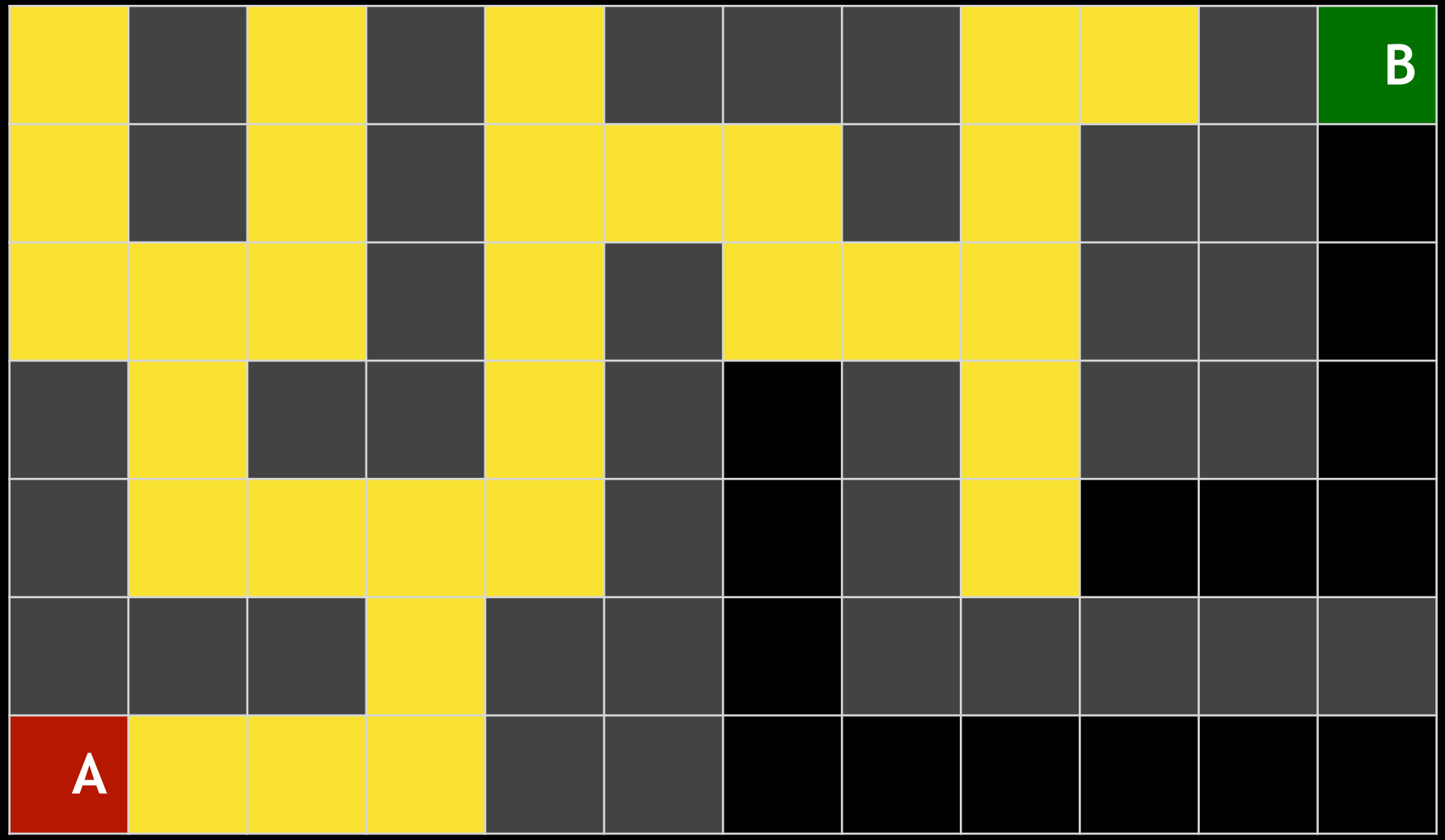

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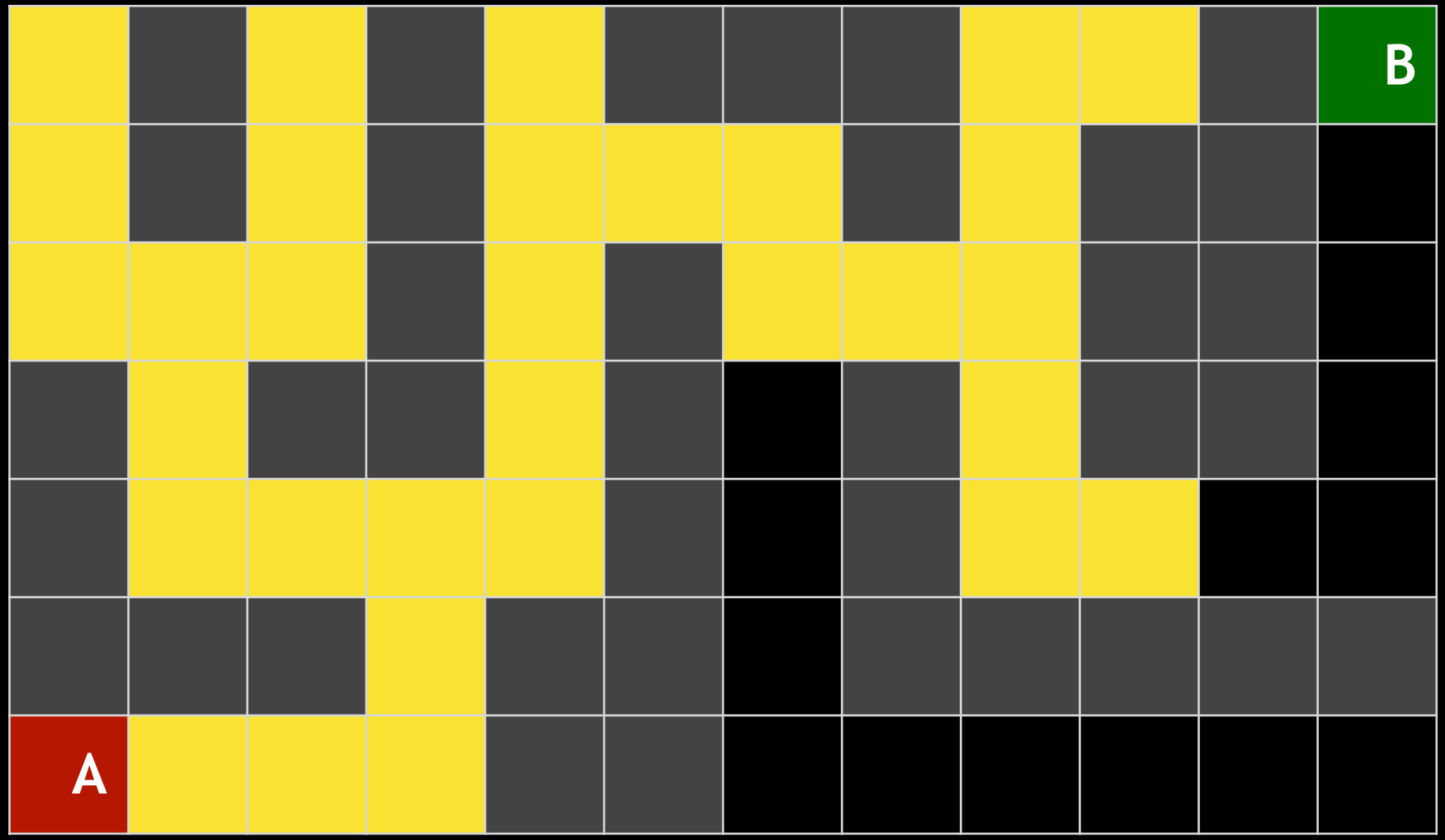

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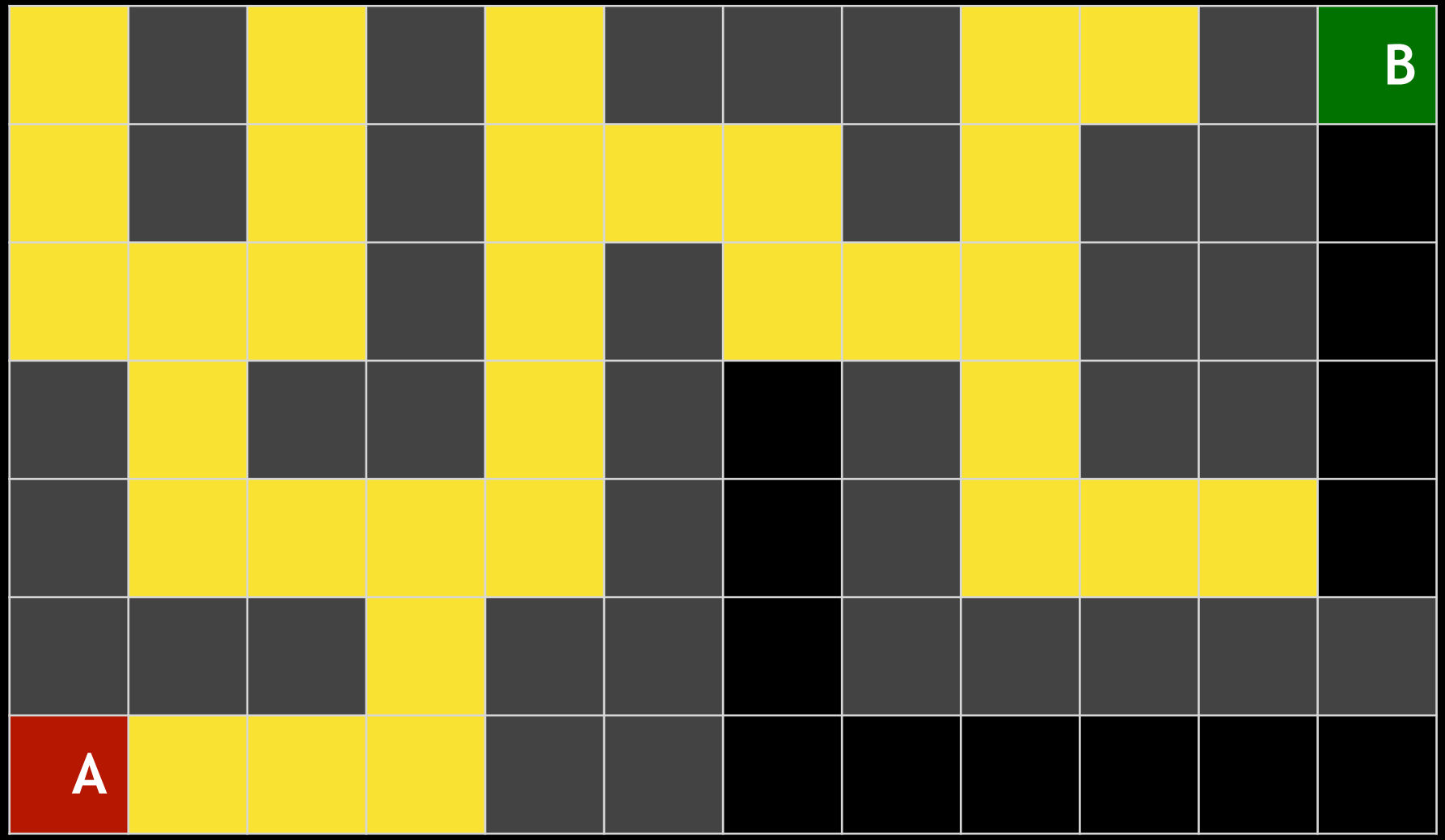

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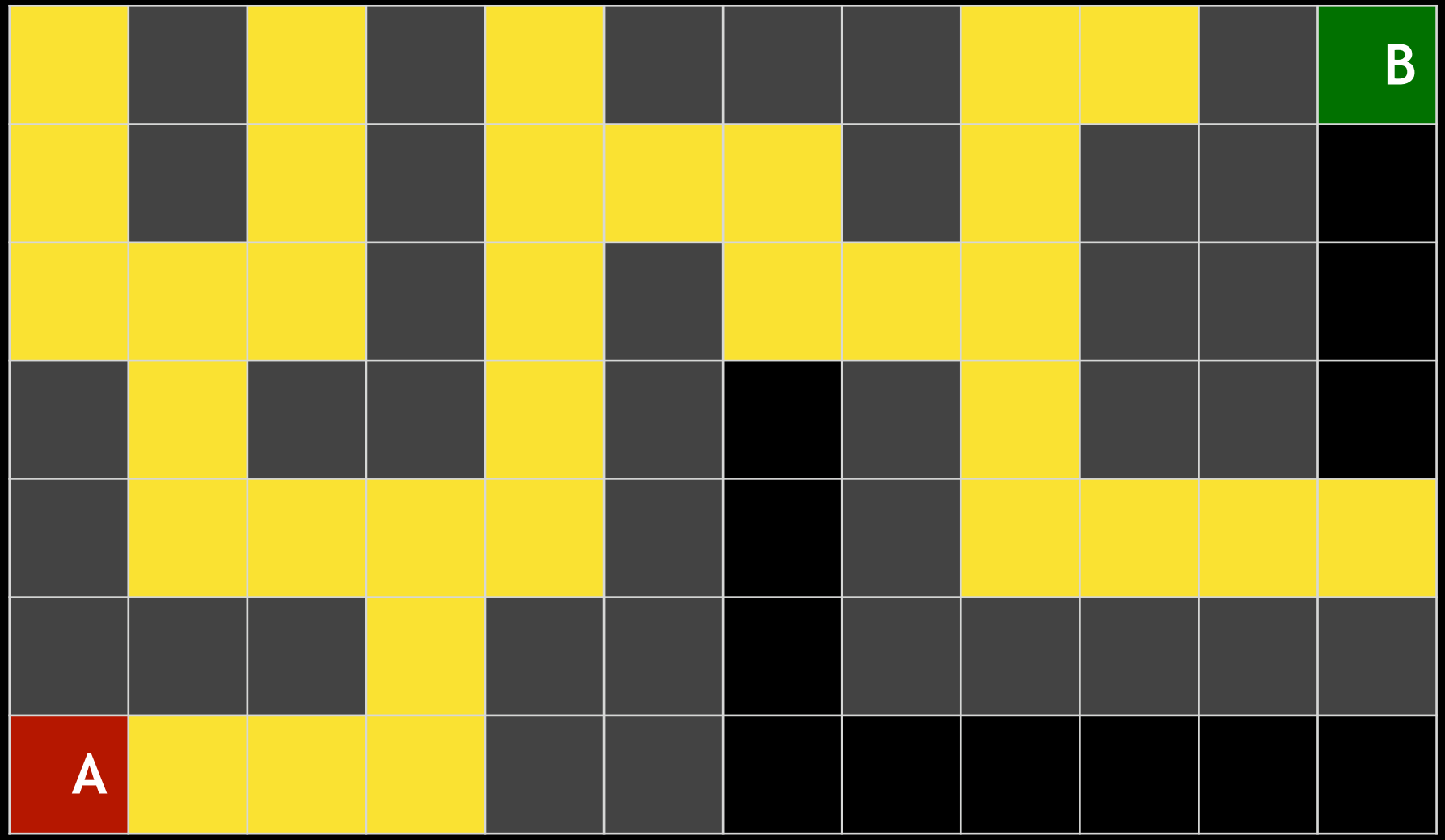

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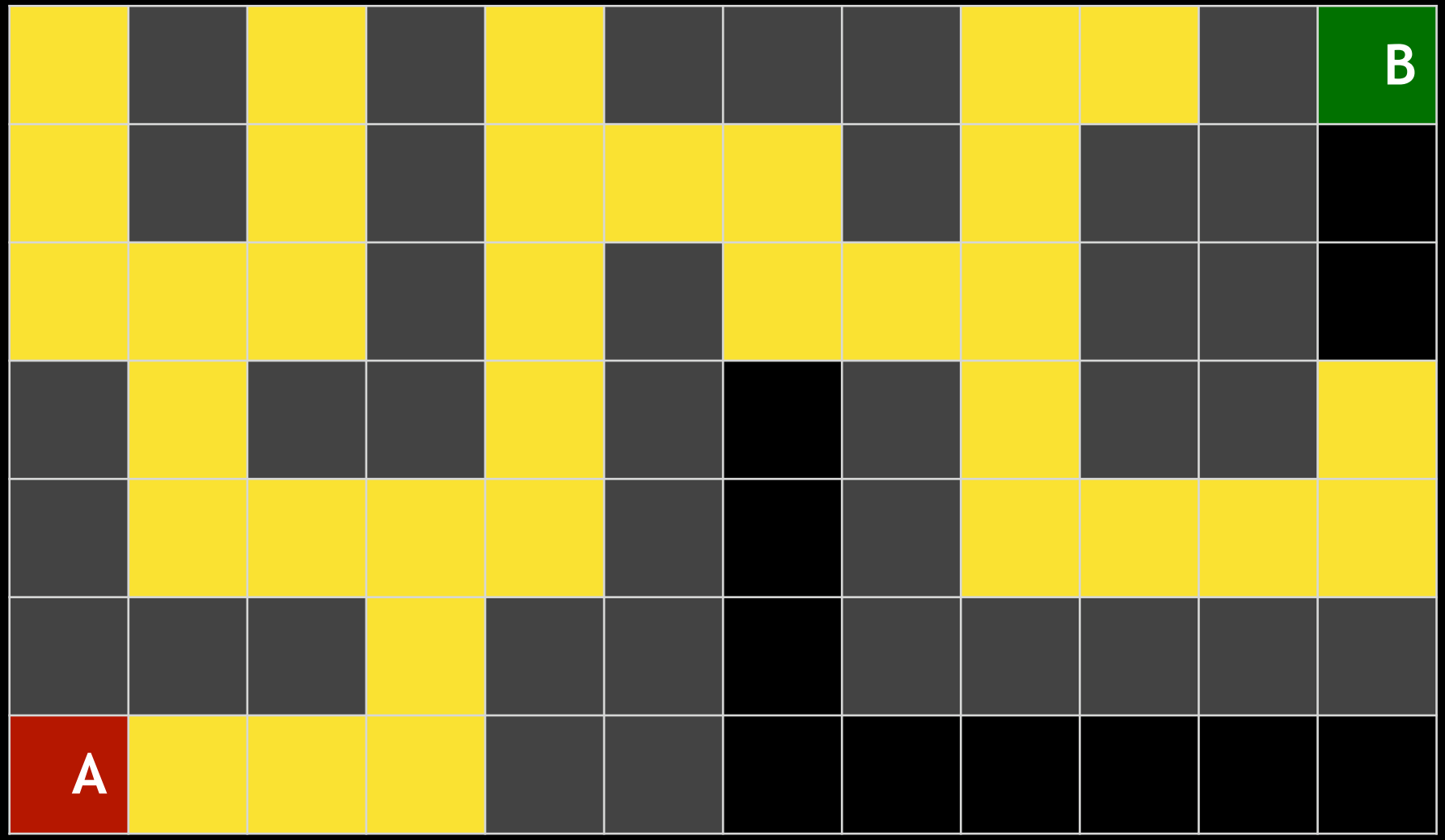

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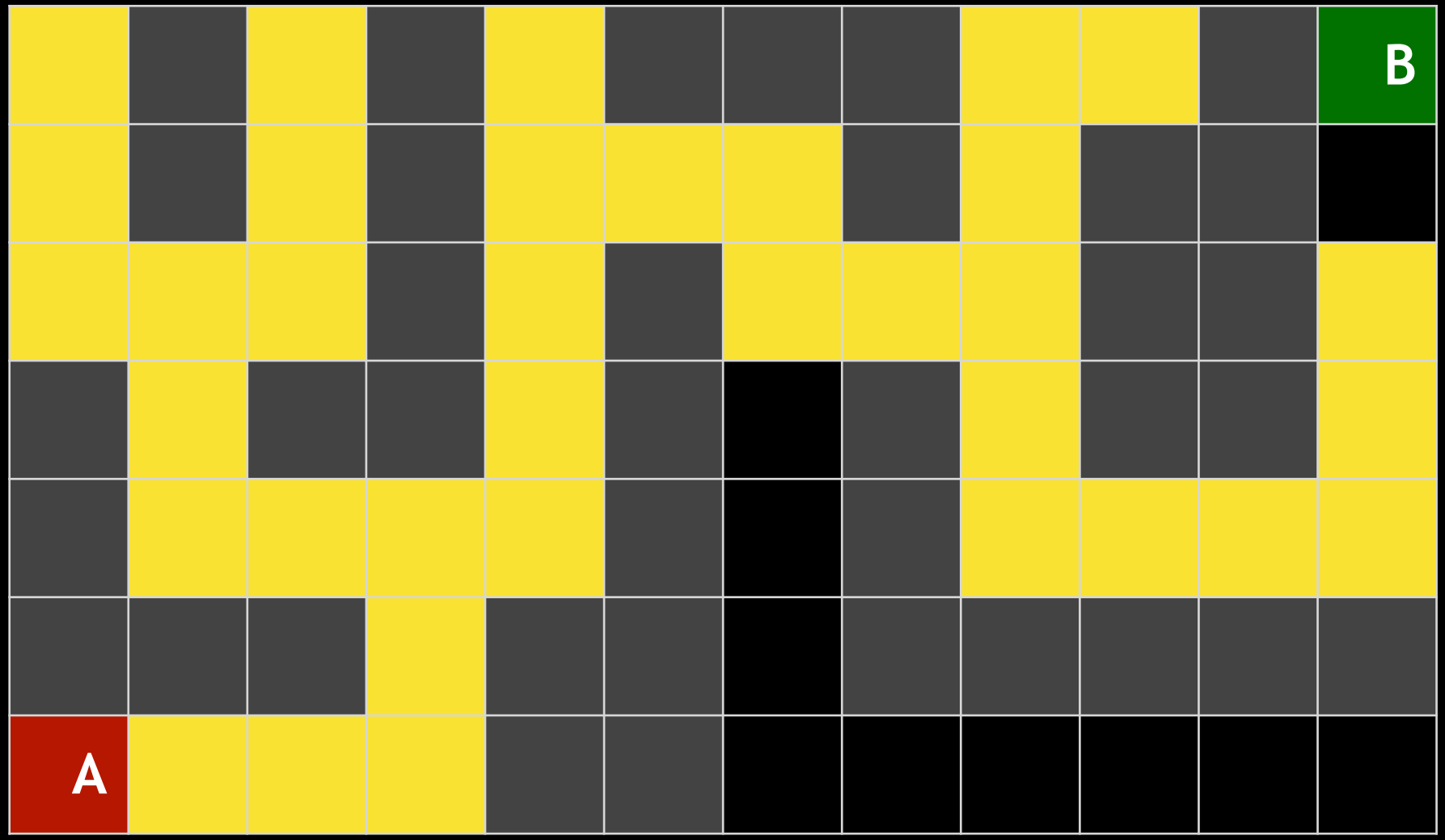

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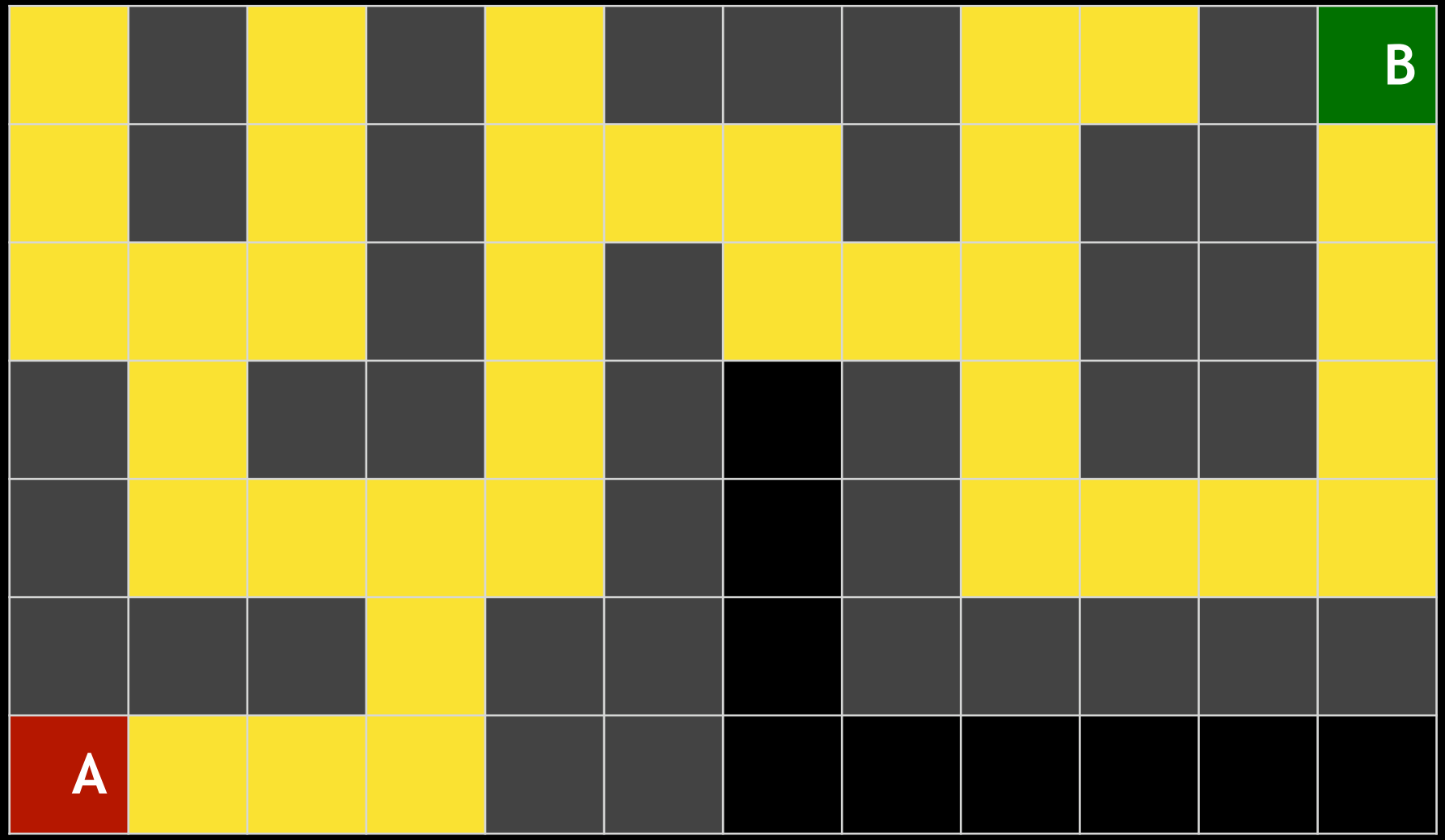
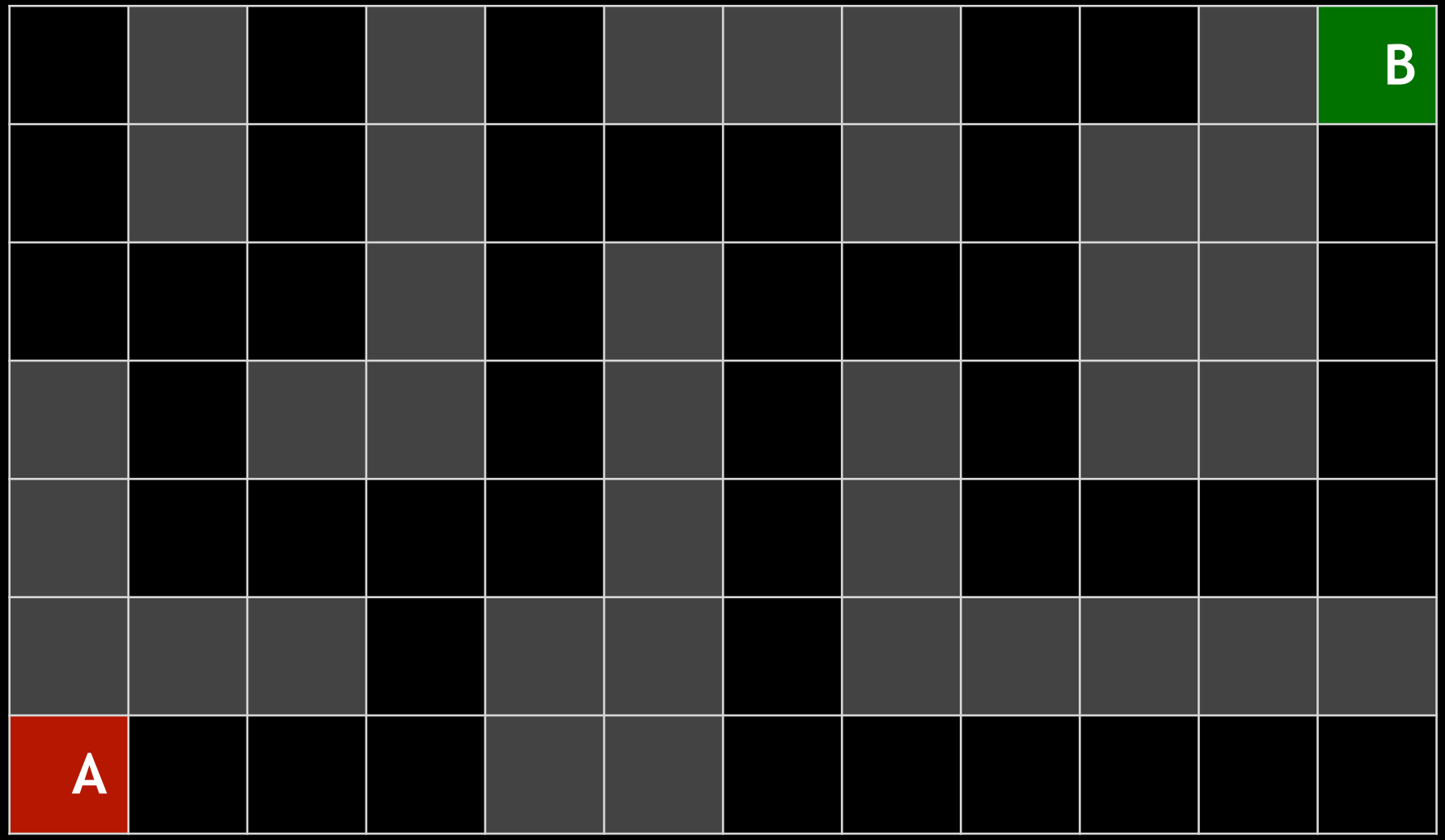


breadth-first search

algoritmo de busca que sempre expande o nó mais raso na margem

Breadth-First Search



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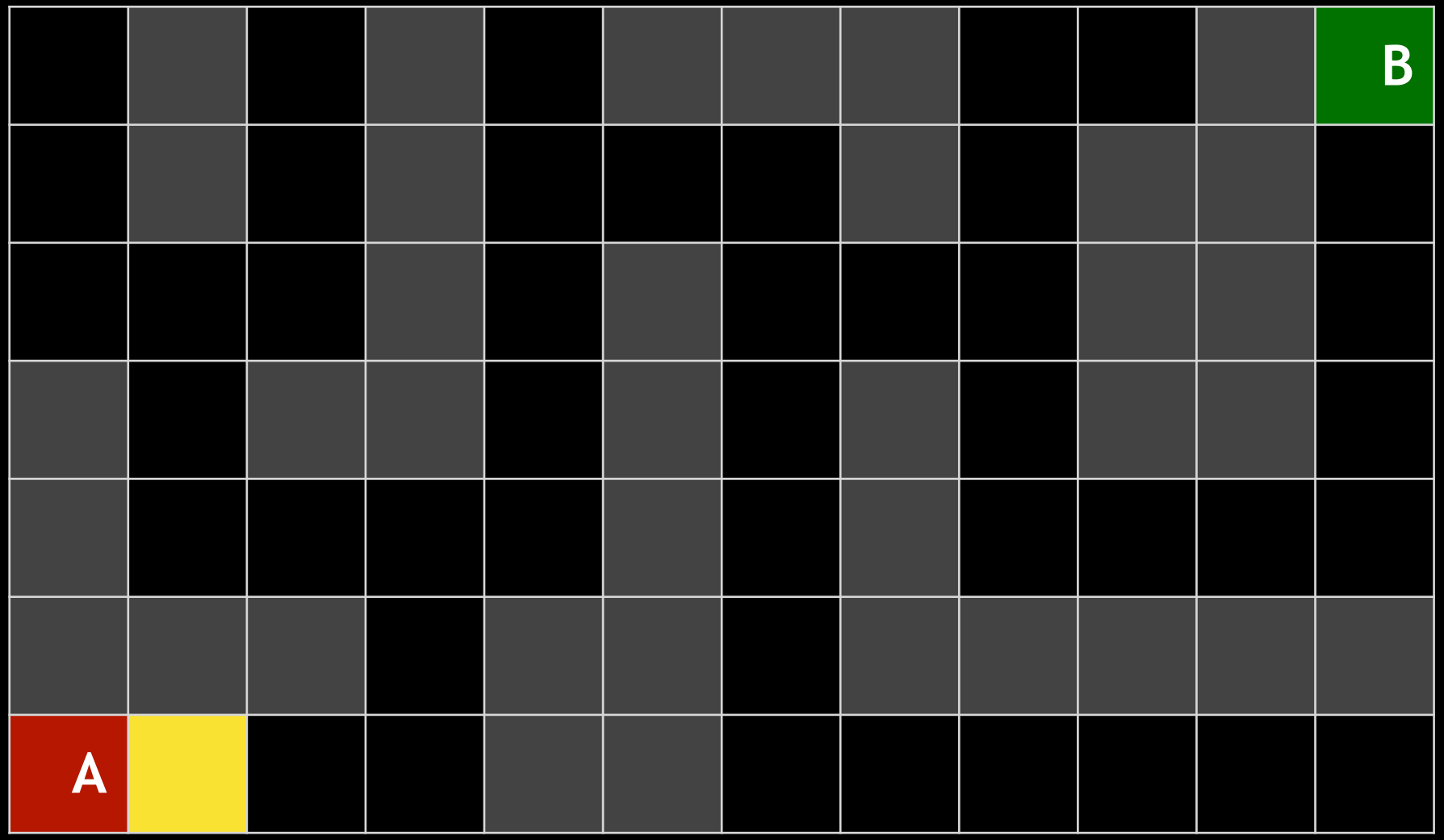

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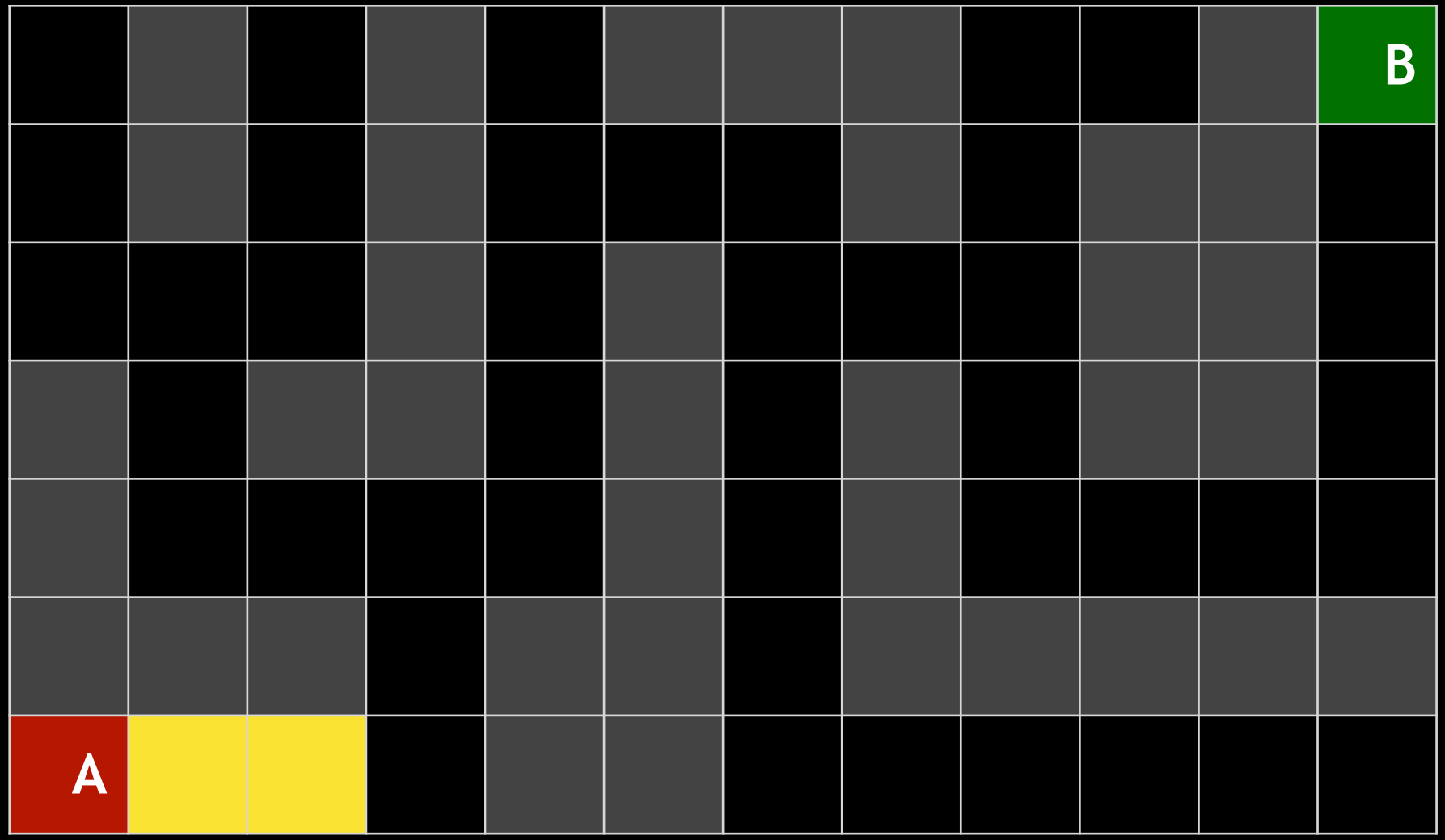

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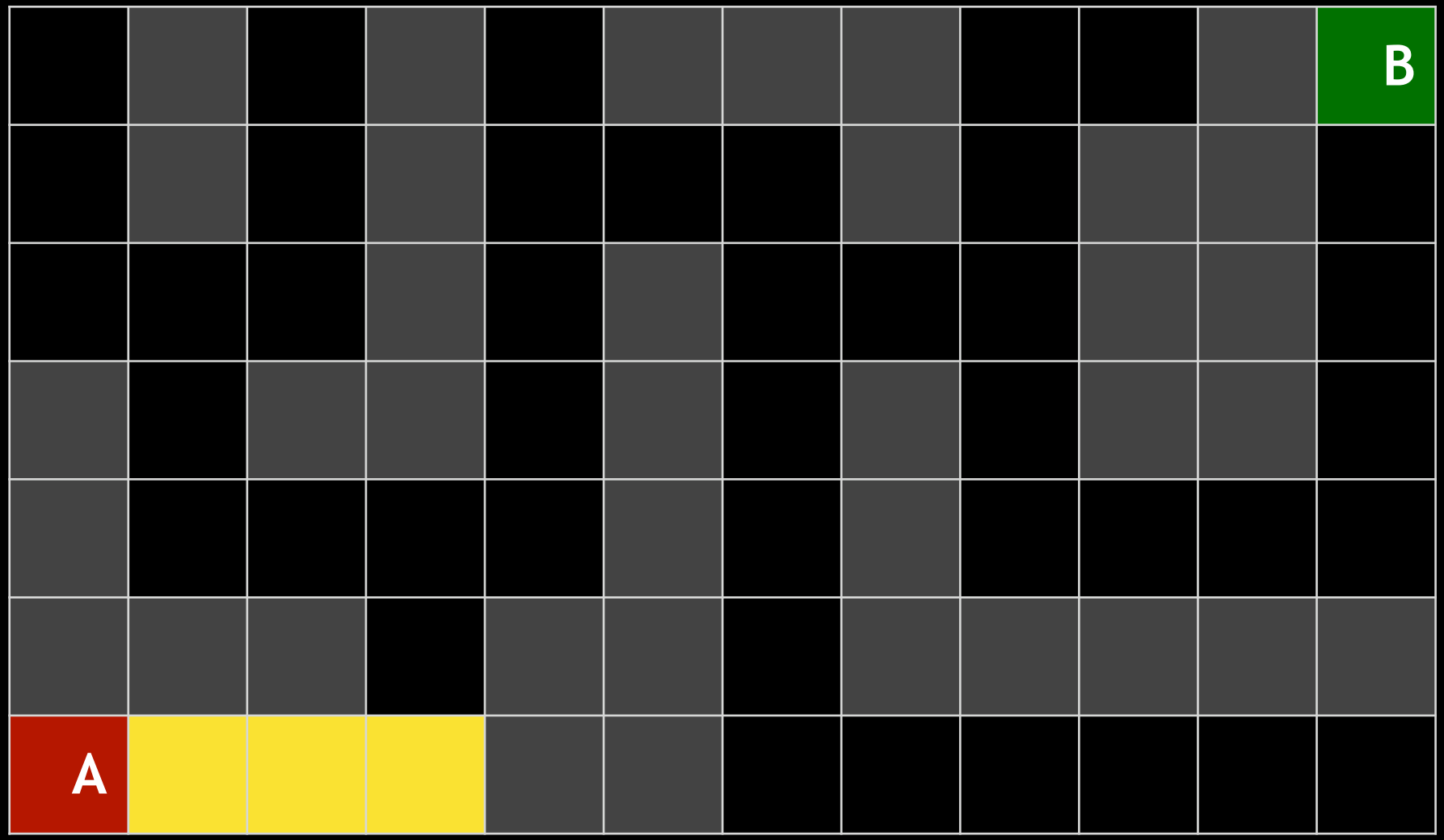

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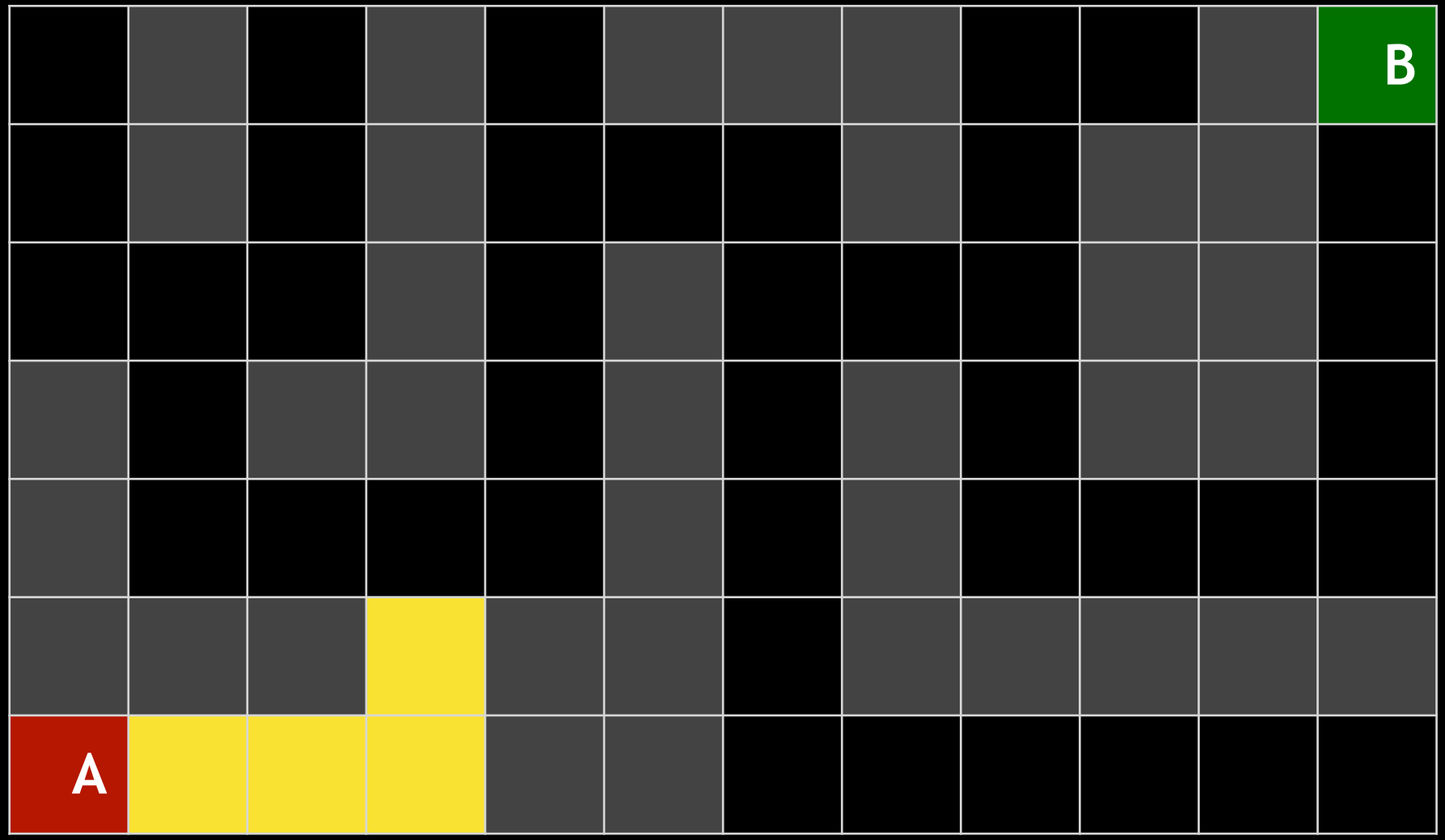

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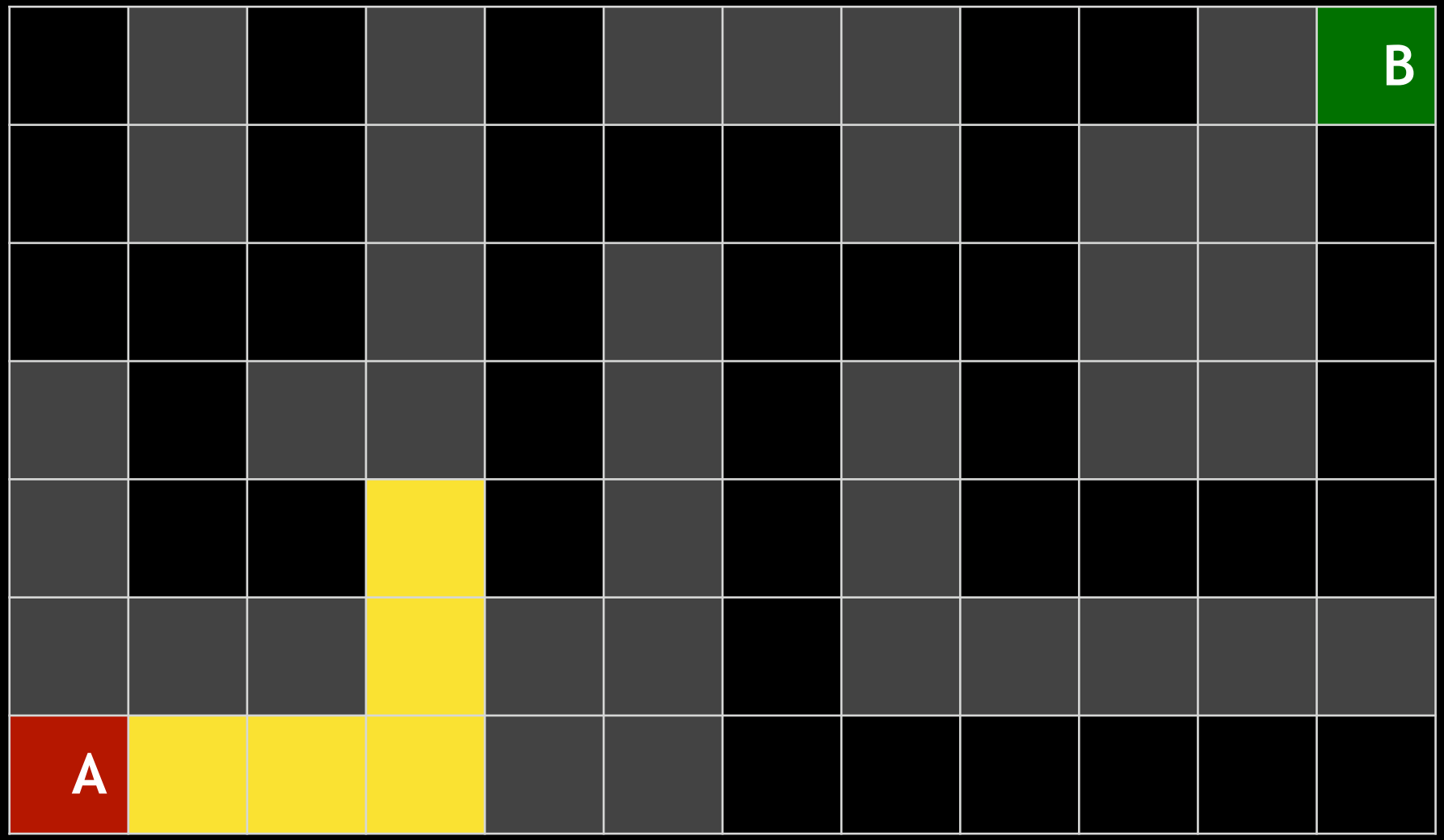

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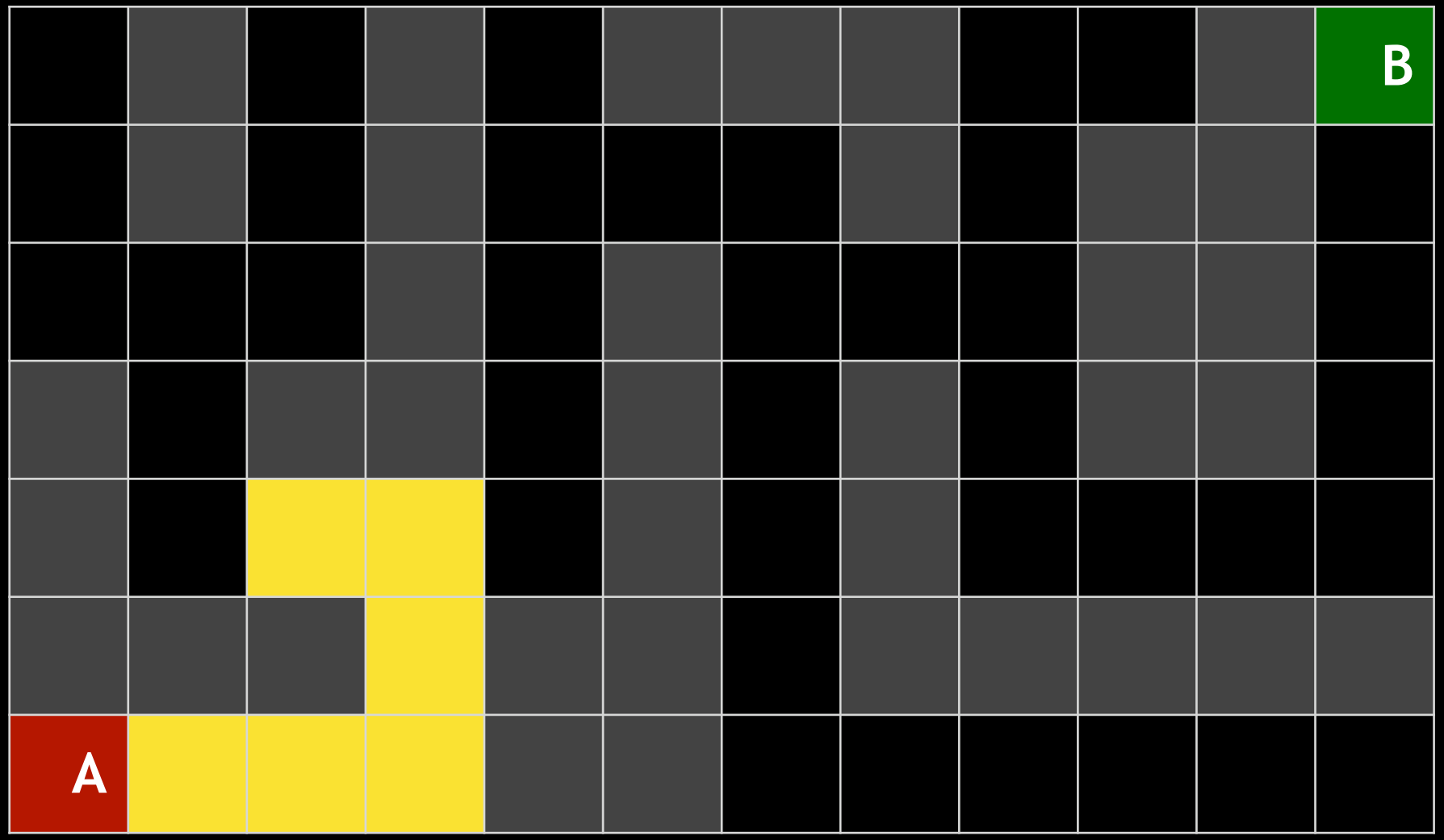

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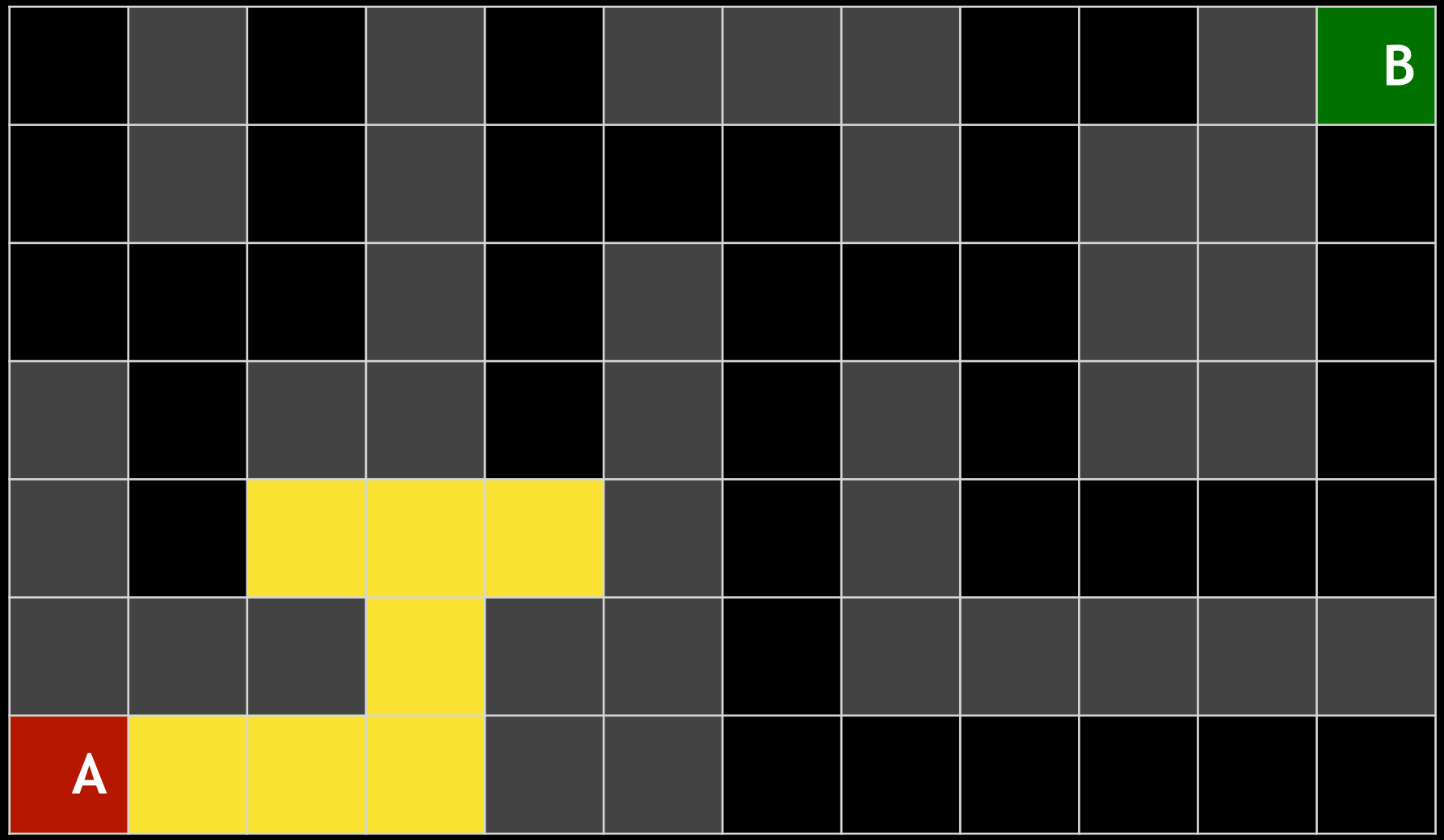

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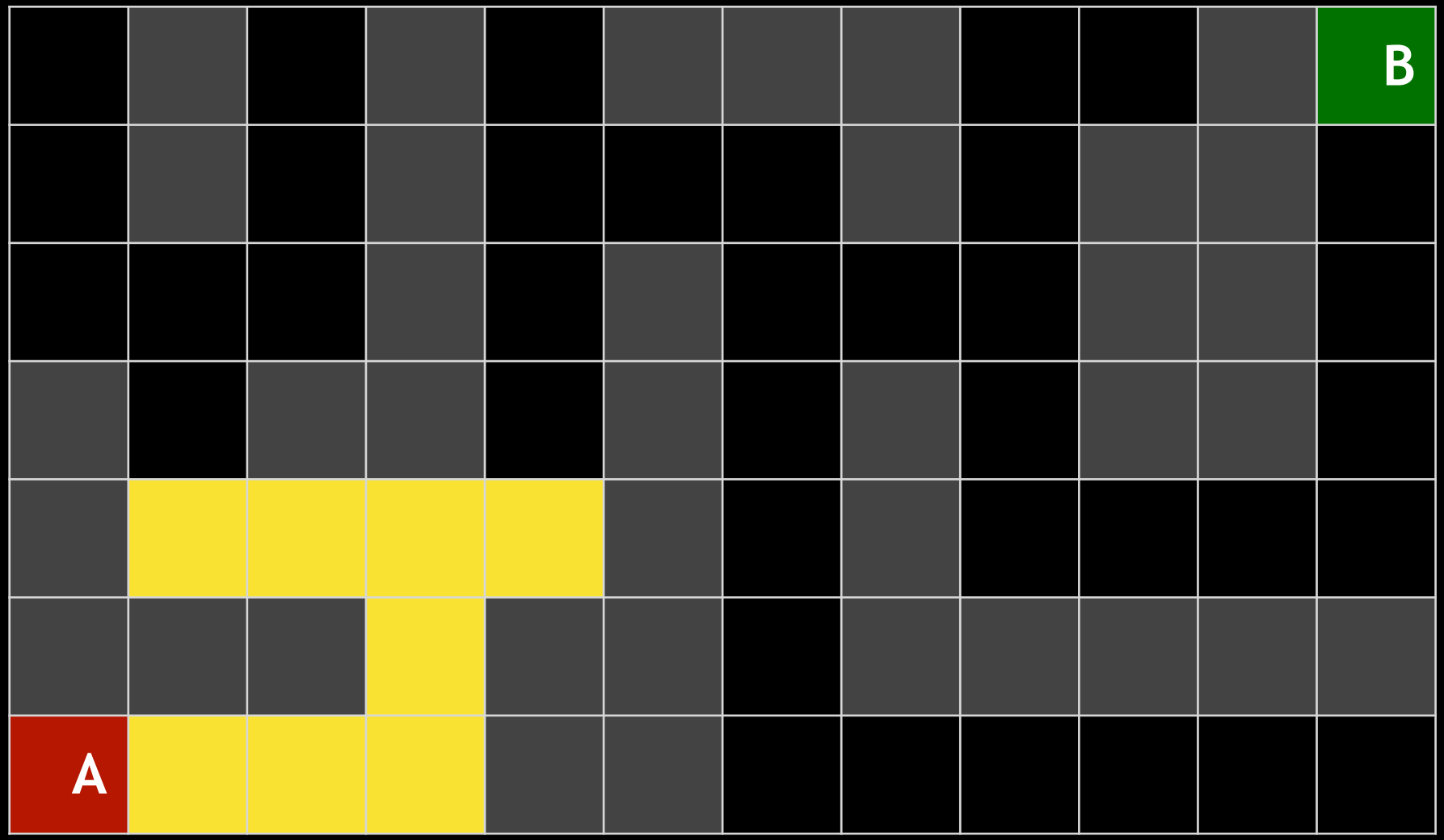

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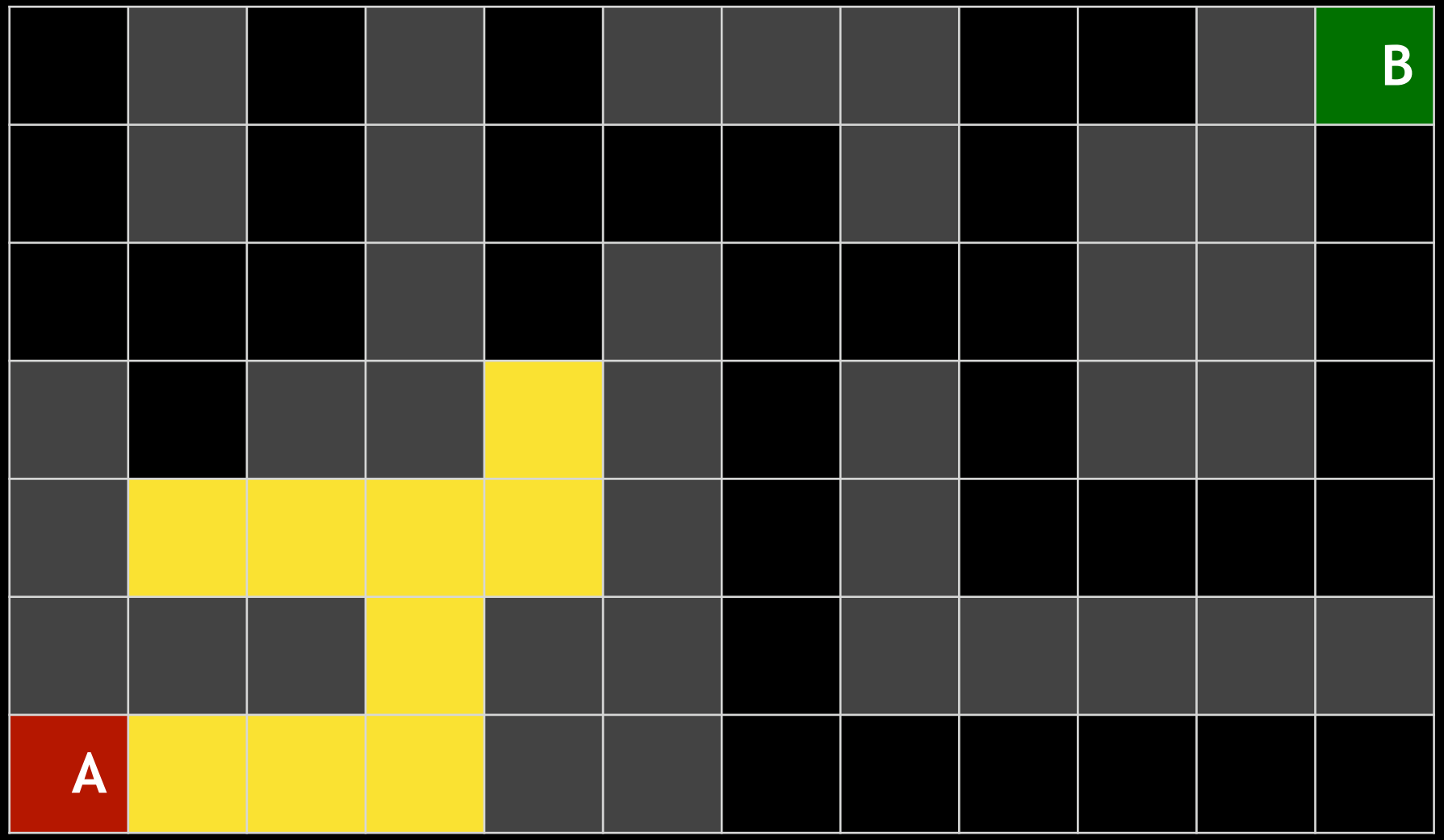

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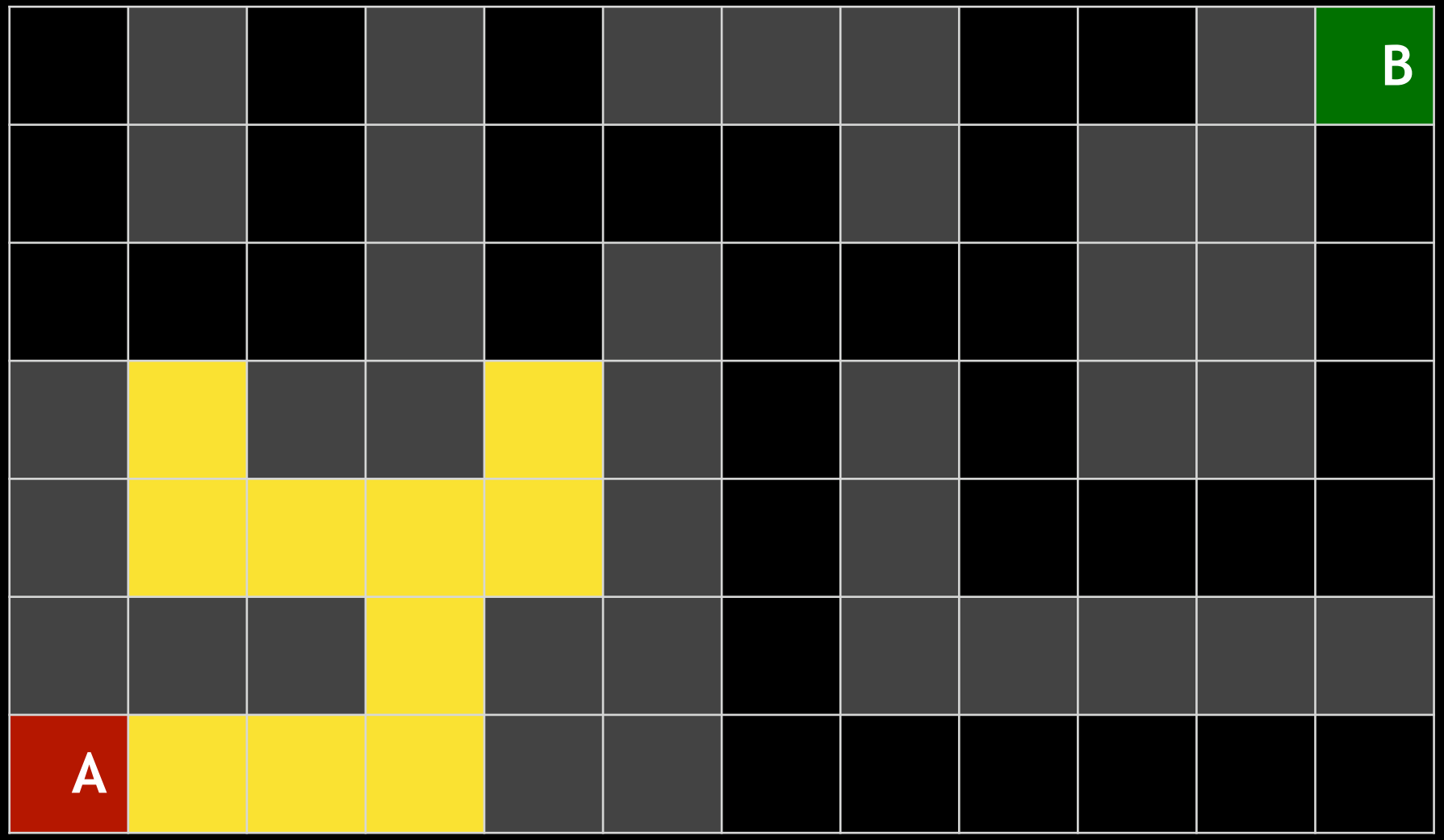

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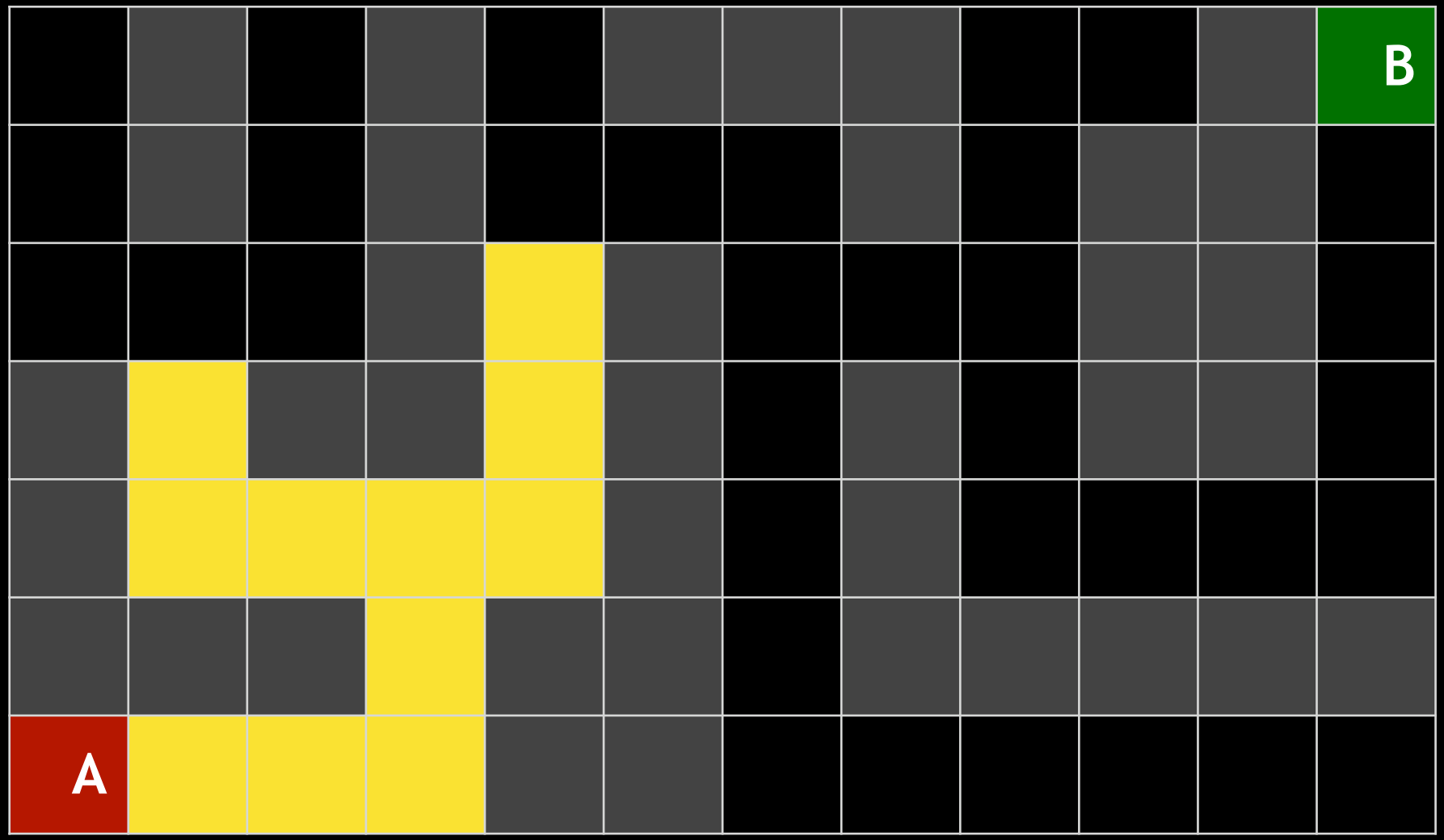

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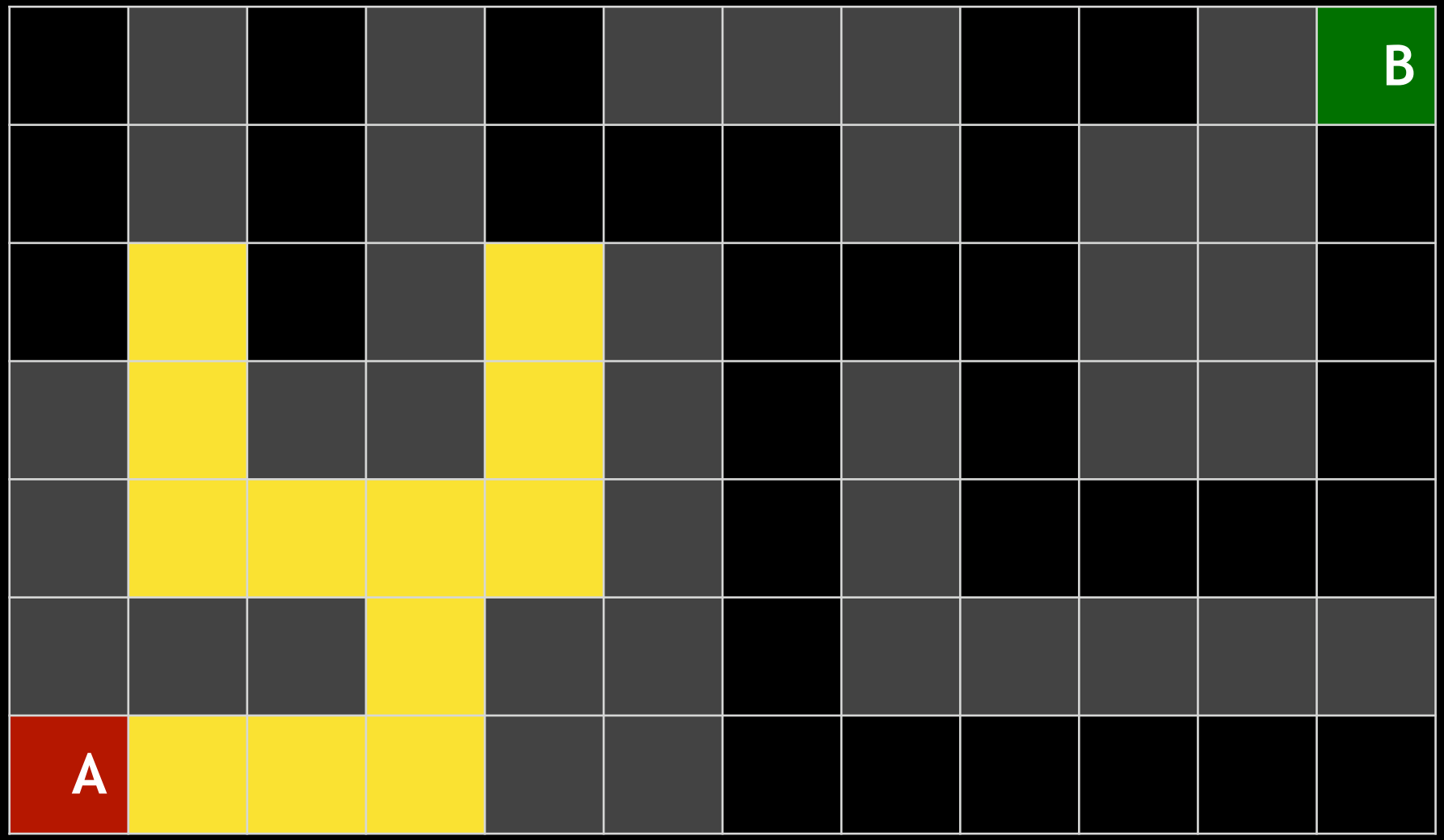

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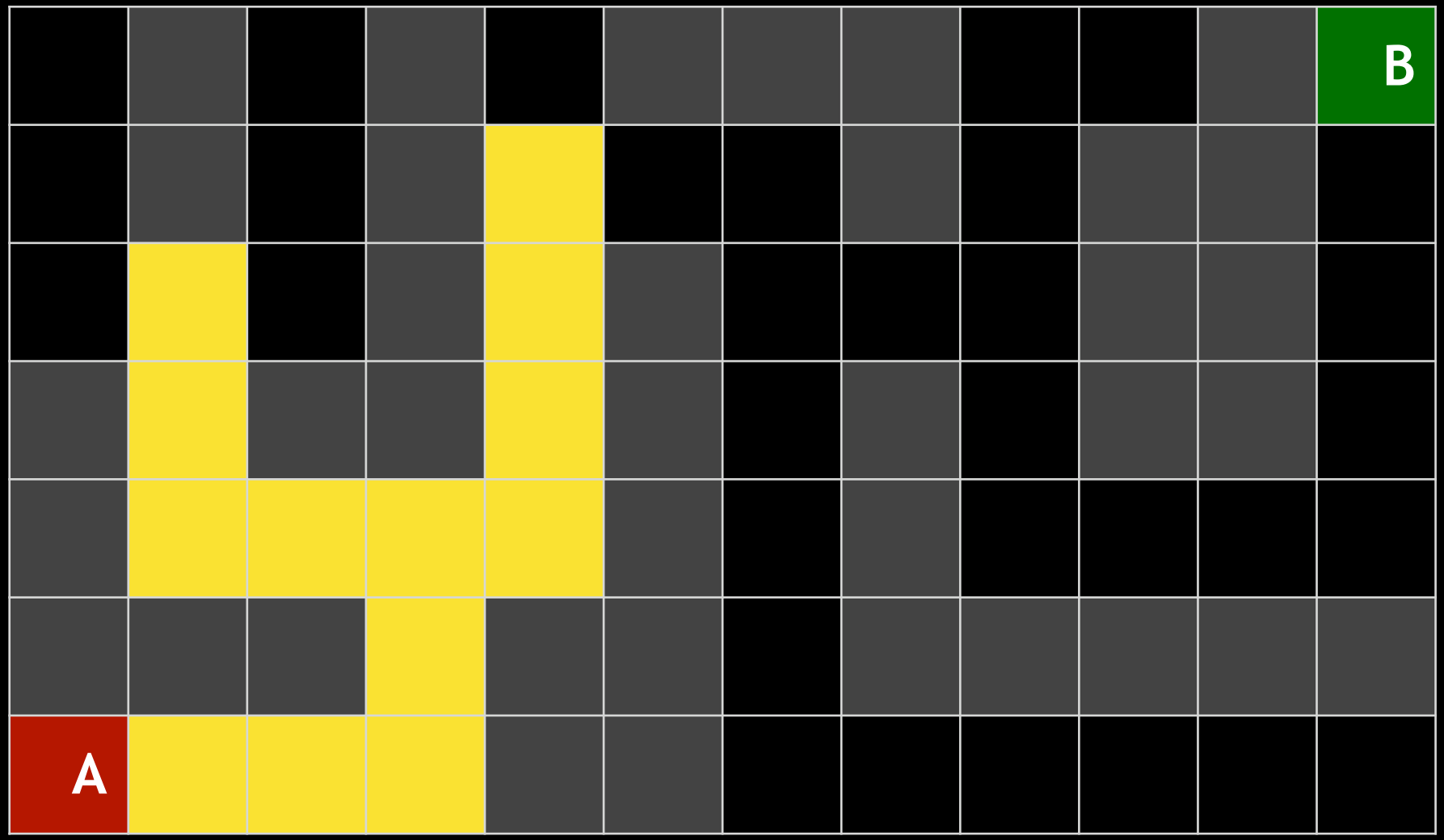

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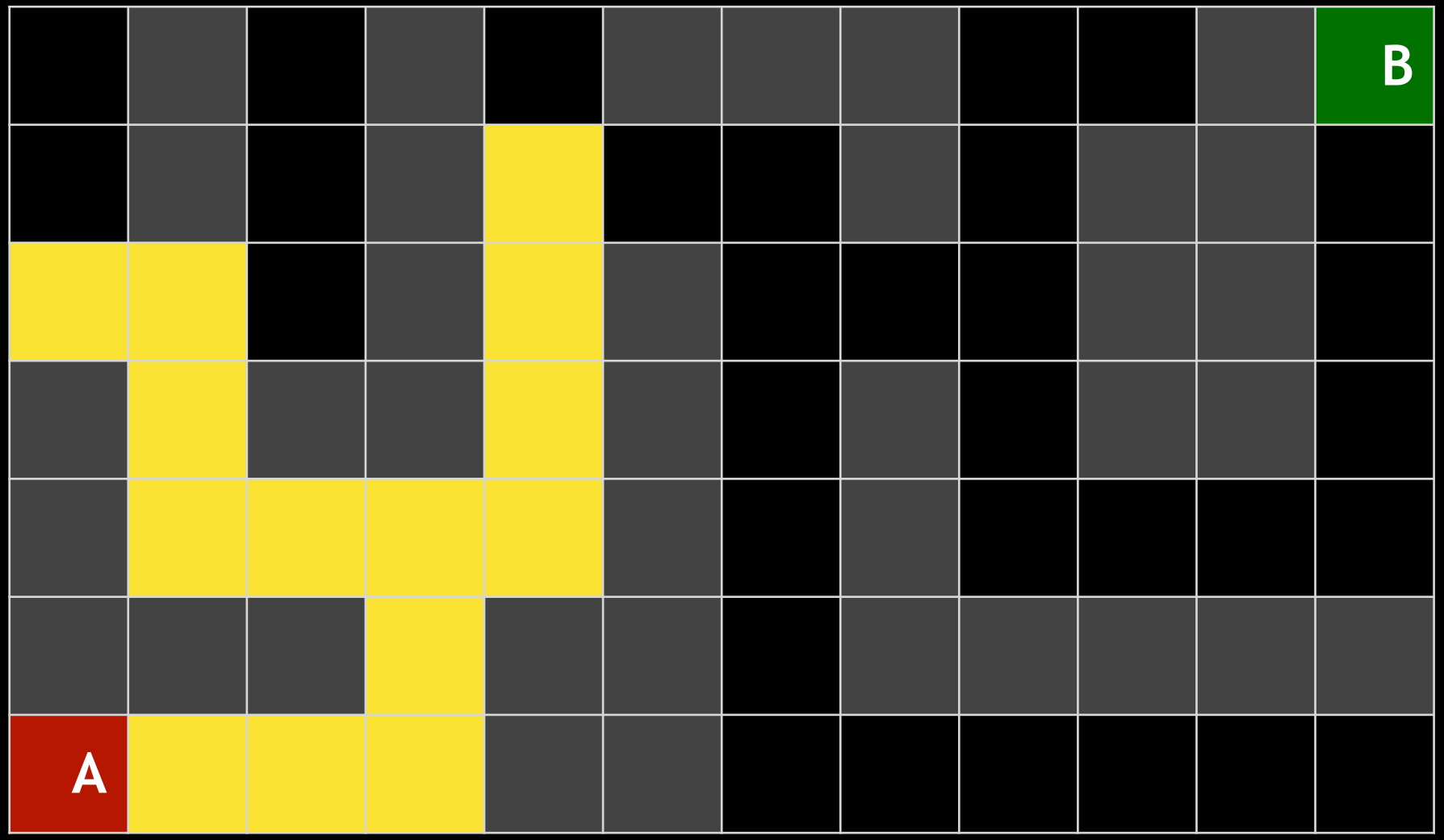

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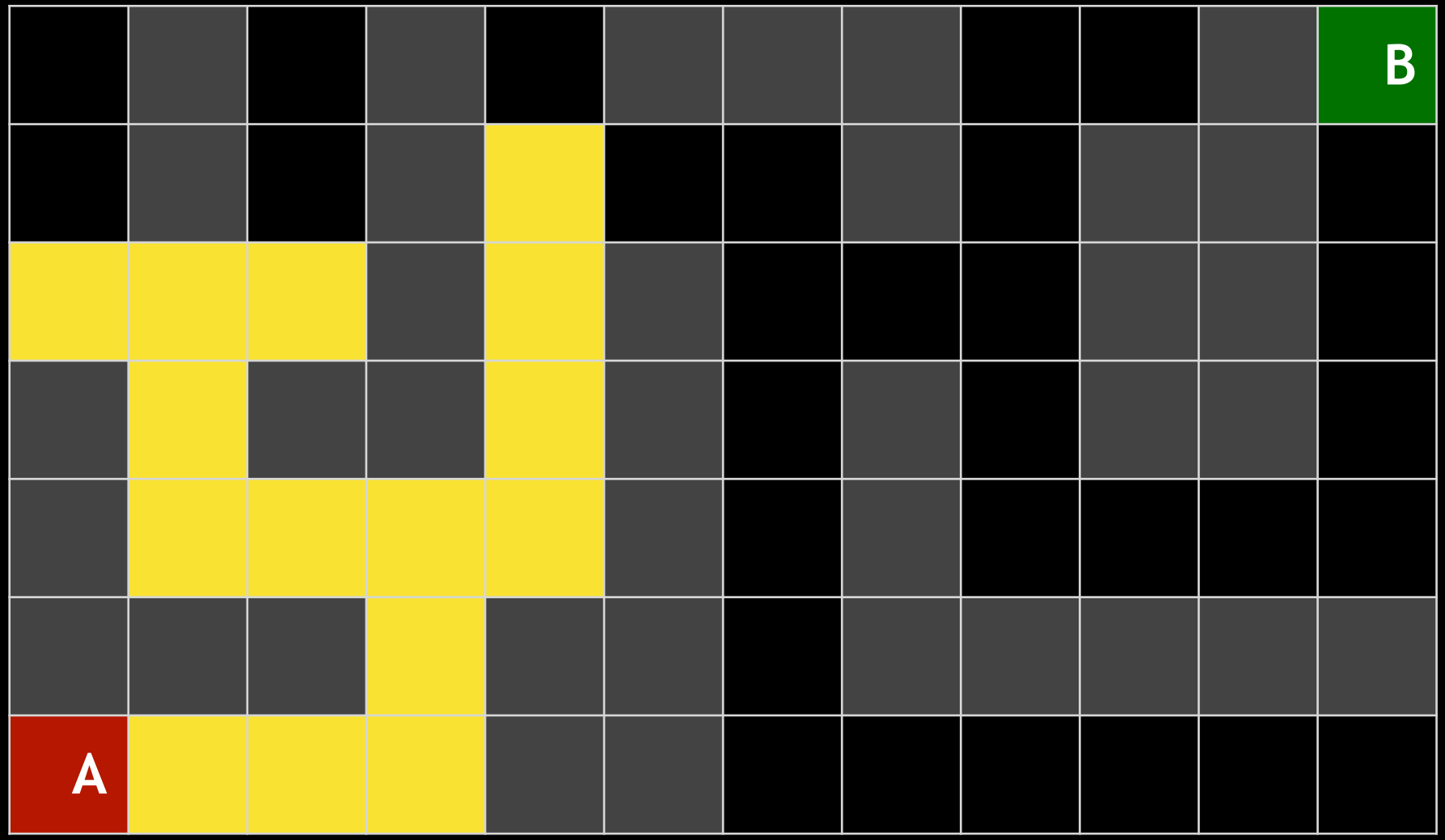

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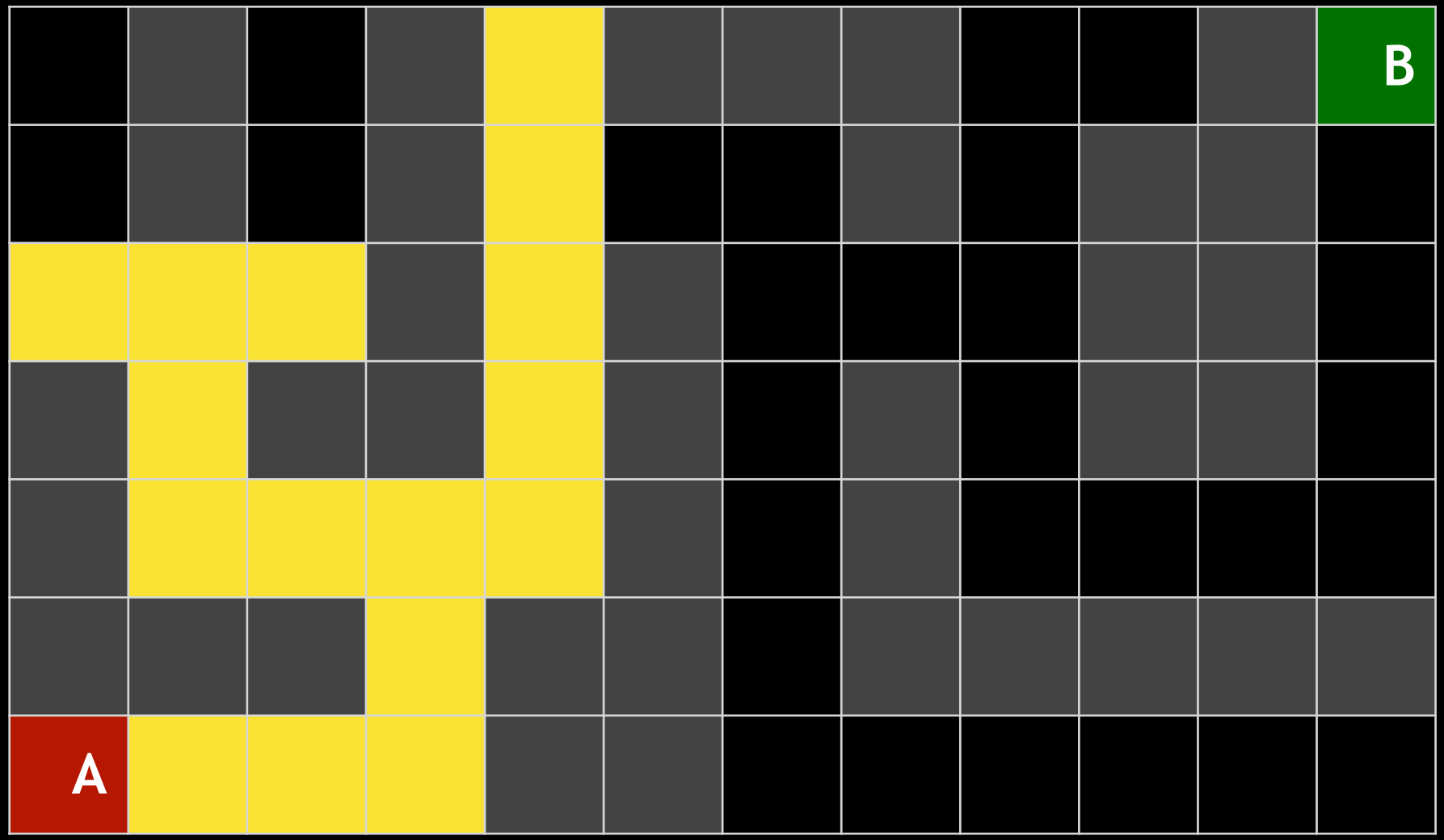

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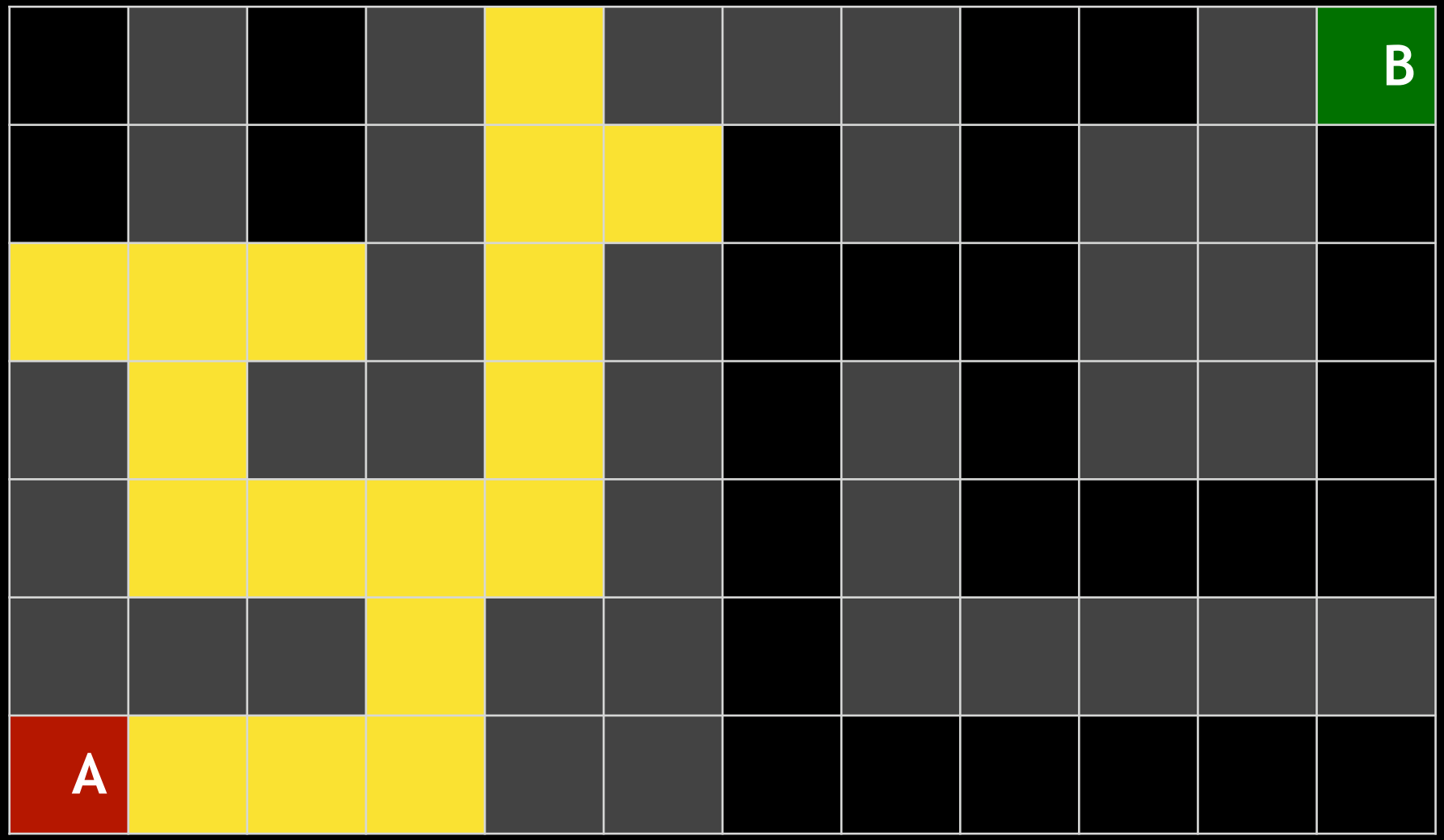

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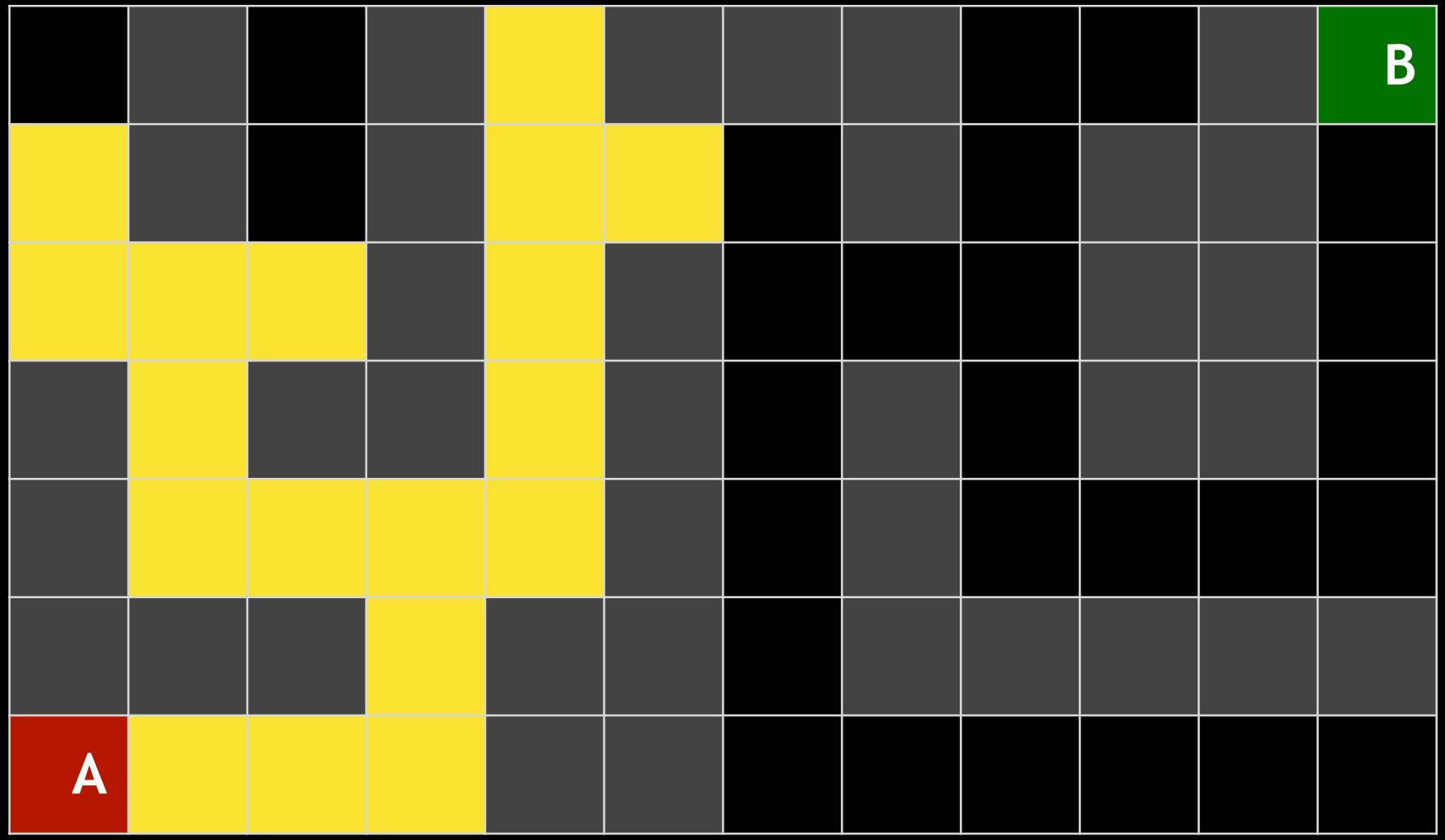

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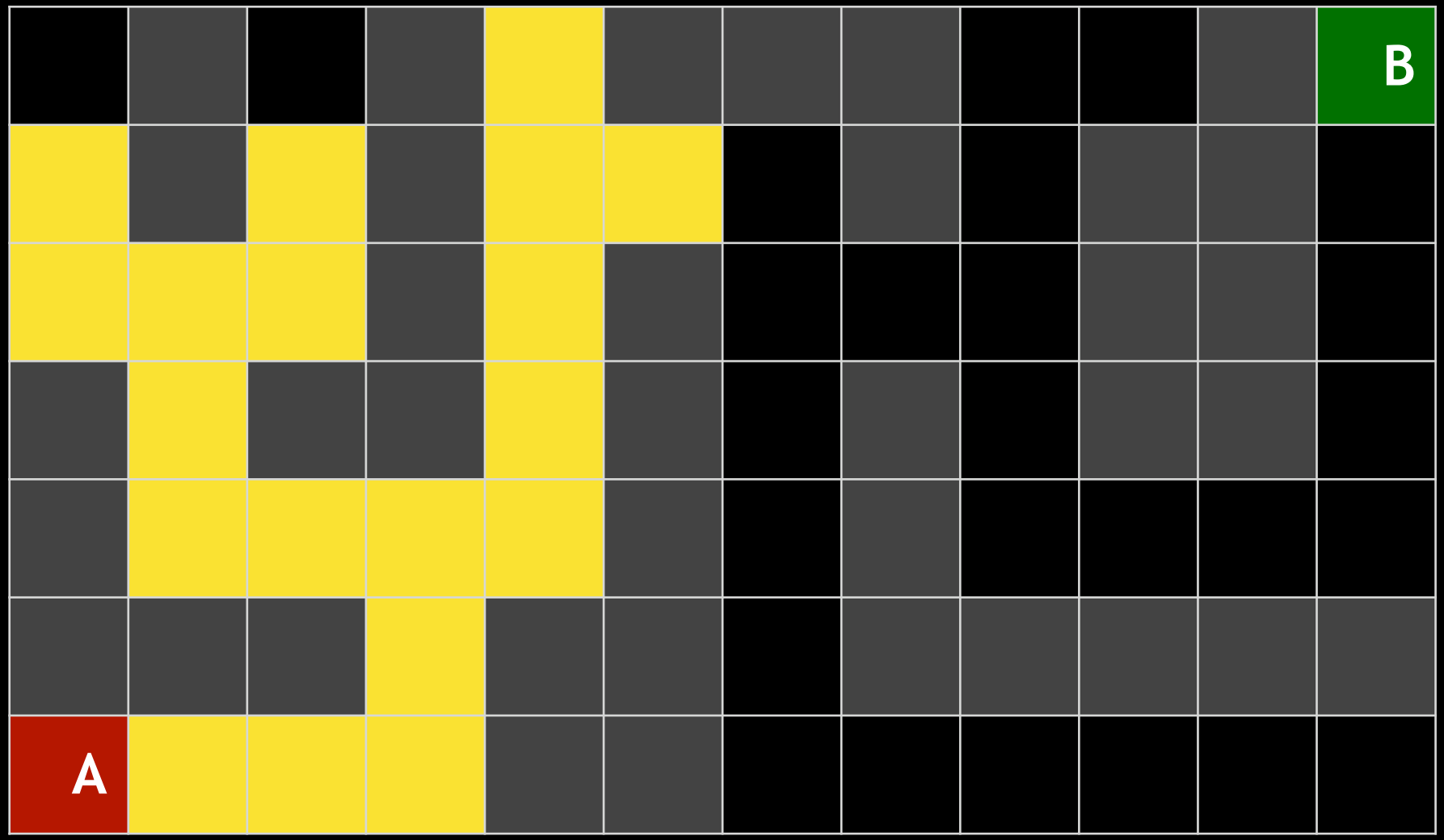

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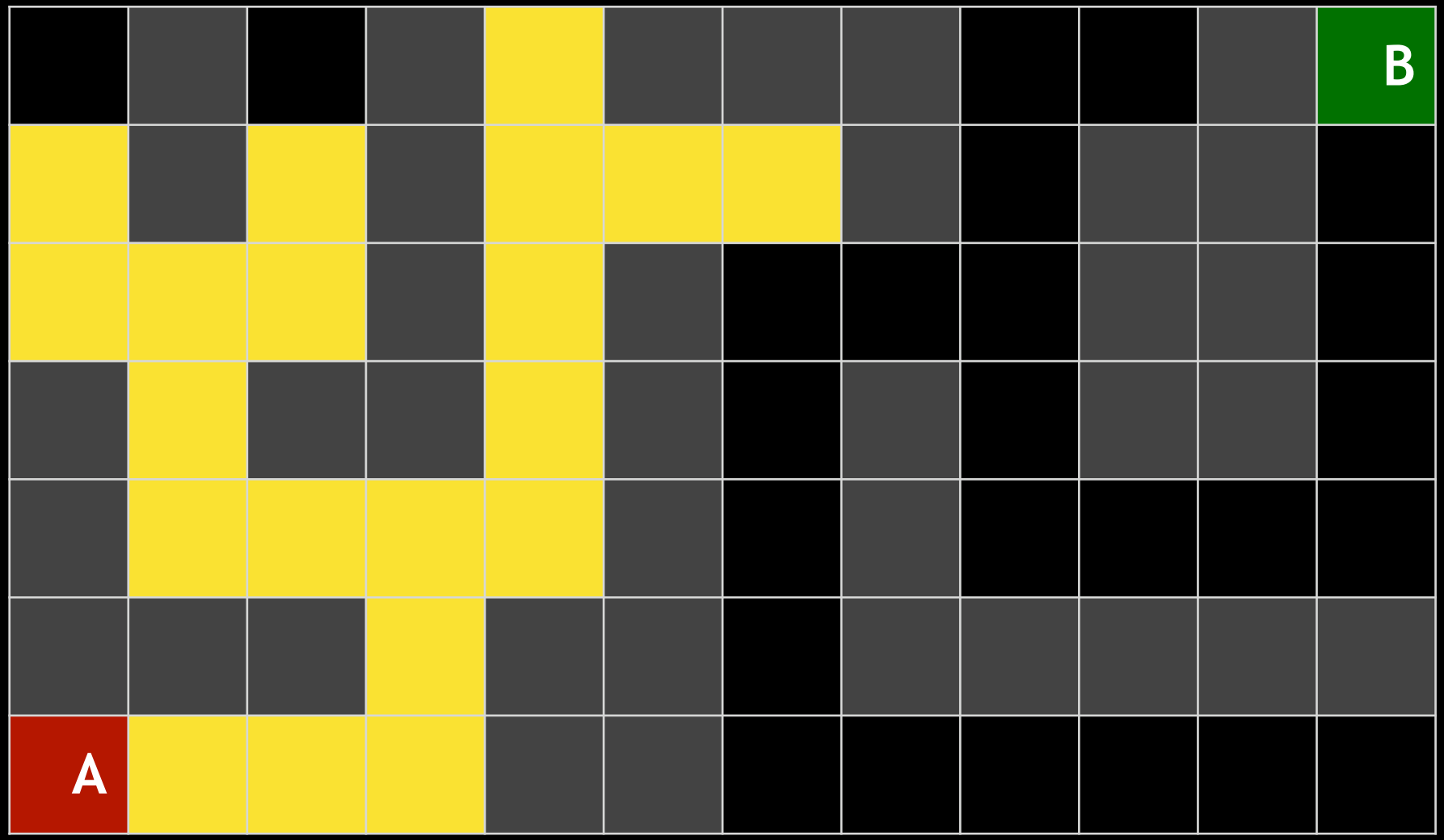

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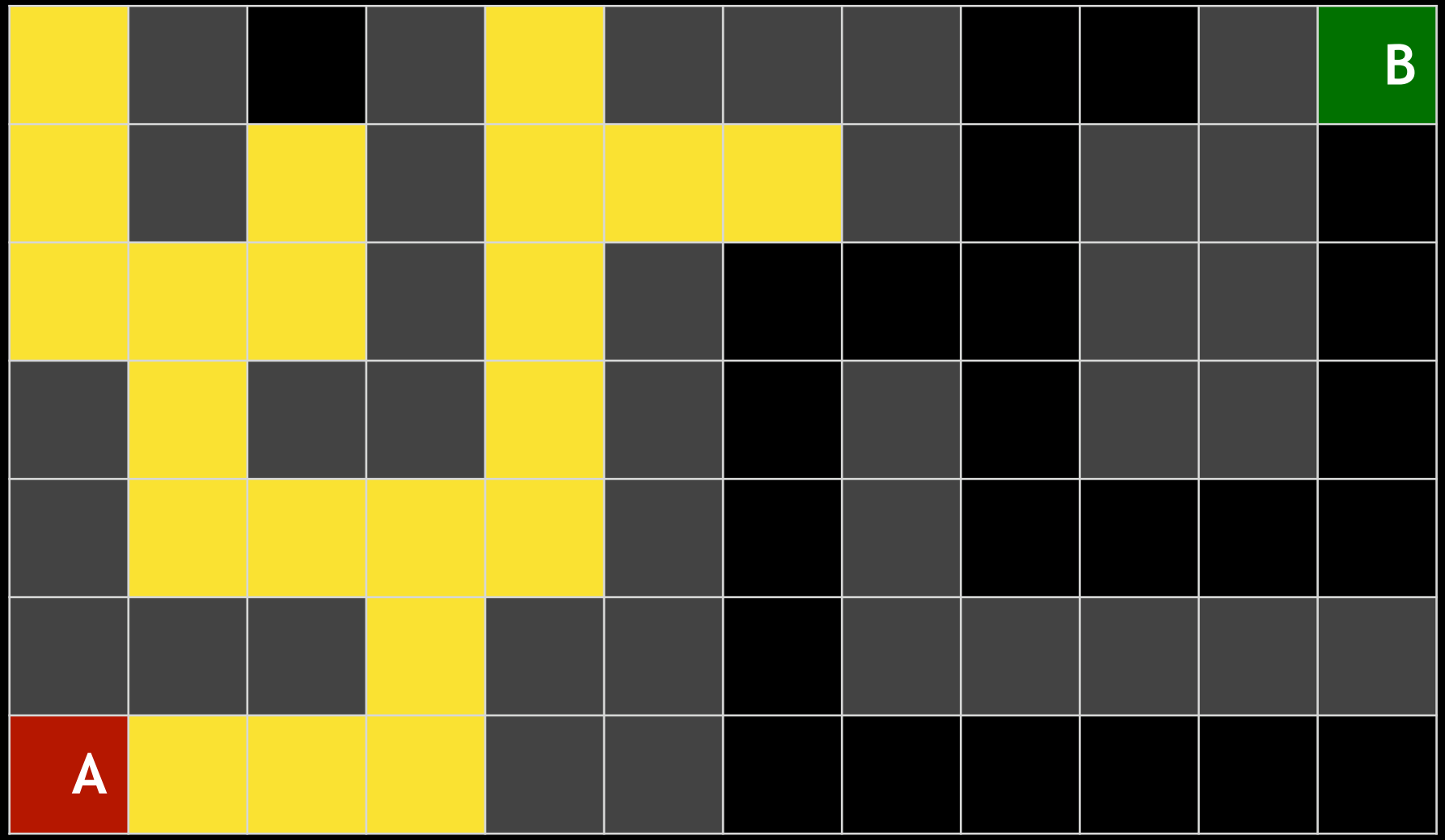

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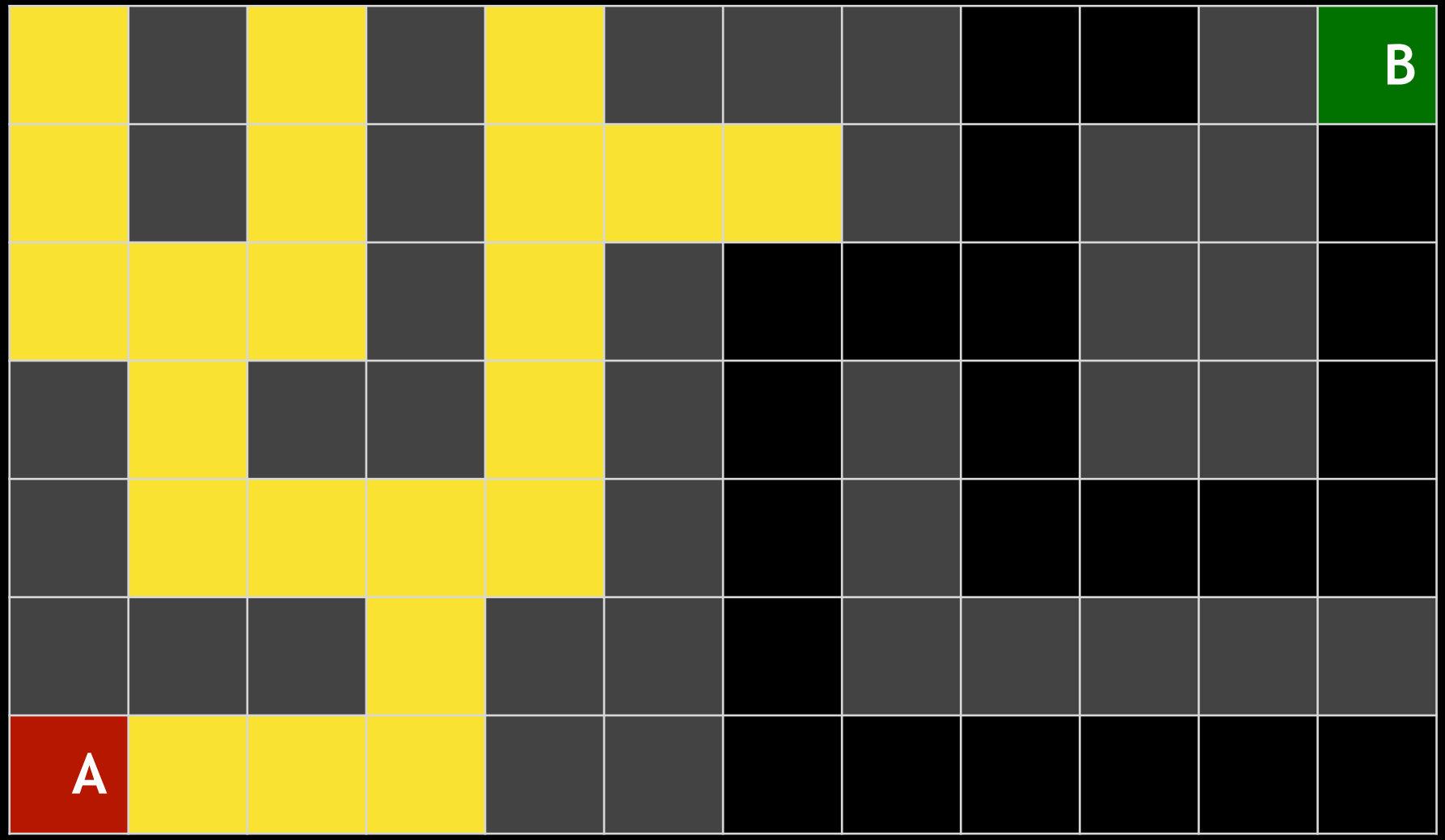

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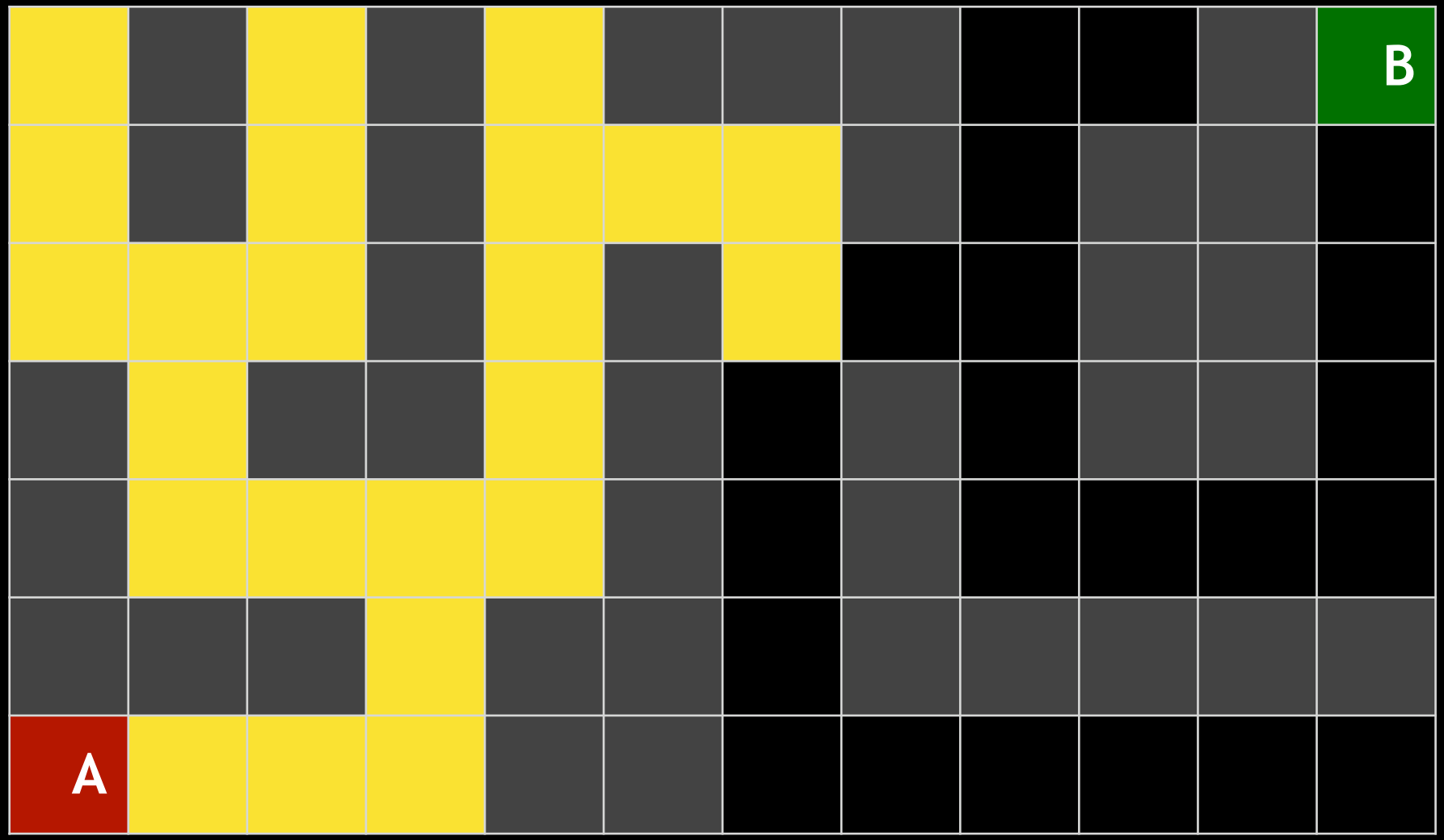

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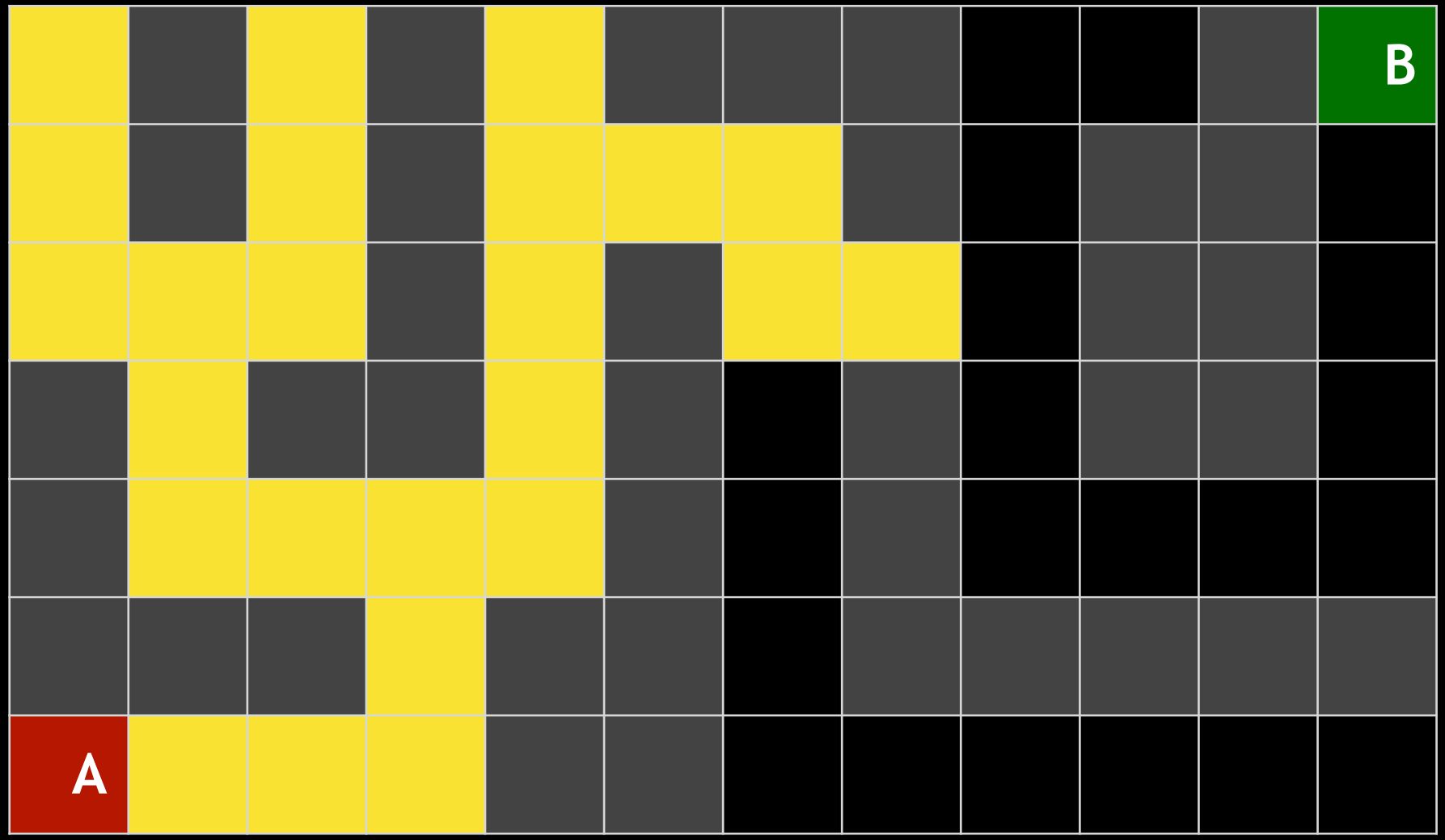

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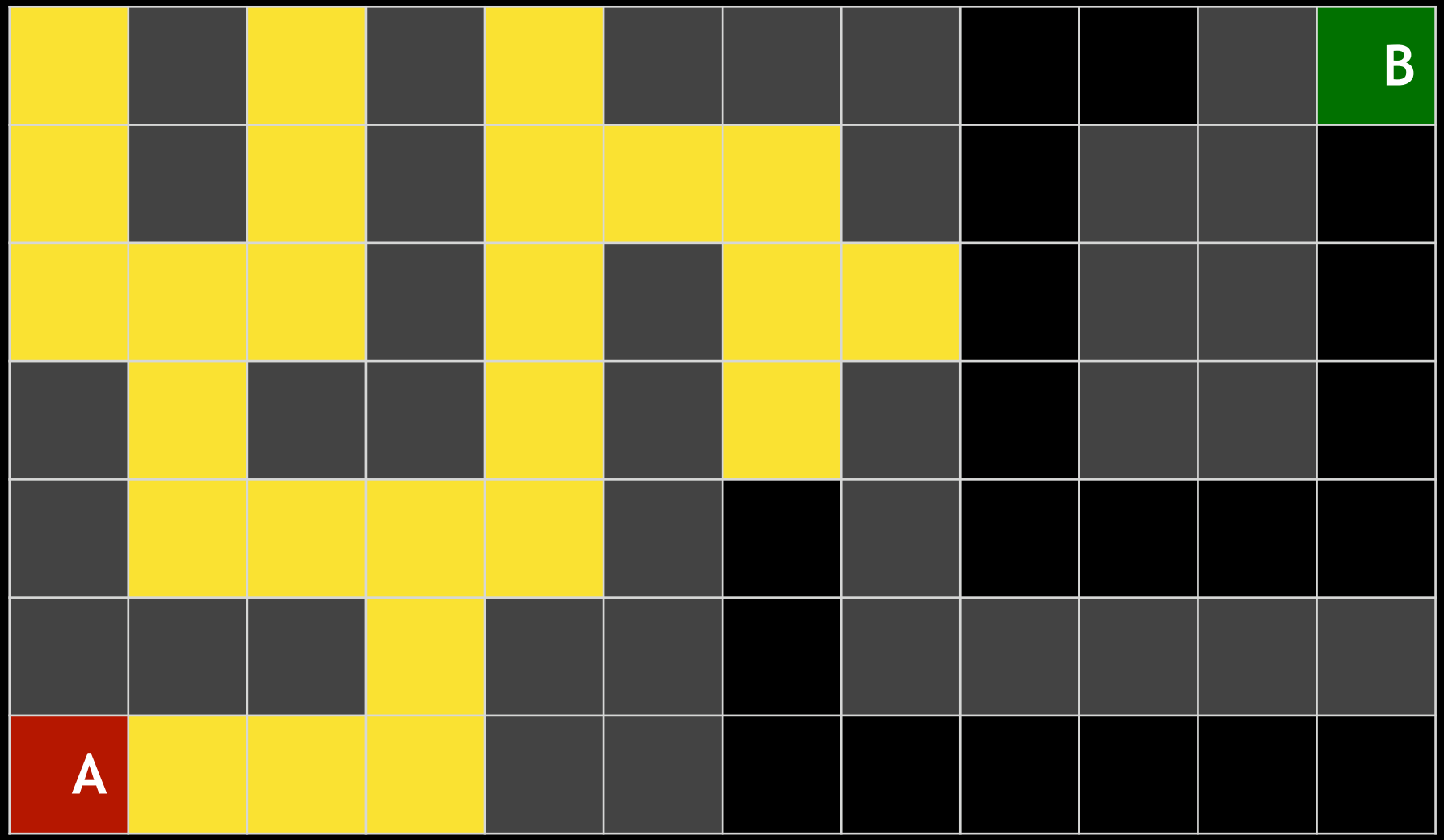

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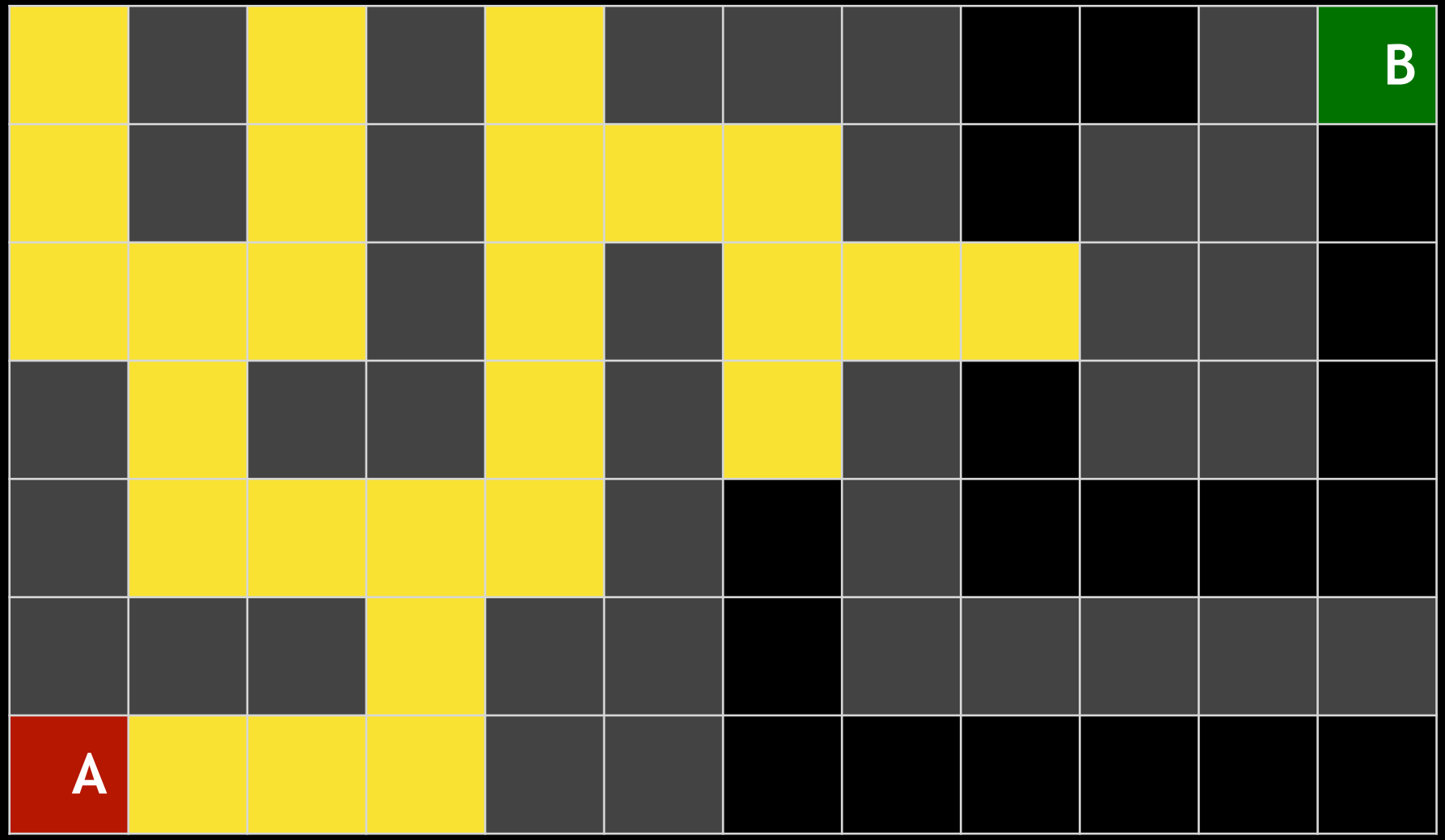

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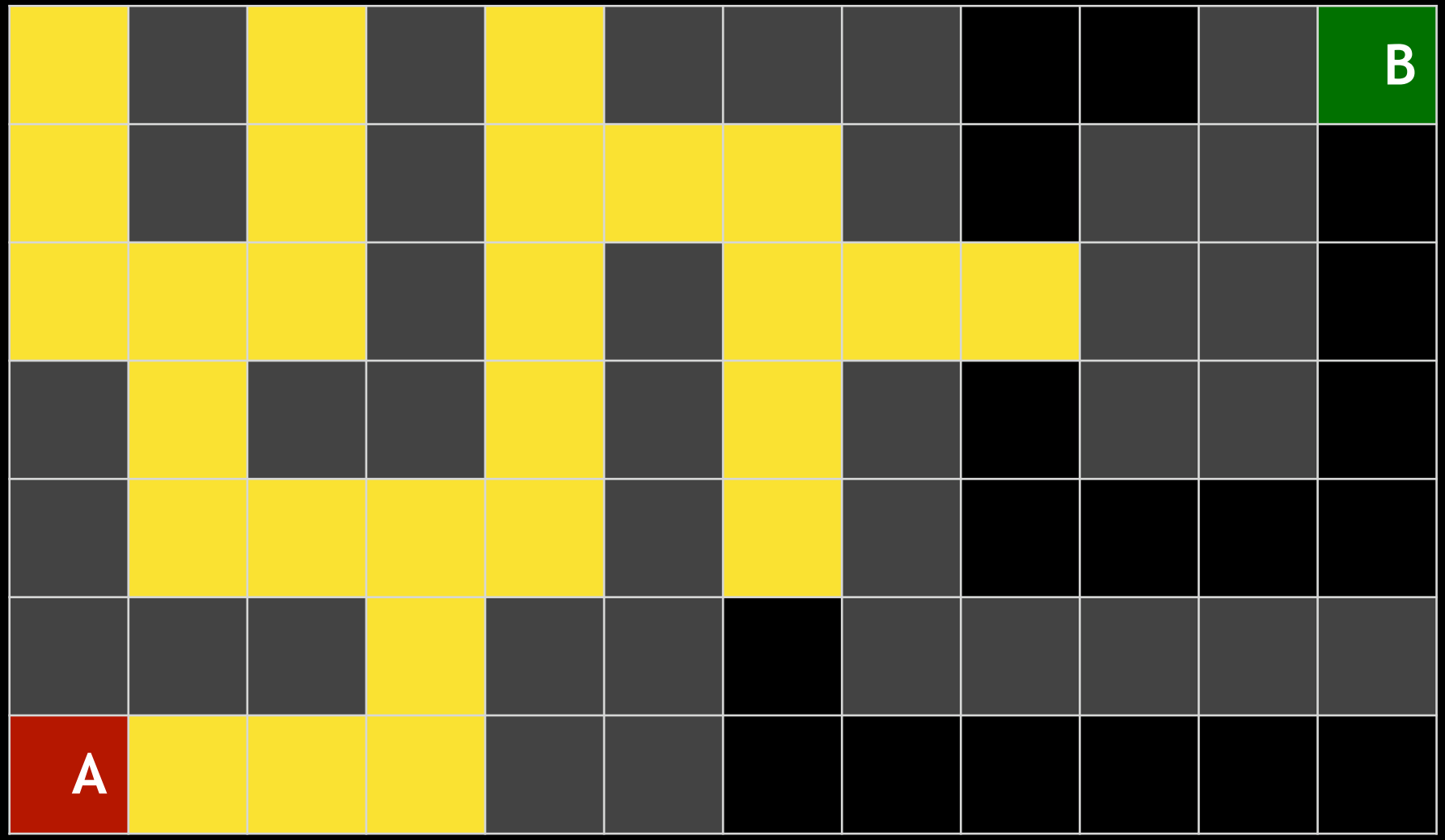

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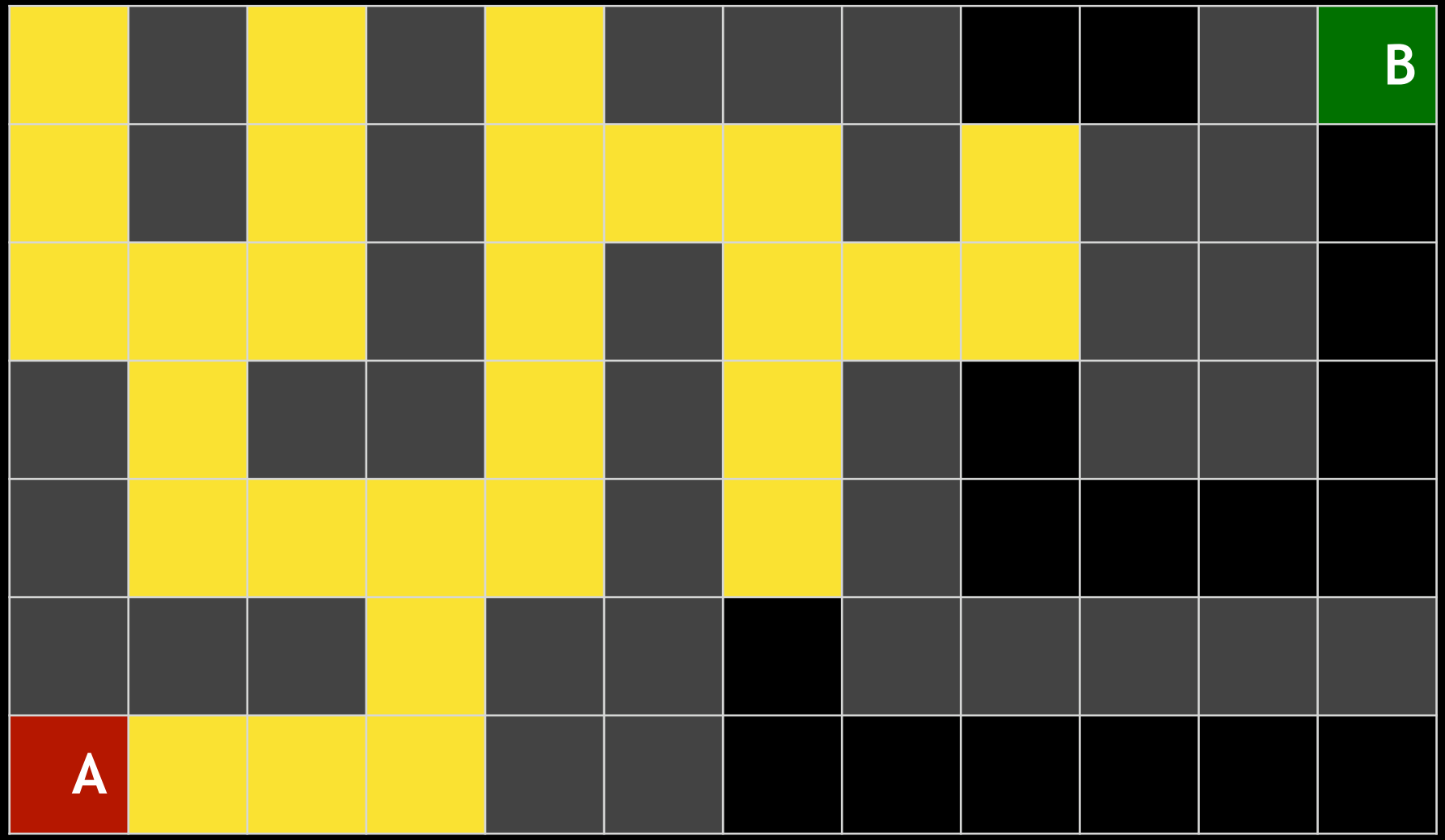

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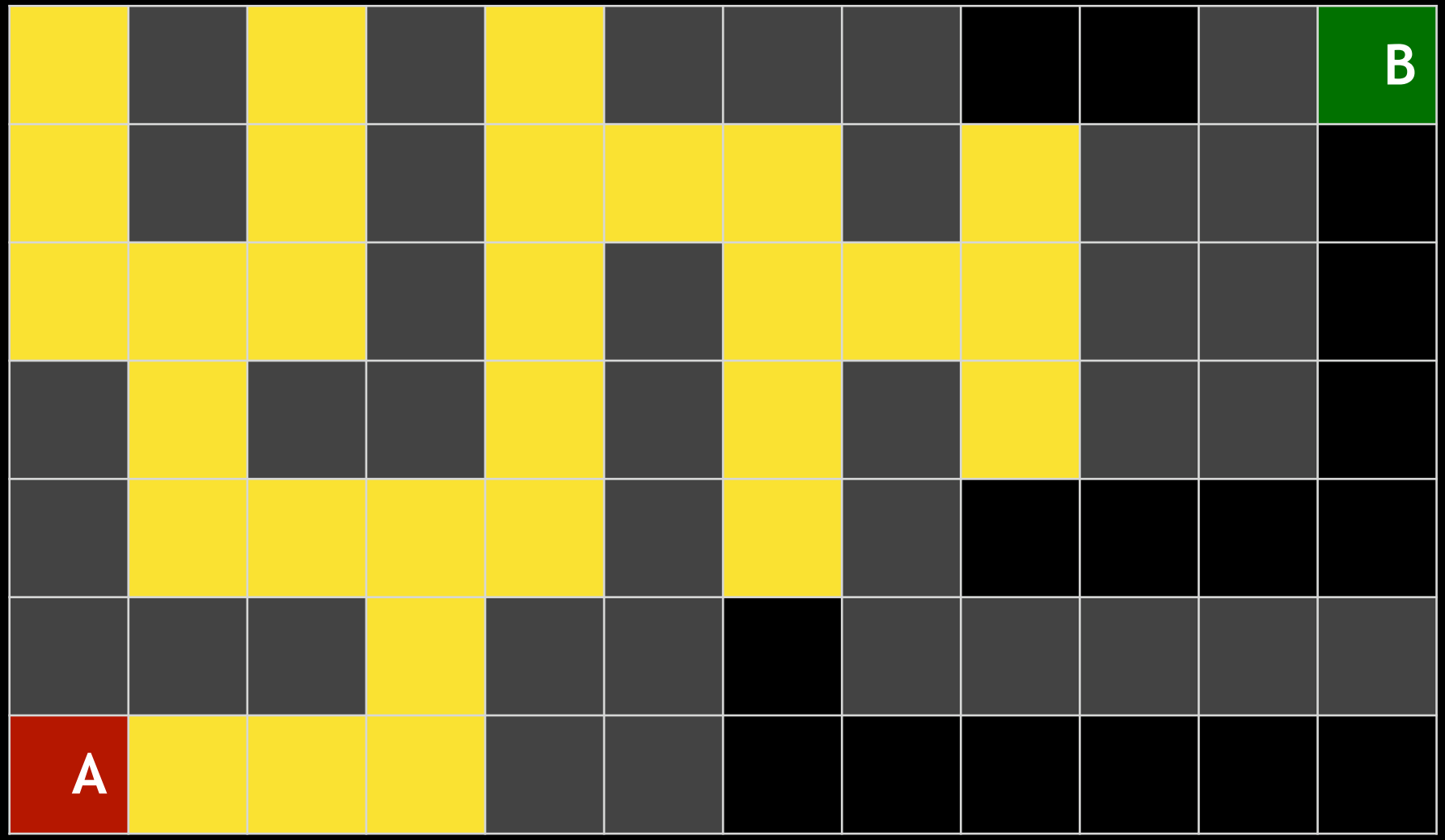

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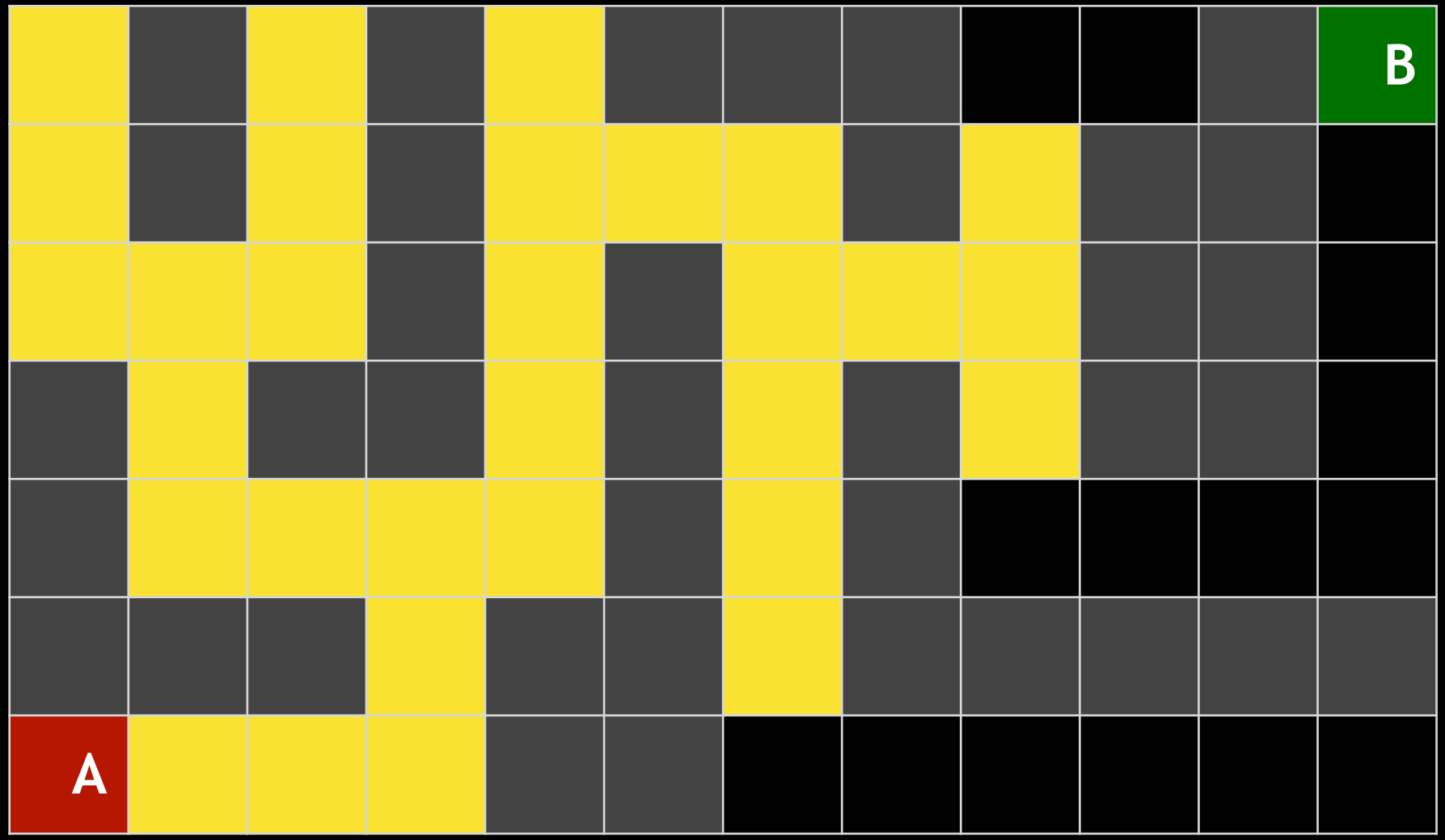

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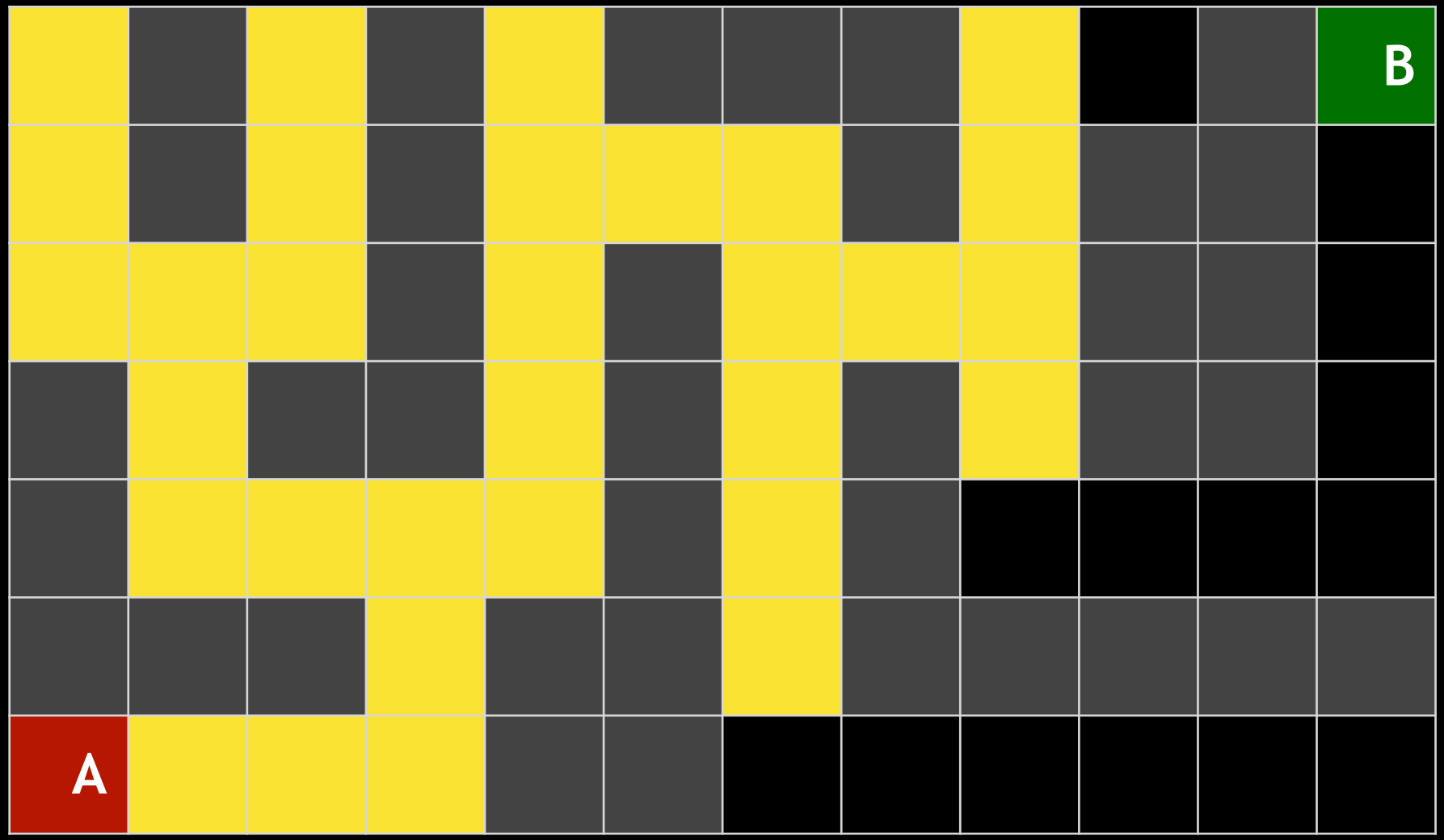

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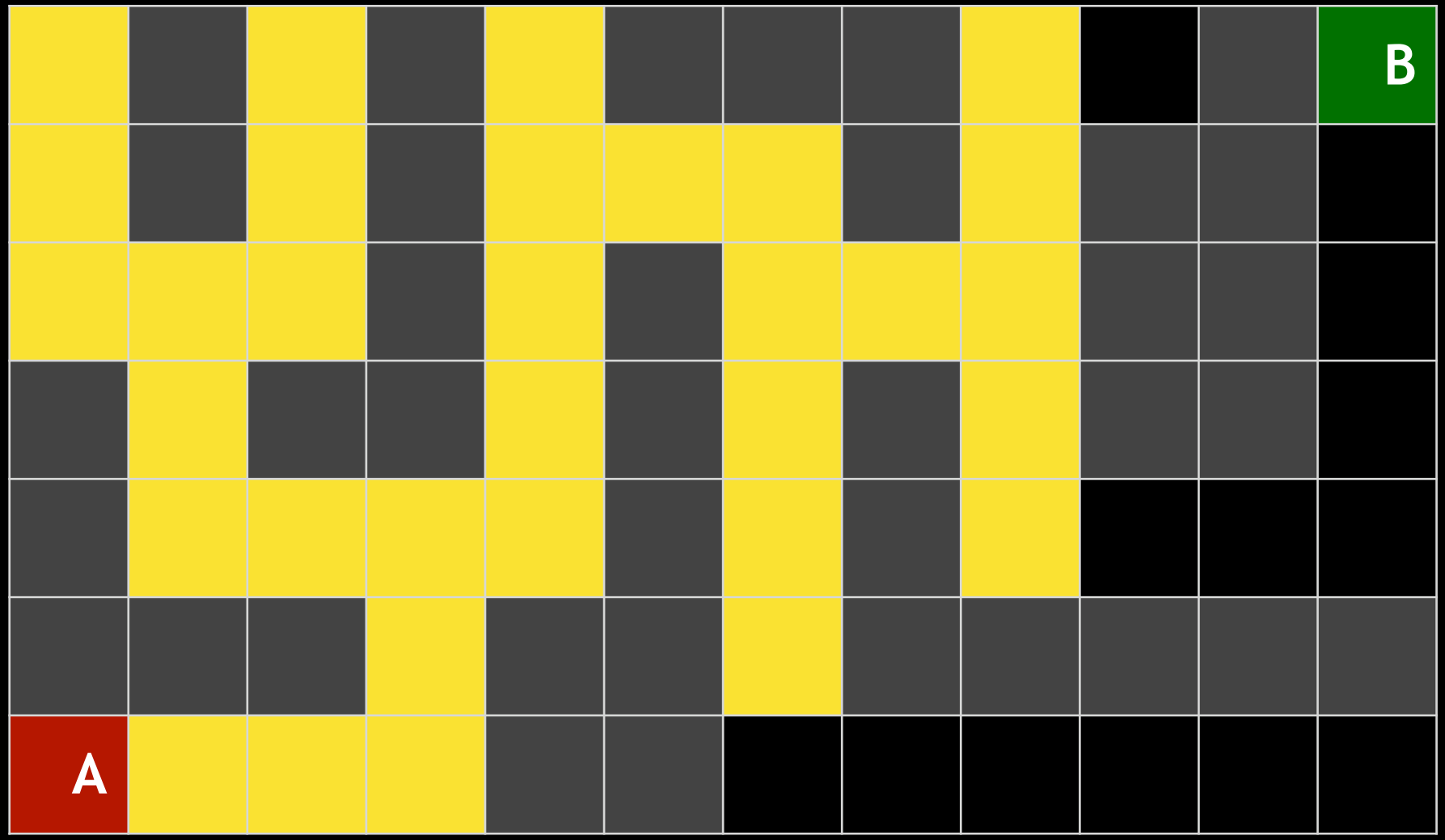

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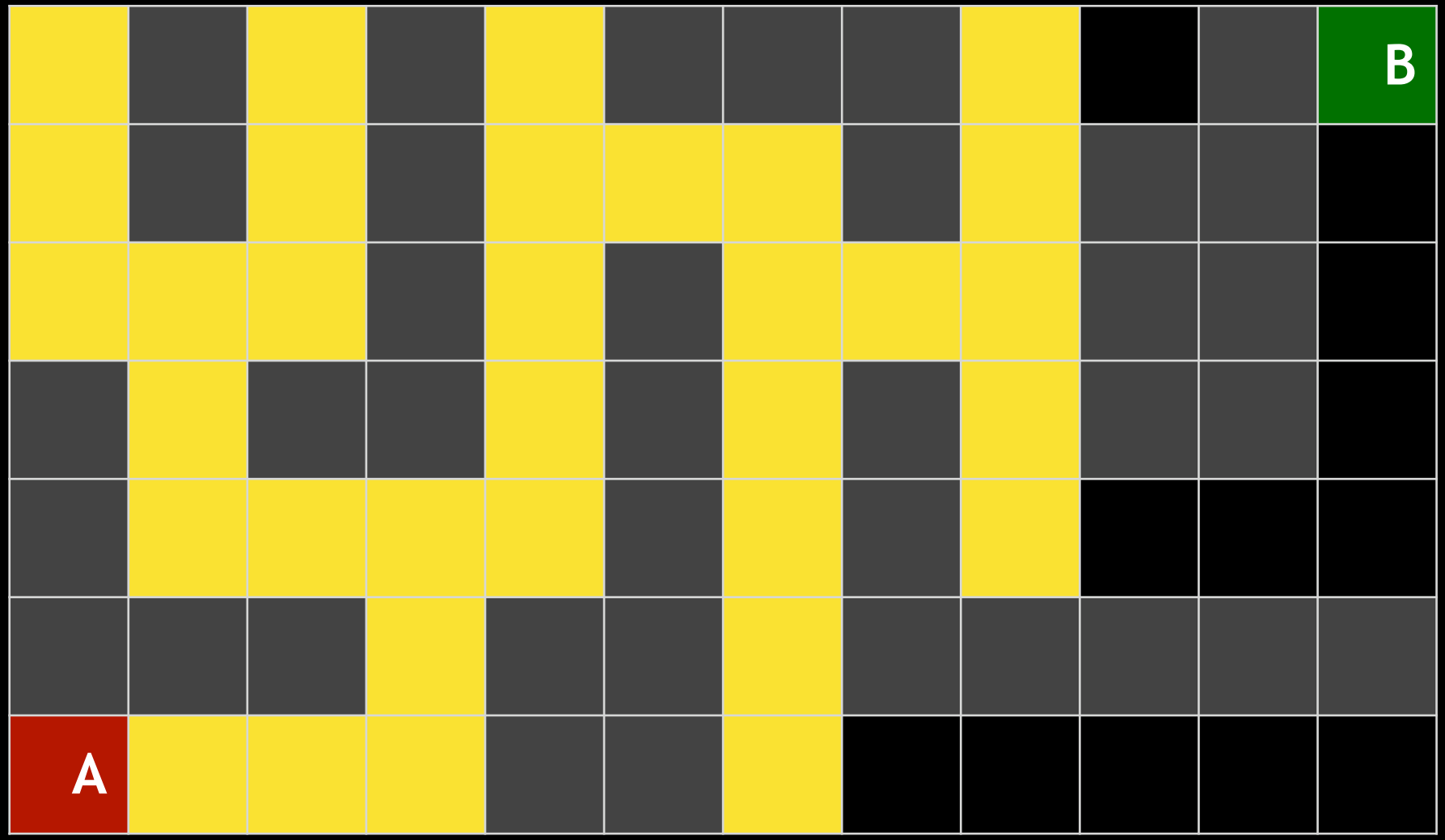

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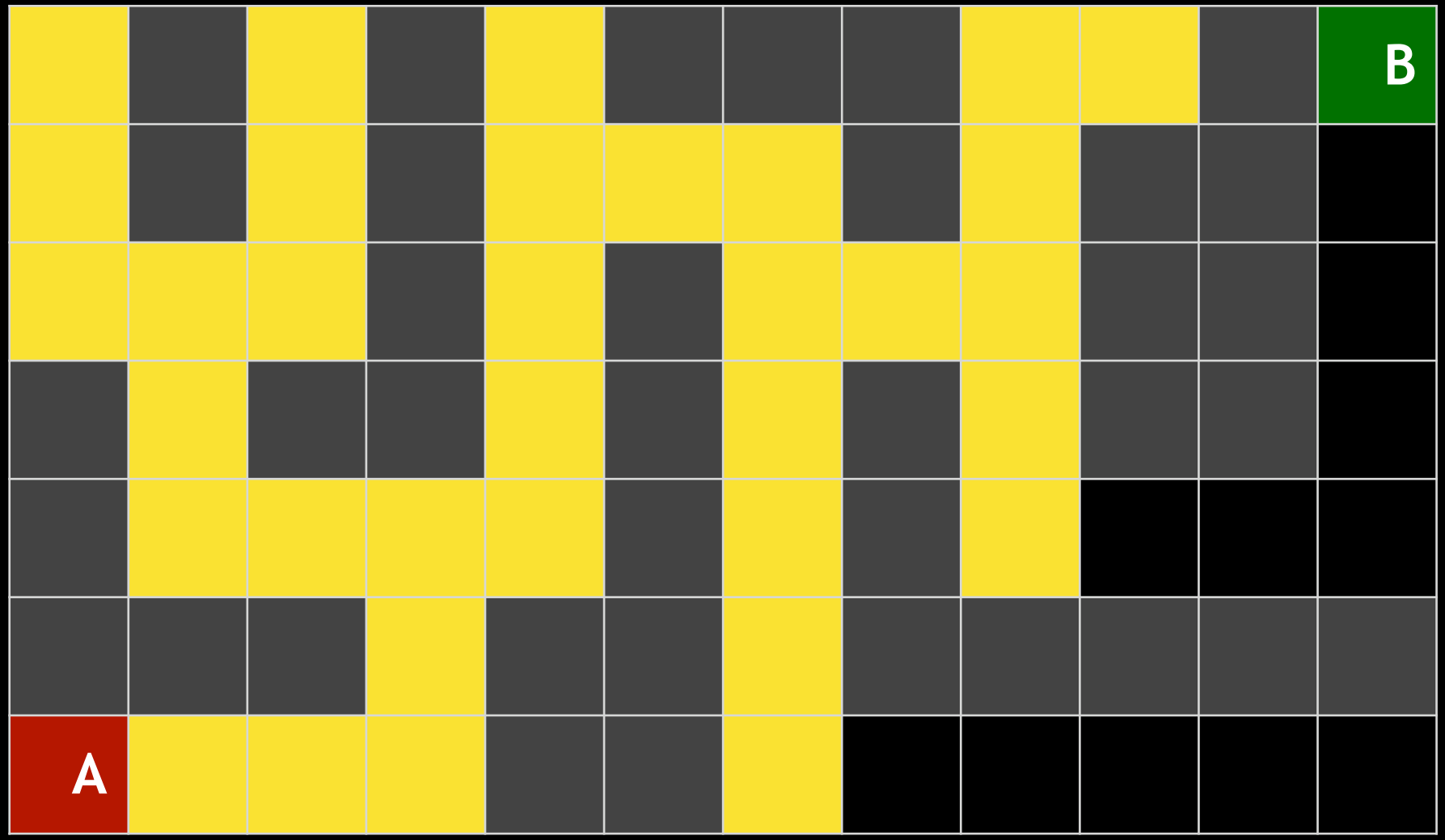

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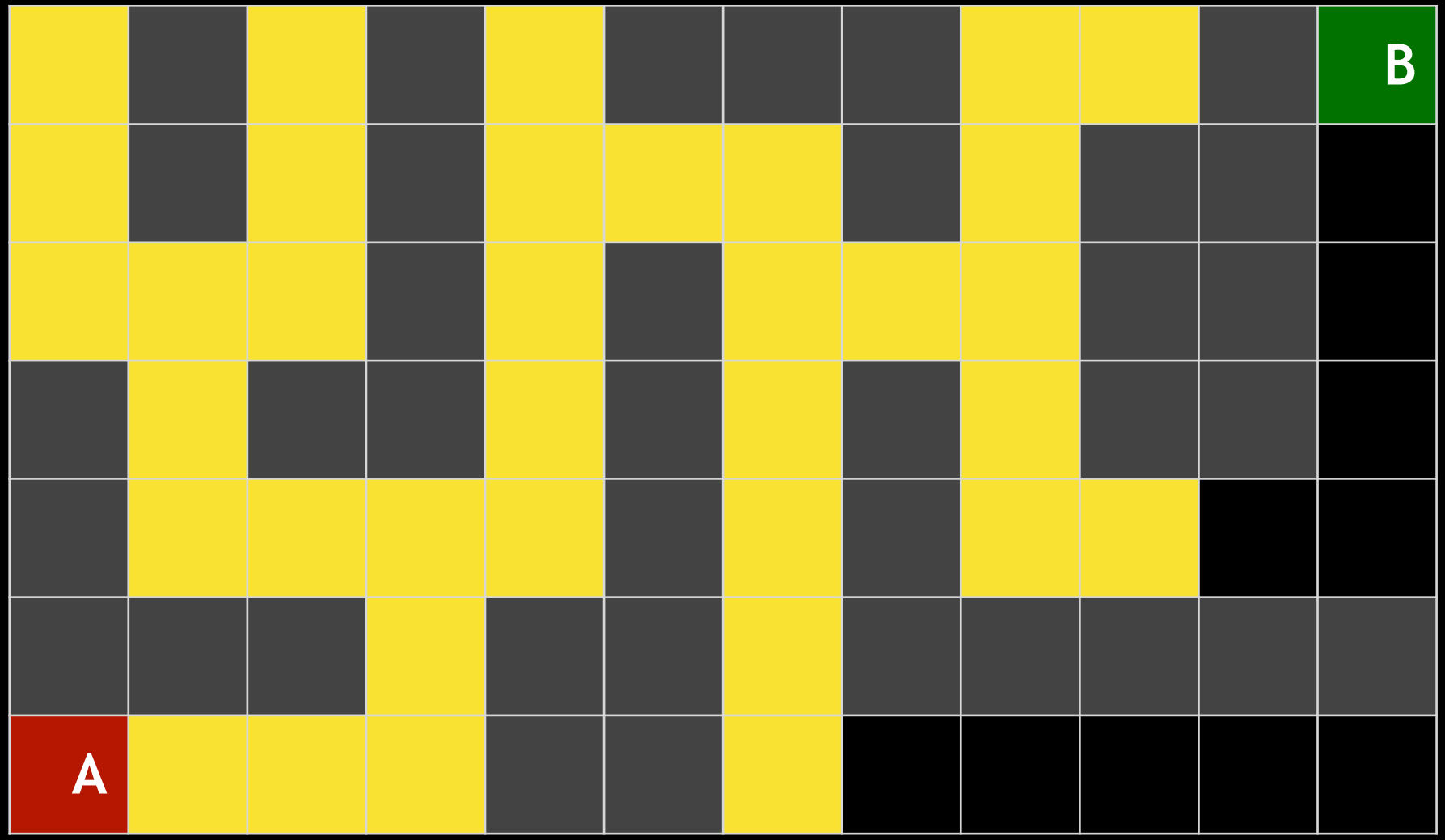

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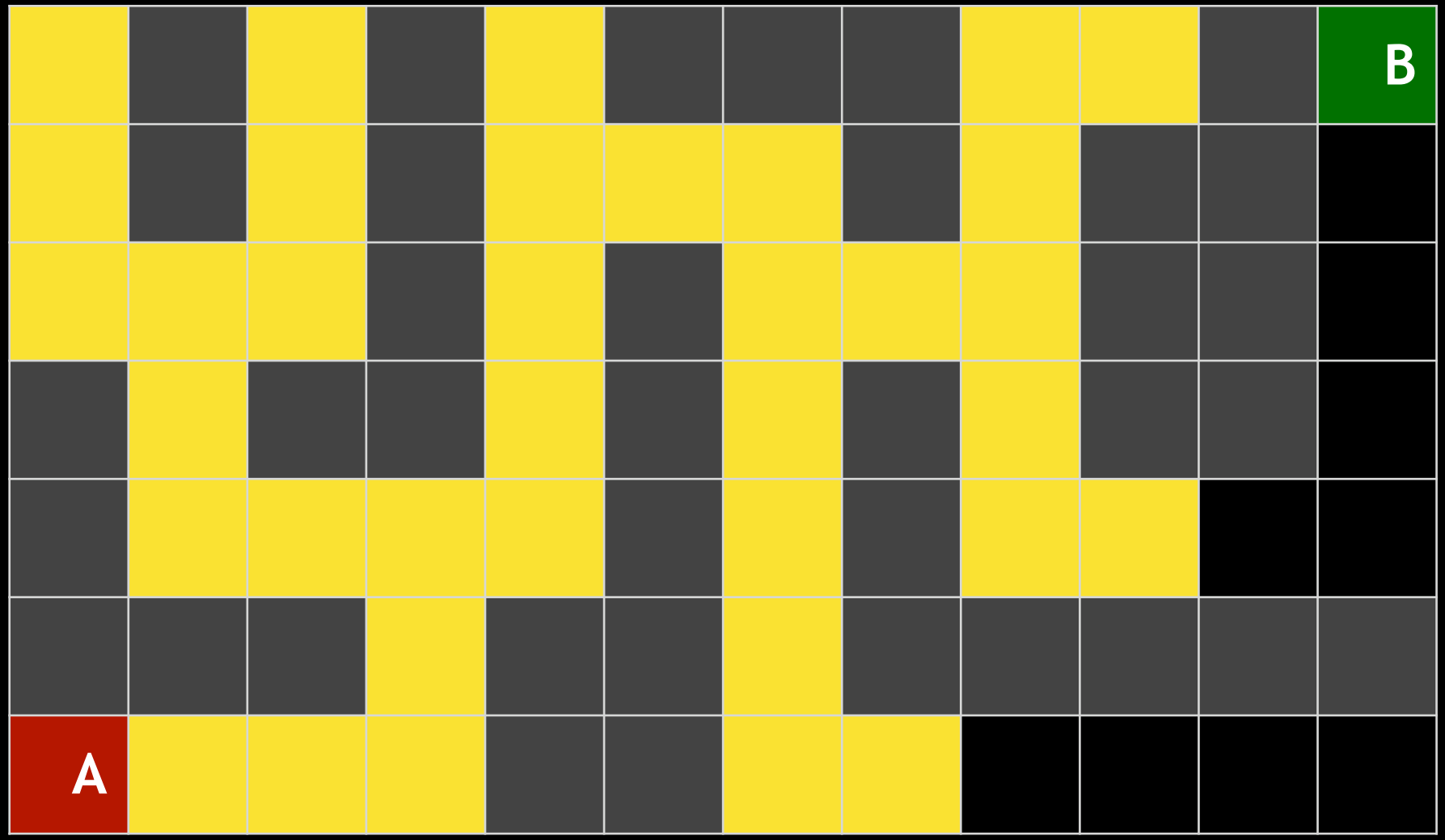

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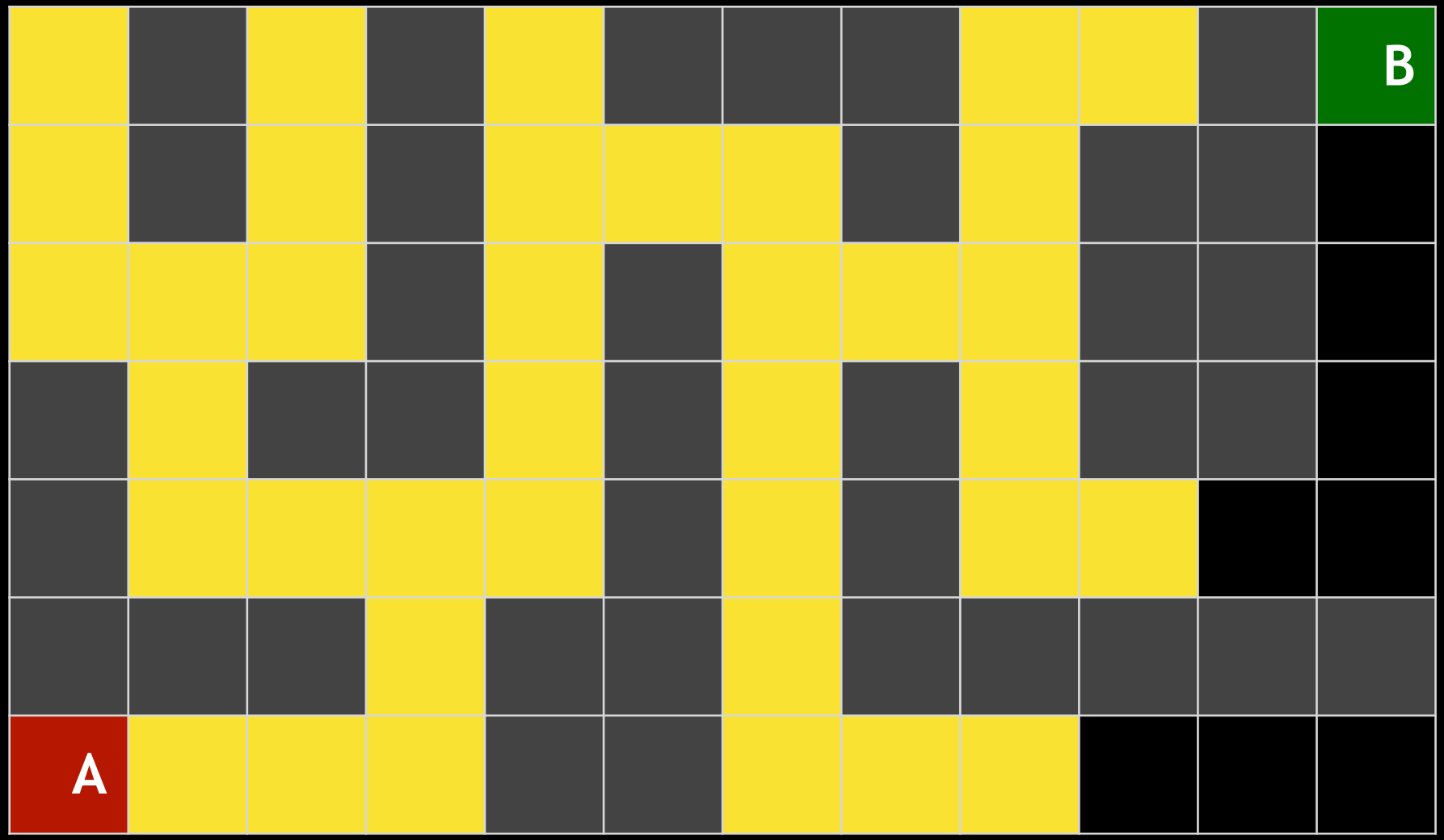

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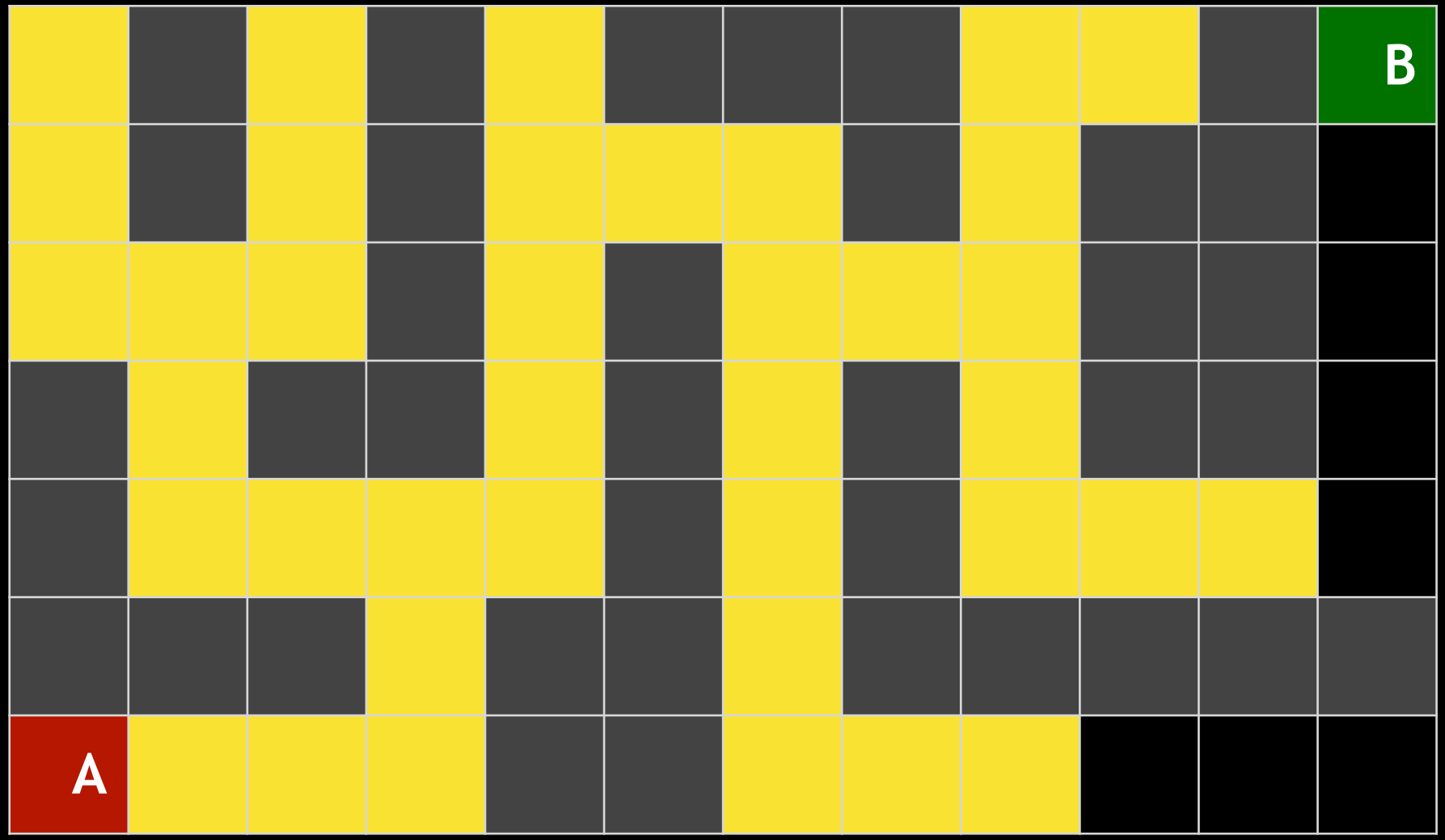

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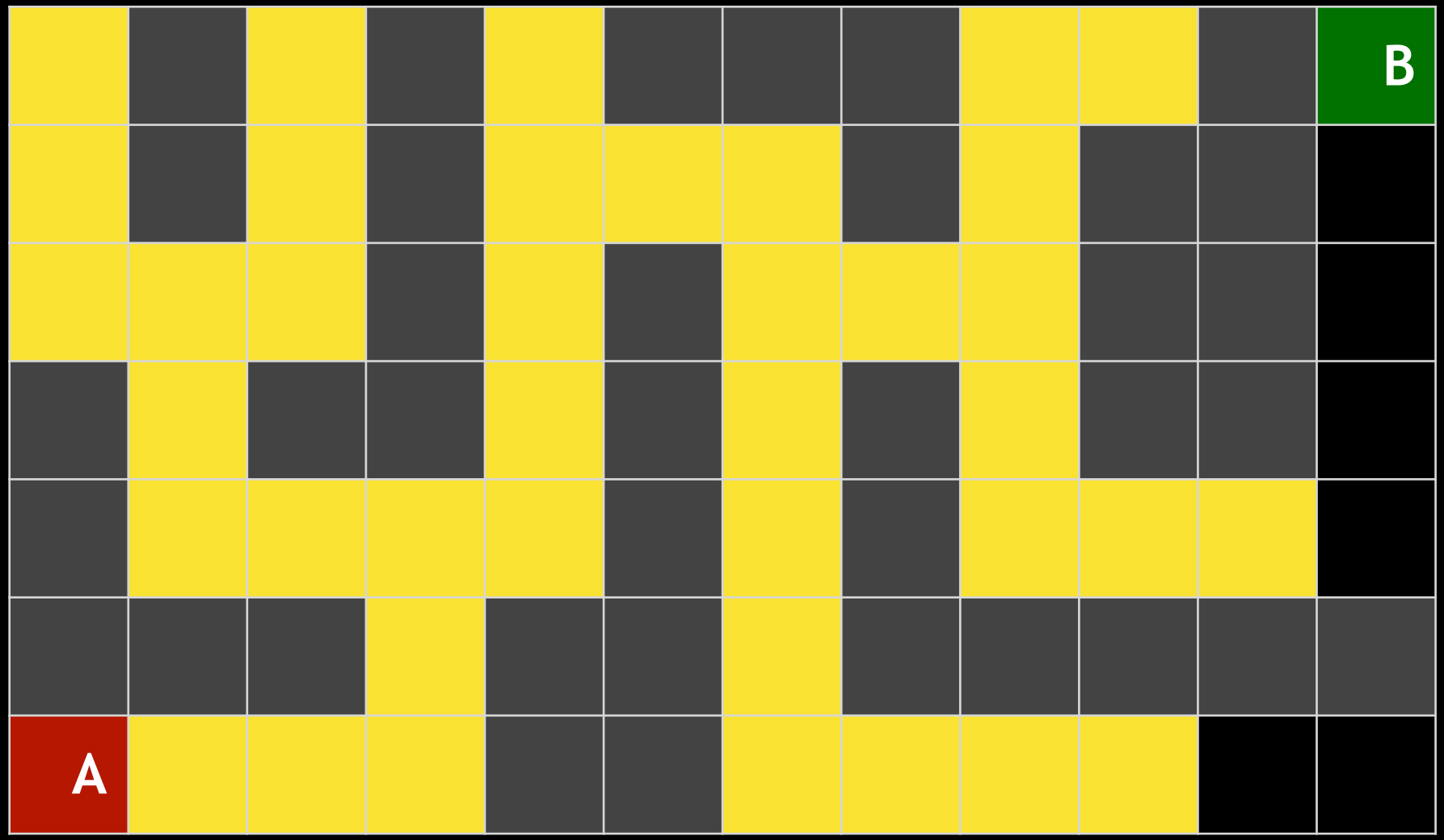

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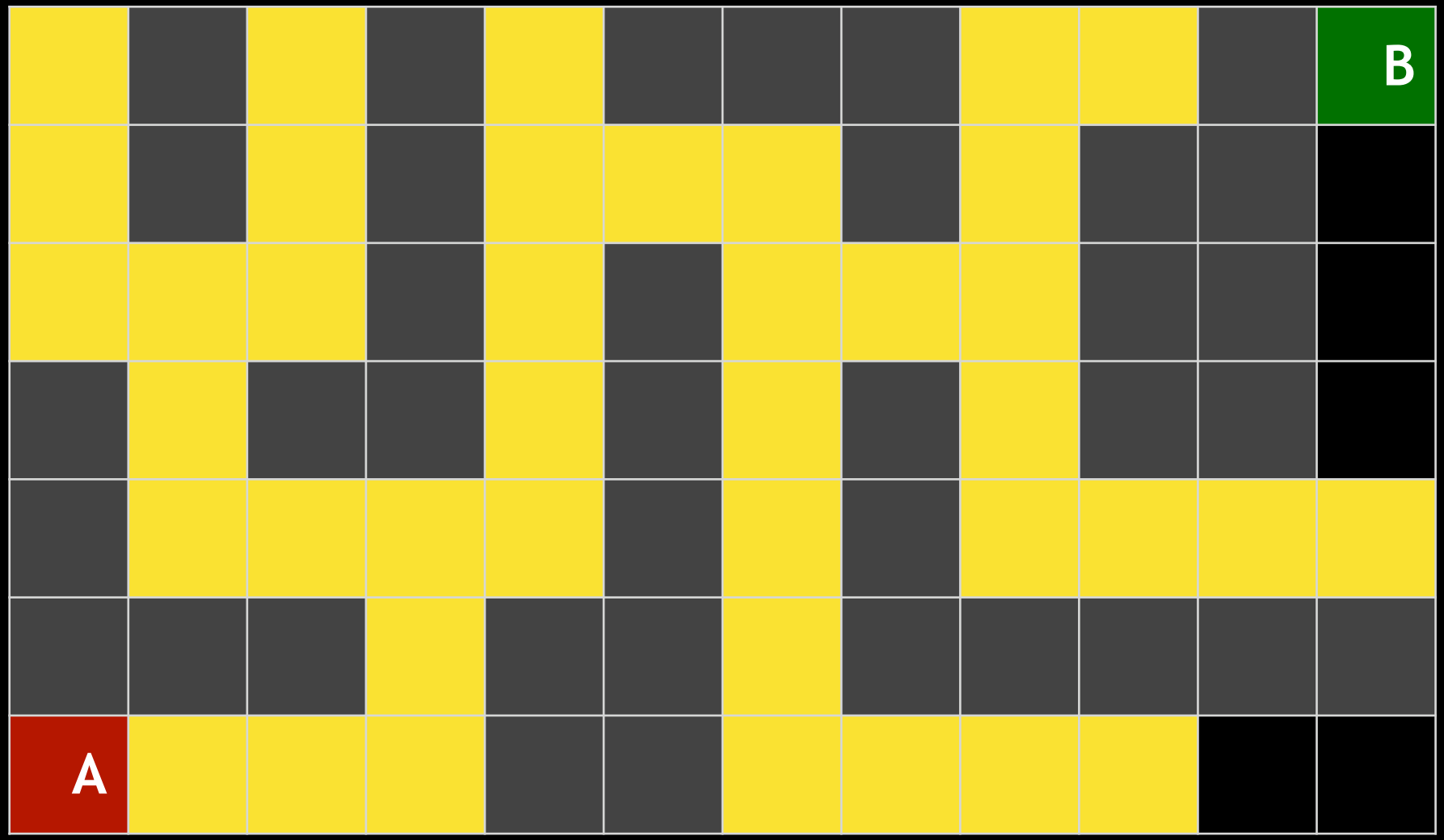

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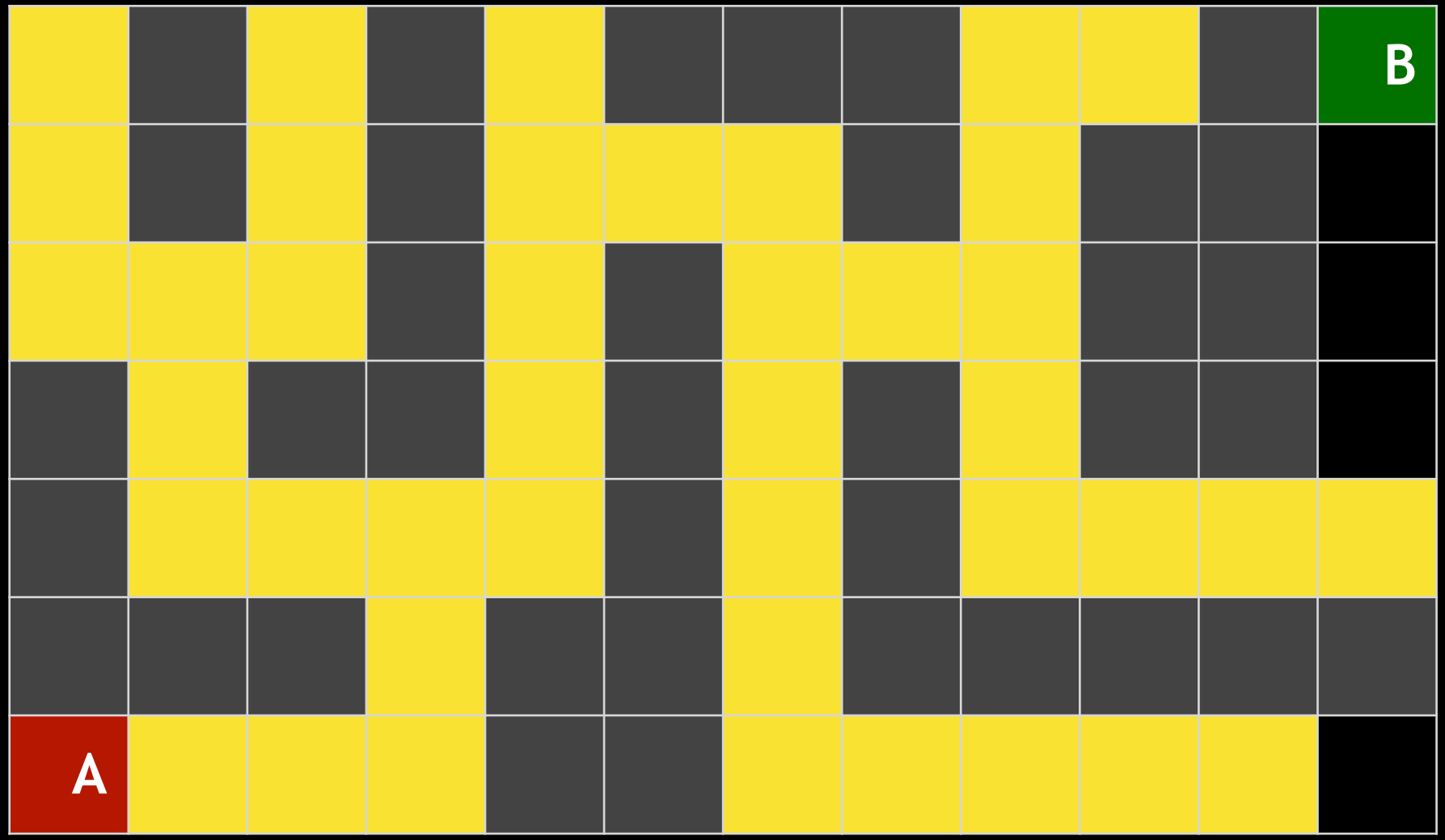

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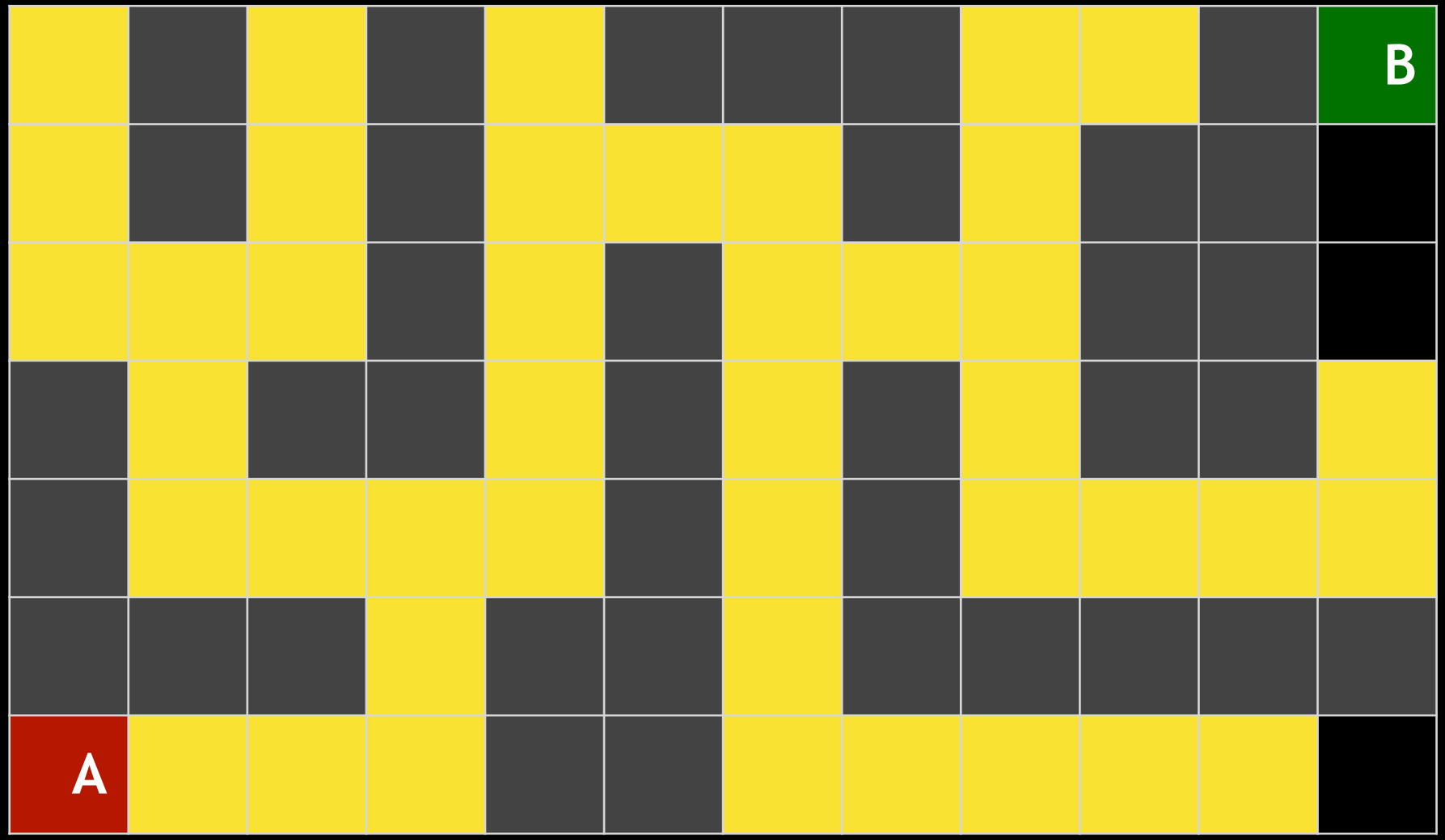

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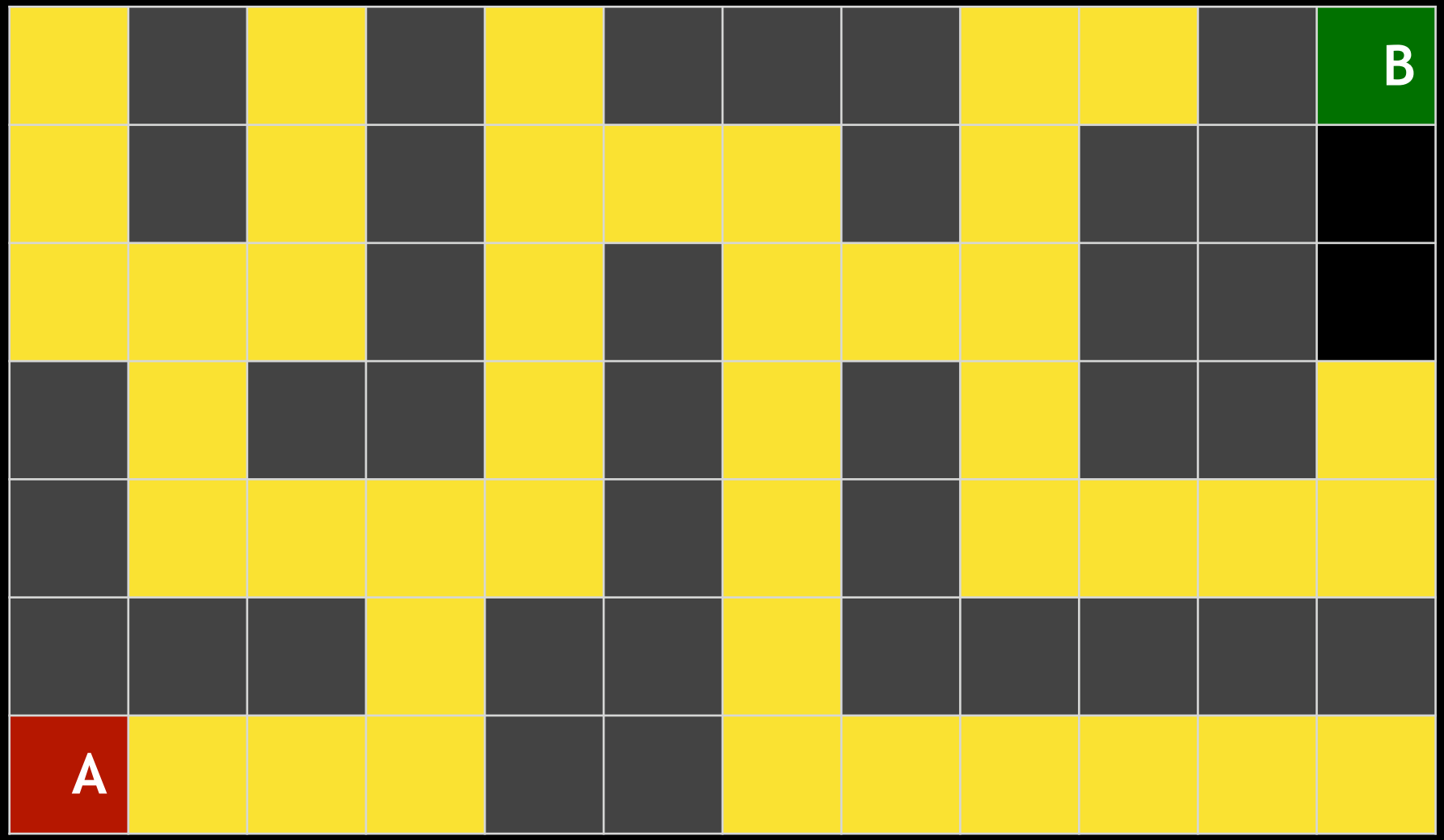

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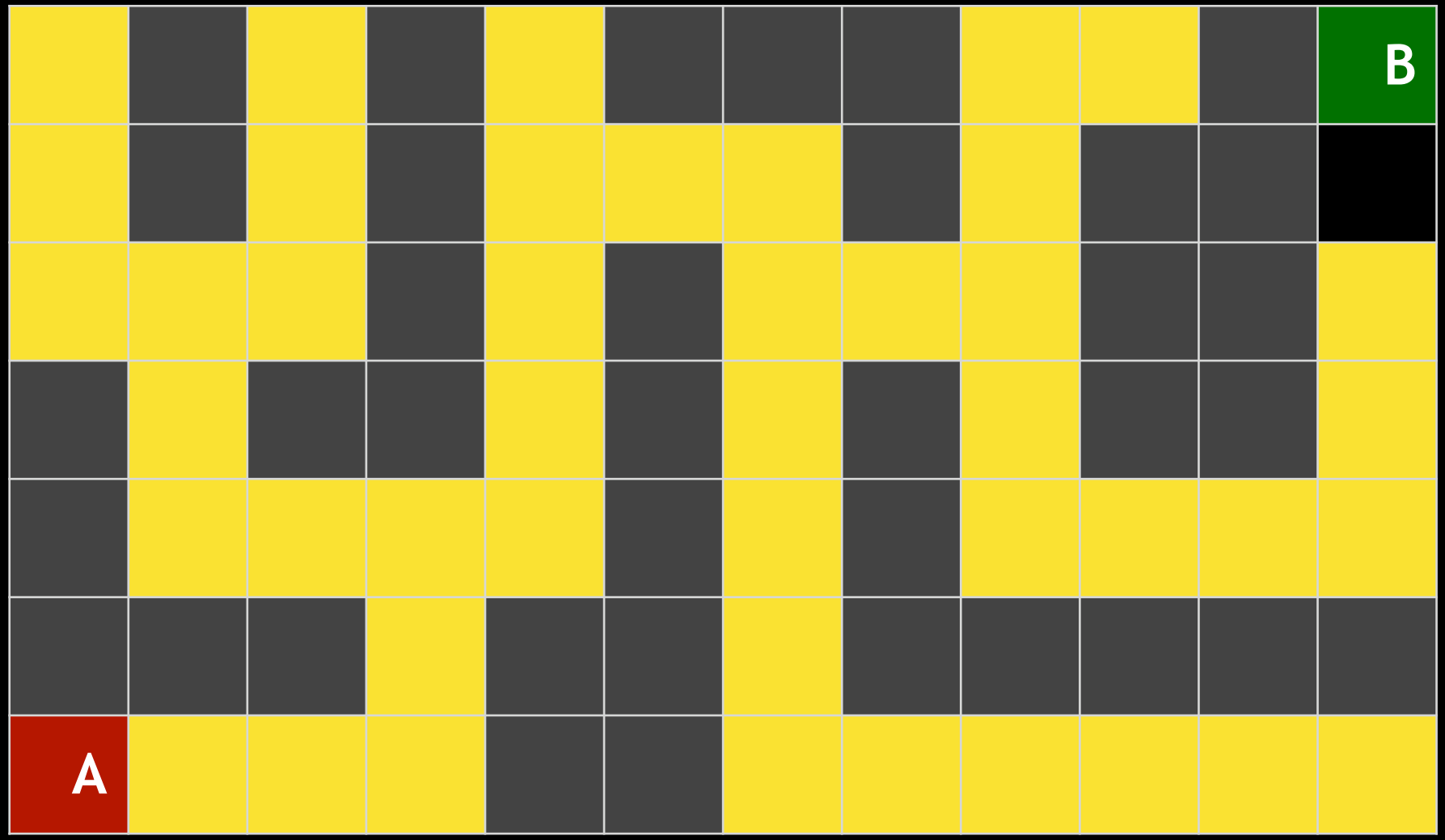

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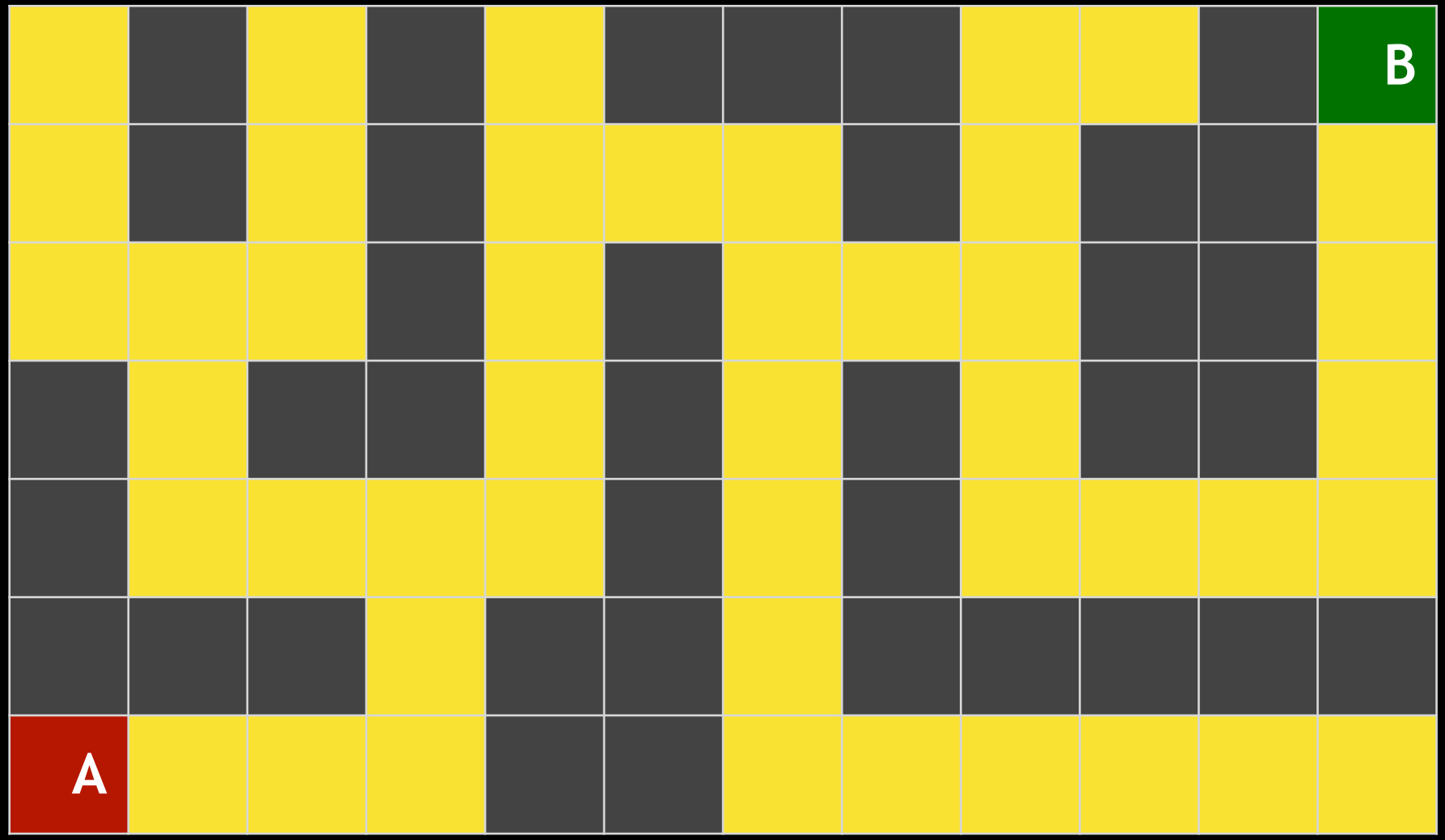

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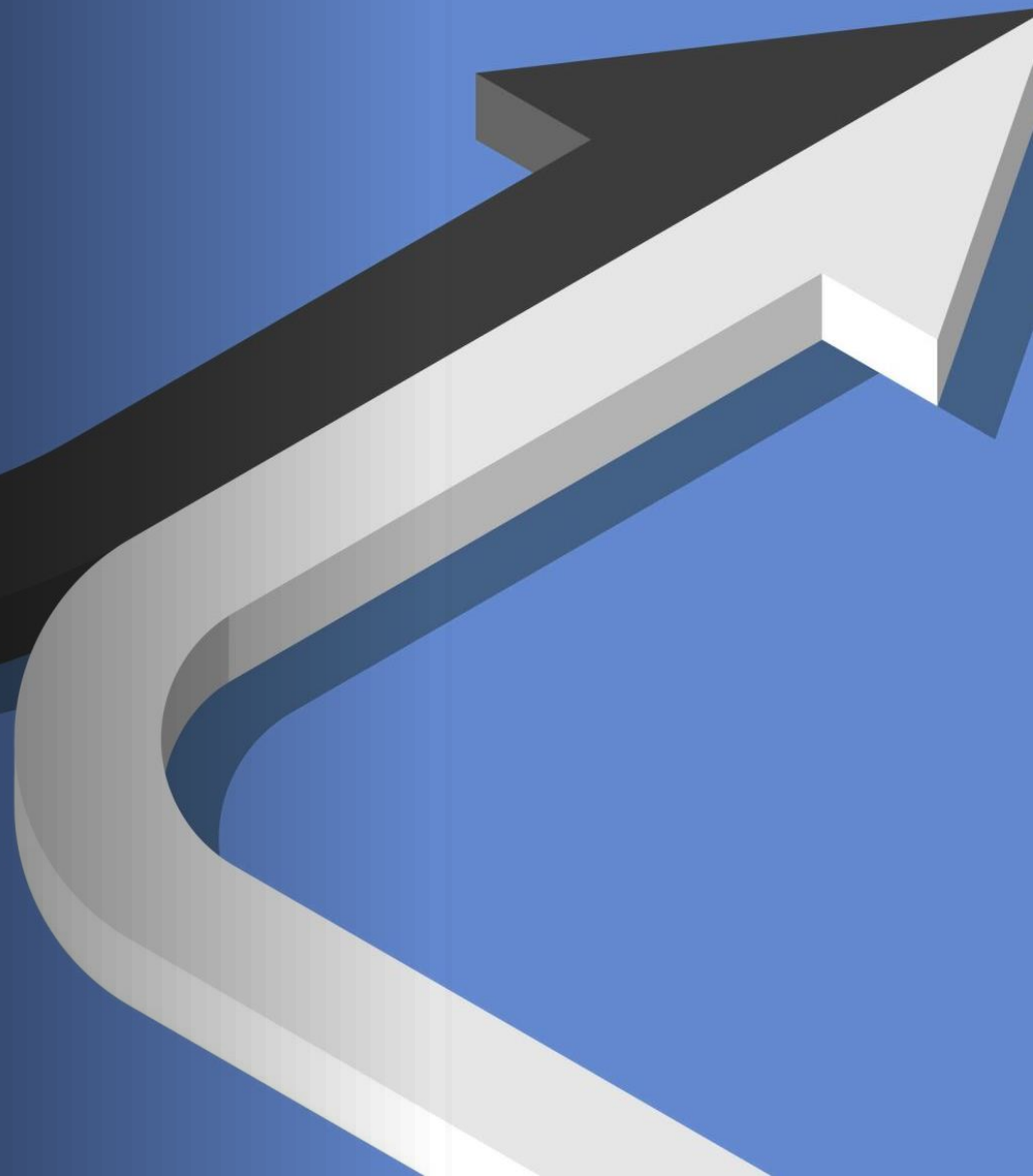
Me. Jean Nunes